edited by
ABBEY K. ELDER, SHANNON M. SMITH, & BRYAN RICUPERO



OPEN MOVEMENTS

Recognizing Challenges & Building Connections

Open Movements

Recognizing Challenges and Building Connections

EDITED BY
Abbey K. Elder,
Shannon M. Smith,
and Bryan Ricupero

Association of College and Research Libraries
A division of the American Library Association
Chicago, Illinois 2026

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Library of Congress Control Number on file.

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ISBNs
9798892553797
9798892553803

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Printed in the United States of America.

30 29 28 27 26 5 4 3 2 1

Foreword

Openness in academic research and scholarly communication has become an increasingly complex goal in the last decade. Or perhaps it's more accurate to say that openness has become an increasingly complex cluster of goals—not just the "open access" with which the movement toward openness began, but also open data, open science, open infrastructure, and more—and the pathways toward these goals have consequently become tangled. Even beyond this proliferation, it's become clear in recent years that "open" is not an unmitigated good: there are artifacts and conversations that for perfectly good reasons should not be thrown open to the world. Navigating all of these complexities requires ongoing conversation among the folks working toward these goals, as well as conversation with the communities affected by them.

My own path in advocating for openness in academic work has been… well, let's say "ad hoc." I first stumbled into it in late 2003 when the manuscript for my first book was declined by the press that had had it under review for ten months; two external reviewers had assessed it quite positively, but the press's board declined on economic grounds.* I was blogging quite prolifically at the time, and of course posted about this situation, to which my friend Matt Kirschenbaum responded, asking why it wasn't possible for me to take the manuscript and the readers' reports and put the whole thing online, where the project would almost certainly end up with a wider readership anyhow? Matt's question was rhetorical, to a significant extent—we all understood why that online project wouldn't be the same as a printed book with a publisher's logo on the spine—and yet the exchange got me thinking about what might make such a thing possible.

My own advocacy, then, for what was being discussed as "open access" was already shaped by open scholarly communities like the network of bloggers I was working with and was driven by thinking about the possibilities for open peer review within those communities. Openness was from this early moment not just about access but also about participation, about community self-determination, and about what the academy writ large would need in order to make the work being done within it not just open to read but also open to publish, open to build upon, and open to full engagement from communities both inside and outside the campus borders.

Since then, I've started several open scholarly community projects, including the largest and most significant thus far, Knowledge Commons. In doing so, I've been led to think a lot about the underlying requirements for such open communities, including a key question about open infrastructure: What would it mean for the goals of all of our open movements if we had trusted, reliable open platforms on which to build our work? But, of course, that question opens up another: What would make those platforms trusted and reliable?

Among other things, open infrastructure requires open, collective governance, enacted through robust cross-institutional and cross-community collaborations and investments. As Deb Chachra has recently argued, "Robust, reliable infrastructure doesn't just happen"; it requires collective care manifested through collective governance.[1] This is, of course, true not just of our technological infrastructure but also for all of the systems of knowledge projection, communication, evaluation, and reward that make up the university environment today—and especially at a moment such as ours when shared governance everywhere is under threat.

* First-time, unknown author plus oddly interdisciplinary approach equals "we don't know how to market this."

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Open, collective governance would allow us to negotiate what "as open as possible, as closed as necessary"[2] might mean for various kinds of research, providing appropriate protections for traditional knowledges, as Jane Anderson and Kim Christen have sought through the Local Contexts project,[3] as well ameliorating the trust and safety concerns that Tara Robertson has explored for the publications of marginalized and endangered communities.[4] Collective governance might also allow us to imagine new modes of quality assurance through new forms of peer review, forms that don't require black-boxed processes and rely on uncompensated labor, instead becoming a mode of mutual aid developed by and for communities of practice. And collective governance would allow us to be more confident that the platforms and services on which our various forms of "open" rely are guided by our communities' goals and values, and that we have the space to engage in the at times challenging process of decision-making about our projects' futures.

Volumes like this one are a crucial contribution to the processes of debate and decision-making on which collective governance must rely. It is my hope that, through exchanges like the ones taking place in these pages, we might find our way through the tangled thicket of ideas surrounding notions of openness in research and communication and think together about how the multiplicity of communities that we are part of, as well as the multiplicity of communities that we serve, might come together in imagining a common future that we all might share.

— Kathleen Fitzpatrick, Associate Dean for Research and Graduate Studies in the College of Arts & Letters and University Distinguished Professor of English, Michigan State University

Notes

1. Deb Chachra, *How Infrastructure Works: Inside the Systems That Shape Our World* (New York: Riverhead Books, 2023), 176.
2. Annalisa Landi, Mark Thompson, Viviana Giannuzzi, Fedele Bonifazi, Ignasi Labastida, Luiz Olavo Bonino Da Silva Santos, and Marco Roos, "The 'A' of FAIR – As Open as Possible, as Closed as Necessary," *Data Intelligence* 2, no. 1–2 (January 2020): 47–55, https://doi.org/10.1162/dint_a_00027.
3. Local Contexts, https://localcontexts.org.
4. Tara Robertson, "Not All Information Wants to Be Free: The Case Study of *On Our Backs*," in *Applying Library Values to Emerging Technology: Decision-Making in the Age of Open Access, Maker Spaces, and the Ever-Changing Library*, eds. Peter D. Fernandez and Kelly Tilton (Chicago: Association of College and Research Libraries, 2018).

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Introduction

Abbey K. Elder, Shannon M. Smith, and Bryan Ricupero

Background

At their core, open movements exhibit a set of shared values centering public access and broad representation. These values and the practices they inform work to prevent the concentration of power, especially as it relates to information and its monetization. Despite these shared values, general discussions about openness have broken off over the last few decades to bring closer attention to more focused topics like open science, open data, open access, and other movements within Open. This has enabled more nuanced discussions to arise, but has distanced some communities from others and, in some cases, it has even distanced communities from the values they purport to hold. As Yoon argues in her deconstruction of the genealogy of open, "A debate that had been about freedom, curtailing of private power, and the user was now contained into a conversation about economics, efficiency, and practicality."[1] Can we honestly argue that the communities in which we engage are working toward the public good, and not an abstract goal of "more open things?" To address this question, we brought together experts from across open movements to share information about the challenges and opportunities they have observed across Open.

Scope of the Volume

The title for this volume, "Open Movements: Recognizing Challenges and Building Connections," posits two things: first, that many open movements have unrecognized or underreported challenges, and second, that among movements, there are connections that remain to be built. These arguments can be seen in the earliest conversations between the editors. The impetus for the volume's development can be traced back to a conversation between editors Shannon Smith and Abbey Elder in 2020. This conversation, a casual discussion about the many intersecting lines between the open education and open access movements, became a recurring theme across their conversations over the years. It would eventually lead to the creation of this book and, more directly, to the penultimate chapter in the volume, "Open Access and Open Education: Missed Connections between Parallel Movements."

To unpack the terms you will encounter in this volume, when we talk about "open movements," we are referring to the larger efforts within (or tangential to) Open that focus on a particular type of information—e.g., data, education, software, etc. Within the volume, we distinguish between these larger social efforts and the institutional projects that are often

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used to push the movement forward. These latter projects we refer to as "open initiatives." The editors hope this distinction helps minimize confusion among readers and authors alike.

In order to capture the full range of topics that might fall under the umbrella of "open movements," we accepted chapters that discuss individual open movements, trends within a movement, or projects that support a specific movement. At the same time, chapters emphasizing the connections between open movements were prioritized and, whenever possible, authors were asked to comment on how the trends and topics addressed within their chapters might apply to open movements outside of their area(s) of focus.

The volume is broken into five parts based on themes, including:

I. Forging and Maintaining Open Communities
II. Exploring the Structures that Support Open
III. Improving Access Points for Open
IV. Navigating Choice and Engagement in Open
V. Reflecting on Our Past to Build an Open Future

Each section contains a mix of theoretical discussions, research reports, case studies, and personal reflections on topics relevant to both practitioners and those interested in learning about Open. The parts are described in more detail below.

Part I. Forging and Maintaining Open Communities

The first part of the volume includes chapters that highlight the importance of community in Open. This is a broad topic, with multiple perspectives on the word "community." Is community-building about developing internal relationships that support collaboration toward a common goal, or is it about bringing in new voices and external partners to catalyze change? How can open advocates support their local communities effectively? These questions and others are discussed throughout this section.

In the first chapter, Jhangiani discusses how his experience in the open education movement could be applied to community-building broadly in support of a shared commons. Next, the chapter by Markus, Hughes, Smith, Elder, and Ricupero takes a more practical approach, examining opportunities that those managing outreach for open initiatives can explore to build connections with the public. Next, the case study contributed by Birkenes, Carney-Nedelman, and Peña-Valencia describes the work that the YouthMappers network has done to connect people to information about their communities through open GIS projects on a global scale.

On a more granular level, the case study by Caizzi and Greer Klein follows the origin story of the open-source software Samvera and how it has evolved over time with support. With a focus on open metadata, BrodeFrank explores how community representation through crowdsourcing efforts can contribute to the development of more inclusive and discoverable collections. Finally, Hoye, Huntsman, and Tobiason's case study delves into questions about how "frequent flyers" are often utilized to support open initiatives to the detriment of the work itself and the communities they purport to serve.

Part II. Exploring the Structures that Support Open

In the second part of the volume, we move from discussions about communities to infrastructures, exploring the theories, practices, and organizational structures that undergird work supporting Open. Sustainable infrastructures and rules of engagement are necessary to support the long-term viability of our movements, and this section reflects the many ways in which these structures present themselves.

In the first chapter of this section, Cressman proposes a framework to encapsulate the many aspects of the open access movement that make it unique. Following Cressman's more theory-based chapter, Marvin's contribution to the volume brings an introspective lens, sharing her experiences with open-source software and how that background affects her understanding of other open movements like open access and open education. Santiago, Fleming, Hertz, and

Miskey's mini case studies review different pathways for supporting open initiatives, highlighting the pros and cons of the approaches that their institutions have chosen to implement.

Building on the practical foundations set by Santiago et al., the chapter by Dawson and McLean presents an overview of emerging collaborations between electronic resource and scholarly communication librarians to support open collections efforts in academic libraries. Next, Sasyk and Heaton examine the relationships that open-source software projects can have with commercial entities and the impact that vendors can have on an open-source project. Finally, Castillo, Key, Ralph, and Boock explore how one organizational approach to strategically aligning work with Open can advance support for open initiatives across an academic library.

Part III. Improving Access Points for Open

In response to the previous sections' focus on community engagement and infrastructures, part three asks the question: How do we make Open more transparent, approachable, and accessible? Chapters in this section confront the challenges that can arise when communicating within and across open movements and discuss potential opportunities for collaboration.

Dubinsky kicks off this section by analyzing the ways in which open access research is often discussed and highlights potential methods for advancing engagement with open access. Building on this metanarrative about discourses surrounding Open, Elder, Smith, Hemingway, and Ricupero explore the roadblocks that those working within open movements may face when discussing their work externally and the ways in which this discourse can affect one's perception of openness.

Next, Barlow's chapter reports on interviews conducted with individuals working across archives and special collections to examine their understanding of the concepts of open licensing and open education. Continuing the thread about issues with communication introduced in previous chapters, Baker explores how the use of the word "accessible" in the open access movement can be misleading and how digital accessibility is nonetheless vital for those advocating for Open. Finally, South, Barker, Hudnall, Feighery, and Scott's chapter outlines pragmatic strategies that open education professionals can leverage when working across regional campuses with varied resources.

Part IV. Navigating Choice and Engagement in Open

Openness is not a one-size-fits-all answer for every individual's needs. Some types of information might benefit from having more restrictive access. Part IV brings forward complications that can hinder the open sharing of information, causing rifts between advocates and those operating outside of Open. The chapters in this section explore the dynamic choices that must be made when considering whether to make a piece of content open.

The first chapter in this section, by Mannheimer and Bayrd, examines how research data is made open, what this means for research reproducibility, and considerations for what constitutes sensitive data where "opening" may not be possible. Next, Dickson and Davis reflect on their experience supporting practitioners who are debating whether to make their work open and the barriers that can complicate the sharing process. Expanding on Dickson and Davis' discussion of barriers and perception, Foasberg examines the language used to describe open access on publisher websites and discusses how these definitions can impact the perceptions that authors bring to open access work.

From a practical perspective, Kirschner, Kamat, and Exner's chapter studies the motivations that bring people into open movements, based on surveys and interviews held with faculty across Virginia Commonwealth University. Finally, Deschamps, Farnel, and Bigelow explore the unique challenges and benefits associated with open metadata, examining the systems, interoperabilities, and collaborations that are needed to make this possible.

Part V. Reflecting on Our Past to Build an Open Future

In the final section of the volume, we highlight chapters that reflect on current norms and practices in Open. These chapters range greatly in their area of focus, from infrastructure and practices to communications and education, but each brings forward unique ideas for how advocates can support a more sustainable open future.

Antwi, Anaya, Boadu, Castro, Credmades, Johns, Hutton, Nguema, and Wilkins kick off this section with a deep dive into open-source hardware (OSH), examining case studies, communities, and motivations for engaging in and supporting OSH. Following the discussion about open-source norms, Halperin explores Creative Commons as a tool for sharing open works and discusses how licenses can and cannot create an actual "commons" within our communities. From a more focused angle, Wojturska examines open access publishing services and their growth in UK libraries in support of diamond open access.

As the section moves from more specific examples to a broader discussion about how we can do more to engage others in understanding Open, Cross, Bonn, and Bolick make a case for educating future LIS practitioners about Open and discuss strategies for how that work might be accomplished. Next, Elder and Smith expand on the connections and unrealized opportunities that are present in the open access and open education movements. Finally, our conversation in this volume is concluded by Tekyl, who offers a thoughtful exploration of the larger movement of Open and its relationship with the technologies and practices within the evolving field of digital humanities.

Conclusion

Open is a continually evolving environment, especially given the constantly changing political priorities surrounding openness and open infrastructure. The circumstances in which this volume was first developed and where it finds itself now are necessarily different. This spans beyond discussions about what our movements are and into practical examples of current workflows and proposals for how we might more effectively collaborate for sustainable change. However, the values that undergird our systems will continue to be relevant well into the future. It is vital that practitioners continue to question why and how we support openness so we can engage more thoughtfully in work that supports not just broad access to information but also public empowerment.

Acknowledgments

There are several people we would like to thank for their support in the creation of this volume. First, we would like to thank Erin Nevius, without whom this project would still be a pitch in a Google Doc. Next, we express our thanks to our families and friends, many of whom served as readers and listeners as we refined our own chapters for publication. The editors would also like to thank the numerous colleagues across the years who supported our development as Open advocates and practitioners. Finally, we extend sincere gratitude to numerous peer reviewers who provided feedback on chapters and helped make the volume what it is today:

Stewart Baker, Martha Barker, Hilary Barlow, Venice Bayrd, Adele Birkenes, Josh Bolick, Maria Bonn, Michael Boock, Jessica BrodeFrank, Mélanie Brunet, Leigh-Ann Butler, Carolyn Caizzi, Colleen Cressman, Will Cross, Michael Cummings, Sabrina Davis, DeDe Dawson, Danielle Deschamps, Ellen Dubinsky, Sharon Farnel, Nancy Foasberg, Robert Heaton, Jordan Hemingway, Marla Hertz, Sherena Huntsman, Sarah Hutton, Rajiv Jhangiani, Abdoul Kafid Toko Koutogui, Jessica Kirschner, Heather Greer Klein, Amber Hoye, Cynthia Hughes, Sara Mannheimer, Helena Marvin, Jaclyn McLean, Zach Motts, Megan N. O'Donnell, Sierra Pandy, Jennie Rose Halperin, Ariana Santiago, Zorian Sasyk, Michelle Shannon, Beth South, Katharine Teykl, Abdulwasiu Tiamiyu, Anders Tobiason, Minglu Wang, and Rebecca Wojturska.

Notes

1. Betsy Yoon, "A Genealogy of Open," *In the Library with the Lead Pipe* (2023), https://www.inthelibrarywiththeleadpipe.org/2023/genealogy-of-open/.

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PART 1

FORGING AND MAINTAINING OPEN COMMUNITIES

CHAPTER 1

Building Open Community: Lessons from the Commons

Rajiv S. Jhangiani

Introduction

At its core, open education is concerned with democratizing education—that is, working to ensure that access to knowledge, and even the creation of knowledge, is not a privilege that is gatekept for the benefit of the few but instead a right that is accessible to all. One might argue that the philosophy of open education is reflected in Article 26 of the Universal Declaration of Human Rights, which asserts that "everyone has the right to education" and that "higher education shall be equally accessible to all on the basis of merit."[1] It is in service of this vision of the future that open educators labour every day to reimagine resources and practices, tools, and technologies and policies and systems. And it is in pursuit of this goal that we work in solidarity and why we intentionally cultivate community.

But what does it mean to build an Open community? What are the design goals? What is the necessary infrastructure? What is the architectural style? What is the social contract? And what sustains it? In this chapter, portions of which were presented during the opening keynote address at the 2024 Open Education Conference, I draw on lessons from years of community-building and community-strengthening in open education to critically consider how we might build an Open future that is designed for justice.

So much of what is a reality of the landscape of open education today—from specialized librarian positions dedicated to supporting OER discoverability to the development of undergraduate degree programs that are free from course material costs as a result of the systematic adoption of OER; from open education research mentorship networks to the growing embrace of participatory open pedagogies by educators; from the 2007 Cape Town Open Education Declaration[2] to the 2019 UNESCO Recommendation on Open Educational Resources[3]—all of this was but a dream twenty years ago. That this is all a reality today is the result of a lot of invisible labour. And while much of it may have been a labour of love, this doesn't lessen the amount of effort it has taken to bring so many visions to fruition. I refer to visions in the plural because, although this chapter focuses on community-building, it would be inaccurate to imply that open educators constitute a single, homogenous community. Instead, as with the different open movements, open education advocates and practitioners are better understood as an ever-expanding patchwork of diverse communities with common threads running through them.[4]

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In what follows, I share five lessons about community-building in open education, learned from working for nearly two decades in higher education with many dedicated people and organizations across the world—from the Open Education Network to the Global OER Graduate Network, from the Open Education Resources universitas to the Open University of Sri Lanka, from BCcampus to the Association for Learning Technology, from the Rebus Foundation to Virtually Connecting, and from Creative Commons to SPARC—who pour their love and their labour into advancing access to education that is equitable, affordable, and inclusive.

Lesson 1: Community is a Verb

My first lesson about building community in open education comes from a dear friend and colleague at Kwantlen Polytechnic University, Arley McNeney.[5] Arley had created a beautiful linocut print featuring a powerful quote by her friend, disabled activist and policy analyst Gabrielle Peters: "Make it accessible or burn it down."[6] To be clear, this slogan is not about being polite while you are being denied dignity. It is not about asking nicely in the hope that you just might be regarded as human. Rather, this rallying cry is about refusing inaccessible spaces, not tolerating half measures in accommodation, and crushing ableism.

Sales of merchandise bearing this motto raised more than $3,000, a sum that enabled Peters and McNeney to buy pizzas for every warming center in Vancouver on December 21, 2022, one of the coldest nights of that year. This wasn't charity, which is when one party with resources and power gives to another party with fewer resources and power, but rather mutual aid, in which people come together as equals to ensure everyone has the resources they need.

This wonderful story was picked up on the local news, and it was in the final line of that story that Peters, incisive as ever, was quoted again: "Community is a verb, not just a noun. It's the act of doing. It's being present and making things happen in whatever way you can."[7]

And there you have it: "community" is a verb. It isn't some mythical place nor a necessary outcome of a group of people coming together. It manifests through intention and action. And when community mobilizes in support of social justice, this relates to praxis, a concept that the Brazilian educator and philosopher Paolo Freire understood as reflection and action directed at the structures to be transformed.[8]

Arley, who sadly passed in March 2023, consistently engaged in praxis and practiced community. For example, her open textbook, *Business Writing for Everyone*,[9] involved collaboration with communicators from diverse backgrounds in order to meet the needs of a diverse student population. And among the acknowledgments, you will read the names of thirty students, almost all racialized, who reviewed early drafts and provided thoughtful feedback.

But perhaps my favourite example of Arley's praxis comes from when she practiced community in the classroom. She routinely empowered her applied communications students to choose their own research question, to pick something that mattered deeply to them. The occasion I am thinking of, however, was when two of her classes chose as their research question, "How faculty at Kwantlen Polytechnic University can reduce student barriers to textbook use." The students then engaged in collaborative research, including surveys of and interviews with other students, and wrote a report.[10] When they went on to recreate the report for different audiences, a group of three creative students elected to develop a board game about textbook affordability, based entirely on the report and the real lived experiences of their peers. The game included everything from decisions about which books they could afford to purchase or rent, or would have to do without, to dealing with random life events—positive or negative—that could influence their financial circumstances and their academic performance. Arley and the students then presented their research and facilitated tabletop play of the game at the Cascadia Open Education Summit held in Vancouver in April 2019.[11] Without a doubt, that student-led session was a highlight of that conference.

Shortly after that conference, Arley and I worked with the tech wizard and open education community builder Alan Levine to adapt the students' game into an interactive web format using open-source ed tech tools.[12] This game, which is openly licensed, is available to play

online and may even be used as a faculty development tool. It remains a shining and powerful example of the practice of community.

Lesson 2: Building in the Commons Doesn't Confer Ownership; It Involves Stewardship

An open educational resource is not just a gift to the world but also a gift that keeps on giving, becoming dearer each time it is regifted. As an OER creator, you derive great joy from the knowledge that you will only ever be aware of the thinnest slice of the positive impact in this world of the OER that you create. This is akin to the understanding, held by many societies around the world, that our purpose here is to plant trees under whose shade we shall never sit. To me, this understanding is part of what it means to be an open educator—to appreciate the richness of generosity. It also helps explain some of the tensions that have arisen when capitalists cosplay open education, engaging in what may be known as open washing or faux-pen.[13]

I still remember when I first heard the term "free riders" being used to describe people who reuse OER without contributing anything back to the commons. It was at the 2015 Open Education Conference in Vancouver, during a session led by one of the major sponsors of the event.[14] I recall feeling both confused and deeply uncomfortable with their characterization, looking around to see if I was the only one who felt like they had tumbled through the looking glass. It was especially confusing to me because, as an educator and OER creator, I want not only my students but also other educators to reuse my work. This is very much a feature, not a bug.

In my mind, there was a basic understanding that was missing here: when you build in the commons, by definition, you do not build for yourself. Building in the commons doesn't confer ownership; it involves stewardship. At the very least, this stewardship involves taking care to provide and support proper attribution (especially when revising or remixing OER), ensuring that you meaningfully adopt an open license by embracing technical openness (i.e., not publishing content in proprietary and/or non-editable formats that frustrate the effective use of the open license) and not enclosing or otherwise fettering the OER.[15]

In their important argument for OER stewardship, Lisa Petrides, Douglas Levin, and C. Edward Watson outline four practices in their CARE framework. In their words:

> *...what all good OER stewards should have in common is a commitment to practices that serve to demonstrate their duty of care to the broader OER movement:*
>
> *Attribute: OER stewards practice conspicuous attribution, ensuring that all who create or remix OER are properly and clearly credited for their contributions; and*
>
> *Release: OER stewards ensure OER can be released and used beyond the course and platform in which it was created or delivered; and*
>
> *Empower: OER stewards are inclusive and strive to meet the diverse needs of all learners, including by supporting the participation of new and non-traditional voices in OER creation and adoption.*[16]

When we create OER, the knowledge we share isn't a private commodity. I sometimes think of this as standing by a flowing river. If you are a venture capitalist, your instinct may be to create a dam and monetize the water by creating artificial scarcity and even drought. But as open educators, our purpose is to roll up our sleeves and clear some debris from the water, ensuring better and clearer access for others downstream. As open educators, it is our social contract to serve as stewards of bodies of knowledge to ensure that they flow, especially to those who have been historically denied access. In short, the issue of "free riders" isn't a problem if you believe in public infrastructure or the commons.

Cable Green, director of Open Knowledge at Creative Commons, touched on stewardship during his keynote address at the 2023 Open Education Global conference when he advocated that "equitable and socially just sharing of public goods should be at the core of any open knowledge model" and further that "publicly funded knowledge should be *stewarded* by public institutions, the Academy, and/or trusted nonprofit organizations [emphasis added]."[17] His point wasn't that one cannot work with commercial organizations but rather that these should be understood as "work for hire" and that the copyright to publicly funded knowledge should never be transferred to such groups. To go back to my river metaphor, a government may support recreational boating and even licensing a commercial ferrying service, but it should never make the mistake of selling the rights to the water itself. Why? Because one cannot fully reconcile a primary obligation to shareholders with a primary duty to the public.

This is also why the infrastructure of the commons needs to be open, to ensure that we are not erecting technical or cost barriers to accessing the water downstream. This is why most of us in open education choose to build with open-source tools that support interoperability and decentralization, such as WordPress (e.g., Pressbooks), Manifold, and Wikis (e.g., Libretext). This is not to suggest that using open-source tools is a panacea, as evidenced by the recent purchase of H5P by D2L, an educational technology company known for its proprietary learning management system. However, using open infrastructure does at least mitigate the risk of being at the mercy of the whims of those who seek to fetter, to control, and to exploit to reinforce and advance their own privilege.

Groups such as Invest in Open Infrastructure[18] are important to this movement because stewardship in open education doesn't involve fettering access to OER or jealously guarding one's strategic advantage. As the authors of the CARE framework explained, "What all good OER stewards should have in common is a commitment to practices that serve to demonstrate their duty of care to the broader OER movement."[19]

Lesson 3: When You Build in Open, Share Your Blueprints

One of the most beautiful and simultaneously one of the most challenging things about the last twenty years in open education is that we had to build the plane as we were flying it. This required vision, but also creativity, collaboration, and frankly even courage, because you were often the only faculty member in your department, the only librarian or educational technologist at your university, or the only program officer in your granting organization who saw and shared this vision. This work still isn't easy today, but over the past twenty years, the constraints have often been even greater.

And these constraints varied greatly. Take public funding as a case in point, which has helped advance open education in jurisdictions as wide-ranging as British Columbia,[20] New York,[21] Poland,[22] and South Africa.[23] In some cases, a government would say, "Here is a tiny amount of funding that you need to somehow stretch to achieve your goals, and if you manage to pull this off you just might get more funding next year." While in another case, a government would say, "Here is a massive amount of funding that you need to responsibly spend in the next eleven months (and if you don't manage to pull this off you shouldn't bother asking us for funding again)."

These kinds of constraints meant the stakes were high for many hardworking folks who were trying to build a model or develop a proof of concept, whether at a national, regional, or even an institutional level. For example, even as I worked with colleagues at Kwantlen Polytechnic University to build OER supports and Canada's first zero-textbook-cost degree programs, I felt a sense of responsibility to the broader community to really try to get it right and some pressure to not let people down, lest our failure were cited as evidence to deny resourcing OER initiatives at institutions with less-supportive senior administrators. And while

mistakes have undoubtedly been made along the way by pioneering practitioners, there have been many iterative improvements and lessons learned.

This is why I especially appreciate the efforts that so many community-builders have made along the way to openly share their thinking and their tools, their motivations and their methods—in other words, their blueprints—in a way that enables others to draw on their experiences and designs.

There are so many wonderful examples of community builders in open education sharing their blueprints, including a *OER: Field Guide for Academic Librarians,*[24] *The OER Starter Kit,*[25] a manual on *Marking Open and Affordable Courses,*[26] a guide on *Making Open Textbooks with Students,*[27] a *Research Methods Handbook,*[28] an *Open Pedagogy Student Toolkit,*[29] and an *Open Textbook Accessibility Toolkit,*[30] to name but a few. Each of these collaboratively developed resources generously shares hard-won expertise, insights, examples, and practical tips with a view to supporting the journey of new and future practitioners.

These projects demonstrate the power of collaborative community-building. Most of them also showcase the consistent efforts of librarians in helping our community to navigate or wayfind this space, even as they deal with academic hierarchies on one hand and vocational awe on the other.[31] But perhaps more than anything, the lesson from these important projects concerns the importance of ensuring that we make it easy for others to build using our designs.

Lesson 4: Democratize Space in the Commons

As we think about open blueprints, we should also reflect on our desired architectural style. After all, advancing open education isn't about building an exclusive gated community.

To borrow from concepts used in the world of open-source software, are we envisioning building something that resembles a cathedral or something that looks more like a bazaar?[32]

The world of legacy commercial textbook publishers is, it seems to me, the cathedral approach, one that is top-down, with a small, elite group of people involved in the design. It is designed to convey authority, to be pristine, and to resist change. It is where the unwashed masses are expected to go to receive enlightenment from the high priests who gatekeep knowledge. And more often than not, this model involves the collection of money while quietly assembling a fortune.

In my mind, many of these precepts are antithetical to the goals of open education, which have a lot more to do with democratizing both access to knowledge as well as access to knowledge creation. So instead of cathedrals, I imagine us welcoming people to engage in the delights of a beautiful, spicy bazaar, one that gives us the ingredients we need to serve our local communities and where we are free to set up stalls ourselves. Or maybe it even means building a community centre, with spaces where people can learn and grow but also just connect with one another. Among other things, this requires considering issues of access (e.g., are there barriers to entry, whether economic, technological, linguistic, or other?), accessibility (e.g., is the space proactively designed to support inclusion?), belonging (e.g., is the space culturally affirming?), safety (e.g., are there protections from harassment and violence?), and governance (e.g., is the space shaped and directed in a manner that reflects the values and priorities of the community it serves?).

Open educators have worked hard to democratize space in the commons. For example, grassroots initiatives such as the Open Faculty Patchbook,[33] Open Learner Patchbook,[34] Open Pedagogy Notebook,[35] and Scholarly Communications Notebook[36] invite educators to share their practices and insights, with an emphasis on diverse representation, inclusive participation, and practical inspiration. Although each of these open-access websites is a space where resources are shared, they are more than passive repositories of valuable content. Instead, similar to the community of practice model that is used to good effect with OER advocacy,[37] the core purpose of these initiatives is to welcome and bring together practitioners who share the values underpinning open education, to act as a hub for sharing and learning, and to reinforce and normalize open practices.

Lesson 5: Design for Justice

My final lesson is about never forgetting what and for whom we are building. As Erin Okuno wrote, "Equity isn't for all. Equity is for those farthest from justice, and if we are working towards true equity, those farthest from justice can define for themselves what they need to be whole, healthy, and in just relations with others."[38]

In open education, we can intentionally orient our work towards three forms of justice: 1) redistributive justice, which involves allocation of material or human resources towards those who by circumstance have less (e.g., replacing a commercial textbook with OER); 2) recognitive justice, which involves recognition and respect for cultural and gender difference (e.g., increasing socio-cultural diversity in the curriculum); and 3) representational justice, which involves equitable representation and political voice (e.g., co-construction of OER about learners of colour by learners of colour).[39]

As we think about these forms of social and epistemic justice, however, we should also be mindful of the difference between responses to injustice that are ameliorative and those that are truly transformative.[40] For example, reducing the cost of education by adopting open textbooks in place of commercial textbooks is a positive, ameliorative response. But transformative responses would do more to address the root causes of economic inequality, such as providing a stable power supply, adequate access to functional computing devices, and affordable and stable connectivity in rural environments. Similarly, translating OER into local languages can be a positive, ameliorative response, but a truly transformative response might include remixing OER critically to engage with and to challenge hegemonic perspectives.

This distinction also applies to broader open educational practices where, despite the best of intentions, the impact may be neutral or even negative,[41] such as if it is done without student agency or choice to opt out of public scholarship, if it reproduces hegemonic knowledge, and if students are not appropriately informed or equipped on how to engage in open practice and its risks, such as when marginalized students are targeted and harassed online. Open educational practices can also be ameliorative if they address economic injustice by producing resources and scholarship that become accessible to populations who would not otherwise be able to afford access. But practices like this can be either instructor-centred or learner-centred, they can be content-centric or process-centric, and primarily pedagogical or primarily social justice focused, and these are the lines along which we will see the difference between OEP that are ameliorative and those that are transformative. In the latter case, marginalized groups are involved in the design of the content and processes, and both epistemologies and structural academic gatekeeping may be challenged. This approach is perhaps best illustrated by two award-winning international open education programs, both of which also happen to focus on the United Nations' sustainable development goals (SDGs). The first, known as Open Education for a Better World (or OE4BW), pairs new OER developers with experienced volunteer mentors in order to build capacity in open education while producing concrete educational materials with the potential for high social impact,[42] whereas the second is a faculty fellowship program for post-secondary educators who work in small, cross-disciplinary groups to design learning assessments that empower their students to serve as agents of change within their own communities.[43] From the development of an online empowerment course that supports pathways for open education influencers (SDG4: Quality Education) to the design of a learning assessment in which students built a scientific-quality database of edible weeds on their campus communities (SDG2: Zero Hunger), projects associated with these programs tend to be learner-centered, process-centric, and social justice focused.

Final Reflections

As I reflect on the significant progress made within open education over the past two decades, I am filled not only with gratitude and pride but also with the strong desire to ensure that lessons learned along the way are remembered and applied across the other open movements.

At its core, open education aims to build a better world, one in which knowledge is—by default—created and shared in ways that are equitable, accessible, transparent, collaborative, and that benefit the common good. As we advance Open in its various forms, however, it is crucial that we do so in ways that remain true to its underlying values. Drawing on the lessons outlined in this chapter, this means we must

1. actively practice community by moving from critical reflection to critical action by mobilizing and providing mutual aid;
2. serve as stewards by not paying lip service to open education, but instead by actively embracing it with practices such as proper attribution, technical openness, unfettering, and inclusive authorship;
3. share our blueprints, to ensure that we make it easier for others to build and improve upon our earlier efforts, knowing also that the field cannot afford to repeat expensive mistakes;
4. democratize the commons, to ensure that we intentionally create inviting spaces where new and diverse practitioners may gather, share, learn, and collaborate because it is the people of this community that are its most cherished resource; and
5. design for justice, to ensure that we are not replicating existing societal hierarchies and that we are pushing to move from open practices that are ameliorative to those that are truly transformative.

I believe it is no coincidence that open education has flourished as the field has grown more diverse, with more traditionally marginalized voices and more lived experiences of exclusion helping shed light on how we can avoid perpetrating harm with the very best of intentions.[44] This growing diversity of perspectives has at times been mischaracterized as divisiveness, and some have even attempted to control it through tone policing.[45] But I will remind proponents of pragmatism that while slow, polite evolution has its merits, and sometimes revolution is the need of the hour.[46] After all, when you find yourself invited to a table that doesn't respect your dignity, it may be time to step away and build a new table.

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CHAPTER 2

The "Public" in Open: Can We Reach an Undefined Audience?

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Introduction: Public Access as Our Purpose

Look at any open movement and you'll see one piece in common: they want to advance access to knowledge. Access for whom, though? Everyone? The Open Knowledge Foundation[1] notes that one of the key defining features of open knowledge is its potential for "universal participation," stating that "everyone must be able to use, reuse and redistribute—there should be no discrimination against fields of endeavor or against persons or groups." However, while we can argue that a piece of content is available if it is free to read, openly licensed, and lacking high technical reuse barriers, it is hard to say that any piece of content is actually approachable to *all* its potential users. Often, the work of Open is driven by academic institutions, as it aligns well with their goals of advancing teaching, learning, and research. It follows that academics are among the more common advocates for open access. How can these advocates be expected to provide "universal" or "public" access to knowledge if they are not engaging with the wider public in ways that build understanding?

Individuals who casually engage with open content often do so through platforms that market their open collections as "free." This is good marketing, but it prevents the public from further learning about the broader open movements that support access to this content. For example, on the popular social media website Tumblr, users extol the virtues of the digital library JSTOR, with thousands of posts sharing articles they've found on the platform and spreading the word about how others can get free access to research. As one user shared, "I've just realised that you can access JSTOR for free without being a university student, actually, and can feel myself standing on the precipice of something AWESOME."[2] Do these users know that, apart from their 100 free articles a month, a selection of JSTOR's collections is actually open access (OA) and can be freely read and used at any time? Probably not all of them. Does it matter that they don't know? Maybe. While you can argue that not knowing what "open access" means doesn't impair a user's ability to appreciate JSTOR's collections, what you can't ignore is that the positive outcomes of OA are being laid at JSTOR's feet and not attributed to the broad support available for OA today. Furthermore, by not understanding that JSTOR's work is part of a wider initiative taken up by many institutions and organizations globally, users can't effectively locate other open collections that might be available through their local library, the Digital Public Library of America, or any of the other repositories collect materials on all sorts of topics.

CC BY SA

This is only one example, but it showcases how marketing can impact individuals' awareness of openness: if the existence of open content is not emphasized directly, you wouldn't know that the information you're engaging with is open at all. And this *is* a problem! Not knowing that open movements exist limits the public's ability to benefit from the resources, services, and communities therein. We will explore this topic in more depth later on in the chapter, but first, let's explore how the disconnect between advocates for open movements based in academia and the wider public occurs.

Public Connections That We Lack

For those of us who participate in advocacy for open movements as part of our jobs in academia, it's easy to say, "We're spreading the word about Open in our communities!" But taking a step back from that perspective, it's important to consider who "our community" is and whether everyone in that space is accounted for in our outreach and advocacy work. Is "our community" the researchers at academic institutions who engage in open research? Or is it the broader societal groups that make up our cities, provinces, and countries? In this section, we'll discuss who we mean when we say "the public" and review a few major barriers to public engagement with open knowledge, as well as how those barriers are built.

We Don't Always "Speak to" Our Audience

Advocates for Open at academic institutions don't always connect with the audiences we want to, either in the public at large or in specific communities. There are multiple layers to this phenomenon that rest on assumptions about what the public wants and what members of the public know about open content. It might help to focus on what we mean by "the public" in the first place. If we mean "society at large," is that a metric we can meet? Or perhaps there is another direction from which we can approach this problem. After all, "public" is not just a catchall term for "society." It is also a term we use to refer to those funds, institutions, places, and infrastructures that exist for the public good.[3] Public libraries, public works, public parks, public schools, and museums; each of these examples is kept up by the government in order to support a specific community. If we are considering how best to reach a public audience, then using a public-serving institution as our partner might be a good idea. However, managing outreach to each of these groups will require a different communication strategy. When academics want to reach "the public" as a whole, we tend to promote our work ineffectively by watering down our message and ignoring the actual interests of each subgroup represented within our communities.

Language and Messaging

We must make more of a conscious effort to use language that is approachable for the audience(s) we want to reach. When speaking to an academic audience that is interested in open science, the tone presented in this chapter would be fine, but the same information might not be approachable for a high school student without additional context-setting. One of the easiest ways to make Open more approachable is to use broad, easy-to-understand language on forward-facing materials, with more detailed information available nearby. When taking this approach, it's vital to directly connect your general messaging to a more advanced description so users can actually benefit from the additional detail you provide. The Metropolitan Museum of Art (The Met) is a good example: one of the topmost pages on their website is a page simply called "The Met Collection," which explains how to access their free online materials (see figure 1). If you want to learn more about those free collections, you can find, linked on the same page, "Open Access at the Met," where information about OA and its importance is provided.[4] However, you don't have to know what OA means or click on that page to learn about The Met's collections being open. You can also learn this in their collections directly, thanks to an Open Access filter with a tooltip description that states: "As part of the Met's Open Access policy, you can freely copy, modify and distribute this image, even for commercial purposes."[5]

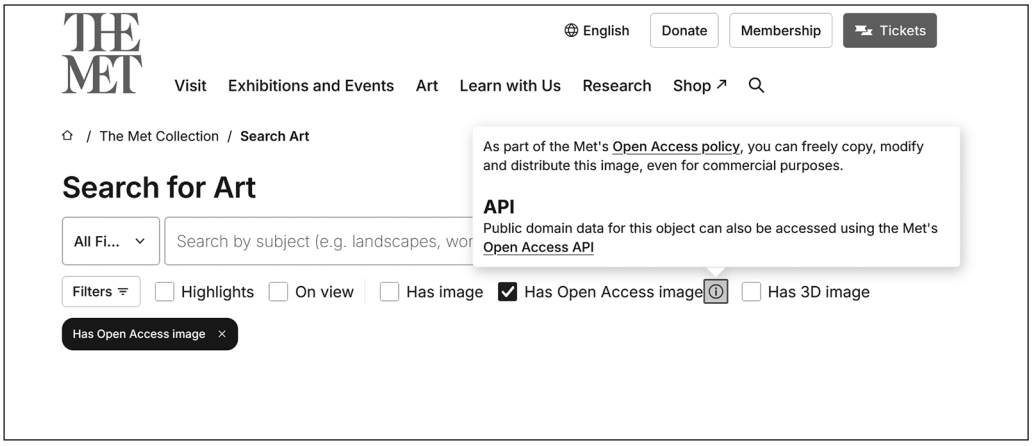


Figure 1. The Met’s Collection search highlights and explains its Open Access materials. Screenshot December 16, 2024, from https://www.metmuseum.org/art/collection/search?showOnly=openAccess.

It’s notable that both of these communication strategies—having a full informational webpage and providing context where users are searching—are being used by The Met, not just one or the other. Only saying that there is “free to access artwork” would obfuscate the impact of making the materials open, particularly the permissions that are available for sharing and reusing The Met’s content online. Alternatively, only providing a detailed description of the Met’s OA policy on a webpage that no one might visit could severely limit who has access to that context. Utilizing both methods—and providing information on high-traffic pages in particular—serves to bring in new users and help them understand what they’re getting access to and how.

Relationship Building

How we speak is only the first barrier to communication. Who we share with also impacts our reach. Local museums, small libraries, state libraries, and extension offices are all examples of organizations where a connection to the public can be made easily. As we try to share information with new audiences, we should consider partnering with community-embedded groups to reach them. An email announcing a new open collection to public schools will not have the same impact on its recipients coming from a university librarian as it would coming from someone situated within the same space; without that connecting point, it could seem random. Having the email come from local leaders (for example, a school librarian or 4-H coordinator) could help the email hold more meaning for the recipients. Moreover, those local leaders may know additional places to share open collections or ways to better promote certain kinds of collections to their specific audience(s).

Discoverability

Finally, an important question to ask when we’re considering why our messaging about Open doesn’t reach our intended audience is: Can anyone find open content once they know about it? We must consider both where we place our open collections and how we promote them. While adding open collections to an institution’s list of databases might make them more visible for some people, there are several other places that should be considered if we want to encourage more people to use these materials. Many extension offices and library consortia maintain online spaces to share resources with their communities.

For example, Indiana University (IU) offers the CRE Open Resource Hub,[6] a resource created by the IU Center for Rural Engagement that curates access to a list of open local, state, and national content tailored to the interests of public users. Through its Discovery Cafe,[7] the CRE

connects with local public libraries to share resources on topics of interest. The CRE Open Resource Hub website includes free selected resources and provides links to wide-ranging content from nutrition to history to technology to music. The materials that IU shares and makes accessible are intended to help librarians and citizens find resources that will promote learning for all ages. Using spaces like this to highlight open collections can boost discoverability beyond the typical academic audience. By leveraging IU Bloomington's celebrated programs in the arts and humanities, the CRE is building vitality in the downtowns and schools of the rural communities that it serves and cultivating a new set of partners on campus.[8]

To Support our Communities, We Need to be Proactive!

Sometimes, the open content we want to share is not intended for easy use by different audiences. For example, while a lot of wonderful open educational resources (OER) exist for colleges and universities* and even for K-12 classes,[9] fewer materials are designed for use by non-profits, public libraries, or other public-facing groups.[10] To build engagement with open content, advocates need to help staff at these establishments—especially those representing rural and underserved communities—by working with them to identify how open materials can be relevant for their individual community's needs.

Sharing with these groups should be a reciprocal effort. To support our communities, our sharing process needs to be more than hitting send on a press release and then patting ourselves on the back. Our goal ought to be building connections within our communities. We want members of the public to know what's available to them, how it can be used, and how it can meet their individual needs. Michelle Reed describes this process of public engagement as "connecting the commons," and gives several examples of how advocates in the open education space could proactively support and reach out to public libraries in their communities, from supporting homeschool programs to providing professional development courses or coding training for local interest groups (see figure 2).[11]



Figure 2. Social graphic from Library Futures, linked at the bottom of the editorial from Reed above, highlighting outreach about the cost of accessing knowledge.

* The Iowa Colleges & Universities Group in OER Commons is one of several state-level initiatives collating and sharing OER for higher ed institutions. "Iowa Colleges and Universities Group Resources," OER Commons, n.d., accessed January 18, 2025, https://oercommons.org/groups/iowa-colleges-and-universities/6023/.

Another major barrier the public faces when trying to access content is when a collection contains a mix of open and closed content. Such occurrences can be incredibly confusing. *You have to sign in for this, you have to pay for that, but those are open!* This was a particularly difficult conversation to have in 2021, after many collections that had been "opened" temporarily during the beginning of the COVID-19 pandemic were suddenly paywalled for many users. As one study on library collections after COVID-19 describes this issue, "…it would not necessarily be apparent to the user that the content they were accessing was only temporarily available, and may once again be paywalled should they return to it later."[12] In some cases, there is a simple and unavoidable reason why different materials within a set might all have varying levels of access: shifting public domain cutoff dates, for example.[13] However, that reason only makes sense to those of us embedded in work with information (e.g., information technology, libraries, publishing, etc.). Because of this, it's important for information professionals to explain why different levels of access are presented to the users across their collections.

One strategy that Open advocates could implement is providing in-house training for public libraries to discuss potential use cases for open resources in their communities. Any such training would need to focus on the needs of that particular audience and not necessarily the intended purpose of the original creator(s). Many training opportunities geared toward an academic audience tend to focus on how a collection was developed rather than how it can be used. Training geared toward other audiences should speak to what the collections contain and provide tips on how to search or use the collections in different ways. For example, the University of Tennessee, Knoxville, has led several Open Access Workshops for Non-Profits[14] to help local groups learn how to locate OA articles and data that they can use for reports and grant proposals, such as the employment and housing data available online through the US Bureau of Labor Statistics.

In addition to training organizations that are positioned to share open materials with the public, we also need to make those materials easier for a general audience to find. This could be done by improving the metadata and linking of our content, spreading awareness outside of academia, and investing in collections that are relevant to local or regional communities. For example, archival collections about local history might be more interesting to people within our communities, as long as we design them with their actual interests in mind. The assumption that open content that is locally situated is interesting for our communities hinges on the idea that we understand the nuances of every person in our various regions and that we know how to get in touch with them. Rather than starting from this assumption, we should work with those in our communities to assess their needs and ensure that new open collections reflect the interests of both scholars and independent users.

We Need to Recognize Academia's Shortcomings

Academia is rooted in a history of exploitation. This means that even when the public is aware of the resources being produced and shared by academic institutions for open use, there are good reasons why the individuals in our communities may be unwilling to engage with us. For instance, as academic libraries seek to legitimize their positions as vital resources for research, they can end up exploiting the people they seek to serve, treating their communities like subjects to extract data from rather than collaborators in sharing information.[15] Repairing relationships with the public and building back trust can take time, especially if the damage done is grounded in deep historical roots.[16]

One area within the open science movement that is often seen as particularly good for public engagement is citizen science, "the practice of public participation and collaboration in scientific research to increase scientific knowledge."[17] However, even this area has the potential to alienate the very public we want to involve. As Fressoli and Arza share, citizen science projects have a tendency to be developed *by* scientists *for* scientists, with limited input from the public on how and where they will be contributing.[18] This doesn't have to be the case. Direct collaboration with the public and determining the scope and focus of a citizen science project can support a more collaborative and engaging project, with longer-term viability.

The Kyrgyz Mountains Environmental Education and Citizen Science Project is one such example. Launched jointly by the NGO CAMP Alatoo and the University of Central Asia, this project engages with people across the remote province of Naryn in Kyrgyzstan, inviting local educators to learn about climate change and working collaboratively with the public to generate and study data on how climate change is impacting their region.[19] In a case study about the project's outcomes, a team reflected on how direct input from the public and teachers helped improve awareness of their work and build a stronger base of interest in studying the environment among young children in the region.[20] This collaboration not only enhanced the project but also ensured a more inclusive and effective approach to environmental education for the public.

Where Else Do We Fall Short?

The public isn't just isolated from Open because of the actions of academic institutions. They are also often isolated from engaging in open science and scholarship because of the way that academia communicates the value of openness. When talking about open movements, we tend to speak as if the only ones listening are those who are immersed in the vocabulary and work of Open. Because of this, we are less practiced at shifting our message to meet the needs and understanding of our current listeners. In trying to reach a broad public audience, we may adopt an overly simplistic tone, "speaking down" to the people we are trying to build relationships with, further alienating them in the process.

As we mentioned earlier in this chapter, listening to and getting feedback from the community is a vital part of the outreach process. Starting from a place of humility and asking for advice from our communities is one way we can build new connections that center on service rather than transactional relationships. One example is the Make-IT Place,[21] which came about from an Institute of Museum and Library Services Grant[22] received by Enoch Pratt Free Library, a virtual makerspace for all ages.

The Make-IT Place uses openly licensed materials for easy science, technology, engineering, and mathematics (STEM) projects, as well as providing resources for educators to develop lessons and programming. This natural integration is a great example of how public-facing groups can leverage open content and practices in support of their communities, based on the needs and interests therein, and not relying on academia's expectations of what our communities need.

How We Communicate Also Affects the International Public

The public is more than just one of "our communities"—they are also those international peers who can utilize Open as a means of connecting with our academic institutions in ways they could not before, although this is not without its own unique challenges. In a 2024 article from *Nature*, the power structure and underrepresentation globally of lower-income countries are highlighted:

> *Another issue is that scientists in low-income countries might choose to focus on research topics that are of interest to editors of prestigious journals based in wealthy nations, in an attempt to advance their career, which further skews the research agenda. It also undermines the knowledge-sharing systems that exist in low-income countries and could prevent strong, relevant research from being distributed to those who need it.*[23]

International participation in open work has also been hindered by limited access to resources for sharing and contributing, as well as restricted capabilities to host and publish research. Moreover, the use of English as the global lingua franca poses a barrier, restricting contributions from individuals who possess valuable insights but do not communicate in English.

In summary, academia's historical isolation from the public has hindered the full potential of open science and scholarship. Yet, examples such as The Make-IT Place and the Kyrgyz Project illustrate that bridging this gap through public engagement and feedback is both possible and beneficial. By fostering inclusive and effective communication, and by addressing the unique challenges faced by international peers, we can ensure that open projects truly serve the needs and interests of the public.

Is Open Access Even *For* the Public?

In the realm of open science, the public is often mentioned in the phrase "the public should have access to publicly funded research."[24] This argument is a high motivator for the development of OA policies and requirements for researchers working with grant funding from the government. The framing of "it's good for the public" that the National Academy of Sciences shares does not advocate for public access for "the common person" but for the sake of research and innovation. The same quote continues: "…the public benefits from open science as new knowledge is utilized more rapidly to improve health, protect environmental quality, and deliver new products and services." In this way, public access to research is less about giving the public access than it is about making research more accessible for other researchers who will utilize it for the sake of the public good. The public may not be the direct audience for all open content, but that is not necessarily a bad thing. Whether individuals are interested in engaging with open content or not, there is no doubt that, as Bolick, Bonn, and Cross state, "You want your doctor and dentist and other health care professionals to be able to benefit from and apply the relevant research being developed at universities."[25]

Academia's prioritization of Open for research, teaching, and service situates the work done by authors and advocates for Open as labor, molding their priorities to fit internal promotion structures. Such structures put less emphasis on public engagement, as it typically falls within the "service" bucket of academic job descriptions and not the higher stakes category of research and creative activity. In a *Chronicle of Higher Education* piece, Kathleen Fitzpatrick summarizes this situation: "If we are saying that our public mission is first and foremost, then what we consider excellence in publishing or other forms of research and scholarly production should have the public-service mission written into it. Similarly, if we're looking for excellence in teaching, whatever it is that we consider excellence should have that public mission inscribed in it as well."[26] When public outreach is rewarded as a bonus rather than an integral piece of our community's efforts, it naturally becomes deemphasized. While numerous open advocates find a path to emphasize public engagement regardless of these structures, the deemphasizing of such engagement is a common thread that should be addressed in our collective organizational structures.

Open impacts more than "our" communities. Throughout this chapter, we have asserted that the public benefits from open movements, whether that is through reading the latest open-access research, learning from open educational resources, or leveraging open-source software in our everyday lives. However, when we talk about engaging with our communities, this limits the public to a much smaller scope, focusing on the locally situated public that we might engage with directly. While that is the focus of this chapter, we want to note here that engagement with the public goes beyond "our" public and should leave room for the many groups that are outside of our communities, those who lack infrastructure, support, and education for what open might mean for them. We cannot reach all of these groups individually, but we can keep them in mind and reach out beyond our local frame of view, when possible, to support wider collaborations.

So, what can we do? We can highlight opportunities to engage in support of Open that are already well known to the public as a starting point. Users know that Wikipedia is both free to access and editable, but it isn't built, maintained, and managed all on its own. There are wider local and global communities dedicated to updating and sharing information on its many pages. One way that local groups can support the ongoing access to knowledge on Wikipedia is through Wikipedia Edit-A-Thons[27] or other Wiki-based projects. Participatory, or crowd-sourced, projects such as this are a useful starting point for open advocates to engage with their community, particularly for those who lack the resources to develop, implement, or manage any open projects themselves.

Closing for the Public Good

Sometimes, in a push to meet a goal or start a strategic initiative, open "by default" can inadvertently become the status quo.[28] Of course, when workers in Open slow down, we understand that not everything should be open by default. This is particularly true for culturally sensitive materials. In the following sections, we discuss several examples of where and why "closing" access can also support the public.

Sometimes Our Motivations Are Flawed

One of the biggest barriers to making culturally sensitive materials available to the public can be the institution (university, museum, etc.) itself when it wants to retain ownership of materials they have put resources into developing and collecting. Sometimes, grant funds that are received by scholars are actually managed and owned by their institution. When we try to support our communities by making them the subject of research, our institution's motives get muddied, with priorities skewing toward acquiring more grant funding rather than working with our communities to legitimately assess their needs.

Just because open content can benefit the public doesn't mean that it will. We can cause harm by making content open without thoughtful consideration of the content itself and the various communities that may be impacted. Let's consider sharing data next, where the "potential for harm can stem from a few things: the content of the data itself, the potential use cases for the data, or unintended consequences of the way data was structured, collected, or interpreted."[29] Data, whether it is numeric, audio, video, text, or something else, may hold a different meaning in the eyes of the reader when it is removed from the context of the project that it was collected to support. Particularly for personally identifiable data and data gathered from underserved populations, it's essential that researchers and academics "think about their data holistically and engage thoughtfully with the power structures and dynamics at play when planning for data collection and use."[30] This holistic-thinking approach is recommended for any decision to make content open. Who has the power and agency in making those decisions? Who benefits from making the content open, and why? Is that who you thought it would benefit? Often, it is helpful to consult with others in these processes, both inside and outside of academia. By doing this, we can create stronger communities, ones founded on mutual care and trust.

Building Trust with the Public Takes Time

When creators or communities who hold knowledge are not consulted and do not hold agency in making knowledge open, we run the risk of perpetuating colonialist ideologies of ownership and Western structures onto marginalized communities. This is particularly true when working with Indigenous communities, student clubs, LGBTQ1 groups, and others who are members of such underrepresented communities. Groups such as these are the most likely to have lost trust in academia due to historical or even current events, and they might question the motivations of institutions pursuing a reparative approach as well. As Ramischwili-Schäfer argues, there is a growing issue with groups appropriating antiracist language and marketing to seem like they are more inclusive for a purely self-serving purpose and not actually doing the work of uplifting and supporting those who need it most.[31]

We can start to repair these relationships by acknowledging that our communities have a right to provide input on our open collections. Marginalized groups often find that the language used in academic publications is not relevant to their lived experiences. We cannot retroactively edit publications, but we can make it easier for our communities to find and engage with our collections by updating our metadata. When the historical metadata describing materials is inaccurate or derogatory, it can hamper our community's ability to see themselves reflected in our content, open or otherwise.

If we want to build trust with the public, we need to connect with our communities directly.† For longstanding collections that were created and curated without the feedback of the impacted community, how can we acknowledge this absence and attempt to make reparations, where possible? Reaching out is the first step, but there are many considerations that impact what comes next, from addressing the priorities of our employers to navigating the legal ownership of the works themselves.

Traditional Knowledge (TK) Labels are one example of a strategy that has been implemented to support Indigenous communities in making their work more open while retaining the rights that are contextually relevant for their groups.[32] This makes TK Labels more restrictive than many licenses leveraged for OA works; however, these extra layers of protection are necessary to support the needs of Indigenous peoples whose cultural knowledge has been taken out of context and altered over time without their consent, or even the ability to revoke consent.[33]

To support truly open and participatory collaborations within and outside our communities, academic institutions would benefit from implementing policies that allow open materials from Indigenous groups and other marginalized communities to be taken down or supplemented by their respective groups.[34] Beyond simple reclamation, demonstrating a willingness to give up control of profitable collections that have caused harm is necessary for building trust in the long run. We recommend that our readers approach these questions with care and consideration for their respective contexts.

Conclusion: Where Do We Go from Here?

Perhaps some of this is about an understanding as advocates for open that "building the future of open access involves challenging current systems and processes—from how knowledge is created and shared to who can participate, both as authors and as readers."[35] When our institutions are built upon a history of oppression, and when our economic drivers are interwoven in capitalism, it is even more essential that we challenge systems and processes.

Every open collection should come with an invitation for feedback. We will make mistakes in moving content to open, so the question becomes: How do we demonstrate to the public that we can adapt and learn with them? While we have examined the perspective of Open for the public and know there are unique challenges to outreach with a broad audience, it is worthwhile to explore some of the ways we can invite the public more fully into our open work.

There are a few strategies that we have identified throughout this chapter:

- **Utilize language that resonates with specific communities.** Share about new collections or training through channels that those communities already utilize. In other words, meet communities where they are. This could look like a place of business, at local city council meetings, or through professional avenues like conference presentations, newsletters, and free webinars.
- **Promote and share materials through regional groups.** Doing this together helps avoid duplicated efforts and provides a single entry point for users rather than requiring someone to be aware of all the individual repositories and collections available. This also allows us to make connections with those outside of the academic sphere.

† An example of a project which handled this well can be seen in https://journals.ala.org/index.php/lrts/article/view/7667/10775.

- **Provide training and support for wider community groups.** This can be done through engagement with extension offices and local non-profits, communications with local citizen science interest groups, and collaborations with public-serving institutions.
- **Advocate for (or, if you have the means, develop) targeted programming** to support key groups that need access to data, research, and other information available through open collections (e.g., non-profits, small businesses, students, and homeless populations).
- **Consider sustainable connections and communication** with your communities about related projects. Keep asking for feedback, have regular check-ins, provide a means for them to get in touch with you, and seek out opportunities to share new collections or changes to existing ones so the community can provide feedback and learn about topics of interest to them.

Although much more needs to be done to expand awareness of how open resources can be used by people outside of academia, there are many ways open resources are being implemented by the public, thanks to clear communication, visibility, and support from local and regional groups, some of which were highlighted in examples earlier in the chapter. We can learn from these positive role models and work with our communities to build more together.

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CHAPTER 3

Mapping Locally, Globally: A YouthMappers Perspective on Open Mapping

Adele Birkenes, Dara Carney-Nedelman, and María Fernanda Peña-Valencia

Introduction: Setting the Stage for the YouthMappers Network

The YouthMappers network is an open community of volunteer mappers who use open geospatial data and tools to address international development and humanitarian assistance challenges. YouthMappers is a subset of the broader open mapping community.[1] To understand the significance of the YouthMappers network's existence, we must first situate it in the history of the geospatial field, the participatory mapping movement, and, finally, the open mapping community.

The Geospatial Field

Geographic information systems (GIS), or computer systems that create, manage, analyze, and map data, are the key analytical tool employed in the geographic discipline. The concept of GIS and its early iterations date back to the 1960s.[2] As geospatial systems have evolved through advancements in technology, the uptake of GIS across diverse sectors has boomed. For instance, geospatial data and tools play a vital role in the international development sector, where they assist practitioners in identifying specific needs across diverse geographies and conducting informed decision-making.[3]

GIS's rise to prominence in the geographic discipline has sparked much heated debate between GIS practitioners and human geographers, particularly in the 1990s.[4] Human geographers, representing the social science side of the discipline, pointed to the potential risks of integrating GIS technologies into geographic practice, including concerns about GIS reflecting certain privileged world views over others (because of design decisions) and excluding some users (because of physical, technical, and social barriers to access).[5] In 1996, a number of GIS researchers and human geographers convened to discuss opportunities for democratizing GIS

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through an approach that they termed "public participation in GIS" (PPGIS).[6] The early developers of PPGIS sought to build a socially and politically engaged practice of GIS that empowered diverse partners to share geographic knowledge that could guide decision-making.[7]

The Participatory Mapping Movement

Since 1996, PPGIS, now commonly known as participatory GIS (PGIS) or participatory mapping, has emerged as a distinct and increasingly popular field within academic research and has been taken up by industry and government actors. Practitioners of its diverse methods and approaches seek to solicit community input on local assets, challenges, and development priorities. Methods generally fall into three categories: PGIS, which refers to participatory mapping with digital tools; sketch mapping, which is paper-based mapping with a basemap supplied; and mental mapping, which is freehand paper-based mapping.[8] Another related methodology that is more focused on mass data generation and less focused on community-based participatory processes is volunteered geographic information (VGI).[9] One example of imagery in response to a disaster.

In practice, participatory mapping practitioners may use participatory mapping terms interchangeably, blend multiple methods, and/or utilize different methods for different stages of data collection. For instance, a project may begin with remote mapping to create a basemap of an area and then leverage smartphone-based surveys and paper mapping exercises to collect additional information about the local context and infrastructure that cannot be gleaned from satellite imagery. Ideally, participatory mapping practitioners would conduct all stages of the project in close collaboration with the community, including by co-designing the project, training residents on survey techniques, and/or engaging the residents in data collection activities that recognize the plurality of perspectives and knowledge systems.[10] Thus, the essence of participatory mapping is to synthesize knowledge about risks, challenges, and resilience in a way that treats community members as experts and decision-makers.

On the one hand, participatory mapping's multidisciplinary and multi-actor nature has been its strength. Its techniques are situated in the geographic discipline at the intersection of critical cartography, qualitative methods, citizen science, and GIS analysis.[11] Its applications span diverse sectors, including urban planning, community development, resource management, and hazards, as well as cover multiple sociopolitical contexts and geographies. On the other hand, participatory mapping's multidisciplinarity has posed challenges to conducting projects and reporting findings in a consistent, transparent, and replicable manner.[12] In the first systematic review of participatory mapping, Denwood et al. note that the field as a whole has multiple incompatibilities with the principles of open science.[13] These include an absence of adequate details about participant demographics, methodological choices, and the influence of mapping efforts on decision-making. They conclude by calling for the adoption of an open science framework within the participatory mapping field, which they state would include a greater commitment to the use and development of open-source mapping software.[14]

The Open Mapping Community

The emergence of the open mapping community in the past two decades has led the way in integrating open science principles into participatory mapping. In 2004, OpenStreetMap (OSM) was created as an editable world map that anyone could contribute to and use, regardless of their expertise in GIS.[15] By utilizing the power of crowdsourcing, OSM rapidly grew into a global community-driven project, attracting contributors from diverse backgrounds and locations.[16] Its open-source nature allows for continuous updates and improvements, resulting in a dynamic geospatial database that reflects the evolving landscape of our world.

Today, OSM stands as one of the most renowned and extensively studied instances of a volunteer-run geographic project, where data is collaboratively sourced worldwide. While commonly perceived as a technological tool or platform, OSM can also be conceived as a community of communities, shaped by a broad understanding of geographical systems.[17] OSM

participants are diverse in terms of both the sectors to which they belong and the manner in which they engage with OSM. There are private enterprises and for-profit entities, nonprofit or civil society organizations, and governments or public institutions that directly contribute by creating and enhancing map data. Additionally, there are data consumers, including individual users who export data for specific uses and companies that integrate OSM data into their navigation or social media platforms.[18] In some cases, data consumers also contribute data back to OSM.

Humanitarian mappers use OSM as a tool to contribute to disaster response efforts by remotely digitizing satellite imagery of disaster-affected communities. "Humanitarian mapping" began in earnest after the 2010 Haiti Earthquake. Over 600 volunteer humanitarian mappers came together around the world to create an OSM map of Haiti, which the United Nations and the World Bank used as the primary map in their relief efforts.[19]

The YouthMappers Network

With all of the developments in the geospatial field, the participatory mapping movement, and the open mapping community, prior to the founding of the YouthMappers network, there still lacked a formal way to bridge groups working on open mapping, academic geography, and international development at scale. Founded with the recognition that young people are among the best equipped to identify development challenges in their communities and address them using digital technology, the YouthMappers network has catalyzed a generation of young people to lead open mapping chapters at their universities.

The Founding, Structure, and Contributions of the YouthMappers Network

A consortium of three US universities—Texas Tech University, The George Washington University (GWU), and West Virginia University (WVU), along with the United States Agency for International Development (USAID) GeoCenter—founded the YouthMappers network in 2014 and launched it on Capitol Hill in November 2015. Arizona State University later joined the consortium and assumed the role of lead fiscal and administrative partner. Between 2015 and January 2025, the USAID GeoCenter served as the primary funder of the YouthMappers program, collaboratively managing it with the consortium of four US universities. The consortium maintains a team of full-time and part-time employees, as well as faculty from the four institutions, who manage network operations, including training, data quality assurance, communications, and technical project implementation. The USAID GeoCenter provided technical support to the network and connected the program with USAID and interagency partners that wished to incorporate crowdsourced geospatial data activities in their youth programming and decision-making processes. In early 2025, the USAID GeoCenter and USAID funding for YouthMappers were terminated as part of broader organizational changes and reductions in foreign assistance funding by the US government. The US university consortium continues to support and implement the YouthMappers network.

With the consortium at the helm, the YouthMappers network comprises a network of over 420 decentralized student-led open mapping chapters at universities in eighty-one countries.[20] Chapters, led by students and guided by faculty advisors, operate independently of formal curricula or degree programs, attracting members from diverse academic backgrounds. These chapters serve as an outlet for experiential learning. They are typically housed in universities' geography departments or related departments such as urban studies, environmental science, or engineering.

Students and faculty jointly develop mapping activities that respond to local geospatial data needs and participate in global or regional remote mapping campaigns. Thus, YouthMappers engages students as prospective participants in the digital workforce and as emerging global citizens within a global society.[21] YouthMappers students map undermapped areas of the world

by digitizing buildings, roads, points of interest, and other features from high-resolution satellite imagery hosted on OSM.[22] Figure 1 shows the cumulative OSM contributions of YouthMappers students, alumni, and collaborators since 2015. These edits reflect data collected during both remote mapping and fieldwork. Figure 2 illustrates the global distribution of all YouthMappers edits since 2015. As a consequence of the 241 million OSM edits that the YouthMappers network has made, OSM has become a more robust "digital public good," or freely available data that underpins modern digital economies and progress toward the Sustainable Development Goals.[23] It is estimated that over 67.8 million people worldwide are covered by the open geographic data generated by YouthMappers mapping campaigns.[24]

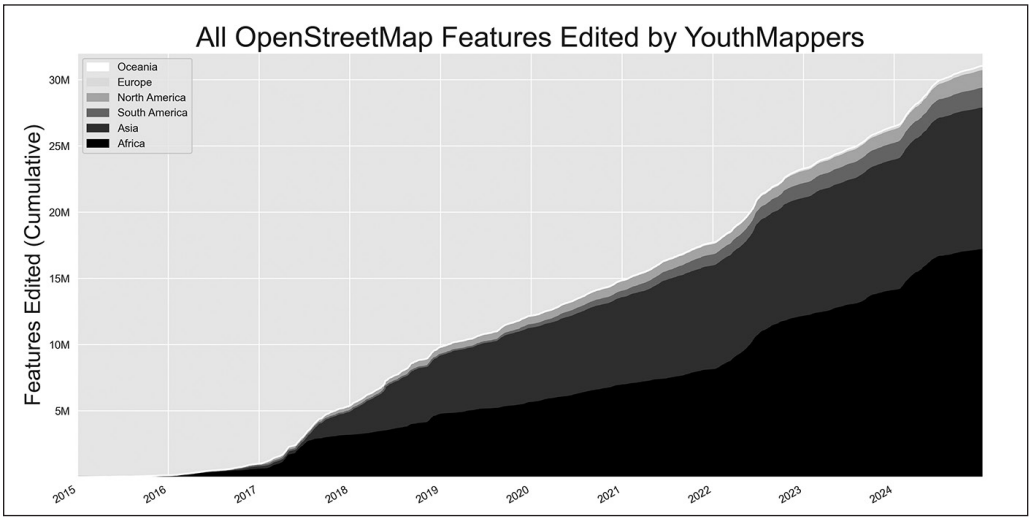


Figure 1. Cumulative OSM edits by YouthMappers members based on the locations that they edited on OSM (2015–2024). Visualization credit: Jennings Anderson.

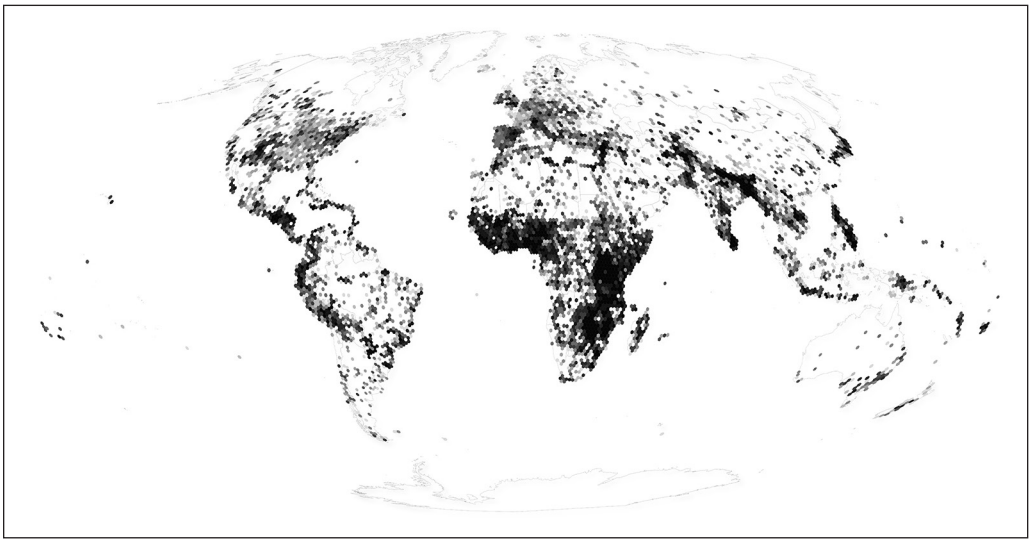


Figure 2. Spatial distribution of OSM edits by YouthMappers members based on the locations that they edited on OSM (2015–2024). Visualization credit: Jennings Anderson.

Universities have increasingly transformed into "third spaces," as proposed by Whitchurch (2008),[25] where activities that transcend traditional learning curricula take place. These spaces connect students with international movements centered on sustainability and justice,

fostering "hybrid movements" that enable students to work simultaneously toward their personal goals, prepare for the work world, and express their identities as global citizens.[26] The design of YouthMappers as an academic-based consortium within OSM has facilitated universities' development as spaces fostering youth engagement and identity formation. YouthMappers has grown into a movement that serves as a practical platform for action and self-expression, thereby advancing OSM communities.[27]

YouthMappers' Unique Approach to Community-building

The impact of the YouthMappers network on the wider OSM community cannot be measured solely by contribution statistics. Through YouthMappers' focus on supporting the participation of university students in low- and middle-income countries, the network is diversifying *who* is contributing to OSM. Since OSM's founding in 2004, its contributors have disproportionately been males from Europe and the United States. While it is difficult to discern the exact gender composition of OSM contributors due to the lack of gender information in contributors' profiles, survey-based studies report estimates of female participation ranging from 2 to 5 percent (studies published in 2010–2013) to 13 percent (study published in 2019).[28] A 2012 study estimated that 72 percent of OSM members who had completed at least one edit in the OSM database were located in Europe, and 12 percent were located in North America.[29] More recent research has found that OSM contributions have become increasingly global since then, with high levels of activity emerging in parts of Southeast Asia, East and West Africa, and South America between 2015 and 2020.[30] Still, the Global South remains underrepresented on OSM in terms of both quantity and quality of data and number of contributors.

For minoritized and underrepresented contributors, such as those from the Global South, who do decide to engage in contributing data to OSM, they may face sexism, misogyny, and racism that discourage them from further participation.[31] "A Call to Take Action and Confront Systemic Offensive Behaviour in the OSM Community," published in 2020, describes an environment in which members of the community are subject to hostile, intimidating, and patronizing communication based on their identities.[32] These issues reflect the participation challenges associated with open data contribution and science. However, when open science is effectively implemented, it can be a tool that promotes human-driven and democratized data creation.[33]

Against this backdrop, the US university consortium that manages YouthMappers has crafted a community-building model that has brought thousands of university students in low- and middle-income countries into the OSM contributor fold. Forty percent of YouthMappers chapters are majority-women, and most chapters have significant leadership and participation from female members.[34] The community-building approach of the YouthMappers network is powered by peer-to-peer mentorship. For instance, the YouthMappers Regional Ambassadorship program situates YouthMappers students and alumni as regional trainers. They provide mentorship and technical support to onboard new university chapters and help them develop mapping projects. Additional opportunities available to YouthMappers chapters are annual leadership and research fellowships, regional "training of the trainer" workshops, training webinars, and the YouthMappers Academy, an online course for building OSM skill sets. Collectively, these activities demonstrate to YouthMappers students that they have a place in the OSM community and that their contributions are valuable. YouthMappers has been shown to have relatively high short- and long-term OSM retention rates when compared to Missing Maps, a volunteer-based humanitarian mapping initiative that does not provide the same degree of community and skills-building opportunities to its volunteers.[35] This may be due to students deciding how their communities are represented on OSM, which is a positive example of democratized and socially driven open science.[36]

The organizational structure of the global network positions students as more than just members. They are the voice of the network. Their stories, projects, and passions are what guide the direction of the community. Students are not only seen as learners within the

network but also the researchers, trainers, project managers, and authors. Students publish their projects on the YouthMappers blog and social media and are encouraged to contribute to research articles. YouthMappers students not only focus on the success of their chapters, but they also create skills and products for the benefit of their communities and humanity. They are creating a digital public good for their communities and communities across the globe.[37]

Case Studies of Success: YouthMappers Students and Alumni

The YouthMappers movement comprises diverse individuals from university communities worldwide, each bringing unique backgrounds, experiences, and perspectives to the network. Together, they collaborate to collect open geospatial data and build maps to support community resilience. Below are quotes from past and present YouthMappers explaining how their network participation has helped them foster skills and prepare for their careers. First are perspectives from the three chapter authors on their journeys with YouthMappers:

Adele Birkenes, YouthMappers Alum, United States

I learned about the YouthMappers program as an intern at the USAID GeoCenter in the summer of 2018. I had the opportunity to attend the YouthMappers Research Fellowship Summit hosted at GWU and WVU. At the time, I was halfway through my undergraduate degree in geography at Vassar College, but I had never heard of OpenStreetMap or humanitarian open mapping. As I listened to María Fernanda and the other fellows describe their community-based research projects, I felt inspired to follow in their footsteps and start a YouthMappers chapter when I returned to Vassar that fall. With financial support from the Vassar Office of Community-Engaged Learning (OCEL), I co-founded and led Hudson Valley Mappers as a joint undertaking between the OCEL and the Department of Earth Science and Geography (see figure 3).



Figure 3. Adele (center), Hudson Valley Mappers faculty advisor Neil Curri (right), and local high school students piloting a smartphone-based survey to assess waste disposal infrastructure in Poughkeepsie, NY. Photo credit: Hannah Benton.

For two years, I planned and implemented volunteer mapping projects in collaboration with the local government and nonprofits in the City of Poughkeepsie, New York. In my final semester, as we transitioned to online learning due to the COVID-19 pandemic, my peers

and I created an interactive web map of food and health resources for residents of Dutchess County.[38] After graduating from Vassar, I joined the USAID GeoCenter as a Geospatial Analyst. For four years, I supported the management of the YouthMappers network. I provided direct technical support to students as they developed mapping projects for their communities, and I led trainings on equity and inclusion in open mapping. For my graduate thesis in geography at GWU, I am studying the impacts of the USAID funding cuts on the YouthMappers network and other humanitarian open mapping groups.

María Fernanda Peña Valencia, YouthMappers Alum, Colombia

I joined the YouthMappers movement as a volunteer university student in a research project. During that time, I realized the potential of maps and geomatics as strategic tools for working in disciplines such as ecology and conservation with great social impact. I also introduced myself to open data principles, which promote the free sharing of information for accessibility and collaboration, and free software environment, which allows users to freely use, modify, and distribute programs. Motivated to continue learning and being part of the community, I had access to various programs, including the Research Fellowship in 2018, and later served as a Regional Ambassador in the 2020 and 2021 cohorts (figure 4). This experience opened doors for me to coordinate research processes involving collaborative mapping, contributing significantly to my personal and professional growth.



Figure 4. María Fernanda (center right) co-facilitating a community mapping workshop in Bocas del Atrato, Colombia. Photo credit: Natalia Da Silveira.

Specifically, through the YouthMappers Research Fellowship program, I had my first significant opportunity to lead, coordinate, and manage projects with the guidance of excellent mentors. My project focused on mapping coastal fishing communities to assess their resilience and vulnerability to floods, highlighting the critical role of mangroves and the importance of their conservation.[39] Additionally, through YouthMappers, I found a supportive community and a methodological platform that allowed me to work within the conceptual framework of

social-ecological systems focused on sustainable development. This path has been a significant learning experience for my professional growth, becoming a cherished chapter in my life's story. Currently, I am a junior researcher in the SDGs center in Latin America and the Caribbean, while still being part of the YouthMappers movement as a global citizen.

Dara Carney-Nedelman, YouthMappers Former Staff, United States and Laos

I participated in a predecessor program to YouthMappers. I was selected for a youth tech camp in high school, where I traveled to La Paz, Bolivia, to utilize mapping to address climate change. During this tech camp in 2014, supported by the American Association of Geographers and the US State Department, I met YouthMappers Executive Director Dr. Patricia Solís and Managing Director Marcela Zeballos. Following college graduation, I reached out to Dr. Solís, seeking to combine my mapping background and new communications expertise. I was hired as the network communications specialist. YouthMappers gave me an opportunity I never expected I would have in an entry-level position, much less reach by twenty-two years of age. Through my role with YouthMappers, I built skills in intercultural communication, webinar hosting, academic writing, social media management, documentary production, and open mapping.[40]



Figure 5. Dara (center) co-facilitating a training during a Sabaidee SVK YouthMappers fieldwork workshop at the Savan Centre with high school students in Savannakhet, Lao P.D.R. (2024). Photo credit: Nicky Sysoukda.

While today I am no longer a direct member of the YouthMappers team, I still refer to myself as a YouthMappers Cheerleader and OSM Advocate. YouthMappers sparked my interest in OSM, and I have continued on as OSM Community Administrator and Co-Founder of OSM Laos, HOT Community Working Group Member, and supported the start of the 400th YouthMappers chapter at Savannakhet University, Laos. During my time in Laos as a Fulbrighter, I connected GIS professionals and recreational mappers in three cities, represented and presented on OSM Laos at State of the Map Asia, and led OSM trainings for university students

(figure 5). YouthMappers gave me the opportunity to build my skills and broaden my perspective on global citizenship and open mapping.

Next, YouthMappers Regional Ambassadors share how they have personally and professionally benefited from their involvement in the network. Each Regional Ambassador has mentored existing chapters and supported other universities in creating new chapters:

Sajeevini Sivajothy, Regional Ambassador 2023–2024, Sri Lanka

Three years ago, my journey with YouthMappers began with no prior knowledge of OSM. Now, I find myself immersed in the vast sea of open source, growing more stable and knowledgeable with each passing day. This YouthMappers Network believed in me, and I believe in the miracles it has wrought. Thanks to the YouthMappers Network and the opportunities it offers.

Tommy Charles, Regional Ambassador 2020–2022, Sierra Leone

YouthMappers exposed me to people from different backgrounds and empowered me to work with them seamlessly. Through the network, I attained all the necessary technical skills to start my career in geoscience. My major highlight as a YouthMappers member is when I was given the opportunity to spearhead the Power Grid mapping project, where I was tasked to make use of my soft and hard skills.

Feye Andal, Regional Ambassador 2020–2024, Philippines

Engaging with YouthMappers has been a transformative experience. It has allowed me to connect with and mentor young mappers, guiding them through the world of open mapping. This journey has also bolstered my professional career, linking me with passionate communities who share our vision and elevating my confidence as an individual. YouthMappers has empowered me to excel as a leader and learner, igniting a deeper sense of purpose and gratitude.

Sawan Shariar, Regional Ambassador 2020–2024, Bangladesh

Being a part of YouthMappers has profoundly impacted my personal and professional growth. Through this vibrant network, I've honed my skills in open mapping tools and geospatial technologies, crucial for various community projects. The global connectedness of YouthMappers has allowed me to collaborate with peers from diverse backgrounds, deepening my understanding of global issues and the pivotal role of data in addressing them. The mentorship and support from the YouthMappers cultivators team and experienced members have been instrumental in shaping my career path and enhancing my problem-solving capabilities.

These stories capture a vital part of the YouthMappers network—it has built advocates for life. The voices represented here are from individuals who have grown from student-leaders at their universities to leaders in the open mapping and GIS communities. By inspiring students not just to build maps but also to become mappers, the network has created a safe and supportive community. The community has evolved into a movement of professors mentoring students, students cheering on peers, and alumni teaching students.[41]

Case Studies of Success: YouthMappers' Support of Open Geospatial Data and Tools in Government Decision-making

The YouthMappers network's growth has unfolded at the same time that governments have expanded their geospatial infrastructure and adopted diverse stances on the incorporation of open data and tools into their systems. While some agencies and governments have embraced the opportunity to integrate crowdsourced data into official datasets and open tools into decision-making processes, others have demonstrated hesitation to deviate from traditional GIS systems that rely on proprietary software and/or private datasets. On the crowdsourced open data front, barriers to adoption include a lack of trust in data quality, data collectors' adherence to standards, and data completeness.[42] Moreover, public institutions generally view open software as less secure than proprietary options and face regulatory and capacity-related challenges in implementing it.[43]

Since the YouthMappers program's founding, its leadership team and chapters have interacted with US agencies and other countries' governments on open geospatial data and tools both directly and indirectly. First, the USAID GeoCenter collaborated with the YouthMappers university leadership consortium and chapters to address gaps in datasets by collecting open, crowdsourced geographic data. Improving the availability of geographic data supports the growth of a country's digital economy, enables governments to better understand and manage their assets, and empowers researchers to investigate and understand our natural and built environment. For example, in 2022, the US Department of Agriculture Foreign Agricultural Service (USDA/FAS) partnered with the USAID GeoCenter and YouthMappers on a pilot project to improve the application of machine learning algorithms (ML) to detect and identify crops growing on the ground in smallholder farms in Africa from satellite images. This initiative trained students in Tanzania on open mapping tools and introduced them to professionals working in the agricultural industry. The training and fieldwork focused on collecting in-season information at the farm level about crop types. The detailed crop data was then made available to improve ML algorithms in environments similar to rural Tanzania, which historically lack sufficient training data needed for crop mapping and forecasting.

Second, the YouthMappers network's roots in academic institutions foster a rich environment for research projects that can assist countries in strengthening their open geospatial ecosystems. With funding from the National Science Foundation and in partnership with the Pan American Institute of Geography and History, the YouthMappers network conducted research that paired US-based YouthMappers students with the National Geography Institutes of Mexico, Belize, Costa Rica, Panama, Colombia, Jamaica, and the Dominican Republic. The students developed action plans for partner countries to increase their use of open, volunteered geographic information in official datasets.[44]

A notable example of YouthMappers students collaborating to create and improve national cartography is the Urban Py (AUPy) project in Paraguay, developed by researchers and students from the Center for Research, Development, and Innovation (CIDi) of the National University of Asunción, and YouthMappers members. Oporto and Villar[45] note that Paraguay's cartography has been dominated by government institutions with a militaristic and extractivist approach, resulting in limited, inaccessible geospatial data. This lack of data restricts public access, academic reflection, and democratic debate based on evidence. In response to this, the AUPy project was initiated in 2020, continuing the efforts of two preceding projects: Infrastructure and Territorial Transformations in Paraguay (6oPY14) and Mapping Paraguay with OpenStreetMap (MapPYOSM). In one of its recent phases, 3,753 building polygons were verified and characterized in municipalities such as Coronel Oviedo and Minga Guazú. These projects leverage free software tools like OSM, Mapillary, and QGIS to produce national cartography, showcasing that organized, open, and collaborative efforts can contribute significantly to the construction of knowledge and development.[46]

Examples like these will increase as the number of universities participating in the YouthMappers program continues to grow. The impacts of the network can also be viewed in terms of its indirect, longer-term relationship with government institutions. YouthMappers students are the next generation of geospatial professionals who will enter government workforces and non-governmental organizations prepared to advocate for open geospatial data and tools in decision-making.

Challenges and The Way Forward

Chapter Dynamics

As with any university group, there are many dynamics that YouthMappers chapters must navigate. Some of those are intrinsic to the university setting, such as challenges with student turnover, balancing student time commitments, and faculty administration. Other hurdles include chapters situated within science, technology, engineering, and mathematics (STEM) campus communities, which may lead to unequal gender representation. Additionally, language barriers—as with many global organizations—make global networking and collaboration difficult. Many of these challenges are addressed through mentorship and chapter ownership. The network encourages peer-to-peer, alumni, and faculty mentorship, which leads to a space for YouthMappers chapters to gradually grow and adapt.[47]

Fieldwork and Local Community Interactions

Although remote mapping on the OSM platform is a common methodological stage in all YouthMappers projects,[48] fieldwork is usually another fundamental step in the process. During this step, the team moves to the study area to make direct observations and collect information. Because a basic principle of the organization is to incorporate local communities in this process whenever possible, fieldwork becomes an important stage that combines data collection with community participation. This methodology has the potential to improve both the quality and utility of maps by including the knowledge of local communities that is not accessible through remote mapping. In addition, involving local communities strengthens the sense of project and data ownership as well as trust with the work teams. This, in the medium- and long-term, can facilitate a more effective and sustainable approach to addressing local challenges.

Both fieldwork and collaboration with communities present significant challenges, but different strategies have been adopted to address them effectively. For instance, respecting local customs requires cultural sensitivity and adaptation, which can be achieved through pre-fieldwork training and consultations with community leaders. Building trust takes time but can be facilitated by transparent communication, actively involving communities in decision-making processes, and consistently honoring commitments. Also, challenges such as accessing remote areas, dealing with adverse weather conditions, and managing health risks often require additional planning and logistical resources. In these cases, it is important to collaborate with local organizations to obtain security advice and contingency plans. Proactively addressing these challenges can significantly improve the data collection process, the quality of the resulting maps, and the overall success of the initiative.

Technology

The OSM database and community originated in Europe, and while the community of contributors has become more diverse, especially in the past decade, the ecosystem of open mapping tools is most easily accessible to contributors with fluency in English. YouthMappers students who are not fluent English speakers face difficulties in becoming proficient in open mapping software. Over time, the OSM and YouthMappers communities have strengthened language support, including by translating tools and accompanying guides into other languages.

Additionally, YouthMappers Regional Ambassadors provide mentorship and training in Spanish, French, and other regionally dominant languages.

The digital divide, or the global gap in technology access, remains a significant challenge. YouthMappers chapters located in low- and middle-income countries are more likely to face barriers in access to internet connectivity, computers, and smartphones.[49] While some mapping tools can be employed in low-connectivity environments, uploading edits to the database requires the use of a device connected to the internet. One way that the YouthMappers network has addressed this challenge is through the Fieldwork Program, which supplies funds to chapters to cover these costs.

Conclusion: The Way Forward

YouthMappers is a success story for how participatory and open mapping can evolve. Within the OpenStreetMap community, it is a beacon for youth-led, localized open data creation. Taking advantage of university workspaces and partnerships with public and community-based institutions, YouthMappers fosters students' personal and professional development and leadership skills through continuous experiential learning. The network has also encouraged GIS professionals and professors to re-evaluate how they are supporting students. Some of the best techniques used have been through inviting students to be co-authors, co-presenters, and co-researchers.[50] By encouraging mentors to value and learn from mentees and inspiring experts in the field to seek out youth as mentors, YouthMappers has created an open collaborative that transcends traditional hierarchical barriers.[51]

Through the student-led approach, the network has created a foundation of trust and confidence in student participants. Students become more active global citizens engaged in finding solutions for local and global challenges, especially in the context of accelerated technological development. They build upon their trust in the YouthMappers community, confidence to be change agents, and innovative open data skills into their roles as regional ambassadors, alumni, and professionals. With this mindset, they pay forward their YouthMappers experience. The YouthMappers open movement has inspired and cultivated open data users, contributors, and advocates for life.[52]

Authors' Note

We began discussing this chapter in December 2023. Between when we started writing the chapter and its publication, YouthMappers was directly impacted by shifting US government priorities. Despite these changes, the founding principles and mission of the consoritum and movement have remained steadfast. Although none of the three authors are currently part of the YouthMappers leadership team or staff, we are confident in the resilience and future of this open movement. The students around the world are its heart. Their stories, innovations, and perseverance are what made creating this chapter possible, and they will lead future generations of YouthMappers in carrying forward the motto: "*We don't just build maps, we build mappers.*"

Acknowledgments

The authors would like to share their sincere thanks to the regional ambassadors who shared their YouthMappers stories and to the YouthMappers Steering Committee members who offered feedback on this chapter. We are grateful to Patricia Solís and Carrie Stokes for supporting our vision for this book chapter from its inception and to Jennings Anderson for creating the OSM contribution visuals.

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CHAPTER 4

Cultivating Community: A Case Study on Samvera's Approach to Open Source

Carolyn Caizzi and Heather Greer Klein

Introduction

The collaborative initiative known as the Hydra Project was launched in 2008, bringing together four organizations: the University of Hull, the University of Virginia, Stanford University, and Fedora Commons.[1] Their goal was to develop open-source software (OSS) components for a common repository framework. This software enabled adopters to customize an open access repository using common components, such as data stores like PostGres and Ruby gems, which is a package of code that extends the functionality of the programming language. Customizing a repository like this is an alternative to developing an entirely new application or purchasing a vended solution.

In 2017, Richard Green and Chris Awre from the University of Hull documented the history of the Hydra Project's transition to the Samvera Community in their article in *UKSG Insights*, "From Hydra to Samvera: An Open Source Community Journey."[2] They describe how responding to the growing interest in the repository's OSS components presented a challenge to the founding organizations. This chapter expands on their work using the framework of It Takes a Village (ITAV) to detail how Samvera became a community by focusing on four areas: governance, resources, community engagement, and technology. ITAV is a guidebook and toolkit developed by Lyrasis with participation from multiple open-source software communities. The guidebook is meant to serve as a starting point for OSS communities as they plan for long-term sustainability.[3] The decisions and challenges of each of these domains show how this OSS project evolved into a values-based community that assesses and adjusts to its members' needs. It will also point to ongoing challenges that call for wider collaboration among open-source communities and those who rely on them for critical digital infrastructure.

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Governance: From an Informal Group to a Nonprofit Board

A small group with an idea cannot know for certain they will grow into a community with hundreds of participants; good leadership and governance structures have to grow along with the community. Initially, Samvera's founding members agreed to a three-year collaboration without dedicated funding, forming the governance for the Hydra Project.[4] They invited additional contributors, called Partners, to help with decision-making, necessitating defined roles and responsibilities. Partners are institutions, corporations, or other groups that have formally committed to contributing to the Hydra community through various means like code, analysis, design, support, funding, or other resources.[5]

The Hydra Project experienced significant growth, expanding from ten Partners in 2012 to thirty-five Partners in 2017. To manage this growth, the founders adopted the Apache governance model and became the Hydra Steering group, working with the Partners in collaborative roadmapping and resource coordination. Apache governance is a meritocratic model where decision-making and project management are driven by community consensus, emphasizing transparency, openness, and collaborative contribution.[6] During the initial growth period, the community decided to trademark its name and logo but encountered trademark barriers to using the name Hydra. The project was renamed Samvera, from the Icelandic term for "being together," in June 2017, emphasizing the importance of community, values, and sustainability in the community's open technology projects.

Alongside the rebranding, the community assessed governance models with a focus on resource growth and transparent processes. A working group researched other open community governance models and articulated desired community attributes, such as a community roadmap, stable communication, and formal Partner contributions.[7] Various organizational models were compared, including those of the American Library Association, the American Red Cross, the Apache Software Foundation, and the Islandora Foundation. The working group ensured the community had opportunities to review and comment on the models by holding open meetings and providing opportunities to provide written feedback via survey tools. The feedback did not present a clear picture of which model was favored.

Since no model was favored, another working group was chartered in fall 2017 to create a governance model with concrete recommendations for the Samvera Partners by April 2018. The recommendations included

1. the Steering Group becoming an elected body of nine members from Partner institutions, with formalized roles;
2. hiring centralized staff to support the Samvera Community, including a technical coordinator and a community manager;
3. developing a contribution model to support centralized staff;
4. establishing a Roadmap Council to align roadmaps and arrange for shared maintenance and development sprints;
5. updating Samvera's bylaws; and
6. assessing changes at the end of 2019 to ensure they meet the community's needs.

These recommendations were unanimously approved by the organizational representatives of the Partners in April 2018, with thirty-two of the thirty-five Partner ballots cast and all in favor.

By 2020, Samvera Steering became a fully elected body with formal roles, such as chair, chair-elect, and treasurer, and three-year terms. This ensured that the community's administration was led by its Partners and that all Partner organizations had the opportunity to serve in leadership roles. The original Steering Group had already begun to invite key community members from institutions that were contributing to the codebase to a yearly event for strategic planning, which evolved into the bi-annual Partners meeting, providing more open processes for decision-making and a more inclusive community.

During the project's early growth, the original Steering Group and Partners required support in administering the community through services such as banking, marketing, and communication. In 2015, the Steering Group entered into an agreement with DuraSpace, which provided these services until its merger with Lyrasis in 2019. Lyrasis is a nonprofit organization that provides services, technologies, and expertise to libraries, archives, and museums, focusing on collaboration and cost-effective solutions. DuraSpace was a nonprofit that supported open-source repository platforms like DSpace, Fedora, and VIVO, enabling institutions to manage, share, and preserve digital content. In 2020, Lyrasis proposed formalizing the relationship in a fiscal sponsorship model. Fiscal sponsorship, while not a recognized legal term, involves a sponsoring nonprofit accepting and administering funds for another project, group, or entity, in compliance with US tax law. This model allows projects to access administrative support without incorporating as a nonprofit.[8] The now fully elected Samvera Steering took this opportunity to evaluate three organizations as potential fiscal sponsors: Lyrasis, Code for Science and Society, and OASIS Open. They identified fifteen criteria for evaluating potential sponsors, prioritizing financial stability, support, and flexibility in the sponsorship model and cost-effective administrative services.[9] OASIS Open's Foundation-as-a-Service program was selected, offering a fiscal sponsorship arrangement similar to the previous DuraSpace agreement but with additional support and legal status as a 501(c)6 Series LLC.

Once the selection was made, Samvera Steering drafted a recommendation report to Partners, including new bylaws and a proposal to rename the Steering Group to the Samvera Board of Directors. Samvera Partners voted to accept these changes, and the OASIS board approved Samvera as a sponsored project in spring 2021. OASIS Open now provides critical services, such as legal review for policies, bylaws, and contracts; complete invoicing/payment processing and accounting infrastructure; routine operations reporting to the Samvera Board; independent auditing services; preparation of a corporate annual report; and tax and corporate filings. They also contract with a third-party HR company for hiring project staff and provide payroll and competitive benefits.

Resources: Aiming for Predictability

From its beginnings as the Hydra Project, Samvera has used a Partnership model. In Partnership, Samvera invites invested institutions, organizations, and service providers to formalize their commitment to the community and commit to greater participation in community governance. A Partner submits a letter of intent outlining why they want to participate and how they plan to continue to participate in and help sustain the Samvera community. Sometimes, this included financial support for community initiatives, and Partners were generous in making financial contributions over the years. However, as Samvera grew and it recognized that adding a staff member required predictable revenue, Samvera Steering wanted to both maintain a Partnership model and include an annual dues contribution from each Partner. The community could pursue other fundraising, but Partnership dues would form the basis of community revenue. This is in contrast to models in which an annual fundraising appeal is made to a wide group of adopters who are encouraged to contribute at a tiered membership level that they select and that can change from year to year.

Defining a contribution model was no small task, and as part of the governance changes in 2018, a working group was established to produce a set of recommendations that outlined the benefits of Partnership and described possible paths forward for financial support. These recommendations were discussed in the fall 2018 Partner meeting during Samvera Connect, the annual Samvera conference, at the University of Utah.[10] One challenge with these recommendations is that they did not explicitly define a financial model that addressed the community's needs. As happened with the governance changes, two working groups were needed, as were multiple opportunities for the community to provide feedback. The second working group was chartered to outline a method to meet the financial needs of the community.[11] It is important to note that when an impactful decision about the community's future must be

made, this process takes time. Impactful decisions frequently take a year or more and require multiple summaries and review at every meeting, since almost every member participates in the community either as a labor of love or as a small percentage of their time at an institution.

In the fall 2019 Partners meeting, the second working group had completed its work and defined a financial model for discussion. The financial model proposed was clear about the financial needs of the community and looked forward to the additional costs of hiring one staff person to start. It recognized there was a diversity of organizations within the community, with both academic, non-profit, or similarly situated institutions, as well as fee-for-service providers. The model addressed both types and acknowledged that the amount of financial contribution should be based on the size of an organization. Through more feedback, the model was formally adopted in January 2020 with the expectation of fully phasing in the model for 2022 financial contributions. The model valued in-kind contributions that Partners always brought to the community, while also being cognizant that Samvera had identified needs that could only be met with a regular and sustainable source of income. The model acknowledged that a Partner could have a financial circumstance that necessitates the need to reduce their financial contribution to the community for a year and allows Partners to request a discount.[12]

Partnership has provided Samvera with predictable income because there is an expectation that specific dues will be paid. However, this model can also mean that inflation chips away at the annual budget in years with few new Partners. Samvera Partners agreed to address the first instance of this inflationary impact by increasing Partner dues by 5 percent in 2025. The Samvera Board, in turn, will work with the fiscal sponsor to create and share three-year cost projections with Partners annually to anticipate when increases are needed before they can cause damage.

Community Engagement: Lowering Barriers

An open-source community requires a clear sense of shared needs and values alongside sufficient infrastructure, communications, and management to meet those needs and ensure that existing contributors are supported and new contributors are welcomed. From the earliest days of the Samvera Community, organizers created opportunities to share information and implemented community infrastructure for virtual and in-person collaboration.

For example, Hydra started holding annual conferences in 2013, called Connect, so that Partners and adopters could meet in person to brainstorm ideas for further development, showcase their work, and collaborate on technical work together in real time. These conferences were hosted by volunteer institutions and relied on one-time contributions from Partners and cost recovery charges for events since Hydra did not yet have a stable yearly income. The year 2016 saw the addition of an annual free virtual conference to make information available to technology adopters around the world. Partners continued to meet in person once or twice per year for decision-making and strategic planning, and there were regional meetings self-organized by interested adopters.

Another important piece of community infrastructure implemented early on was a lightweight framework of working and interest groups to organize work across the community and ensure work was not being duplicated. Two of the requirements of forming a working group in the community, still in effect as of 2024, are that 1) there must be an open call for participation and 2) that three Partners must commit to providing resources to scoped work and deliverables, with no more than three Partners objecting to the work. These requirements solved a challenge that open-source communities struggle with, ensuring work the community recognizes as a need is generalized enough for multiple adopters and that openness and sustainability are built into the process. If the work is not supported by three Partners, the chance that the work proposed is too specific to local needs is much higher.[13]

A structure for working groups allows community participants to move forward with large-scale proposals while ensuring sufficient community participation. As a growing Samvera underwent governance changes, there were community discussions about an emerging new

purpose and a need to articulate a vision statement. A working group was chartered and used Peter Senge's *The 5th Discipline Fieldbook* to conduct a visioning process for the Samvera community. The process sought to build shared meaning through a series of questions about stakeholders, providing value, assessing impact, setting technical direction, organizational structure, and signs of sustainability. This vision statement was created and adopted in 2019: "Samvera is a vibrant and welcoming community of information and technology professionals who share challenges, build expertise, and create sustainable, best-in-class solutions, making the world's digital collections accessible now and into the future."[14]

Additionally, the Samvera Community has collaborated with and looked to other OSS communities for new ideas to make participation easy and to make contributors of all kinds feel welcome and supported. One example of this was the development of a Samvera Code of Conduct and Anti-Harassment Policy, developed by a working group that began in 2019, building upon existing codes of conduct used by the Digital Library Forum and Code4Lib conference for their events. These communities generously made their documents available with Creative Commons licenses, and Samvera in turn created robust Community Guidelines and an Incident Response Guidebook. This work further benefited the Islandora Community when they developed their own guidelines in 2022.[15]

Communication channels and community infrastructure developed early and remained vital over fifteen years. Hydra began with email lists, IRC, a wiki, and a website in 2011. Samvera in 2024 uses email lists, wiki, Slack, and website. The primary change is that the channels moved from external management to community management. For example, the first wiki space was part of the DuraSpace-managed wiki space, but the community moved to their own community-managed wiki in 2021. Samvera also began paying for Google Workspace to organize documents in one location to ensure documentation is not lost when community members come and go. This change required a commitment to pay for some of these services, while others are available for free via special OSS community licenses.

As the community grew and communication and community facilitation grew more complex, the Steering Group and Partners recognized the need for centralized staff. While the community worked to find a financial model to support hiring staff, Richard Green from the University of Hull was contracted as a part-time operations advisor beginning in 2019. A working group was formed to define community needs and write corresponding job descriptions. The group learned from other OSS and related communities about their staff roles, such as the community manager and technical coordinator roles found in the IIIF (International Image Interoperability Framework) community. Since the Partners could more clearly define the needs and the role of hiring a community manager, this role was prioritized. A search committee was created to define and recruit for the role, and Heather Greer Klein was hired in fall of 2020 as Samvera's inaugural community manager. The hiring of a dedicated community manager has been key to sustaining, coordinating, and engaging the community. However, the community still has an identified need for a coordinating technical role but would require additional Partner dues to support a second staff position.

Technology: Balancing Flexibility and Cohesion with Multiple Projects

The organizational work of open-source communities—enacting governance processes, generating resources, and building a welcoming community—is all in service of the ambitious goal of creating and maintaining software with developers and stakeholders from different institutions around the world. Samvera, as a community and builder of software, works to "deliver common core repository functionality in a way that can be flexibly applied to a range of use cases."[16] This collective had to create processes for planning, executing, and receiving feedback on products and for handling disagreements and making decisions. The community also has to account for maintenance, documentation, and developer onboarding. Technical

leadership maintains the software that institutions rely on and accounts for new technology options and changing institutional requirements. This work grows in complexity, particularly in a community like Samvera, where there are multiple software components developed and maintained within one community.

The Hydra Project made many early decisions in the management of the software that have served the community well over the past fourteen years. All code produced in the community is distributed under an Apache 2.0 open-source license. The code versioning, reviews, and releases are all managed in a Samvera GitHub organization to simplify asynchronous development work.[17] Named code components (Ruby gems) and solution bundles (collections of software such as Hyrax or Avalon Media System) each have a volunteer product owner who alerts the community to maintenance and other needs. The solution bundles also have volunteer tech leads to act as a point of contact for development questions and decisions and to lead work sprints for maintenance and feature development. Since 2010, there has been a weekly virtual meeting, the "committers call" or "tech call," for open conversation and questions about all aspects of Samvera technologies.[18]

Another technology management process is Samvera Labs. Labs is a GitHub organization for experiments, works-in-progress, and beta versions of gems and apps for the community to review and contribute to. Having this space ensures that Samvera's technology stack can evolve, and developers can experiment with different approaches and languages to solve use cases and still benefit from cross-institutional collaboration. After meeting code, documentation, and use requirements, gems and apps can be promoted out of Labs to the full Samvera Github repository by a discussion in a monthly tech call. Promotion out of Labs signals to the community that a gem or app is ready for other organizations to adopt and reflects that the promoted software is no longer an experiment and the organization has committed to maintaining it.

One outcome of governance recommendations in 2019 was the formation of a Roadmap Council, tasked to "facilitate communication among the product owners of the Core Components as well as other Samvera solution bundles and Interest Groups. The Council strives to maintain a holistic view of the work being done by the Samvera Community and ensure coordination of that work."[19] This group further evolved into a Roadmaps Alignment Group, which ensures that volunteer community leadership regularly meets with the community manager and each other to identify upcoming maintenance priorities, cross-technology efficiencies, and other opportunities to coordinate and remove barriers to development progress.

Reliance on volunteers for development has long been identified as a challenge that sits at the heart of Samvera and many other OSS communities. Inconsistent resourcing has led to release delays, maintenance backlogs, delayed technical decisions, and burnout affecting the most dedicated developer volunteers.[20] One way the Samvera Community has approached this is by contracting with service provider Partners to help with maintenance and feature development. This happens in two ways, either through direct contracts or as part of contracts with adopters who require the service providers to contribute to community code. While Samvera Partners continue to identify a technical coordinator staff role as an unmet need that would help with this issue, the community has also tried new processes to manage this perennial challenge. Beginning in 2021, an annual pledge drive has asked institutions to pledge developer hours to maintenance work. This successful approach allows both institutions and product owners to predict when development resources will be available and how to make the best use of their time. Lastly, beginning in 2022, Samvera instituted two-year terms for product owners and tech leads to prevent burnout and ensure equitable distribution of community leadership responsibilities. Previously, community members would serve in these roles for years at a time, without recognition of the time and effort it takes to perform them.

Additional Challenges

The changes and evolution discussed in these four areas of governance, resource, community, and technology were not without their challenges. In terms of governance, even with thirty-two Partners in 2024, there have been challenges to maintaining a fully elected board. Board members sometimes move from their Partner institution to other jobs before the three-year term ends, and the practice has been to hold a special election to fill the vacancy. There have been multiple special elections over the past four years of having a fully elected board. As with most non-profit boards, engagement changes as dynamics shift depending on expertise, experiences, and personalities. Ultimately, the board is tasked with maintaining the sustainability of the community and is critical to ensuring that the community finds value in participation. Having an elected board requires board leadership and the community manager to directly invite potential board candidates and to make sure that individuals invested in the community are aware of the opportunity to run regardless of their job title at their institution.

In terms of both resources and community, there have been challenges with the Partnership model. Entering into a Partnership is time-consuming, while leaving only requires providing notice. Rather than casting widely with an annual membership drive, community leadership has to be in touch with newly invested adopters to get them started in considering Partnership. It is critically important that the most invested users are on the path to Partnership because other Partners may need to leave Partnership for any of a variety of reasons, which can significantly impact the budget. The community manager must develop relationships with all Partner contacts to remain aware of local circumstances. There are still some challenges to community participation that continue even with a full-time staff member and strong infrastructure. The bulk of the work of creating, documenting, and maintaining community work falls to volunteers, and the energy and availability of volunteers are subject to local institutional and individual needs and competing priorities. Often, more needs and ideas exist than there are available participants to address them.

Lastly, there are always additional challenges around sustaining technology. Throughout those initial years and growth as the Hydra Project (2008–2017), active development was not limited to those core components. Partners collaborated to develop specific initiatives and obtain grant funding. These grants resulted in a few institutions taking the lead to develop fully featured software applications to solve specific use cases, like creating access to time-based media and the creation of a cloud-deployable, multi-tenant repository that smaller and larger institutions alike could more easily adopt and manage as an "out-of-the-box" repository solution.[21] Grant-funded software initiatives meet goals but face sustainability challenges after funding ends due to changing institutional capacity and user needs. Partnership requires commitment to the community and software, but the terms of this commitment are not explicitly defined. This discussion of technical sustainability will never end and will be ongoing as Samvera persists.

Conclusion

The Samvera community's journey from the Hydra Project's inception in 2008 to a mature open-source initiative was made possible by a deep commitment to adaptive governance, sustainable resource models, inclusive community engagement, and responsive technological development. Each of these pillars has evolved through persistent effort to address challenges and enhance the community's resilience and impact. Other open-source communities can take note of Samvera's example in prioritizing these core areas to cultivate a strong foundation that can adapt to changing needs and scale over time.

Looking forward, Samvera stands at a pivotal juncture. By continuously evaluating its governance practices, the community can ensure it remains responsive to members' diverse needs while preserving a strong commitment to open, values-based collaboration. Developing

a robust framework for resources, including flexible financial models, will be essential to sustaining the skilled staffing and volunteer support that drive Samvera's initiatives. Open-source projects with similar goals can adopt Samvera's approach by creating scalable governance structures and inviting regular community feedback to evolve in response to growth and emerging requirements.

Equally, enhancing community access and participation through streamlined infrastructure and communication channels will strengthen Samvera's inclusivity and expand its influence. Other communities may benefit from investing in straightforward, reliable communication tools that foster active engagement, allowing contributors to collaborate effectively across diverse backgrounds and geographic locations.

Samvera's technical roadmap reflects both innovation and practical solutions, but achieving sustainable technology management requires ongoing investment in human resources and collaboration with service providers. Efforts like the annual developer hour pledges and the introduction of term limits for technical leadership help to distribute responsibilities, though a community-funded technical coordinator would be the most effective solution, and funding such a role remains a community goal. By embracing similar strategies, open-source communities can encourage shared responsibility, prevent burnout among volunteer leaders, and establish consistent development timelines that align with the community's evolving needs.

Samvera invites new institutions, stakeholders, and partners to actively contribute to this evolving project, building on the community's achievements while collaboratively shaping its future. Now more than ever, Samvera is poised to make a significant impact in the digital repository landscape, and community participation will be crucial to its continued success. For other open-source initiatives, fostering an open invitation to new collaborators and ensuring clear paths for participation can drive greater adoption and community growth, setting the stage for long-term success.

Acknowledgements

The authors would like to thank all current and former Board members and Community members who participated heavily in the stewardship of Samvera. In particular, the authors thank Rosalyn Metz and Chris Awre for reading a draft of this chapter and offering comments and edits.

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CHAPTER 5

Metadata Openness in Cultural Heritage Collections: Adapting Metadata Crowdsourcing for Transparency, Accountability, and Access

Jessica BrodeFrank

Introduction

The world is vastly different now than it was even five years ago, and though this chapter is written from a Western and mostly American perspective, impacts are seen throughout the globe. Access to cultural heritage collections changed from onsite exhibitions and reading rooms to online catalogs and social media. As society has evolved alongside racial justice reckonings, climate change, artificial intelligence, and ethical concerns,[1] the ways in which the best practices of describing collections have stayed regrettably stagnant.[2] Access to collections is a nuanced and multiplicitous problem, one that is beholden to the professionalization of language production. This professionalization leaves many questions, including: Who is included in the description process? What language is used to describe collections? How are collections described?

Examining who holds authority in the description process reveals a major issue in the professionalization of cultural heritage description. The cultural heritage sector, from its inception, has been grounded in an ambitious mission to preserve for posterity and for all of humanity.[3] However, in light of cultural heritage's expressed mission and purpose to preserve for everyone forever, this very advocacy for posterity means that heritage professionals cannot be the only ones to define and manage their collections.[4] As Graham states, "If heritage is so important, if it is about all of us and all dimensions of our lives, then it has become clearer and clearer that significance cannot be known without directly involving people who know and care about a particular place."[5]

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Moving from who is included in the description process, the question must also include the language used in the process. Access problems centered around language are symptomatic of larger issues related to the history of power in institutions, the history of whose story is recorded, and how it is recorded to obfuscate what is purposefully excluded.

Here, it is critical to introduce this larger question within the disconnect between the language of the professional and that of the public. Within cultural heritage, a disconnect presents itself in language choices from the use of specialized language, such as scientific names (*Rhopalocera* instead of butterfly), technical names (folio or volvelle in rare books), artistic names (tintype, melainotype, or ferrotype instead of photograph), or outdated language that has not kept relevant with the times and that may perpetuate racist, homophobic, or other problematic issues.[6]

Professionally created metadata establishes an empowered group, the metadata creator or academic researcher, while the public becomes disempowered due to a lack of knowledge and transparency of the metadata language and structure. A mutual understanding hinges on the use of normative codes for meaning-making that both the *creator* and the *user* can understand—a shared vernacular or transparency in the language selected.[7]

Further obscuring the breadth of access to collections is the way in which collections are described, not just in what objects and narratives are selected to be described, but also in *how* they are actually described. Cultural heritage professionals often focus on describing what an object *is,* versus what an object is *about.* This may seem like a matter of semantics, but it is not. Unlike text-based forms of information, cultural heritage objects and their images featured online have not traditionally been described according to subject matter—the *aboutness*.[8] Descriptions often focus on things like who made the piece, what it is made of, what size it is, when it was made, and what materials it is. Though helpful, this information does not help when a person is querying a database looking for a butterfly but the record is titled *Rhopalocera* or looking for a Civil War-era photograph that is labeled as a *ferrotype*.

Not only does the language of museums, the erasure of certain narratives, and other previously mentioned issues with bias impact discoverability, but this difficulty in finding what the public believes GLAM (galleries, libraries, archives, and museums) institutions hold also has further affected their trust in these institutions. David Nicholas has analyzed the ways in which the born-digital generations determine the trustworthiness of information on the internet:

> *They appear to rely on cross-comparison, asking friends on Facebook or Twitter, reading reviews, relying on Google. For some, the traditional view that organizations which have been established for a large number of years are inherently more reliable seems to work in the opposite way now. Moreover, younger people are inherently less likely to recognize traditional brands.*[9]

Nicholas's analysis is a concern for the cultural heritage sector that has often relied on name recognition and historical standing to elicit the public's trust: with the emerging generations, this institutional reputation will not be enough. This is of particular importance to cultural heritage institutions, many of which expressly state within their missions that they are committed to engaging the public with their specific brand of cultural heritage. By reimagining the description process as one not conducted solely by professional catalogers but one in which professional descriptions are enhanced by those of the public via crowdsourcing, it is possible to create an openness and a transparency that enhances discoverability, while exposing participants to the importance of language for search.

Description and metadata production can be incorporated into crowdsourcing most effectively when viewed as an engagement technique through public projects that transparently lay out the lack of accessibility in professional documentation practices and the semantic gap present, calling for users to help fill that gap and learn more about the collections and the bias of information retrieval sites in other parts of their lives.[10] This mentality of including communities is reflected in the emerging field of critical heritage studies.

Background

Critical Heritage

Beginning in the 2000s, critical heritage studies attempted to consolidate the disparate interests in heritage studies across libraries, archives, museums, archeology, and more. It describes a body of scholarship that moves beyond heritage studies as a field traditionally focused on technical issues and practice and into one that emphasizes the political, cultural, and social phenomenon of cultural heritage itself.[11] Critical heritage studies changed the emphasis of heritage studies from one of conservation and preservation of heritage to complex questions of the power that heritage entails and produces.[12]

During a time when there is unrest and introspection not only in academic communities but also coming from the public, the adoption of critical heritage studies has a clear moral impetus.[13] Following the rapid expansion of cultural heritage institutions that came after the 1972 World Heritage Convention, new industries and professions were spun up, including the professionalization of museum, library, and archival practices.[14] Many of the processes enforced by this professionalization focused on the top-down decision-making traditionally used in institutions that fuels inequalities and injustices by consuming and using cultural resources[15] of communities while also excluding these communities from their own heritage-making.[16]

By thinking of cultural heritage institutional mandates not just as preserving material culture or learning about the past, but instead as a tool to challenge contemporary issues,[17] it is possible to see how critical heritage studies can be used by institutions to directly tackle issues of transparency and trust by creating a critical consciousness that the public can engage with.[18]

Cataloging Standards

In the 2020 publication *Cataloguing Culture: Legacies of Colonialism in Museum Documentation*, Hannah Turner tracks the history of cataloging and of access to museum collections from the 1800s to today.[19] This work asserts that the decisions grounded in best practices of the past still affect tasks like naming, standardizing, classifying, and excluding.[20]

Turner documents the earliest forms of cataloging, referencing the lists and guides created by curators for use in the field during acquisitions and later used by cataloging teams divorced from the curatorial process. These guides emphasized attention to detail in the cataloging and labeling of objects because they were often the only written record for an object.[21] In creating these lists, there was an attempt to structure and standardize the field of collections cataloging, but this establishment of fields of information also worked to value some information over others, placing primacy on physical description over contextual information.[22]

As an example, pulled from Turner, museums may classify a guitar and a ceremonial Native American rattle as both being "musical instruments" because they have the capacity to produce music.[23] The narrow scope of this classification, in turn, denies the rattles' actual use as a ceremonial healing object and essentially dismisses the community and their practices, a dismissal that is then replicated and cemented in the catalog for posterity.

As collections increased in size, the catalogs became more valuable for research.[24] And as the second half of the twentieth century dawned, computerization in museums brought an increased awareness of the organization of information and its importance in facilitating research and access to collections.[25] Staff at institutions investing in computerization and digitization of card catalogs understood that too much information would lead to excessive record sizes, which would be insufficient for researcher needs and technological functionality, and thus they made decisions about what to include and what to exclude.

Owing to these technical difficulties and other constraints, including the cost of computerization, data storage, and staff time, institutions needed to determine what constituted the indispensable minimum of information that the catalogs had to include. This decision was

complicated because different users, like collections managers and curatorial research staff, had different needs with respect to the early computer tools, and different research questions and institutional uses required different kinds of information about objects to be recorded in the catalog.[26] As computerization led to databases, and, specifically, out-of-the-box databases like TMS (The Museum System), Emu, etc., it became acutely clear that the "boxes" and categories provided in the database were ineffective in holding the complex and robust information about collections, but as they were necessary components of the systems, most staff learned to work within the constraints by making difficult decisions on what is and is not included in the recordkeeping.[27]

In the early 2000s, these databases became online public access portals, in effect sharing access to collections with the public via the internet, though with little to no change in how this knowledge was documented from the days of the initial index card and descriptive catalog.[28] This made it possible to search large amounts of data on a collection from anywhere in the world, but for the most part, these portals adopted the same strategies implemented in the earliest catalogs, bringing with them the same choices, decisions, and biases of over 200 years of description and cataloging.[29]

This is an issue because cultural heritage public access portals are often built with search in mind. Much like one would use Google, these sites rely on the user to know what they want to find and provide a search term from which results are returned based on whether the staff added that exact language to items. Even though portals have added browsability functionality through clickable keywords or thematic carousels of objects, these are dependent on the use of catalog terms to facilitate this experience.

In 2021, in response to the digital pivot many institutions were forced to make due to COVID closures,[30] *Rolling Stone* magazine even took jabs at the inability of cultural heritage institutions (and museums in particular) to provide meaningful experiences online for their guests.[31] This example, in particular, shows how these conversations are moving outside the field and into the public purview, and it is no longer just an inability to meet guests' expectations online but also a realization by the public and the media that missing these expectations is akin to missing out on the mission-critical purpose of cultural heritage. With these expectations in mind, it becomes clearer that, in much the same ways language and control are imperative to equity, trust, and accountability for museums, archives, and libraries, the methods of the production for these collection descriptions are similarly imperative.

Public Trust and a New Definition of Museums

The American Alliance of Museums (AAM) Code of Ethics for Museums states that "Museums in the United States are grounded in the tradition of public service. They are organized as public trusts, holding their collections and information as a benefit for those they were established to serve."[32] Though this specifically references museums in the United States, universally, museums rely on the public and are one of the most trusted institutions in society; therefore, they need to maintain the highest level of accountability and transparency.[33] Many museums in the late 1960s to the 1990s redefined their role and image from a temple for passively viewing already-interpreted objects to a forum for interaction, learning, and open-ended interpretation.[34] This switch from expecting guests to be passive receptacles for knowledge to a place where knowledge was built in tandem was one of the earliest attempts to mitigate the loss of trust in the curatorially controlled narrative, similar to open pedagogy. If museums aren't seen as being responsive, then museums (and cultural heritage institutions in general) will lose not only trust but also relevance.[35]

In a digital space, cut off from the authority and reputation of the onsite institution, what it presents can garner as much distrust and present biases, but with the disadvantage of being removed from a real person to defend the issues, biases, or inconsistencies. On site, a docent could explain a specific curatorial decision or even steer a guest toward additional exhibitions when context is needed. Online, there is no one to facilitate this explanation if that information

is not included. As online spaces are designed for users, the ideas of trust must also be incorporated, perhaps the most important being "trustworthiness of the [information] depends firstly on the inclusion of all relevant information … disclosure of the underlying data and the designers' decisions about data, representation, and interaction can increase trustworthiness."[36] In this way, transparency is key, especially for institutions already losing trust due to singular authoritative narratives and perceived biases.

Right now, the public only sees a biased end product—the act of metadata creation and cataloging choices by invisible labor—leading to databases that can be difficult to navigate, with objects and narratives obfuscated by these choices. One way trust is being lost among the public is by a lack of transparency about how institutions work, why they work in these ways, and what their goals for this work are.[37]

The International Council of Museums (ICOM), in the wake of the 2020 dual upheaval of the COVID-19 pandemic and social justice movements, proposed a new definition of museums in 2022. The new definition of a museum now reads:

> *A museum is a not-for-profit, permanent institution in the service of society that researches, collects, conserves, interprets and exhibits tangible and intangible heritage. Open to the public, accessible and inclusive, museums foster diversity and sustainability. They operate and communicate ethically, professionally and with the participation of communities, offering varied experiences for education, enjoyment, reflection and knowledge sharing.*[38]

This new definition stresses a new aim for museums to facilitate diversity and sustainability, with the museum's mission and reason for being shifting from the previous definition's use of the word "study" to the new definition stating museums exist to be places for "reflection and knowledge sharing."[39] This subtle shift signals that the cultural heritage sector is moving away from a hierarchical position of privileged authority and toward a more level network of collaboration.[40]

By reimagining the cataloging process from one dictated solely by institutional authority to one of shared co-creation with the public, it is possible to create a process of description that also responds to the call of the public to make institutions more visitor-centered, places of enjoyment, leisure, and participation—even if it is impossible to reach every member of the public.[41] This will help meet what a digital-age audience expects and how institutions can better align their content not only for accessibility but also expectations, leading to a more transparent process of digital engagement whereby the discoverability of collections is aided by representative language added through transparent systems such as crowdsourcing.

Crowdsourcing in Cultural Heritage

As a new acceptance of history and culture as ambiguous, contested, nuanced, and open to change and contribution continues to spread in cultural heritage, the establishment of users adding language and ownership to materials should follow.[42] As Cameron and Kenderdine state, "Museums have been given a greater social mandate and responsibility to combat social exclusion, disadvantage, and discrimination, and to promote social inclusionist practices in all areas of museum work,"[43] and this has been key to the use of user-generated metadata in the past and is the key to its future uses.

In 2005, crowdsourcing entered the public zeitgeist when James Surowiecki published *The Wisdom of the Crowds*.[44] Within the introduction of the book, two points are made that are directly relevant. First, that relying solely on experts is a costly mistake, as often a crowd will include experts as well as individuals with novel perspectives, and second, that diversity and independence are important and integral to collective decisions as disagreement and contest fuel the best decisions, not consensus and compromise.[45]

In much the same way that a cultural heritage institution's authority is based on the trust of the public, so too is the "wisdom of the crowds" model. If the public does not trust that there

was a diversity of thought, an independence in working and thinking, and a decentralized system, they will not provide their honest opinions, nullifying the purpose of this model. Looking to science as one of the best examples of the "wisdom of the crowds" model of collaboration, it is important to note that this collaboration is imperative because of the specialized nature of scientists now, who often know one thing extremely well but find it difficult to know everything they may need for a specific project. In much the same way, specialization in cultural heritage means that a curator may not know the needs of the public in the ways an educator might or not know the terms being searched online, like a website manager might track. Collaboration, for both the scientist and the cultural heritage institution, allows an incorporation of many different kinds of knowledge and an ability to do so in an active and engaging way.[46]

The Spirit of Volunteerism

Though these terms took off in the mid-2000s, the roots of engaging with the public to enable research or production can be seen as early as the 1880s, when the *Oxford English Dictionary* invited educated members of the public to supply Oxford's lexicographers with variations in spellings, etymologies, and definitions, purely on a voluntary basis.[47] From citizen science projects in fields like ecology, conservation, and habitat studies in the seventeenth century[48] to cultural heritage institutions' similar use of volunteers through docent programs, participation by the community has been seen as integral to the work and continual growth of institutions for centuries.[49]

Trevor Owens of the Library of Congress states, "Most successful crowdsourcing projects are not about large anonymous masses of people. They are not about crowds. They are about inviting participation from interested and engaged members of the public. These projects can continue a long standing tradition of volunteerism and involvement of citizens in the creation and continued development of public good."[50] By focusing on crowdsourcing as a "set of processes, not just a set of outcomes, or new resources,"[51] it is possible to bring the same focus on engagement with volunteers online that museums create for in-gallery experiences.

In fact, this also falls in line with the ICOM's new definition for museums, referring to "democratising, inclusive and polyphonic spaces" responsible for safeguarding "diverse memories" through "participatory and transparent" practice.[52] It is time to begin changing the way these institutions work to deliver on this promise, creating transparent processes to expand and enrich metadata through crowdsourcing projects that particularly address the bias of cataloging and how language affects search across the public's daily use of the internet.

Opening Up Metadata Production

Early advocates for crowdsourcing metadata projects often saw the expansion of language in catalog records as the key to increasing discoverability. Proponents of these projects, specifically in the mid-to-late 2000s and early 2010s, advocated that each additional entry to the catalog, each new term, would bring in new and novel viewpoints. It was contended that there could be no superfluous or irrelevant metadata when serving a public audience—it was, in fact, a matter of context.[53] Alemu and Stevens[54] advocated that the quality of metadata should not be weighted only on its mere persistence to rules and standards but also on its effectiveness in making resources findable, accessible, and usable, aligning with the early goal of many projects: increasing access points to data through involving users as proactive metadata co-creators.[55]

Cho, Yeh, Cheng, and Chang noted in 2012 that museum professionals were designing websites that used controlled vocabularies unfamiliar to non-museum professionals, and that as discovery and retrieval were both the goals of museum professionals operating their websites and a demand of the wider internet audience, retrieval issues were a major problem of this semantic gap.[56] Whether viewing cataloging data, and metadata in particular, as a

tool for internal discovery and organization for staff or for external discovery and access for users, it's clear that the main purpose of metadata is to enable discoverability, and, as shown by Alemu and Stevens, this prioritization of discovery then means that metadata must be reflective and iterative in its service to discovery. Terminologies, language choices, and even context change over time, and therefore the metadata should be able to reflect these changes.[57]

Cultural Heritage Metadata Projects—Examples

One of the earliest examples of a cultural heritage institution attempting to decentralize its classifications using crowdsourcing in this way was the steve.museum.[58] The steve.museum, also known as the steve tagger, was a collaborative project started in 2005, eventually being awarded a $1 million grant from the United States Institute of Museum and Library Services. It saw staff from the Indianapolis Museum of Art, Guggenheim Museum of Art, Cleveland Museum of Art, Metropolitan Museum of Art, and San Francisco Museum of Modern Art working to explore the possibilities of user-generated descriptions of art.[59]

One of the main goals of the steve.museum project was to investigate the role of crowdsourcing in augmenting museum documentation with unique access points not found in traditional cataloging.[60] Results of the steve.museum project signaled that crowdsourcing could augment museum documentation with unique access points not found within the traditional cataloging structures, initially supporting a bridge for the semantic gap between cataloguers and the public.[61]

A 2009 analysis of metadata crowdsourcing projects by Jan-Erik Bråthen demonstrated a fundamental semantic gap between traditional classification and crowdsourced terms, and the ability of tags to bridge this gap when it was shown that 50 percent of tags entered in metadata tagging projects were deemed valid classification that had previously not been included in traditional metadata.[62] Susan Cairns believed that social tagging projects were a tactic to bridge the divide between experts and non-experts in the online museum space, increasing access to online collections and providing museum staff with insights into their users.[63] Mathilde Pulh and Rémi Mencarelli looked at crowdsourcing projects hosted in the 2010s at the Philadelphia Museum of Art and the Brooklyn Museum.[64] They overall saw the increase in access points and perspectives as an intuitive form of navigation, helping to support serendipitous discovery previously limited by curatorial voice.[65]

Max Evans believed that crowdsourcing of metadata was so effective in expanding access and discovery because it went beyond a semantic gap and addressed an aspect of description that cultural heritage staff could not do alone, that of granular item-level description. While archivists and catalogers are bound to classification styles and rules that look at minimal processing data of collections, volunteers and crowdsourcing participants had the ability to describe at an item-level, expanding and enhancing how collections were described and thus increasing their discoverability.[66]

Crowdsourcing as Participatory Experience

Today, there is an understanding that there is a marked value in the participation of the public in expanding access points, but, in many ways, the value is dependent on devoting proper staff and resources to guide and support the project participants.[67] By viewing crowdsourcing of metadata as a modification in the way institutions catalog, it is possible to again address the need to diversify narratives and contexts to address inherent biases that threaten trust and relevancy in museums while providing participatory and engaging experiences for the public that yield new metadata that can be used in tandem with the professionally curated metadata of the cataloger.

This reconceptualization of crowdsourcing as a participatory experience that expands diversity, representation, and context has also been advocated by Cameron and Kenderdine as a way for museums to fulfill their goal of enlarging audiences to include underserved

populations and novice learners by providing experiences that actually accommodate them in ways visiting galleries alone cannot.[68] This focus also demonstrates the reciprocal nature and benefits these projects should, and can, have.

Limitations

An important limitation and criticism of modern crowdsourcing in cultural heritage is that many platforms and audiences for crowdsourcing lack diversity in themselves. Demographics of users can be difficult to ascertain, as this information is not always readily available in statistics and often is collected by individual platforms voluntarily with participants; however, there are noted issues of these platforms being dominantly white, male, and highly educated spaces.

For example, in 2019, user demographics for the citizen science platform Zooniverse[69] depended on the subject type of the projects. Projects centered on astronomy saw a demographic split on average of around 30 percent female to 70 percent male, whereas projects centered on ecology saw a demographic split on average around 60 percent female to 40 percent male.[70] In a survey of citizen science projects run by the National Academies of Sciences, Engineering, and Medicine in 2018,[71] participants who reported their race showed a demographic breakdown of 88.6 percent white, 6.1 percent Hispanic, and 4.6 percent Asian, with no projects reporting majority participation by Blacks or African Americans or Indigenous people. Similarly, a publication in 2020 published by "Theory and Practice: Citizen Science" reflected that in a survey of the RiverWatch citizen science project, participants were "disproportionately white, highly educated, and affluent compared with the Illinois [site of RiverWatch] general population."[72]

A key limitation and critique of crowdsourcing metadata in museums is the need to attract and engage a large, diverse audience.[73] Institutions have experimented with various strategies to motivate diverse participants to participate in crowdsourcing projects and to publicize their collections as part of the transition toward engagement and participatory models.[74]

Motivation through Gamification

As a way to build engagement and incentivize participation by diverse audiences, many projects that launched in the 2000s and 2010s were centered around gamification.[75] As Jenkins, Clinton, Purushotma, Robison, and Weigel reported in a MacArthur Foundation report, play and gamification were deemed one of the best ways to create learning experiences, as no matter the frustrations or determination expended in the process, participants often only remember the fun.[76]

One of the earliest studies on user tagging of image-based collections was done by Von Ahn and Dabbish with the 2004 ESP Game, an application designed to encourage tagging of images through a gaming interface with the express mission to improve searchability on the web.[77] The ESP Game technology was deemed successful, with findings that 85 percent of the words for each image would be useful in describing it, and only 1.7 percent did not relate to the image at all. The ESP Game technology was later adapted to underpin the Google Image Search algorithms, further establishing interest in tagging by cultural heritage institutions.

In 2009, the Brooklyn Museum launched *Tag! You're It!* with the goals of making the data collected available to researchers and scholars, providing a way for a casual user to jump in and start visually navigating and creating an interface that would be mission-driven and community-oriented.[78] A secondary game called *Freeze Tag* was added to remove inaccurate or inappropriate tags created in *Tag! You're It!* But both games were discontinued due to low participation rates, hinting that gamification alone was not enough motivation for crowdsourcing project participation and success.[79]

From 2012 to 2019, Metadata Games, the National Standard open-source crowdsourcing game platform, was used with over forty-five collections representing eleven institutions. These forty-five collections consisted of tens of thousands of media items that generated over 315,000 tags, with even the least-tagged images receiving over 200 tags each.[80] The results of this

suite of games helped reinforce earlier steve.museum findings that gamifying tagging provided an initial motivation of fun, which participants flocked to. The project was run by TiltFactor at Dartmouth University. Project leads Mary Flanagan and Peter Carini pointed to issues found by the Brooklyn Museum's foray into gamification, namely that gamification itself was not the final answer to user motivation questions and concerns and that the true challenge involved in making a metadata game was incentivizing players in ways that encouraged participation and high replay potential while still creating accurate metadata tags.[81]

The draw that gamification of tagging projects offered to avid gamers and traditional participants of crowdsourcing platforms made it a particularly interesting evolution of crowdsourcing projects in the mid-2010s.[82] An array of projects outside of the Metadata Games suite also reported increased participation and metadata tag creation, including: Digitalkoot, a gamified crowdsourced project in Finland,[83] gamified design principles within citizen science platforms like Zooniverse.org,[84] and Game Jams[85] at the British Library.[86] These projects lauded gamification for the ability to take difficult and time-intensive tasks of traditional crowdsourcing and, in creating a more engaging and fun design, increase the rate of participation and number of participants involved.[87]

Possibilities of AI Integration

Whether due to the promise of machine learning* to automate the description process,[88] to provide simple subject level descriptions,[89] to provide semantic metadata for audio visual materials,[90] or to create a more robust, documented, and discoverable collection,[91] machine vision, automation, and artificial intelligence (AI) have become a more prevalent part of the conversation on crowdsourcing in cultural heritage in the last decade, with the majority of practitioners and scholars using these methods to replace human participation. Machine vision and AI are still rather new and novel in museum cataloging and metadata projects, but it has been deployed in multiple institutions so far. Several museums, including the Metropolitan Museum of Art,[92] the Barnes Foundation,[93] Massachusetts Institute of Technology,[94] Philadelphia Museum of Art,[95] and Harvard Art Museums,[96] have employed machine vision to analyze, categorize, and interpret their collections images, and although the application of AI to museums is still emerging, the results being reported show promise for enriching collections data across collections types.[97] However, limitations and subsequent critiques for the effectiveness of machine learning on its own have also begun percolating in the field.

As machine learning in cultural heritage advocates Villaespesa and Murphy noted, "At present, museums rely on third-party algorithms, or off-the-shelf tools to utilize these technologies from technology companies such as Microsoft, Google and IBM. These sophisticated computer vision tools have been trained using millions of images to create an algorithm that can identify visual trends and patterns."[98] However, though these platforms have created algorithms that can detect subject, color, and physical elements[99] well, they can still struggle with contextual information—the *aboutness* at the core need to include human participants as shown thus far.

Not only are these AI models limited in their ability to process complexity, but they are also still trained by humans based on cataloging practices of museums, which have noted issues with bias.[100] Padilla[101] and Rea[102] were early critics of the bias being trained into AI models and emphasized the need to recognize the continued marginalization this could have. Padilla warned in an OCLC Research Position Paper published in 2019 that the historic and contemporary biases in cultural heritage that had created the dominantly represented and underrepresented/marginalized communities could be perpetuated and amplified with the adoption of algorithmic methods.[103]

* Machine learning is a subfield of artificial intelligence that gives computers the ability to learn without explicitly being programmed. Machine learning starts with data—numbers, photos, or text—the data is gathered and prepared to be used as training data, programmers supply this data and let the computer model train itself to find patterns or make predictions. Human intercessions can help with specific task training. "Machine Learning, Explained," MIT Sloan, accessed November 25, 2024, https://mitsloan.mit.edu/ideas-made-to-matter/machine-learning-explained.

Machine learning and AI have the promise to enrich metadata and collections but also the threat to continue perpetuating biases clearly prevalent in cataloging by professional staff. However, machine learning and AI could be rethought, not as a replacement for metadata crowdsourcing projects, as traditionally framed over the last decade, but instead as a motivation and incentivization for crowdsourcing projects. In much the same way gamification has proven to be a motivation for engagement with metadata tagging projects, so too could machine learning and AI models be complemented, not supplemented, within these projects.

Conclusion

A key discrepancy in online public access portals being searchable and usable by the public is that the metadata they are looking for to search a collection by its thematic and contextual aspects is likely not a part of the professionally created metadata. When it comes to using descriptive metadata to discover cultural heritage materials, users want it all. Users want to search names by keyword, discover subjects by browsing, and browse by keyword or name, too.

It becomes clear that a reworking of metadata presented to the public versus that used by staff internally is not only needed to increase searchability but also to keep cultural heritage institutions relevant and necessary. Part of the solution proposed to bridge the gap should include the use of crowdsourced metadata.[104] By creating projects that transparently expose the public to the process of metadata creation, it is possible to not only increase access to cultural collections but also foster discussions around algorithmic biases, digital literacy, machine learning, and more culturally impactful issues that cultural institutions can help the public tackle as part of their missions to serve the public.

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CHAPTER 6

Open Skies or Merely Frequent Flyers? Chartering Open Opportunities for All

Amber Hoye, Sherena Huntsman, and Anders Tobiason

Frequent flyers (noun) are individuals who are regularly called upon to contribute to committees, projects, and decision-making processes within higher education institutions. They are valued by committee organizers and project managers for their demonstrated ability to show up and deliver on tasks and their experience working on other initiatives.

Introduction

In recent years, institutions have turned to DEI (diversity, equity, and inclusion) initiatives and programs to uplift underrepresented voices. However, they have often missed the necessarily expansive view of diversity needed and have often been pitted against dominant, traditional methods rather than working together to create new approaches. It is our goal to use an expansive definition of DEI to inform how we approach the value of open movements and, more explicitly, open education.[1] We approach this DEI perspective in its broadest sense, including diversity of languages, locations, ages, abilities, economic backgrounds, expertise, position, experience, and positionality in addition to race, ethnicity, and gender. With this more expansive definition of DEI in hand, we seek to recognize whose representation and perspectives are often left out of open initiatives and programs across our campuses and create environments that invite and welcome them to the discussions and decision-making.

This broader understanding of diversity, paired with the values of open movements, forms the necessary foundation to promote an authentic culture of openness. This foundation generates genuine collaboration and transparency in ways that value the mixing and remixing of understanding, perspectives, and knowledge to create fresh insights. However, these practices often come into conflict with institutional inertia and power dynamics in higher education contexts.

Open movements demand different voices, but traditional systemic methods embedded into the fiber of higher education support, sometimes invisibly, the use of frequent flyers. Frequent flyers are individuals who are frequently asked to participate in open work throughout

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institutions; they are valuable because of their experience and knowledge, but using these flyers also creates limitations to an open movement (see figure 1). Frequent flyers are the path of least resistance because they have institutional knowledge and experience that expedite program development, even if that development is built with the same bricks and mortar that has worked to exclude others. This is why open movements require the use of the intentional inclusion of different perspectives, knowledge, and understanding.



Figure 1. The value and limitations of using frequent flyers in an open movement. ***Frequent flyers*** are individuals who are regularly called upon to contribute to committees, projects, and decision-making processes within higher education institutions. They are valued by committee organizers and project managers for their demonstrated ability to show up and deliver on tasks and their experience working on other initiatives. When filling roles for a committee or starting a new project, it's easy to think, "I had a great experience working with x, let's bring them in again!" However, this convenience and perceived lack of risk directly conflict with the goals of open movements, which seek to eliminate barriers to the creation, adaptation, and sharing of knowledge. When we invite the same individuals time and time again, we smother innovation by excluding new ideas as well as diverse voices and perspectives. Further, relying on these same individuals to participate over and over can lead to overtapping and, worse, cause burnout.

In this chapter, we discuss the current environment of the open education movement and illustrate a gap between observable applications, including current scholarship, and the effective application of open principles. From the perspectives of an academic librarian, faculty member, and instructional designer, we explore how we might expand opportunities to include voices across the academic environment in the work of creating and supporting open education initiatives on college campuses. We conclude with a description of recommendations that individuals and administrators may use to progress the application of open principles across institutions.

Open Education Initiatives at Higher Education Institutions

Seaman and Seaman's 2023 report provides detailed data on the faculty awareness of open educational resources, surveying 2,483 faculty and 641 administrators from 1,427 institutions across all fifty states and the District of Columbia. The majority of faculty are aware of open educational resources (OER), with 64 percent of instructors familiar with OER and over 50 percent of faculty actively using OER materials in their courses. However, many may still

confuse the values and concepts of "open" with those of "free" materials. Nonetheless, OER use is no longer considered fringe and has become mainstream. It is here to stay, making it crucial to unite all voices across our campuses to create, support, and sustain open education initiatives.

A brief review of the literature surrounding open education reveals a plethora of articles about how Open benefits students, but a dearth of articles addressing how to successfully create an open education initiative that includes diverse voices. While the few articles that exist on developing open educational initiatives do briefly mention how they formed campus-level committees, no attention is paid to how this formation often is ad hoc, exclusionary, or less than open.[2] The assumption is often that there is a group of librarians, maybe someone from the provost's office, often someone from the bookstore, and some folks working in information technology, who form the core of the committee. Just exactly how this committee is formed is often described as voluntary or appointed, but the details are rarely discussed.

One article that specifically mentions how to deliberately craft committees for inclusion is "Reimagining Leadership in Open Education: Networking to Promote Social Justice and Systemic Change."[3] Cangialosi et al. write that "so that many of our incipient leaders, like the students we wished to serve, were also people with lived experiences of being underserved and underrepresented, we worked to recruit participants especially from Hispanic-Serving Institutions (HSI), Minority Serving Institutions (MSI), Historically Black Colleges and Universities (HBCU) and from institutions with lesser economic resources. We also built our advisory team with an intentional focus on diversity and inclusivity."[4] While campus-level task forces will be different, the lessons here are that those in leadership positions should look to deliberately include members of underrepresented groups as well as look to non-traditional parts of campus to make sure individuals with diverse lived experiences and non-traditional groups are included in this work. The key takeaway is that the formation of the committee must be deliberate and must specifically strive to include all voices.

The other major gap in the literature is that there is no discussion of how the leadership or formation of committees functions in the context of other major front-facing initiatives at the university. Open education initiatives and affordable learning materials task forces are often written about as if they occur in a vacuum.* When partnerships are mentioned, which many articles discuss as important for success, these partnerships are specific to open educational resources. In other words, how the power structures on campus impact open educational initiatives is not unpacked. Morgan et al. write about how institutional culture and leadership are intertwined and suggest that there can be pitfalls regarding the leadership of open educational initiatives when they run into institutional cultural barriers, but they do not delve much into how to mitigate this or how to ensure that the information and leadership structure do not become siloed.[5] For our purposes, siloed means information that is held by a small group of individuals that is not shared outside of selected other individuals. This information or leadership structure is not made public; to find out who is in the leadership structure or who has the knowledge, one must know who to ask. There is little mention in the literature about how leaders are selected, about who they are in terms of campus culture, or about how they can really influence (both positively and negatively) how open educational initiatives are perceived by other potential stakeholders on campus. Many articles show that grassroots efforts at OER promotion are often the beginning of open educational initiatives being formed, but there is little discussion of what happens after the structures become more formalized.[6]

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* Open education initiatives and affordable learning materials initiatives are related but not exactly the same. Open educational initiatives are often part of affordable learning materials initiatives, but affordable learning materials initiatives often also include relying on or promoting materials that are not open. These non-open educational materials include library-licensed materials, inclusive access bookstore materials, and instructor-created materials that are not shared openly. The goal of open educational initiatives is broader than affordable learning materials initiatives (whose goal is only cost-savings), but they often get lumped together. This can contribute, especially on campuses where this occurs, to the sense that the only purpose of open educational resources is to save students money. This does a disservice to the many other benefits (e.g., diverse perspectives and up-to-date information) students using OER experience.

Table 1 details the current landscape for open education or affordable learning initiatives, associated groups and committees, and available support grants for ten peer institutions, as listed on the Boise State Peer Institution page.[7] Frequent flyers appear in many of the campus committees, with many institutions adopting closed committees and appointed positions that limit participation. As a potential alternative to these restrictive groups, several peer institutions (including Boise State, Cleveland State, Portland State, The University of Texas at San Antonio, and Wichita State) are turning to state-led groups that can promote a wider exchange of ideas and practices across institutions. Many campus-level initiatives are hosted by libraries (including Cleveland State, Portland State, University of Texas at San Antonio, University of Nevada- Reno, Wayne State University), while others are led by campus administrative units and centers for teaching and learning.[8] While many initiatives include other units as campus partners, there is a lack of representation from groups like student government, cultural or diversity centers, gender equity centers, first-generation student programs, international student services, educational access centers, academic coaching and educational support, English language support, veteran services, career services, and more. For example, student government representatives could advocate for student concerns, first-generation student programs could help identify barriers faced by students without a family history of higher education, and cultural or diversity centers might highlight the importance of culturally relevant resources.

Peer Institutions	Open Education or Affordable Learning Initiative	Host	Other Partners	State-Led Group	Campus Committee or Group	Open or Closed Participation	Support Grants	Grant Restrictions
Boise State University	Yes	Provost Office	Library, Center for Teaching and Learning, eCampus, Learning & Technology Services, Bookstore	Yes	Yes	Committee: Closed, State-led group: Open	Yes	Yes, available only to faculty teaching online courses
Cleveland State University	Yes	Library	Center for eLearning, Center for Instructional Technology and Distance Learning, and Center for Faculty Excellence	Yes	No	Open	Yes	Yes, visiting professors and graduate students not eligible
Indiana University Purdue University – Indianapolis	No	N/a	Library	No	No	N/a	Yes	No
Portland State University	Yes	Office of Academic Innovation	Library, Bookstore	Yes	Yes	Committee: Closed, State-led group: Open	Yes	No
The University of Texas at San Antonio	Yes	Library	Academic Innovation, Office of the Registrar, Student Government, Campus Bookstore, Advising	Yes	No	Open	Yes	No
University of NV – Reno	Yes	Library	Office of Digital Learning, Nevada Online	No	No	N/a	Yes	No

Table 1. The landscape for open education or affordable learning initiatives, associated groups and committees, and available support grants for ten peer institutions. Collected June 2024. [9] (*continues*)

Peer Institutions	Open Education or Affordable Learning Initiative	Host	Other Partners	State-Led Group	Campus Committee or Group	Open or Closed Participation	Support Grants	Grant Restrictions
University of South Alabama	No	N/a	Library	No	No	N/a	No	N/a
University of Wisconsin – Milwaukee	Yes	Center for Excellence in Teaching and Learning	Library	No	No	N/a	Yes	No
Wayne State University	Yes	Library	Computing & Information Technology, Office for Teaching and Learning	No	Yes	Closed: working group of librarians	Yes	No
Wichita State University	Yes	Academic Affairs	Library	Yes	No	Closed: university-appointed representatives	Yes	No

Table 1. The landscape for open education or affordable learning initiatives, associated groups and committees, and available support grants for ten peer institutions. Collected June 2024. [9] (*continued*)

Reflecting on Personal Experiences

We offer our experiences not as complaints or simple observations of what is missing, but as a springboard into how we might work around the systemic glorification of expediency in order to promote a culture of openness in our educational communities.

Amber Hoye, Instructional Designer

My open education journey began in 2014, collaborating with faculty to develop textbooks using iBooks Author. While this early work exploring fair use and OER principles felt rewarding and purposeful in reducing textbook costs and empowering faculty, it was also isolating as my colleague and I experimented with new tools and concepts. Later, after launching a larger initiative to develop openly licensed materials with students and faculty, the eCampus Center's OER coordinator reached out about our work and introduced me to the campus Open Education Group. Despite working with OER for over a year, this was my first connection to the community. Joining the group brought welcome collaboration and a sense of community, yet I couldn't help but wonder why it took so long to find them.

I suspect that the group initially invited the same "frequent flyers" from various campus groups, assuming word would spread over time organically. However, relying on limited communication channels often leaves out those who might benefit most. The Open Education Group was open to everyone, but the chances to find out about it were limited to those who stumbled upon it. Adjunct instructors, new instructors or staff members, or students—individuals with unique perspectives—may have never heard about the group at all. Eventually, meeting information was added to the campus OER webpage, and attendance grew, but the question remained: who else might still be missing?

Embracing open means more than making resources available; it means opening the door to others who might find the same connection and meaning in this work, bringing open education to them, not waiting until they find it. This requires hosting workshops, distributing information in common spaces, and intentionally collaborating with diverse campus groups. It also means celebrating the many reasons why we value open education, from reducing costs to fostering creative autonomy for instructors.[10]

We know that bringing like minds together to support and collaborate is a best practice, yet at large institutions, where departments or units operate in isolation from one another—often referred to as "siloed"—finding that community can be particularly challenging. To address this, we need structured and proactive strategies to ensure equitable access to participate in open education initiatives. Sage Crump, a culture strategist, fittingly states that "nothing happens in isolation. There is always a squad, collaborators, a body that supports change occurring."[11] Making an impact requires building that squad—intentionally and inclusively.

Later, as the leader of the Open Education Group, I took steps to broaden our reach by merging the group with the larger state-led Idaho Open Practitioners, Advocates, and Leaders (OPAL). This move connected faculty, staff, and students across institutions, amplifying diverse voices and fostering statewide collaboration.

Through this work, I've learned that while OER may be free to use, it is not free to create. If we want to increase the diversity of voices present in our discussions and decision-making, we must provide sustainable paths forward. As a professional staff member, my involvement was often considered "off-the-desk work"—not part of my official responsibilities. Instead, I creatively integrated it into my existing role or took it on as an additional commitment. This highlights the need for clearer institutional support and recognition of this work. While my department had grown increasingly supportive of open education, not everyone has this advantage. How can we support staff members wanting greater involvement, junior faculty needing tenure-worthy projects, adjunct instructors who do not have service as part of their role, and student creators with valuable perspectives?

To truly foster a thriving open education community, sustainable funding and institutional recognition are essential to address these challenges. A frequent trend at my institution, and others, was the reliance on one-time funding opportunities. While well-intentioned, they fail to account for the ongoing care that open education projects need to thrive. Without sustained, recurring support, we risk limiting the diversity of voices represented. We need long-term funding models that provide equitable support for all participants—ensuring projects can evolve, scale, and continue making an impact over time.

Sherena Huntsman, Assistant Professor

As a faculty member, my experience with open movements is strongly connected to the classroom and consideration of access to classroom materials specifically. For me, open is a lens to investigate how the classroom environment is welcoming and inviting to students from all identity backgrounds and across the ability spectrum. This lens asked me to consider how materials are accessible and representational of any student.

I first tiptoed into what "open" means as I developed an OER textbook for my department's introduction course. Because all of our majors and several majors across the campus required this service course, the impact of an OER textbook would be significant. I participated in a small institutional grant process to obtain additional funding to support my effort. What I found in this journey was a lack of fully open materials available. Although the materials were open for use without cost, they were closed in many ways to various voices because the materials lacked attention to digital accessibility features and representation of a variety of student identities and experiences. Without additional support from my institution, I was left to my own limited experience and resources to integrate more authentic open materials that are legally and ethically accessible.

Because of this experience in my classroom, I was appointed to a task force assigned to consider the affordability of course materials across campus. As a committee, we discussed this assignment as an opportunity to recognize Open more broadly than just the consideration of cost as a barrier. I was specifically assigned to a subcommittee tasked with including potential ways the institution could promote the use of legally accessible course materials. We investigated and proposed various methods the institution could use to build accessibility training, accessibility monitoring, and methods for incentivizing attention to access through the tenure and promotion process. Finally, the task force compiled a plan that included these methods for considering how course materials could promote a culture of openness across instructional environments. After the plan had been endorsed by the full committee and the faculty senate, the provost reassigned the plan to frequent flyers, the same small group of individuals who were most often appointed to decision-making committees, and the plan was reduced to a mandate to track only the cost of course materials, within the purview of this same small group of individuals. The variety of different voices and the perspectives amplified through the original plan were easily dismissed because the institution wanted to expedite the process. Accessibility and other considerations that might have established an open view of course materials were relegated to the cutting room floor, systematically removing the welcome mat for different voices.

As a faculty member, it is my responsibility to ensure my students have opportunities to encounter and to participate in open environments, including through the materials I bring to class. It is also my responsibility to question policy and to question decision-making voices. I may not have the power to broadly change the whole institution, but I can impact my office colleagues, my committee members, and my community partners.

Anders Tobiason, Assistant Professor/Librarian

So, what happens when you end up being a potential frequent flyer? My personal reflection will explore how this happened to me, some lessons learned for folks in this kind of position, and what I think can be done to mitigate the harms of a frequent flyer model.

My background as a librarian is in critical information literacy studies. I work with multimedia development and the user experience of our online spaces and I am part of the Instruction and Research Services unit. Open educational resources appear nowhere in my job description, shared job duties, or anywhere within the scope of my position. And yet, I have ended up working with OER in direct and tangential ways.

I seem to have gotten asked to be a frequent flyer for no good reason that I can think of, other than maybe being in the right place at the right time and saying something about an interest or commitment to open principles.† I was initially part of the open educational resources initiative at my current university because I was simply willing to help out. But when we lost our OER librarian, my willingness to help out turned me into a potential frequent flyer. I have resisted becoming a fully fledged frequent flyer in the open educational space on our campus because I know there are others far more qualified who have been left out of the conversation. Being asked to be a frequent flyer is a very odd place to be when you are fundamentally critical of the institutional gatekeeping that is why frequent flyers exist in the first place.[12] I sit with a fundamental paradox: How do I try to do good work, while also trying to resist the system that put me in that position?

I have been a full-time librarian at two institutions and have seen committee work at the institutional level work very differently. At my first university, a strong union and very strong faculty ensured that what one was doing was kept truly within the scope of one's job description. Now at my second, the opposite is nearly true. Without a strong union or a history of a strong faculty, job scope creep is very real. Thus, frequent flyers are often operating outside of the defined scope of their job. I know I certainly am. This goes for other initiatives as well, but it is striking to see it in open educational initiatives.‡ By not making membership explicit, using job descriptions or titles, individuals with the correct political connections can form the core of the initiative.

The ad hoc nature of this type of work ends up lending itself to gatekeeping and political fiefdoms. In formal committees, membership is set by rule, by volunteering for an open position, and through explicit means.§ By keeping affordable learning and open educational committees ad hoc, the rules for who can be included are vague at best. At my institution, ad hoc is a pretty good term to use for the campus-level initiative that encompasses open educational resources. And, as the ad hoc definition from Baylor notes, the rules for who is on the committee are set by the committee itself.[13] A prime way to make sure frequent flyers rule the roost.

As someone who could potentially be a frequent flyer, I feel it is my responsibility to say no and to suggest others who may not have been invited by someone from the inside. I am privileged, and this is a call for people like me—potential frequent flyers without this in their job scope, job descriptions, etc.—to check that privilege at the door and ask who isn't getting a seat at the table.

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† I seem like a safe bet to include, at least on the surface, because I am a cisgender, credentialed white male in academia. But my goal in my scholarship, service, and teaching has been to use this unearned privilege to promote critical librarianship, diversity and equity, and work to dismantle systems of oppression.

‡ Other initiatives often include things like student success initiatives, EdTech initiatives, etc. The difference being that usually at least a few folks have this kind of work directly in their job descriptions. But the gatekeeping still often applies. What is striking in open initiatives is just how much this flies in the face of the supposedly open ethos of inclusion, diversity, and sharing.

§ I find Baylor University's definitions of these different kinds of committees, Standing and Ad Hoc, helpful in understanding the difference. But one also immediately notices that things like affordable learning committees really should be standing committees, not ad hoc committees. "Purpose," Baylor University, https://university-committees.web.baylor.edu/purpose.

A Call to Action

Management and Administration

Managers and administrators have incredible power to shape how inclusionary or exclusionary an open initiative can be. They often mandate that these initiatives be formed, but rarely do they use their authority to ensure that power structures on campus are not reinforced through who gets to participate. The following are a set of recommended actions that leaders can take to break down those barriers to inclusion and dismantle problematic power structures to create more "open skies."

- Create open calls for participation and circulate them widely.
- Remove barriers to participation through compensation, recognition in tenure and/or promotion, time release for staff, or other incentives.
- Ensure that every role has an expiration date and a clear method of appointment or re-appointment.
- Consider varying meeting times and provide flexibility for how members attend (face-to-face or virtually).
- Write Open work into job descriptions instead of assuming that it can fall into "other duties as assigned."
- Write all of the above into policy so that these actions are difficult to dismantle by future administrations.

Actions as Individuals

As individuals, we may not always have the explicit power and positionality—unless we are frequent participants in decision-making processes—to directly create a more inclusive environment for diverse voices. However, we can still find small ways to reconstruct and reshape institutional processes from within (see figure 2). Reconstruction begins in our classrooms, offices, and labs, and then extends beyond our immediate sphere of influence. Whatever committee we sit on, whatever voice we are given, it is our task to question the decisions and the decision-makers. We question the actions and the policies. When given a microphone, we share it. We disrupt our spaces to create a place for the silenced to be included and heard. The actions listed below offer individuals, regardless of rank or positionality, opportunities, even in small ways, to open space for others to be heard.

- Intentionally develop ways for our peers to actively engage with open, accessible materials.
- Ensure diverse representation across our committees by asking who is present and being heard, and who is not present.
- Pay attention to who is not speaking/participating in our committees and make space for them by encouraging and valuing their experience and expertise.
- Question how open movement policies impact across identity backgrounds.
- Expand our circle of influence and broaden our own exposure and understanding of open movements across our communities.

Creating a culture of openness requires individuals, administrators, and institutions to embrace difference and welcome missing or underrepresented voices. By moving beyond the concept of simply selecting "frequent flyers" and instead providing "open skies," our committees, groups, and meetings must actively seek, invite, and reward participation from people with diverse expertise, positions, and experiences. We should also allow frequent flyers to step back or suggest others for roles. Rooted in open principles, this commitment to inclusion will both enrich open movements and create a more sustainable and vibrant future.



Figure 2. A reference card of questions and suggestions for frequent flyers attending meetings, committees, or groups.

Notes

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PART 2

EXPLORING THE STRUCTURES THAT SUPPORT OPEN

CHAPTER 7

Open Access as Label and Rallying Cry: A Framework to Unify Two Senses of OA

Colleen Cressman

Introduction

Open access is both a label and a rallying cry. As a label, it minimally refers to the removal of price barriers from digital online scholarship. Often, though not always, permission barriers are also lowered or dropped.[1] When price barriers are in place, access to scholarship is conditioned on payment in some way. Academic journals, for example, have traditionally been made available to individual readers or their institutions upon payment of subscription fees. So, removing price barriers entails removing such conditional payments as subscription fees or other consumer costs. When there are no so-called paywalls regulating this reader-side access, anyone can read the scholarship free of charge.* Permission barriers, on the other hand, pertain to activities that require the appropriate rightsholder's approval to overcome. These activities include the public display, redistribution, reproduction, and adaptation of the scholarly work beyond what would be allowable under fair use or another applicable copyright limitation or exception.[2] Removing permission barriers, then, enables anyone to reuse the scholarship in the allowable ways, beyond fair use, and without need for further approval. Since these barriers are multifold, however, their removal can vary in kind and by degree.[3]

As a rallying cry, open access signals a commitment. The Budapest Open Access Initiative calls itself "a statement of principle, a statement of strategy, and a statement of commitment."[4] Similarly, the Bethesda Statement on Open Access Publishing, and the Berlin Declaration

* Publishers have increasingly started to introduce fees to the author side in the form of article processing charges when providing publisher-mediated open access. See Leigh-Ann Butler et al., "The Oligopoly's Shift to Open Access: How the Big Five Academic Publishers Profit from Article Processing Charges," *Quantitative Science Studies* 4, no. 4 (2023): 778–99, https://doi.org/10.1162/qss_a_00272; Shaun Yon-Seng Khoo, "Article Processing Charge Hyperinflation and Price Insensitivity: An Open Access Sequel to the Serials Crisis," *LIBER Quarterly: The Journal of the Association of European Research Libraries* 29, no. 1 (2019): 1–18, https://doi.org/10.18352/lq.10280. Importantly, however, open access publishing need not require any author- or reader-side fees. See Jeroen Bosman et al., "OA Diamond Journals Study. Part 1: Findings," Zenodo, March 9, 2021, https://doi.org/10.5281/zenodo.4558704.

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on Open Access to Knowledge in the Sciences and Humanities have each made the case for open access as important, necessary, and enabling a public good.[5] Just so with the newer Delhi Declaration on Open Access and São Paulo Statement on Open Access, which resolve respectively "to accelerate the progress of research and to address real societal challenges" and to "provid[e] universal, unrestricted, and immediate Open Access to scholarly information, including use and reuse by humans and machines."[6] Rights retention-style open access policies adopted by institutions worldwide can express commitments, too. Harvard's policy, for example, cites a commitment to "disseminating the fruits of its research and scholarship as widely as possible."[7] Commitments to open access, by word and deed, also figure into the open access publishing landscape among editors, publishers, networks, and indexes. The Directory of Open Access Journals (DOAJ), for instance, "is committed to being 100% independent and maintaining all of its primary services and metadata as free for everyone."[8] There are many more examples within open access and other realms of open; these are just a handful that have endured long-term or have come about more recently.[9]

My aim is to unify the label and the rallying cry under a single framework. In doing so, I draw on Bernard Suits's analysis of gameplay in the philosophy of games. Although open access and its associated scholarly activities are decidedly *not* games, the way Suits defines 'playing a game' will be useful within our very different domain. Just as Suits's definition consists of three elements, so too does the framework I propose. The first two elements—the goal and the rules—yield the scholarship to which we apply the open access label. The third element is a commitment. When combined, all three elements describe open access as the rallying cry, or what it is to engage in persistent, intentional activity to bring about open access scholarship.

This three-element framework brings together the label and the rallying cry without conflating them; it demonstrates how the two senses of open access are related, due to the interrelatedness of the elements themselves, while preserving their essential differences. Finally, the framework reminds us why fostering a commitment—adopting the rallying cry—is so important: It's not the case that open access would otherwise be too elusive or fleeting. On the contrary, the rallying cry matters precisely *because* the label is now so attainable, enduring, and embedded in scholarly practices. In other words, open access is ubiquitous and resilient, not rare and fragile, and a commitment keeps it in check. For the value of open access lies in the way we bring it about, not simply in our ability to apply a label to scholarship.

The Label

Consider the definition of playing a game. Imagine you are downhill skiing or playing charades, Tetris, or basketball. Does playing a game require some level of sustained physical exertion, as with carving a turn on skis, pantomiming a clue, rotating a shape by remote control, or dribbling a ball? Perhaps gameplay entails the exercise of skill, physical or strategic. Or maybe playing a game involves competition between players, from the amateur to the professional. Yet, the *way* we play games—effortfully, skillfully, or competitively—doesn't tell us what gameplay *is*.

Bernard Suits offers a definition of gameplay whose stepwise structure is instructive for analyzing open access, even though open access is not an example of playing a game and does not involve gameplay at all. Suits's definition is both simple and robust. It is simple because it comprises just three elements: a goal, rules, and an attitude toward the rules.[10] It is robust because it captures the rich variety of these activities. According to Suits, playing a game is to

> *engage in an activity directed towards bringing about a specific state of affairs [goal], using only means permitted by rules, where the rules prohibit more efficient in favor of less efficient means [rules], and where such rules are accepted just because they make possible such activity [attitude].*[11]

Beginning in this section, and continuing in the next, I unpack Suits's definition element by element. In tandem, I present a modified version of this structure that distinguishes the two

senses of open access, the label and the rallying cry. Below, I introduce the first two elements, the goal and the rules, which together constitute the open access label.

Achieving the Goal

It is natural to think the goal of playing a game is just to win. Suits takes a different tack. He distinguishes the in-game goal of winning and the out-of-game goal. The out-of-game goal comprises the first element in the definition of playing a game. It does not refer to winning or, indeed, any aspect peculiar to gameplay at all. As Suits describes it, this goal is simply "a specific achievable state of affairs." Since this goal is not internal to the activity of gameplay itself, "it can be described before, or independently of, any game of which it may be, or come to be, a part."[12]

Imagine again that you are downhill skiing. The out-of-game goal of the activity is to *travel from the top to the bottom of a slope*, whereas the in-game goal is to *ski down a slope, following a marked course, to get the fastest time*. With the out-of-game goal, the way you bring about the state of affairs is unspecified; there is no mention of the mode or means and, likewise, no mention of what would constitute winning. Instead, what *is* specified is a basic success condition: to get from the top to the bottom of a slope. As someone who doesn't downhill ski, I can achieve the same out-of-game goal just by virtue of walking downhill as part of a trip to the store. Yet, when I go out for groceries in my hilly neighborhood, I am neither skiing nor playing any sort of game. So, the goal alone is insufficient to distinguish playing a game from engaging in some other type of achievable activity.

This level of generality is deliberate, however. By adopting the out-of-game goal, Suits does not foreclose what *could* count as playing a game in the first place. After all, "any achievable state of affairs whatever could, with sufficient ingenuity, be made the goal of a game," Suits says.[13] Constraints will come later, with the remaining two elements in Suits's set, so that all three work together to give the definition its shape.

Let's now turn to the goal of our domain of interest. Adopting Suits's formula, it too is an achievable state of affairs, albeit narrower than the corresponding goal of gameplay.† Call this goal ACCESS:

Goal (ACCESS): Make digital scholarship available online.

ACCESS need not serve as the *ultimate* goal in a succession of activities; there may be goals to achieve beyond it, and ACCESS may serve as a means to those end goals. For example, a medical researcher who posts a dataset in an online repository has achieved the goal of making the scholarship available online. But she may also have further and loftier goals she hopes posting the dataset will help to achieve, such as promoting the progress of global scientific knowledge and contributing to advances in medicine and thus the welfare of future patients. Just so with any goal that satisfies the parameters of making digital scholarship available online, although the further goals may differ.‡

ACCESS is also general enough to accommodate a variety of scholarly activities—spanning articles, books, datasets, educational resources, and everything in between—as long as those activities involve making digital scholarship available online in some way. Notice it makes no mention of price or permission barriers, meaning ACCESS is independent of open access itself. I could achieve ACCESS, for example, simply by placing an article on my website—and charging a fee to all those who try to read it. Conditioning the availability on the payment of a consumer fee, however, is disqualifying for *open* access even though it is consistent with

† It is possible Suits's broader and purely structural goal could be more fitting here if it were important the definition of open access encompass activities beyond those pertaining to scholarship or the digital, online environment.

‡ This formulation is consistent with the Budapest Open Access Initiative's "20th Anniversary Recommendations," which makes the point explicitly: "[Open access] is not an end in itself, but a means to other ends."

ACCESS.§ Given that it covers both open and non-open access scholarship, ACCESS alone is not enough to define what open access is. Enter the rules.

Following the Rules

With the introduction of the rules, Suits starts to narrow the definition of gameplay. The rules, according to Suits, function to "prohibit use of the most efficient means for reaching [an out-of-game] goal."[14] In this way, the second element refers to the first: the rules aim at the goal, and the goal is constrained by the rules. Return to the example of downhill skiing. To get from the top to the bottom of a slope, the rules forbid using a snowmobile instead of skis and skipping gates instead of following the course.

"Once these rules are in place," notes Thomas Hurka in his commentary on Suits's analysis, "success in a game typically requires achieving the [in-game] goal as efficiently as they allow."[15] Compare the sluggish skier and the swift skier. Both travel down the slope on skis, do not miss any gates, and cross the finish line. So, both skiers bring about the same state of affairs, and both take the less efficient means as required by the rules. Yet, the swift skier is more successful because she goes faster, and part of how she goes faster is by carving her turns more efficiently, not less efficiently, than the sluggish skier. Thus, the rules are a constraint, but they do not *rule out* the effortful, skillful, or competitive ways players operate within their bounds in order to win.[16]

Likewise, the second element in our own framework covers the rules. When paired with the goal of ACCESS, these rules—call them OPEN—make possible the open access label:

> **Rules (OPEN):** Achieve the goal by removing either price barriers or price and permission barriers.

The rules are not a surprise. At the same time, they depart from the Suitsian rules, in that they do not limit the efficiency by which the goal may be achieved. It is worth discussing in greater detail what the rules describe and how the components are related. Price and permission barriers are often found together, but they do not perfectly overlap. Peter Suber presents the terms 'gratis' and 'libre' to sharpen this essential distinction:

> *Gratis [open access] is free of charge but not more free than that. Users must still seek permission to exceed fair use....Libre [open access] is free of charge and also free of some copyright and licensing restrictions. Users have permission to exceed fair use, at least in certain ways. Because there are many ways to exceed fair use, there are many degrees or kinds of libre [open access].*[17]

When digital, online scholarship removes only price barriers, it is gratis open access. And when it removes those barriers, in addition to at least some copyright and licensing restrictions, it is libre open access. In the above example, I uploaded my scholarship to my website. If I were to do away with the paywall so the content may be read for free, I will have provided gratis open access. Alternatively, if I were to remove the fee and confer reuse permissions—say, by applying one of the standard, public licenses in the Creative Commons suite—then I will have provided some form of libre open access to my work.[18] Either way, my achieving ACCESS by way of OPEN has resulted in an open access contribution. This is what it is for scholarship to merit the open access label.

As a label, open access encompasses all forms and models that bring it about: for-profit and not-for-profit; publisher controlled and scholar controlled; with and without author-side fees; individually funded and community funded; publisher mediated and repository mediated; and so on.[19] It is even possible to achieve the goal, according to the rules, without the contributor's having directly aimed at producing an open access contribution, specifically. This last

§ This is not to say that open access scholarship may not ever be provided for sale. But for scholarship to be open access, it must be accessible free of charge even if that is not the *only* way to access it.

point needs emphasis: ACCESS and OPEN are silent on just how intentionally or incidentally they are met. All that is required is that they *are* met. Put another way, a contributor need not *deliberately* apply the rules in order to achieve the goal according to them, much less endorse the rules as particularly choice worthy or important. A contributor simply may be complying with the policy of her funder or institution, accommodating the journal that asked her to apply a Creative Commons license to her forthcoming article, or following the norms and practices of her field. If the policy didn't exist, if the journal didn't ask, or if the norms were different, she may or may not attempt to achieve ACCESS or do so by way of OPEN. In sum, the goal and the rules together are sufficient for her scholarship to merit the open access label, but the label cannot tell us about the stance she takes toward her activity or what commitment she has, if any, to sustaining that activity over time.

This is a feature, not a bug. Insofar as open access is a label applied to scholarship, we need only assess the relevant facts of the contribution rather than any facts of the contributor. Otherwise, we risk applying the label too narrowly. At the same time, contributors who are less deliberate in the ways just described could elect, absent external obligations and nudges, not to take up the task reliably and repeatedly over time. They may also be apt to engage in practices that could frustrate the further end goals to which open access is a means, even as scholarship continues to be produced and continues to merit the open access label.[20]

Among such end-frustrating practices is the publishing model that shifts price barriers from the reader side to the author side in the provision of open access scholarship. Increasingly popular among large, commercial publishers of journals, this pay-to-publish approach privileges well-resourced authors while crowding out authors who cannot afford the fees.[21] The vulnerability here is not that the open access label could become scarce or result in a net deficit over time. But without more attention to its creation, including the forms and models that ought to be endorsed, we risk unfairly limiting *whose* scholarship can merit the open access label in the first place.

In the next section, I introduce the third and final element in our framework. It combines with the goal and the rules to constitute the rallying cry of open access, and it is this rallying cry that can safeguard against the vulnerability I described above.

The Rallying Cry

To conclude his analysis, Suits adds a final element. As a condition on the attitude of the players, it consists of "acceptance of [the] rules just so the activity made possible by such acceptance can occur."[22] Notice the attitude refers to the previous element, just as that element referred to the one before it. And this attitude, while minimal, performs a critical limiting function so that not just any activity taken up voluntarily and satisfying the goal and the rules will count as playing a game. When the sluggish and swift skiers compete in the downhill race, they "accept the restrictions the rules impose because [they] want to engage in the activity of pursuing the goal in only the ways [the rules] allow."[23] This is gameplay.

Contrast the skiers with my running errands. When I fetch my groceries on foot instead of by bicycle, I am attempting to bring about a state of affairs, and I am voluntarily using less efficient means to do so. What's different about my activity compared with that of the skiers? In my case, and unlike in theirs, I do not accept my mode of travel deliberately *because of* the restrictions imposed on me as I run my errand. I lack the attitude the skiers adopt toward the rules, even though I do take them to be rules that constrain my activity. They want to get to the bottom of the slope, and they want to do so with less efficient means because that's what the rules dictate; this is what it is to adopt the attitude on Suits's analysis. As for me, I just want to get my groceries. So, they are playing a game, while I am not.[24]

With Suits's third element in place, the set is complete, and 'playing a game' emerges. Suits summarizes the full definition provided at the beginning of the last section with what he calls his "more portable" statement: "playing a game is the voluntary attempt to overcome unnecessary obstacles."[25]

If we were to adopt this same element and construe it in our framework the same way Suits does in his, we would not get as far as we need. After all, our goal is to unify the two senses of open access, the label and the rallying cry. Whereas a Suitsian attitude could provide the appropriate condition on contributors so as to *rule out* incidental instances of open access scholarship, and thereby *rule in* only intentional contributions, the rallying cry demands much more. So, the third element in our framework will depart from Suits and, instead, incorporate a stronger, more persistent species of intention: a commitment.

Fostering a Commitment

In addition to the goal and rules, adopting the rallying cry of open access requires a commitment. Advocates, scholars, and practitioners use the term frequently in association with their aims, motivations, and beliefs about open access. But what, exactly, is a commitment? In the remainder of this section, I will employ Cheshire Calhoun's account of commitments to define open access, in the second sense, as a rallying cry. Upon incorporating this final element into the three-element framework, we will have unified both senses of open access.

Commitments are "species of intention," distinguishable from mere intentions and provisional plans.[26] When I form a mere intention or make a provisional plan, I can change my mind or decide not to follow through without as much need for careful consideration, and the bar for reasonably changing my mind need not be as high. Say I set a goal of distributing the accepted manuscript of my journal article from an online repository. I locate the journal's self-archiving policy and judge it at a glance not to be too long or complex. I have the time to read it, but I would rather use that time to grab a cup of coffee. So, I abandon the goal altogether. In this case, and under these circumstances, my reconsideration is not a problem.[27] It would pose a problem, however, had I *committed* to the goal. In that case, my reason for not following through—I preferred a coffee break instead—would not be good enough. Stronger reasons from me would be called for: that I did not understand the policy even after reading it carefully or consulting my colleagues or that I was pressed for time with other deadlines to meet. These reasons could be enough to discharge my commitment temporarily—at least until I received clarification from the editor upon my asking, got guidance from colleagues, or took care of my tasks with soon-to-arrive deadlines. At that point, my commitment would dictate that I resume my pursuit of the goal and get ready to do the same for the next article.[28] Thus, what distinguishes a commitment from a mere intention or provisional plan, according to Calhoun, is that "a commitment is *both* an intention to engage with something … *and* a preparedness to see to it that the intention to engage persists."[29]

The rallying cry of open access, then, is to intend to make digital scholarship available online in a manner that removes barriers and to be persistent in this pursuit. More precisely, the rallying cry takes the following form:

> **Goal (ACCESS):** Make digital scholarship available online.
>
> **Rules (OPEN):** Achieve the goal by removing either price barriers or price and permission barriers.
>
> **Commitment:** Intend to follow the rules in pursuit of the goal and be prepared to see to it that the intention persists.

Persistence is key. In the alternative case above, in which I had formed a commitment, I didn't just give up on pursuing ACCESS by way of OPEN because I faced certain difficulties. The experience also set me up to expect similar difficulties in similar future cases and to be prepared either to work through them or to take another tack in service of the goal and according to the rules. That is, I could take the same approach again in the future, knowing what ended up working for me this time, or I could choose another path to ACCESS that OPEN permits, say, by publishing an article in an open access journal. Likewise, I am just as prepared to close off avenues that would contravene my goal, or to do what Calhoun calls a "readiness to

engage in a set of refusals."[30] For my commitment to be a commitment, I also need to persist in "refraining from cultivating activities, attitudes, and ways of life that are incompatible with sustaining one's commitment."[31] So, I cannot be committed to open access if I am choosing—to the extent I do have the choice—to engage in activities that would be disqualifying for the open access label. To the extent that I have the opportunity to live up to my commitment, to adopt the rallying cry of open access, I *must* choose the avenues that are consistent with this commitment and my sustaining it over time.

Now that all the elements are in place, the framework unifies the two senses of open access without conflating them. In the first sense, open access serves as a label and comprises just the first two elements, the goal of ACCESS and the rules of OPEN:

> **Goal (ACCESS):** Make digital scholarship available online.
>
> **Rules (OPEN):** Achieve the goal by removing either price barriers or price and permission barriers.

It is flexible enough to accommodate both intentional and incidental contributions to open access. In the second sense, open access serves as a rallying cry and binds all three elements by adding a commitment alongside ACCESS and OPEN:

> **Goal (ACCESS):** Make digital scholarship available online.
>
> **Rules (OPEN):** Achieve the goal by removing either price barriers or price and permission barriers.
>
> **Commitment:** Intend to follow the rules in pursuit of the goal and be prepared to see to it that the intention persists.

In contrast to the label, the rallying cry is more rigid. It requires a contributor's intentional and persistent uptake of the rules in pursuit of the goal.

Hurka points to the upshot of this "nested" structure, observing of Suits's analysis that it "explains why it has the parts it does" all on its own. In the case of gameplay, Hurka writes, "[I]f it contains the … attitude, it has to contain the rules, and if it contains the rules, it has to contain a … goal."[32] Just so for our own framework: if it contains the commitment, it has to contain the rules, and if it contains the rules, it has to contain the goal.

Next, I discuss how a commitment to open access—adopting the rallying cry—can safeguard against the vulnerability I raised in the previous section concerning the unfair exclusion of some scholars from opportunities to contribute scholarship that merits the open access label.

Safeguarding with a Commitment

Recall the vulnerability. As a label, open access covers all forms and models. It applies to a given contribution due to facts about that contribution rather than those of its contributor—whether author, editor, publisher, or some other individual or entity. In this sense, open access is neither fragile nor rare. In fact, it is increasingly more attainable and widespread, on the whole, than in the past.[33] But some models of open access are, in fact, less attainable than others, particularly the pay-to-publish scheme in which costs shift from the reader side to the author side. So, the vulnerability is not that the open access label risks becoming scarce or less prevalent overall, but that specific models under its banner will—and already do—unjustly crowd out contributors for economic reasons alone.

The rallying cry can safeguard against this vulnerability even if it cannot shore it up entirely. There are two main ways it can do this. First, the rallying cry requires a commitment. Contributors must be intentional and persistent in their open access pursuits; incidental and wavering contributions will not be enough for the commitment to take hold. So, the rallying cry can enhance the choosiness of contributors just by virtue of obliging their care and attention in the uptake of the rules in service of the goal of open access. Second, our framework doesn't

specify *what* the commitment ought to be, just as the rules make no dictates against *forbidding* author-side fees. Since the commitment operates on the rules, contributors can adopt commitments to fairer models of open access whose rules prohibit both reader- and author-side fees.

Consider a case of safeguarding along these very lines. Many en masse resignations of editors from commercially published academic journals cite unassuaged concerns with publishers' business models and prioritization of commercial interests. Several resignations have centered on disagreements with steep increases in subscription prices and the imposition of high price author-side fees as mechanisms for publishing open access and generating revenue.[34] Recently, the editors and board members of the longstanding *Philosophy and Public Affairs* revealed that they had found themselves caught in a "conflict between purpose and business model." They chose to "embrace the purpose" instead of continuing with the status quo: "We take this step," they explain in their letter of resignation, "because we believe that scholarly journals—including our own—serve important purposes, and that these purposes are not well-served by commercial publishing."[35]

Following their unanimous resignation, the editors have now founded a new journal with the not-for-profit publisher Open Library of Humanities. Aptly named *Free & Equal: A Journal of Ethics and Public Affairs,* the journal will be fully open access and charge no fees to publish or to read.[36] In their own words,

> *Our name, Free & Equal, expresses our two founding commitments: to free inquiry on equal terms about matters of public concern; and to the free dissemination and equal accessibility of the products of this inquiry.*[37]

In Cheshire Calhoun's parlance, the now-former editors of *Philosophy and Public Affairs* engaged in a set of refusals so that they could, with *Free & Equal,* persist in their commitment to academic publishing and dually foster and safeguard a commitment to open access. And they have chosen a way of pursuing open access that alleviates, rather than reinforces, the conditions that had prompted their initial resignation and motivated them to start a new journal. Put another way, the editors of *Free & Equal* have adopted the rallying cry of open access.

Conclusion

In this chapter, I introduced a framework to unify two senses of open access, the label and the rallying cry. Inspired by Bernard Suits's analysis of gameplay, this framework contains just three elements: a goal, rules, and a commitment. As a label, open access comprises a goal and rules. The goal is to make digital scholarship available online, and the rules state that the goal is to be achieved by removing price barriers alone or price and permission barriers together. When the conditions specified by these elements are met, a contribution merits the open access label. As a rallying cry, open access incorporates all three elements. In this way, a commitment attaches to the rules in pursuit of the goal. And when this occurs, a contributor will have adopted the rallying cry of open access. This framework has several advantages besides unity. It resists conflating the two senses of open access so that their essential differences become and remain clear. It also demonstrates why both senses of open access are important and explains how adopting the rallying cry can safeguard against a particular vulnerability that the label cannot attenuate on its own.

Acknowledgements

I thank the editors of this volume, peer reviewers Anders Tobiason and Rebecca Wojturska, and Peter Suber for their helpful comments. I also wish to express my gratitude more broadly to Peter for his work in open access, which has been pivotal to my own, and to Ian Dunkle, whose encouragement has helped me cross the finish line.

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CHAPTER 8

FOSS: Foundational to Open

Helena Marvin

Introduction

I am fascinated by the power of the internet and of technology more generally and have been since I was in middle school in the late 1990s. I could build a computer from parts, but, as is true for most children, I had no concept of intellectual property. From the point I figured out how to use a computer to do so, I happily shared music with strangers on the internet. When the Recording Industry Association of America started its war on Napster, I was horrified to learn that what I had been doing was illegal. For the first time, I wrestled with concepts of intellectual property and its ethical implications. I switched from piracy to free culture by switching to free and open-source software (FOSS) and legally shared music, exploring the world of digital rights and ownership of materials online. In this chapter, I provide some history of FOSS and its impact on open culture, the origins of license agreements, and the technologies that underpin the digital infrastructure open movements need to thrive. My personal anecdotes are interwoven into this chapter to ground it in lived experience.

During an online open education conference, a panelist went on a tangent to advocate passionately for the need for programmers to create open educational software. Their apparent lack of experience with open source forced me to confront my unfounded belief that anyone familiar with open educational resources (OER) would be familiar with FOSS. The free software movement is what led me to other open culture movements, and realizing there are folks who are working in other open movements without knowing about FOSS sparked my desire to write this chapter. My aim is to raise awareness of how FOSS is foundational in spirit and as infrastructure for other open movements, because how it is related and distinct from other open initiatives is fascinating to me. The differences between open movements are most evident when participating in multiple open communities but often invisible to those familiar with only one open community of practice.

"Open" does not have the same meaning across all contexts. The "open" of open access is not equivalent to the "open" of open educational resources, nor do these movements have the same "open" as the "open" in open-source software. The traditions I work with are both distinct and related, and these will be explored. The internet provided a means to share digital objects easily without cost or impact to the original work, but many people don't consider the legality of sharing files. Open movements exist in response to the rise of easily shared digital materials, and they address the legality of sharing for different communities. There are disparate needs, motivations, and expectations for sharing between different open communities. To begin, I explain what software, and specifically open software, is and how it provides the underlying technical foundation for our digital ambitions.

CC BY

Where I'm Coming From

Thanks to my mother, an early adopter of new technologies, I spent my childhood online in our family's "computer room" enjoying high-speed cable internet at a time when few of my friends had dial-up, if they had home internet at all. After installing Napster, I made friends with university students in England who asked me to seek out specific songs for them because I had a fast upload speed. Connecting with people, especially cool college students on the other side of the ocean, was exciting to a kid in Kansas. This virtual interaction was what the internet promised: a thriving cyber community forging digital bonds by sharing music on the information superhighway. The very act of downloading and uploading music was exhilarating.

When the Recording Industry Association of America took Napster to court in 1999,[1] I thought it made me a criminal. At thirteen, the fear that I was going to be arrested for sharing music prompted me to learn about intellectual property rights. While learning about licensing, I realized that it wasn't just music I was pirating but also the software I had found on warez* sites. This revelation inspired me to look for legal free alternatives, and that is how I stumbled into the world of open-source software. I was using Linux in 2001, when it was much less user-friendly but still a fully usable operating system with a suite of free and open-source software available at my fingertips. Indirectly, Napster led me to abandon the Microsoft Windows operating system and become a free, open-source software Linux user before I could drive. The path took me from an unintentional Napster pirate kid to a teenage Linux user to a philosophy undergrad whose capstone was titled "Software and the Philosophic Foundation of Intellectual Property Rights." Now I am a librarian at an American public university supporting faculty in creating open educational resources and sharing open access articles.

Software and Free Software

When it comes to computers, the tangible things are hardware. Hardware includes everything from the motherboard and the chips to the monitor and the mouse. By contrast, software is intangible and includes everything from your operating system (Windows, Mac, and Linux being the most common) to your word processor and web browser. Computer hardware can be kicked, but software can only be cursed at. Software is crafted by individuals, companies, and organizations of various types. Whether alone or in a group, programmers invest their time, their resources, and their mental energies into the production of software.

Software is written as instructions, called "source code," which is compiled† into software. Source code is generally "human" readable (though it makes little sense if you don't know the language, similar to how English is a human-readable language but makes little sense if you are only familiar with Chinese). The source code for software can be hidden from view in a binary, or it may be accessible.

How Software Becomes Free

Intellectual property rights are meant to aid in the advancement of the arts, science, and technology. Developing something, be it a poem, a book chapter, or a piece of software, is costly in terms of education, labor, time, mental energy, and necessary technologies. Recouping the cost of development is one justification to defend proprietary rights. Just as an author can waive or transfer the rights to their work, so too can a programmer, but the rights to the intellectual property they create are initially their own, though this can vary between legal jurisdictions (countries). It can take considerable time to write even seemingly simple

* Warez (said like wares) was a common shorthand for pirated software that has been illegally copied and made available for download online.

† Or in some computer languages, interpreted.

programs. Programming is akin to writing a series of instructions in a logical form where all definitions are non-ambiguous and everything follows without contradiction. Unless complicated by being a "work for hire," authors own the rights to their words and programmers hold the rights to the code they create.

The social and legal infrastructure protecting intellectual property was established long before the internet, back when material could be copied no faster than a printing press could be cranked. Many different elements of intellectual property protection and legal frameworks, such as copyright, patent, and trademark law, matured slowly over time in an attempt to balance the protection of creators and serve the public good. Applied to the much more massive scales and speeds of the digital world, these legal systems can feel cobbled together. Richard M. Stallman, a problematic[2] and challenging character who is often described as the father of free software, said,

> *These laws originated separately, evolved differently, cover different activities, have different rules, and raise different public policy issues. Copyright law was designed to promote authorship and art, and covers the details of a work of authorship or art. Patent law was intended to encourage publication of ideas, at the price of finite monopolies over these ideas—a price that may be worth paying in some fields and not in others. Trademark law was not intended to promote any business activity but simply to enable buyers to know what they are buying.*[3]

Programmers can protect their software by hiding its source code from the user, or it can be protected legally by making users enter contracts with the creator, agreeing to additional terms for use. When a creator shares only the binary version of the software without the code being accessible, it prevents people from changing the program easily. When licenses, commonly called end-user license agreements (EULAs), are included with the software, the user must agree to the terms before they can legally use the software. In the early days of software, neither EULAs nor software binaries without source code were common, and when Stallman first encountered them, they troubled him both technologically and philosophically.[4]

EULAs often protect the software copyright holder and hamper the user, preventing the user from editing or sharing the code without the permission or compensation of the original creator. However, a license agreement can also expand the rights of the user and keep the code free. Early in the history of computing, there were software EULAs developed in academia, referred to as permissive free software licenses, which provided code for free but did require that the resulting code remain free. The Berkeley Software Distribution (BSD) made it possible for software to be freely shared and included in other non-free works as long as recognition appeared with the developed software.

Copyleft

An early example of what is now called open source is the source code for Li-Chen Wang's Tiny Basic for Intel 8080 published in the December 1976 issue of *Interface Age* magazine,[5] which was published with the copyleft notice stating "ALL WRONGS RESERVED."[‡] In the 1970s, copyleft was not yet the defined movement requiring continued sharing that it would later evolve into. The explicit expectation of continued sharing began with Stallman's Emacs software (an abbreviation of "editor macros") in 1984. Emacs was a text editor initially developed and used in the MIT Artificial Intelligence Lab that is now known for its powerful features, devoted users, and steep learning curve. As interest in Stallman's program grew, he started sharing the source for the software[6] but made an "informal rule that anyone making improvements had to send them back" to him.[7] This sharing of source code and improvements is at the heart

‡ The original uploader was Swtpc6800 at English Wikipedia, *Copyleft All Wrongs Reserved*, August 23, 2007, August 23, 2007, transferred from en.wikipedia to Commons, https://commons.wikimedia.org/wiki/File:Copyleft_All_Wrongs_Reserved.png.

of Stallman's digital practice and philosophy. The requirement of continued sharing is how copyleft differs from permissive licenses.

To codify the right to share, Stallman released Emacs with the General Public License (GPL), which later became the GNU GPL. GNU is a recursive acronym—"GNU is Not Unix,"—as the goal was to build a Unix-like operating system with only free software. Stallman clarified that GNU was not in the public domain, as public domain would allow for proprietary modifications to be made, while Stallman's goal was to ensure all versions of GNU remained free.[8] He leveraged the popularity of Emacs as the flagship software for his GNU project, and GNU started a domino effect.

GNU software was foundational to Linus Torvald's Linux kernel. Ian Murdock began the Linux distribution Debian (referred to as GNU/Linux) in 1993, and along with it published the Debian Social Contract, which served as the basis of the Open Source Definition published by the Open Source Initiative in 1998. Debian Linux is the free, open-source operating system I began using in 2001 as a teenager. Its free, accessible, and hackable nature is why it, and other flavors of Linux, were adopted by many technology enthusiasts who run the underlying infrastructure of the internet, which likewise powers the open movements I work with.

Information Wants to Be Free

Emacs was developed during a pivotal moment in the history of computing. The realities of digital sharing were just starting to be understood broadly. The expression "information wants to be free" has been attributed to Stewart Brand (co-founder of the Whole Earth Catalog), who said it to Steve Wozniak (co-founder of Apple) at a Hackers Conference in 1984. Stewart Brand had been describing the tension between information as an expensive and valuable resource and a resource that was becoming easier and faster to share due to advancements in technology.[9] Stallman responded to the notion that information wants to be free by saying, "I believe that all generally useful information should be free. By 'free' I am not referring to price, but rather to the freedom to copy the information and to adapt it to one's own uses.... When information is generally useful, redistributing it makes humanity wealthier no matter who is distributing and no matter who is receiving."[10]

When Linus Torvalds began working on the Linux kernel[§] in 1991, it was initially protected by Linus's copyright, but, as he wrote in his book about the history and development of Linux,

> *To make Linux usable, I had relied on a lot of tools that had been distributed freely over the Internet—I had hoisted myself up on the shoulders of giants. The most important of these free software programs was the GCC compiler. It had been copyrighted under the General Public License.*[11]

Linux worked with GNU software, and so the creator adopted the GPL for the Kernel operating system, and it is now inseparable from it.

Gratis and Libre

The term *free* is confusing because it means multiple things at once. Free software is often both free in that it requires no payment (*gratis*) and that it allows for the freedom (*libre*) to change and share the source code. Stallman defines four essential freedoms of software:

- The freedom to run the program as you wish, for any purpose (freedom 0).
- The freedom to study how the program works, and change it so it does your computer as you wish (freedom 1). Access to the source code is a precondition for this.

§ A kernel is technology that is a bit like the heart of an operating system which allows all of a user's software to work together.

- The freedom to redistribute copies so you can help others (freedom 2).
- The freedom to distribute copies of your modified version to others (freedom 3). By doing this you can give the whole community a chance to benefit from your changes. Access to the source code is a precondition for this.[12]

Open-source software powered 61 percent of publicly accessible web servers by August 1999,[13] and the amount of FOSS infrastructure has continued growing. Free software was and is shared on FOSS-powered servers, and so too is the vast majority of content shared by other open movements. There is no need for someone working in open access (OA), OER, or any other open movement to necessarily know what a server is, but our digital materials and open communities are powered under the hood by free and libre technologies.

Copyleft is the practice of leveraging the rights of copyright with a license agreement to secure libre freedoms and the continuation of those freedoms. It is predominantly for software. The Free Culture movement emphasizes the Creative Commons (CC) for cultural works other than software.[14] Lawrence Lessig initiated the Creative Commons and presented it as "some rights reserved" instead of "all rights reserved," normally paired with a copyright notice. There are many different Creative Commons licenses, which is helpful since the goals of a copyright holder often differ in different open movements and even within them. Open-source software includes copyleft, permissive licenses and public domain works, and the Creative Commons itself has so many permutations that it can be used to license works as public domain, permissive, copyleft, and for noncommercial use (see table 1).

Type of License	Public domain & equivalents	Permissive license	Copyleft (protective license)	Noncommercial license	No Derivative license	Proprietary license
Description	Grants all rights	Grants use rights, forbids almost nothing (allows proprietization, license compatibility)	Grants use rights, forbids proprietization	Grants rights for noncommercial use only. May not be combined with copyleft.	Grants rights for sharing but no rights to make derivative works	Traditional use of copyright; no rights need be granted
Software	Public domain, CC0 (Creative Commons 0)	BSD, MIT, Apache	GPL,AGPL	JRL, AFPL	Shareware software is proprietary software that is distributed freely to users	Proprietary software, no public license
Open-Source Software	Any of these licenses can be for open-source software				Not open source	
Open Educational Resources	Any of these licenses can be for open educational resources				Not open educational resources	
Open Access	Any of these licenses can be open access works					Not open access

Table 1. CC BY-SA Helena Marvin (adapted from Wikipedia Permissive software license Comparison table https://en.m.wikipedia.org/wiki/Permissive_software_license#Comparison_table 2024).

Open source was established as a term for free software in 1998. It was a rebranding effort to make commercial companies more comfortable with free software and open development practices.[15] This change was not embraced by Stallman, but was adopted by the more pragmatic Linus Torvalds.[16] Also in 1998, the Open Content License was released by the Open Content Project (OCP) as one of the first licenses for copyrighted works other than software that worked similarly to the GNU GPL. OCP was superseded by the Creative Commons Project.[17] In the early 2000s, around the same time that the Creative Commons launched, the term open access (OA) came into common parlance. The power of digital sharing, robust and powered by the growing internet, technologies and infrastructures were happening concurrently, impacting different communities simultaneously, but they were in conversation with one another.

Open Access

The *open* in open access (OA) leans more toward providing gratis access to academic works that are otherwise hidden behind paywalls rather than toward libre freedoms. Peter Suber says in his seminal book about open access that there are two kinds of OA, the sort that removes price alone and the sort that removes permission barriers, "gratis OA and the latter libre OA."[18] It was software creators who codified the terms of free, both *gratis* and *libre*, as well as the software technology they created, and laid the digital foundation for other open movements. For both the scholarly works of open access and the source code for open-source software, the creator affirmatively waives their rights.[19] Suber explicitly states he is "borrowing the gratis/libre language from the world of software," but he does not mention open-source software in his book. The foundation of free and open-source software is the same as in the open access movement, and the term "open source" was culturally established before open access. The open access focus of gratis before libre is a point of departure between the two movements. Similarly, the OER movement focuses on libre before gratis, and that distinguishes it from the broader open access movement.

Open Educational Resources

Open educational resources (OER) consist of copyleft works created for educational purposes. There are five foundational principles in OER, known as the five Rs, that underpin the work of creating and sharing OER. The Rs refer to the words *retain*, *reuse*, *revise*, *remix*, and *redistribute*, which are the libre rights associated with OER content: to retain, or keep forever; to reuse the work for your own purpose; to revise, adapt, modify, or translate the work as needed; to remix and combine with another resource to make a new work; to redistribute and share the work with others.

These five Rs are protected by most, but not all, of the Creative Commons licenses.** Though there are other licenses used (including public domain), the Creative Commons is the most common type[20] adopted by creators of OER. Not all Creative Commons licenses protect the five Rs. The major exception, No Derivative or ND (which is more common for open access works), prohibits revising or remixing the material in "derivative work" (see figure 1).

** For an in-depth guide to Creative Commons licenses, I recommend the guide from Wikimedia at https://meta.wikimedia.org/wiki/Open_Content_-_

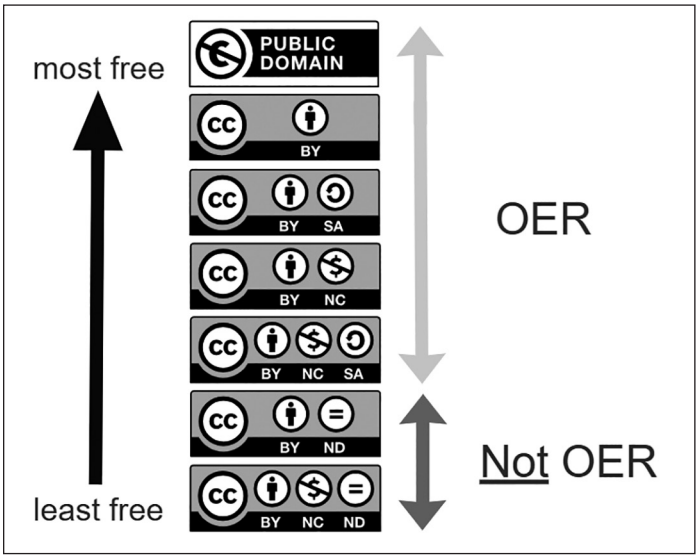


Figure 1. OER least to most free CC BY Cable Green. Adapted from http://www.slideshare.net/cgreen/updated-keynote-slides-october-2014?qid=e49d7e3b-3a0a-47a4-992-309e2f30574&v=qf1&b=&-from_search=13, published July 27, 2015.

The five Rs of OER are reminiscent of the four libre freedoms of FOSS. This connection was made clear to me while chatting with a high school chemistry teacher at a mutual friend's birthday party. He brought up an "open source" chemistry textbook he had found. As I work with OER in a professional capacity, and I'm familiar with open-source software, I understood he was talking about something more commonly called an OER. He was not working with OER regularly, and he was more familiar with open-source technologies from his hobby with GIS technologies, but he rightly associated the two movements. There are textbooks that refer to themselves as open source. The trouble with word choice is that if he used *open source* as the search term when looking for suitable textbooks, he may fail to retrieve relevant OER works that would better serve his needs.

In our conversation, I talked about how I work with open educational resources as an academic librarian, and how they are distinct but related to open source. He was thrilled that educators had embraced an open-source mentality. Similarly, while working with faculty at my university, I encounter professors who refer to OER as open access, and I have likewise stumbled over such words myself. Even when I do it, I worry that the confusion between these terms might prevent people from finding the resources that would best serve them. Misunderstanding the difference may inadvertently impede their future work.

The different ambitions and aims of different communities in open movements impact those within and outside their communities of practice. The open access movement continues to have a split focus between free and libre, with a primary goal of preventing paywalls from blocking readers from scholarship. Open educational resources tend to have an emphasis on the revising and remixability of their content, as the community centers on the practicalities of educational practice. FOSS software, which encompasses both permissive and copyleft works, is focused on software. In all these movements, creators are freely choosing to surrender some rights that would otherwise be theirs to empower others, but how they do this differs.

Who is being empowered and why the creator is choosing which open license often also differs. In my experience, an open access creator is motivated by a desire to expand access to scholarship and focuses on the reader. In the open educational resources movement, there is a focus on empowering educators, and in the open source movement, there is a focus on programmers. The motivation for these choices does not limit who these movements empower, but by boiling down who the licenses are crafted for and adopted by, we understand how different movements center the notion of open differently.

The Internet Runs on Open and Open Runs on the Internet

License agreements developed for code informed the development of the open licenses that came after them, both for software and for other open materials. The history of EULAs is one way that FOSS is foundational to other open movements, but FOSS also supports the digital tools and environment that power the creation and sharing of open materials. The lack of understanding of how enmeshed FOSS technology is in our virtual lives is as prevalent for open movement practitioners outside of FOSS as it is for the general population:

> *Most people don't realize it, much of the technology we rely on every day runs on free and open source software (FOSS). Phones, cars, planes, and even many cutting-edge artificial intelligence programs use open-source software such as the Linux kernel operating system, the Apache and Nginx web servers, which run over 60% of the world's websites.*[21]

Open movements are supported by the backends of websites hosted on open-source-powered servers and many of the software tools that support the creation and sharing of digital works. One such tool is Pressbooks, which I've used extensively as an OER librarian. Pressbooks is a fork[††] of the open-source blogging tool WordPress, itself an example of how software can be built upon and a tool to build open materials to be copied and shared. If you use Google Chrome, Microsoft Edge, or Opera, you are using software powered by Chromium, which is open source, or if you use Firefox as a browser, you are using software that is powered by open source. Much of the infrastructure that supports digital materials in a virtual environment runs or is supported by FOSS technologies.

Digital Sharing Gratis and Libre

In my Napster days, I was enchanted by the notion of gratis MP3s. As an academic librarian, I am enchanted by the notion of libre resources. The term *open* is used, like *free*, to describe both things, and this overlap can be very confusing. I have met many incredible people working in different open movements and doing amazing work, creating vital resources, who do not share my fascination with copyright or licensing. Few people know how FOSS, in its capacity as essential technologies, materially support their open movement and the broader internet, or how the history of software licenses informed the development of Creative Commons.

What distinguishes the meaning of "open" between the three movements discussed is the aims and audience of the created works. The audience for software is the end users, but the license on a piece of software is of most concern for programmers. It is the license on the code that provides a programmer the libre freedom to manipulate the code. The student learning from an OER is benefiting from the license, but it is the educators who are creating and ensuring libre freedoms with the materials. While there are those in the open access movement concerned with libre freedoms of academically published works, there is a greater focus on the gratis access for the readers.

The free and open software community codified a framework for defining free as both libre and gratis before the internet was accessible to the masses. The popularity of open movements and the embrace of the term *open* bookend the rise and fall of Napster, as open source was coined in 1998,[22] and open access "was introduced by the Budapest Open Access Initiative in February 2002,"[23] the year Napster filed for bankruptcy. Napster was a piece of proprietary software that facilitated piracy, but it introduced me to the power of the internet for sharing

[††] A fork happens when developers take a copy of source code and start independently developing it into something new. Pressbooks is designed to make books (both online and printable) but it was built out of WordPress source, which is a blogging platform.

digital copies of music. The publicity of its downfall in court helped bring awareness of how digital technologies were impacting the distribution of copyrighted works, but it also illustrated the power of digital sharing.

The open movements I work in, open access and open educational resources, rely mostly on Creative Commons licenses to protect the rights of scholars to share works. This work is related to the copyright corner of intellectual property rights. By contrast, open-source software includes many types of licenses and has intellectual property concerns beyond copyright. The permissive open license Apache includes patent grants,[24] while the term Linux itself is protected by a trademark[25] for which the Linux Foundation freely provides sublicenses. Such concerns—patent, trademark, and code—are generally beyond the interest of OA and OER practitioners, but an awareness of such, and of the history of FOSS more generally, empowers practitioners in other open movements. By establishing and promoting notions of free as both gratis and libre, developing the frameworks of open license agreements and sharing open-source software, the free open-source software movement laid the foundation on which other open movements thrive.

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CHAPTER 9

How To Be Open About Open Initiatives: Case Studies on Organizational Structures and Their Impact on Open

Ariana Santiago, Rachel Fleming, Marla I. Hertz, and Christina M. Miskey

Introduction

In this chapter, four academic librarians discuss case studies based on how their own libraries position open initiatives. Each of the authors is at a public research institution in the United States but is situated in different organizational contexts, both in terms of staffing and in how each institution focuses on open initiatives. The points of view our chapter represents include: a librarian who is responsible for all open-related services; two librarians with specialized roles (research impact and research data management) located within scholarly communication units; and a librarian whose role is focused on open education at a library with a department dedicated to that area.

In each case study, the authors discuss how open initiatives are supported at their institution. They explore the various approaches to organizational structure, staffing levels, proximity to other library units, and the resources available to those units. The authors also discuss how they address challenges to cross-coordination, such as navigating siloes, and the benefits of physical and/or organizational proximity to other library departments or divisions. In the last sections of the chapter, the authors collectively reflect on the benefits and drawbacks of various organizational models and their impacts. They conclude by drawing from their experiences to provide actionable recommendations for institutions looking to build their own units focused on open initiatives.

CC BY NC

Case Studies

Supporting Open as a Solo Practitioner in the Library

The University of Tennessee at Chattanooga (UTC) is a growing comprehensive regional institution with an enrollment of 11,000 primarily undergraduate and professional graduate students.[1] At UTC, open initiatives are headed by the scholarly communications librarian, housed in the Special Collections department. Special Collections is, in turn, part of the UTC Library's Collection Services division. The Scholarly Communications position was created in response to growing campus needs from both students and faculty. The position has a mandate to manage and grow a course affordability program advocating for open educational resources (OER)[2] and to consolidate and develop other faculty-focused initiatives, including open access, research data management, and campus open publishing. The creation of the scholarly communications librarian position advanced open initiatives, which previously had no formal oversight. With a position in the library dedicated to supporting open initiatives, the community has a place to look to for advice and to advocate for additional services.

Benefits of This Approach

In addition to providing a public-facing office, not available in other Collection Services spaces, being affiliated with Special Collections provides opportunities to collaborate on open pedagogy projects.[3] Co-location creates a strong awareness of collections and services, highlighting opportunities for faculty referrals and collaborations. For instance, the *Global Contemporary Artist Analysis Archive*[4] is an open pedagogy project begun by an art history faculty member who was made aware of publishing services through collaboration on a student exhibition catalog, *Ancient Latin American Objects in the Archive*.[5] After the art historian, who had begun the *Global Contemporary Artist Analysis Archive*, left UTC, Special Collections played a role in continuing the project by sharing information about the project, which resulted in a referral and adoption of the project. Deep understanding of services facilitates effective promotion and referrals within the department.

Consolidation of open services into a single position allows coordinated development of services in response to campus needs. Because of the combined services, the scholarly communications librarian is able to pivot time and energy toward emergent needs without staffing or assignment changes. For instance, in the wake of the COVID-19 pandemic response, UTC saw a dramatic increase in interest in authoring open educational resources, and the scholarly communications librarian was able to respond by developing stronger services in those areas. As the demand for open campus publishing and service structures stabilized, the scholarly communications librarian was able to re-allocate resources to develop more robust research data management and open access support services in response to changing federal mandates supporting open science.[6] As new services are formalized, it is essential to provide clear boundaries on what is provided to avoid "scope creep" and guard limited resources supporting open initiatives.

Challenges of This Approach

As a sole practitioner, the foremost challenge is balancing a broad range of services and expanding services to meet new needs without overcommitting. Limited time and many demands always threaten to undermine outreach opportunities and ongoing engagement with open movements, which are essential to keeping these academic library services relevant. Maintaining a state of readiness to meet new challenges and support nascent projects can mean not implementing new services without expressed needs. Similarly, maintaining and developing project leads until both parties have the resources needed to execute a project is a continual focus of outreach. For instance, an ongoing collaboration with the UTC Arts-Based

Collaborative[7] has provided several opportunities to educate area primary school teachers about OER but has not yet resulted in projects housed at UTC.

When managing a broad range of services with limited capacity, one must be careful not to over-promise to campus partners. Developing clearly defined services can reduce administrative overhead by spending time creating frameworks to support services offered on an as-needed basis. Service definitions can also guard against "scope creep" when supporting projects. Maintaining awareness of developments across many open movements requires an ongoing dedication to professional development. UTC leverages one-time support, such as graduate assistants for developmental and time-bound projects that combine program evaluation with advocacy. Projects like student surveys, campus needs assessments, and cost analyses can serve as assessment and as outreach tools to promote the value of our work. Time-saving strategies allow for prioritizing outreach across the institution, which can result in highly successful collaborative projects, word-of-mouth support for services, and overall campus buy-in for open initiatives.

Supporting Open as a Department within a Library Technical Services Division

At the University of Nevada, Las Vegas (UNLV), the Scholarly Communication and Data Services (SCADS) department is housed within the Collections, Discovery, and Scholarly Communications (CDSC) division. The institution is situated centrally in the Las Vegas metropolitan area and has approximately 32,911 students enrolled as of 2024.[8] The CDSC division mainly houses technical services-focused departments that are responsible for cataloging, collection management, acquisitions, and serials. Initially, the SCADS department was focused primarily on the management of UNLV's institutional repository. Since late 2022, however, the department has expanded to include new positions focused on more public-facing areas, such as research impact, research data, and geographic information services (GIS)/data visualization. Thus, the department now comprises both public service/outreach positions and technical services staff that continue to focus on our institutional repository. Each position within the department incorporates some aspect of Open into our work, such as advocating for open science, openly sharing research data, utilizing open-source tools, and more.

Benefits of This Approach

The benefits of incorporating what is now primarily a public services department into a division centered around technical services are numerous. Our scholarly communication librarian is on the library's Collections Committee, which allows her to advise the committee toward selecting more openly available resources for the library's collection. She also works closely with the library's acquisitions team to educate our campus community about topics in open access, such as read and publish agreements[9] and changing federal mandates.[10] Our institutional repository manager and team are able to consult regularly with the Discovery Services department to ensure that the institutional repository is findable and accessible in the library catalog, but also to ensure that the repository follows established metadata standards, including incorporating standards such as Dublin Core[11] and OAI_PMH[12] to maximize the discoverability and search engine optimization of the repository. Our department also frequently works with the Acquisitions team to process payments for our article processing charge fund, which supports faculty and postdoctoral scholars by paying up to $2,000 for article processing charges (APCs) to make their scholarship open access. The physical proximity to the other departments within our division, as well as the organizational proximity that has allowed for easy relationship-building, has reduced potential barriers that would have been difficult to surpass if our department was situated in another division. These connections and ongoing collaborations likely would not exist as they are without the SCADS team being housed within the same division as the technical services departments.

Challenges of This Approach

There are also several challenges that our department has had to face when evolving to incorporate public-facing services, when the department was previously primarily technically focused. The biggest of these challenges has been finding our voice as a cohesive department when developing new service areas and communicating to others within the library and on campus how our team has evolved. To combat these challenges, the department as a whole has begun regularly sharing the functional expertise that we now offer to researchers on campus. We also have focused on collaborating across divisional lines by working with subject liaison librarians to support researchers in their areas of expertise; we've hosted events such as themed workshop series in data management and GIS, a research impact challenge, and more.[13] Individually, the public services-focused members of the SCADS department have built strong relationships with campus stakeholders to solidify our status as functional experts. Our team has demonstrated that we can work together with faculty and graduate students one-on-one, as well as with campus offices (including the Office of Research, the Graduate College, and the Office of Faculty Affairs) to support and guide campus open initiatives ranging from open research infrastructure, encouraging use of ORCiD,[14] and changing federal guidelines as outlined by recent Office of Science and Technology Policy (OSTP) memos that affect both the campus and researchers alike.[15] Overall, we are able to share these successes that help to define our department and our functional expertise within the library and on campus.

Supporting Open as a Unit within a Library Reference and Instruction Division

The University of Alabama at Birmingham (UAB) is a research-intensive public institution with over 22,000 students.[16] For years, limited copyright support was provided by liaison librarians. But without a formal unit, it was a challenge to educate and connect library users with this service. The Office of Scholarly Communication (OSC) was born out of the need for a defined service point for scholarly communication within the library. Our office has three members: a director who specializes in copyright, a research data management (RDM) librarian, and an institutional repository and outreach librarian. The OSC was added to the Reference division, which was renamed the Reference, Instruction, and Scholarly Communication (RISC) division. The rationale for this organizational structure was primarily based on convenience to minimize restructuring and administrative burdens. It was also a natural fit, given that at the time of its inception all OSC members were heavily involved in instruction.

Benefits of This Approach

Positioning the OSC within the reference division yielded multiple benefits. It increased contact between scholarly communication librarians and subject liaisons in cross-training, cross-promotion of services, and collaboration. OSC librarians work broadly with liaison librarians on topics such as copyright, research data services, journal selection, avoiding predatory publishers, open access initiatives, and, most recently, the responsible use of artificial intelligence in scholarly writing. For example, the RDM librarian and a STEM liaison co-authored an article providing data management and visualization tips for tabular data.[17] To ensure that the OSC does not overstep its bounds, it maintains a policy to involve liaisons when planning programming in their subject area. Finally, the choice to group the instructional units reduces administrative burden. For instance, the same survey form is utilized to capture instruction statistics from the two reference departments and the OSC. These data are centrally maintained to simplify the process of compiling instructional output reports.

Challenges of This Approach

An effective scholarly communication office should collaborate with other library units. Our main organizational barrier to collaboration stems from our library being spread across multiple buildings, a common challenge to any institution uniting a dispersed workforce. Our library has two main branches: a health sciences library and an academic library. All OSC employee offices are located at the academic branch, making it more difficult to connect with coworkers at the health science branch—namely, the health sciences reference unit and the Technical Services department. In contrast to the scenario at UNLV (as described in the case above), our OSC is separate from the library's Technical Services division, both in the organizational and physical sense. Thus, careful planning and oversight from library administration were necessary when we launched our website and when we established an institutional repository (IR). Now that the IR is operational, it functions more independently and is maintained primarily by the IR librarian with support from technical services as needed. On the other hand, physical proximity to the Acquisitions division facilitated our collaborative efforts to broker and maintain read and publish agreements and other collection development activities.

Overall, grouping the OSC within the reference and instruction division was a functional pairing that facilitated outreach to the different schools on campus and aligned best with the mission of our office. This case study demonstrates that a conscientious effort to bridge both physical and administrative distances was necessary to operate across a diffuse ecosystem.

Supporting Open as a Specialized Department in the Library

The University of Houston (UH) is a large, public research institution with enrollment of over 46,000 students.[18] The Open Education Services (OES) department is situated in the Research and Student Engagement (RASE) division of the Libraries and consists of two full-time librarian positions: the head of Open Education Services and the open education librarian. The department focuses on expanding the use of open educational resources and practices to support student learning and innovative teaching. Services include supporting the adoption, creation, production, and publishing of OER, and advancing open pedagogical practices.

The OES department was established in 2022, following over four years of the OER program being led by a librarian whose role was situated within other departments. The program was initiated from student advocacy, followed by buy-in from the former provost and former dean of libraries; with support and funding from these key administrators, the librarian position was created to develop and lead an OER program at the university. Initially, the OER program was housed in the Liaison Services department, which primarily consisted of subject liaisons; after an organizational restructure and a shift away from the subject liaison model, it moved to the Collections Strategies and Services department. Then, shortly after the addition of a second librarian to support open education, the OES department was formed. Expanding from an "OER program" managed by a single librarian to a dedicated department in the Libraries was a natural next step toward ensuring sustainability for open education at the university.

Benefits of This Approach

A significant benefit of this approach is the increased capacity that came with the addition of a second full-time position dedicated to open education. With expanded staffing, the department is able to be more intentional about planning and developing services, has bandwidth to explore emerging needs and opportunities, and benefits from regular internal collaboration.

A mainstay of the library's OER efforts is an incentive program,[19] which awards and supports instructors in their adoption, adaptation, or creation of OER. Since becoming a specialized department, OES has refined and better articulated the library's scope for supporting OER creation and improved internal workflows (such as for consultations and OER production),

formalizing services in ways that were previously limited by staffing capacity.[20] These are now integrated into the incentive program, providing enhanced communication and support for faculty. Additionally, increased capacity enabled the department to pilot and develop open pedagogy services.[21] The incentive program, which historically had textbook affordability as its primary focus, now also awards faculty for implementing renewable assignments,[22] expanding its impact on student learning through innovative teaching methods. These significant advancements in services would not be possible without the dedicated resources and institutional support for open education.

As a department with a scope specific to supporting open education, OES is able to focus on the specialized needs of Open for teaching and learning environments. Additionally, being positioned within the RASE division allows for easy collaboration. For example, OES shares open publishing knowledge and tools with the Research Services department and collaborates with the Teaching and Learning department on synchronous and asynchronous instruction to support open pedagogy.

Challenges of This Approach

Having a department fully dedicated to focusing on open initiatives has clear benefits that enrich the quality of support for open education and impact on student success. However, it is imperative to continually evaluate the department's capacity and the level of services it can provide, even with the recent growth and increased staffing. This is especially true when developing new services while maintaining existing services. With additional resources, it may be tempting to test or launch new services or ideas, though this should be done carefully to prevent employee overcommitment or burnout. For example, while there have been improvements to support for OER creation and publishing, expansion remains intentionally slow, as any step taken toward expanding services needs to be evaluated for feasibility and sustainability.

The organization of the library is largely based around functional specialization, with various aspects of Open distributed to different departments. While there are benefits to this approach, it also results in a lack of a unified vision or approach around Open. Housing open educational services in a single department has resulted in improved services related to those areas, but the department does not support all open initiatives. On a practical level, members of the campus community wishing to engage with Open may need to connect with different library staff depending on the task at hand.

Reflections

The four case studies present a few examples of what open initiatives look like within different institutions. We acknowledge that our experiences are shaped by our environments, and readers should consider the advice given within this context and take away what resonates with their institutions and specific situations.

Each of our institutions has addressed emerging needs differently; however, capacity and sustainability are significant concerns across institutions, no matter the size or resources available. Therefore, we believe that it's important to consider scalability and growth when creating new services. Pursuing open initiatives without factoring in scalability has the potential to overwhelm and overwork staff, stretch services too thinly to be effective, and create a problem of unrealistic growth that may undermine the long-term success of programs.

Support for open initiatives in individual positions and units has tended to emerge organically according to specific local contexts in each of our own institutions. These positions and units are generally organized based on an institution's needs and heavily influenced by the personnel involved in its establishment; thus, a range of organizational placements and service portfolios naturally results. Because of this organic evolution of services, we've also noted that the services provided within each of our case studies do not fit neatly into traditional divisions

of library services. There are benefits to both proximal and organizational alignment with a variety of different library divisions, such as those described in each of the case studies. Our roles within our libraries are unique and often deal with a variety of stakeholders in highly collaborative environments. Our broad mission to educate, inform, and advise our institutions becomes fragmented and difficult to accomplish in isolation. We have found that intentional communication and collaboration within the library and across the institution are key to meeting the needs of our constituents.

We also note that there are several benefits to being both in a larger organizational structure and operating independently. For example, a larger structure (i.e., a unit within a department or division) reduces administrative burden and provides additional support within the unit for open initiatives. On the other hand, independent practitioners of open initiatives often wear many hats and thus develop broad expertise, giving them the flexibility to adapt to organizational changes. There is no one right way to organize positions or units that support open initiatives. The amount these initiatives are prioritized by the library and institution can often dictate how much time, money, and dedicated person-power is provided. Ultimately, careful consideration should be paid to the benefits and challenges presented by these case studies on how various organizational structures may impact service provision, relationships within the library, communication channels, and any existing or potential partnerships.

Recommendations

Throughout this chapter, we've provided examples of four case studies that show how different institutions organize units related to scholarly communication. In this section, we offer recommendations for other institutions that are looking to (re)organize their own services, support, and personnel in the areas of open initiatives.

- **Stand out.** Create a unified voice and defined services using mission, vision, or value statements by preparing an elevator pitch or another method. This will help you and your unit to stay focused in a strategic direction and easily convey to others what you do and how you contribute to the larger mission and vision of the institution.[23]
- **Plan proactively.** There may be a tendency to overlook planning work and focus on individual tasks or services, especially when growing under-resourced units. To conceptualize and execute sustainable growth, use a framework that you are comfortable with (examples include SWOT analysis, strategic planning, or technology roadmapping). Be mindful to incorporate flexibility in the planning process.
- **Formalize services.** Even if you're a small initiative or a single person responsible for an entire suite of services, defining your scope can assist you with communicating your mission and vision. This can include developing a strategic plan, tracking outreach and interactions with patrons, and/or creating a web presence, such as a landing page connected to the library website or a physical space for people to go to find resources and support.
- **Create community.** As solo practitioners and even when working in a team, finding a supportive space is important to sustaining services, combating feelings of isolation, avoiding burnout, and keeping up to date with best practices. Connect with others who support open initiatives in their institutions—like-minded individuals in your own institution who can support you and your work. Outside perspectives can help you overcome challenges by offering a new perspective and aid in creating a supportive community of practitioners.
- **Prioritize outreach.** The time spent on tailored, personalized outreach (both in the library and on campus) to relevant stakeholders is an investment in the success of open initiatives. Building strong relationships can lead to referrals, partnerships, and positive outcomes.
- **Grow sustainably.** Utilize resources available to you to help create sustainable growth, both for strategic planning and professional development. For example, institutes and

certificate programs often develop "action plans" designed to accelerate engagement and adoption of open educational resources. Examples include the SPARC Open Education Leadership Program,[24] AAC&U Institute on OER,[25] FORCE11 Scholarly Communication Institute,[26] and OEN certificates in Open Education Leadership and Open Pedagogy.[27]

Conclusion

Open initiatives, and scholarly communication in general, evolve quickly. This causes inherent tension within the broader effort to modernize academic libraries, which themselves are grounded in another traditional enterprise, the university. Analysis of these case studies brought to light the common themes of intentionality and advocacy to help pave the way forward.

Be intentional about which movements to follow and avoid jumping onto every new trend that appears. Instead, focus on using the tools and resources you already have access to and what you can accomplish with them. This is a balancing act, as taking on new projects will keep the unit fresh and relevant. Before adopting a trending service, carefully consider whether it is aligned with institutional needs and potential effects on the overall service portfolio of your unit.

Similarly, be intentional with staffing and organizational structure. Whether the personnel focused on open initiatives are concentrated in one or two individual positions, organized within a larger unit, or function as an independent unit, it is imperative to consider where best to situate these positions, both physically and within the organizational hierarchy. In doing so, consider the benefits of alignment with other library units and be amenable to re-arranging services for better synergies, especially during periods of growth. This approach actively fights against the inertia of the "we've always done it this way" mentality.

One question you may be asking is, When is the best time to implement the recommendations set in this chapter? We recommend assessing and advocating for your goals each time there are major staffing changes, growth in services (scaling up), and during leadership or strategic planning transitions. Taking the time to conceptualize, plan, and communicate the mission and services offered can result in significant returns.

There is no one way to offer support for open initiatives, and the four examples we've provided are focused on a few US-based institutions, which may limit our own perspectives. Ultimately, it is up to each institution to gauge what support they can or are willing to offer and how programs that support open initiatives will work best in their institutional context. We hope that by sharing our own experiences, others can mindfully employ our recommendations in a way that sustainably makes sense for their institutional needs. Greater awareness of how organizational structure affects service provision will allow practitioners to leverage benefits and mitigate challenges.

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CHAPTER 10

Shifting Fronts in the Open Access Movement: Open Collections Work in Academic Libraries

DeDe Dawson and Jaclyn McLean

Academic librarians have long been at the forefront in advocating for the principles and values of open movements and in supporting authors or instructors in making their outputs open. To date, much of this open work has centered on the areas of open access (OA), open education, or open data and has fallen to librarians with public service roles such as scholarly communication librarians (SCLs) or those in related positions (such as repository librarians or scholarly publishing librarians, etc.). However, open work is rapidly emerging as an area of interest in collection development as well. Conventional publishers are confronting the fact that OA is the inevitable future of scholarly publishing and are experimenting with new business models to support this transition, such as read and publish (R&P) agreements and Subscribe to Open (S2O) deals. It is a dynamic and fast-growing area that is now drawing in collections librarians, such as those with acquisitions, licensing, and other responsibilities related to electronic resources. Although electronic resources librarians (ERLs) have extensive experience with conventional subscriptions, these new kinds of open collections agreements require them to upskill in scholarly communications topics and become more aware of the principles of the open movement that SCLs have long advocated for. One such guiding principle is the imperative for the transition to an open publishing system to be equitable and financially sustainable.[1]

Since ERLs typically negotiate licenses and administer large collections budgets on behalf of the library, they are in the ideal position to operationalize the aspirations of such an equitable transition. To respond effectively to this quickly shifting front in the open movement, close collaboration between SCLs and ERLs is necessary. In this chapter, we discuss this shifting focus in the OA movement, open collections work, and the importance of libraries continuing to fund scholarship in new ways, and we provide examples of how SCLs and ERLs can work together to help achieve an equitable transition to OA publishing.

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Shifting Fronts in the OA Movement

The Budapest Open Access Initiative (BOAI) unofficially launched the OA movement in 2002 with its declaration that peer-reviewed journal literature is a public good that should be completely free for any curious person to read.[2] Supporting equity through the removal of access barriers is the foundational principle of OA. For over two decades now, SCLs have worked hard to advance this principle primarily by encouraging changes in author behavior and providing technical infrastructure and other supports to facilitate these changes. The most obvious examples of this are the outreach efforts to support author rights retention and to make research outputs OA through repositories—a route known as "green OA." This route enables authors to make their papers OA without paying article processing charges (APCs);[3] despite the considerable investment of time and resources toward these efforts, however, libraries have had limited success. Institutional repositories remain largely underutilized,[4] and authors still typically transfer exclusive rights to their works over to publishers without negotiation.[5]

On the other hand, publishers have been extraordinarily successful in convincing authors that the only route to OA is through paying a fee to publish in gold or hybrid OA journals. In our experience, authors commonly assume that all OA publishing involves paying such an APC. From a publisher perspective, charging authors a fee is a relatively straightforward solution to implement and, as they've found out, incredibly lucrative. Butler et al.[6] estimate that, globally, authors paid $1.06 billion in publication fees to the oligopoly of five commercial publishers (Elsevier, Sage, Springer Nature, Taylor & Francis, and Wiley) from 2015 to 2018. Many of these fees were to hybrid journals that continue to also charge libraries subscriptions, a practice known colloquially as "double-dipping"[7] and considered unethical and financially unsustainable by many librarians.[8] These big five commercial publishers control more than half of all the scholarly literature produced and are well-known to generate excessive profits.[9] Through APCs, these publishers have discovered an additional revenue stream and seem to be developing ways to further maximize profits associated with it.

An APC model incentivizes profit-driven publishers to publish more papers. And indeed, there are reports that such publishers are now pressuring journal editors to continually increase the number of papers accepted.[10] This is in obvious conflict with journal quality and the mostly volunteer editorial and peer reviewing labor upon which the system depends. These conflicting values and competing priorities, in addition to the inequities of unreasonably high APCs, are driving a recent spate of mass resignations of editors and editorial boards from major commercial journals.[11] Another way to maximize profits is through price increases, and indeed, APC prices are increasing at rates higher than inflation,[12] and data also indicate that publishers set APC prices according to the prestige of the journal and the perception of funding available to authors in each market.[13] There are costs associated with publishing, of course, but these exorbitant prices do not reflect reasonable fees for publishing services. Instead, they are composed primarily of profits and "non-publication costs," such as marketing, public relations, lobbying, innovation, acquisitions, and erecting and enforcing paywalls.[14] Privileged authors with funding are able to pay such "prestige taxes" and advance their careers more easily than those without such funding.[15] This means that unfunded authors, particularly those based in low-income countries, are at risk of being even further marginalized from the research conversation.[16] While the original goal of the OA movement was to reduce barriers for readers, these barriers are now shifting to authors due to the strong preference for the APC model by profit-driven publishers. As such, in addition to being deeply unpopular with authors, APCs are also now widely considered to be inequitable.[17]

Open Collections Models

Instead of (or in addition to) charging authors directly, publishers are now developing a new variety of "open access big deal"* based on the APC model to sell to libraries. These are known as read and publish (R&P) or transformative/transitional agreements (TAs). R&Ps bundle together "read" payments, which provide the institution's readers with access to the traditional subscription content from the publisher, with a "publish" amount, which pays for the APCs of corresponding authors of the institution in the publisher's journals. It has been argued that there are enough library collections funds currently invested in journal subscriptions globally to fully fund OA publishing instead, transforming all existing subscription journals to fully OA models.[18] This is the origin of the term "transformative agreement." But there are growing criticisms that these R&P models are actually not leading to such a transformation; they are instead perpetuating the problems in the current scholarly publishing system, further entrenching powerful commercial interests and high prices and exacerbating inequities for authors at institutions that cannot afford to join an R&P agreement.[19]

Mission-driven publishers have developed viable alternatives to the profit-driven model of APCs and their successor, R&Ps. While it is understandable that, in a capitalist society, companies will seek to maximize profits, we argue that this is an unethical and inefficient use of the public funds entrusted to universities for the purpose of supporting research. Alternative OA journal funding models exist that are mission-driven (as opposed to profit-driven), do not charge fees to readers or authors, and are more equitable and financially sustainable for the library community to invest in. Subscribe to Open (S2O) is one such promising model that is being adopted by non-profit publishers such as Annual Reviews.[20] S2O converts a publisher's journals to OA on a yearly basis contingent upon sufficient participation by subscribers.[21] Other initiatives include sponsorship models (e.g., Lyrasis' Open Access Community Investment Program)[22] and collaborative funding and infrastructure models (e.g., Open Library of the Humanities).[23] These and other funding mechanisms support an estimated 29,000 diamond (no APC) OA journals now, many of which are published by scholar-led and mission-driven organizations.[24] This vibrant and growing area of publishing demonstrates that although there are still costs associated with OA publishing, they have the potential to be significantly reduced if institutions are not obligated to fund the non-publication expenses and extreme profit margins of legacy publishers.[25]

All this activity seems to be shifting the main front in the OA movement in libraries from outreach and training approaches (i.e., focused on changing author behavior) to collections approaches (i.e., libraries directly funding OA publishing through new models). Academic libraries are not just purchasing content for their campus communities anymore, but redirecting collections funds towards OA publishing, whether through APC-based R&Ps or innovative alternative models to support diamond OA journals and open infrastructure. There is much more at stake in these new circumstances, and librarians have a professional obligation to use our knowledge of the inequities and inefficiencies in the profit-driven system to influence change in an ethical, equitable, and sustainable direction that is more in alignment with the values and missions of the academy. From decades of advocacy in the trenches, SCLs are intimately aware of these challenges and can partner with ERLs to continue this work on the rapidly growing new front of open collections.

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* There is a growing trend of libraries cancelling these big deals as tracked by SPARC (https://sparcopen.org/our-work/big-deal-cancellation-tracking/). "Big deal" is a colloquial term that refers to an online collection of most or all of a publisher's journals sold to libraries as a subscription package for one price (Kenneth Frazier, "The Librarians' Dilemma: Contemplating the Costs of the 'Big Deal,'" D-Lib Magazine 7, no. 3 (2001), https://doi.org/10.1045/march2001-frazier), similar to cable television packages. Big deals have been controversial due to unsustainable price increases and decreasing value since many titles in the packages are not used (Cindy Sjoberg, "E-Journals and the Big Deal: A Review of the Literature," SLIS Student Research Journal 6, no. 2 (2017), http://scholarworks.sjsu.edu/slissrj/vol6/iss2/3/; Fei Shu et al., "Is It Such a Big Deal? On the Cost of Journal Use in the Digital Era," College & Research Libraries 79, no. 6 (2018): 785–98, https://doi.org/10.5860/crl.79.6.785).

Open Collections Work

At first glance, one might think that if a resource is openly available, there is no role for the library to play in acquiring it or managing it in the collection. But in fact, there are costs associated with publishing open scholarship, maintaining open infrastructure, and managing metadata for discovery[26] that require support from somewhere. Academic libraries have traditionally funded scholarly publishing through their role as purchasers of content, and as this content transitions to OA, it makes sense that the role of libraries also shifts to supporting the production side. And as mentioned, in recent years, publishers and vendors have been rapidly innovating in this area, introducing new open collections models for their content to library customers.[27] As a result, acquisitions work in academic libraries has become exponentially more complex.[28]

New Workflows and Skills for ERLs

ERLs typically negotiate and administer licenses for access to electronic resources such as literature databases, ebook packages, and journals on behalf of the library and the campus community. E-resource management and licensing have changed significantly over the last twenty years, from digitized microfilm and scans of print copies to streaming videos, three-dimensional interactive models, and other born-digital items. The sales and pricing models have also diversified, from one-time purchases with annual access/hosting fees to demand-driven acquisition (DDA) and evidence-based acquisition (EBA) for ebooks,[29] and the aforementioned big deals, with complexities to manage, such as core lists, add-on titles, and limited perpetual access. These transitions seemed chaotic with steep learning curves at the time, but looking back now, they feel like the halcyon days. Work was changing and new learning was required, but the foundation of the work remained the same: ensure user access through complex systems configurations, enforce paywalls, and contain costs.

Most conventional e-resource licensing follows similar frameworks[30] so ERLs can streamline their work, but newly emerging OA collections models like R&Ps typically have different workflows or structures that require new knowledge, skills, and ongoing management by specialist library staff.[31] Emery and Stone's[32] inclusion of the new category of "open access" in their Techniques for Electronic Resources Management (TERMS) framework acknowledges that this is a new and important area of e-resource management work. The growing complexity of managing these models suggests that large R&P agreements are likely to take up significantly more staff time than smaller S2O, diamond models,[33] or traditional licensing. In some cases, new OA collections models follow conventional processes, so they are easier to adopt. For example, one of the significant benefits of the S2O model for libraries is that it uses existing subscription systems and established workflows.[34] This means that libraries do not need to create special processes and train staff to administer them. Similarly, in our experience, financially contributing to diamond OA initiatives or supporting open infrastructure typically involves one-time or annual payments that are also reasonably straightforward to administer under current acquisitions frameworks.

In contrast, R&P licenses and administration outstrip even the most complex conventional license, requiring significant staff time to manage both the access, or "read," elements of paywalled content (if any) and in administering the "publish" elements of the license for the campus community.[35] There are practical aspects of R&P deals that ERLs and collections staff need to manage now that were not part of their responsibilities in the past. Such things include verifying author affiliations for the "publish" portion of the deal, which can be challenging since the definition of affiliation is often unclear in the agreements.[36] But perhaps the biggest challenge is clearly and concisely communicating with campus authors about these deals because the details of agreements are complex, differ from one publisher to another, and are subject to change after each renegotiation. In our experience, authors strongly associate open access with paying expensive APCs, so some may immediately reject any OA offer

without scrutiny. This is one area where SCLs and liaison librarians can help, but the work is time-consuming, commonly requiring multiple communications with individual authors to explain the deal and alleviate author skepticism.

These new licenses require ERLs to significantly upskill in understanding different OA models and how to negotiate for favorable license terms within each model. A basic level of knowledge of each model is the minimum requirement, but a deep understanding of OA publishing is what is ideally needed. These include an understanding of the issues inherent in the APC-based hybrid and gold OA publishing environment upon which R&P models are based, but also adjacent topics such as author rights issues, copyright, and Creative Commons licenses.

Assessment is also changing. Gone are the days when evaluating collections could be done based on the number of article downloads or cost-per-use data alone. Assessing the value and performance of these deals will require new techniques in monitoring the publishing activity of affiliated authors and the fluctuating APC prices of publishers. Although this kind of data seems like it should be readily available from publishers, that is rarely the case.[37] Instead, librarians and other information professionals are developing elaborate methods to track institutional outputs (e.g., see Langham-Putrow and Enriquez's *Analyzing Institutional Publishing Output: A Short Course*)[38] and investing massive amounts of time in tracking APC prices across publishers and over time (e.g., see Butler et al.'s open APC dataset).[39] Usage data is still relevant, and the COUNTER standard,[†] developed to more consistently measure online usage of scholarly content across different vendor platforms, has evolved significantly over the last several years. In its latest release (5.1), it attempts to address concerns about how OA usage is measured and reported (though it is also important to note that COUNTER can only measure usage by authenticated library users—and the nature of OA is that it is free to read and does not require institutional affiliation or login).[40] In short, ERLs now need to know more about the entire publishing process and incorporate it into assessment work, bringing in more data sources beyond COUNTER usage data and cost-per-use measures.

Collaboration with SCL colleagues can help support ERLs in this upskilling. SCLs can (and do) also contribute to the management, communications with authors, and assessment of these new licenses. These collaborations are dependent on breaking down the traditional silos between these roles and on SCLs and ERLs having the capacity and willingness to engage in this new way.[41] Building the capacity in both roles to take on open collections work requires either reducing or deprioritizing other work or building larger teams to spread these responsibilities. For example, there are opportunities for engaging liaison librarians in handling the communications to authors in their areas. We are also anecdotally aware of a growing trend of library administrators, such as associate or assistant university librarians for collections, having SCL responsibilities added to their roles. This perhaps signals the strategic importance and complexity of this new area of work.

Values-Based Collecting

Historically, collections work in academic libraries has focused on cost containment, financial management, and the responsible stewardship of funds for the purchase of print resources and access to online (paywalled) content to support the teaching and research needs of campus. Managing those funds is a central responsibility for most ERLs. The expanding focus from thinking primarily about containing costs to now also thinking about the values and issues involved in *how* the money is spent is significant. This change of focus also requires librarians to think more innovatively and strategically about library collections. For example, how collections funds can play a role in increasing bibliodiversity[42] and in facilitating the transition to

[†] The COUNTER standard strives to create consistency in usage data of online published literature so that publishers, aggregators, and technology providers can deliver credible, consistent, comparable usage metrics to libraries and consortia around the world (https://www.countermetrics.org/).

equitable, sustainable, mission-driven publishing models owned or controlled by researchers and in line with the values and interests of the academy.

Changing the direction of the library's collection budget or creating new funds for open initiatives requires the ability to clearly articulate the issues with the current system and to advocate for the diversion of (often limited) funding in a new direction. MIT Libraries have been leaders in thinking more strategically about collections spending. In 2016, they incorporated their collections budget into their scholarly communications program in order to invest more systematically in transforming scholarship toward openness and democratized access.[43] One way they found to do this was to invest in local, unique collections that other academic libraries would not have the opportunity to collect—a concept Lorcan Dempsey calls "inside out" collections.[44] Such values-aligned investments can also support the equity, diversity, inclusion, and accessibility (EDIA) goals of the library and university.[45] Lindsay Cronk refers to this practice as "operational scholarly communication"[46] and argues, "We have used our investment in collections as a proxy investment in scholars, but what if we actually directly invested in scholars?"[47] Investing with a values-based lens has the potential to promote EDIA perspectives and reduce the whiteness and colonial focus of current library collections while also supporting bibliodiversity in the publishing ecosystem.

This disinvestment in traditional scholarly publishing and redirection of funds toward building a more equitable scholarly publishing environment for the future requires advocacy but also needs to be considered within the current paradigm. It's unlikely that libraries can divest completely from journals and books produced by profit-driven publishers. But we can perhaps consider more principled approaches in our negotiations with legacy publishers. Massachusetts Institute of Technology (MIT) Libraries has also been a leader in this realm. Their *Framework for Publisher Contracts*[48] has been endorsed by dozens of other academic libraries that also want to advance open scholarship and the public good.

Taking a principled approach to decisions on what new models or opportunities to invest in is a natural path for librarians. As the open collections models continue to shift to funding publishing instead of access to read the literature, the question is raised about the future of the library's role. We are no longer solely purchasing the outputs at the end of the scholarly publishing cycle but are now involved in funding the earliest stages of its production. This raises some questions. Should the library involve research offices and other stakeholders when considering open collections agreements? Does the role of covering APCs for authors even belong to the library? These conversations, at least for now, circle back to the roles of ethical stewardship of funds and access to information that libraries have always had and many ERLs strongly believe in.[49] In contemplating these sorts of questions, ERLs could benefit significantly from collaboration with SCL colleagues, who have been steeped in similar existential discussions for a long time.

SCL and ERL Collaborations at the University of Saskatchewan

The emerging importance of open collections work within the open movement is blending some of the previously disparate roles of SCLs and ERLs and requiring new learning and collaboration. SCLs need to learn about collections models, license negotiations, administration of models like R&Ps that require various levels of support for authors and new kinds of assessment techniques. ERLs are upskilling on scholarly communications topics, such as the various routes to OA and models for funding it, especially the inequities of APCs and collections models based on APCs (i.e., R&P models). ERLs also need to assess these new open collections models in ways beyond usage and venture into new areas of work, such as communicating with campus authors and advocacy. There are many opportunities for collaboration between SCLs and ERLs on all these topics, and our resulting work will benefit from the combination of our expertise and experiences.

At the University of Saskatchewan (USask), we have been exploring such collaborations for several years now. Early on, when publishers began offering APC discounts for USask authors, we collaborated to document and communicate to campus authors about this. As we began participating in various R&P deals, this collaboration ramped up with the increased need to communicate more complex information to authors and administer various aspects of the deals. In 2022, we formed an Open Collections Working Group with several other colleagues to formalize this and other work related to values-based collecting. One such project involved the creation of a document, *Library Criteria for Investing in Open Scholarship*,[50] which we are now in the process of applying to vendor offers. Academic libraries are increasingly producing these kinds of documents outlining the principles and values involved in assessing potential open collections investments,[51] and SPARC[52] has created a list of many examples.[53] As open collections and values-based collecting continue to grow in importance, we anticipate many more opportunities for collaborations between SCLs and ERLs.

Conclusion

SCLs have been at the forefront of the open movement in libraries, mostly through supporting authors and instructors in making their works open. However, open collections is a rapidly growing new area of the open movement that is now drawing in collections librarians such as ERLs. ERLs typically negotiate licenses for e-resources and administer large collections budgets, and they are now being confronted with making decisions about new kinds of open collections deals offered by publishers. There is much more at stake in these new circumstances: while OA was originally intended to increase equity of access to scholarly literature, the focus on the APC model by profit-driven publishers is increasing inequity for authors. Open scholarship is not free to produce, and it needs to be funded, but the transition to an open publishing system also needs to be ethical, equitable, and financially sustainable. Academic libraries can work toward this ideal by thinking more strategically about how they invest their collections funds. This requires innovative, values-based collections decisions. It also requires deep knowledge of the issues and advocacy. SCLs can be natural allies in engaging with ERLs to operationalize these aspirations. The emerging importance of open collections work is shifting the front within the open movement and making close collaboration between SCLs and ERLs necessary.

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CHAPTER 11

Better Together: Exploring Vendor Roles in the Koha, Evergreen, and FOLIO Open-source Communities

Zorian Sasyk and Robert Heaton

Introduction

At the 2016 Code4Lib annual conference, Index Data* co-founder and president Sebastian Hammer presented a then-nameless concept for a new, open-source library services platform[1] (LSP; see table 1 for a comparison with the prior-generation "integrated library system" [ILS] concept). That system became FOLIO,† which now, almost a decade later, is used for day-to-day workflows at hundreds of libraries. The rapid development and adoption of FOLIO widens the long-term industry presence of open-source software (OSS) solutions for libraries, building on the historical presence of Koha (the most widely adopted ILS in the world)‡ and Evergreen. Because of their strong communities of users and developers, these open projects thrive even in a consolidated market that includes Clarivate's proprietary Alma, Sierra, and Polaris systems as well as OCLC's WorldShare Management System. This chapter investigates the intersections among core library values, the values of the OSS movement, and the business interests of library vendors, highlighting how various vendors participate in and shape the respective communities that maintain Koha, Evergreen, and FOLIO. Emphasis is placed on the history, governance, and development models of these projects, illustrating that library vendors play a unique and important role in the healthy functioning of these three OSS communities.

* Index Data is a library software company that provides open-source software services.

† Styled as an acronym for "The Future Of Libraries Is Open."

‡ Used at over 4,200 libraries, according to Marshall Breeding, "Library Technology Vendors in the United States and Canada," Library Technology Guides, 2022, https://librarytechnology.org/vendors/.

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Integrated Library System (ILS)	Library Services Platform (LSP)
Term popularized	
1980s	2010s
Functional characteristics	
Library workflows—e.g., material acquisitions, resource description, user and circulation management—are conducted in one "turnkey" system.	Electronic-resource-management (ERM) functions previously handled in assorted external systems are integrated alongside other library workflow areas.
Technical design	
The software is accessed via a locally installed client application.	The software can be deployed over the cloud and accessed via a web application or programmatically via application programming interfaces (APIs).
The program is modular, containing discrete areas supporting individual workflows.	Groupings of microservices can be customized to include those relevant to specific workflows.
Example systems	
Koha (open source) Evergreen (open source) Voyager (Ex Libris) Symphony (SirsiDynix)	FOLIO (open source) Alma (Ex Libris) WorldShare Management System (OCLC)

Table 1. Technical and conceptual comparison: ILS vs. LSP. *Sources*: Adapted from Marshall Breeding, "Smart Libraries Q&A: Differences between ILS and LSP," *Smart Libraries Newsletter* 40, no. 10 (October 1, 2020): 3–4; Marshall Breeding, "2024 Library Systems Report," *American Libraries Magazine* (May 1, 2024), https://americanlibrariesmagazine.org/2024/05/01/2024-library-systems-report/.

Free and Open Source Software (FOSS)

Within the larger OSS movement, there are various functional definitions of what it means to be "open-source software," based on the software license's permissiveness. As a general principle, according to Richard Stallman's "GNU Manifesto," free software means "software that comes with permission for anyone to use, copy, and/or distribute, either verbatim or with modifications, either gratis or for a fee. In particular, this means that source code must be available."[2] The important further distinction is in how "copyleft" the open-source license is. Copyleft licenses like the GNU General Public License[3] require all repurposed or adapted software to retain the same license, effectively prohibiting proprietization. Individuals or companies can modify the software code and even charge to distribute it, but they do not retain any exclusive right to their additional code. In contrast, non-copyleft open-source licenses like the MIT License or the Apache 2.0 license allow their code to be used in proprietary solutions as long as the license terms are included for the original code. This does not bind additional code to be released with open-source licensing.

Balancing corporate needs with those of the community is not unique to library OSS projects; many large OSS projects face this same reality. Many of the best-known projects in the

OSS movement have significant corporate involvement. The Android mobile operating system is driven by Google, which has a vested interest in the project as the basis of its mobile phone business. The Linux operating system received approximately 61 percent of its code contributions for the latest Kernel release from vendors such as Intel, Google, Red Hat, AMD, Oracle, and Microsoft.[4] There are many other examples of successful OSS projects with similar vendor participation, such as Apache, Mozilla Firefox, and LibreOffice. It is not surprising that libraries, with their core user base of public institutions, such as municipal libraries and academic institutions, have also developed OSS both with the guidance of the OSS community and with the structure and material resources of software vendors.

FOSS, Libraries, and Vendors: Values and Services

The library profession and the OSS movement have broadly similar values and ethics. Compare the excerpts in table 2, one from the ACRL's "keystone principles" for how library communities should work, the other from the GNU Project's "four essential freedoms" regarding how FOSS communities should work.

ACRL's Keystone Principles	GNU's Four Essential Freedoms
Public good/community benefit	
"Scholarly and government information is a 'public good.'"	"By [improving and redistributing code] you can give the whole community a chance to benefit from your changes."
Broad freedom from external influences	
"[Scholarly and government information] must be available free of marketing bias, commercial motives, and cost to the individual user."	"The freedom to run the program as you wish, for any purpose."
Systems innovation	
"Libraries are responsible for creating innovative information systems for the dissemination and preservation of information and new knowledge regardless of format."	"The freedom to study how the program works, and change it so it does your computing as you wish."
Intellectual community through sharing	
"The academic library is the intellectual commons for the community where people and ideas interact … to expand learning and facilitate the creation of new knowledge."	"The freedom to distribute copies of your modified versions to others."

Table 2. Comparison of library and OSS values. *Sources*: "The Keystone Principles," American Library Association, *College & Research Libraries News* 61, no. 2 (February 1, 2000): 103–4, https://doi.org/10.5860/crln.61.2.103; "What Is Free Software?," Free Software Foundation, GNU Project, January 1, 2024, https://www.gnu.org/philosophy/free-sw.en.html.

Both libraries and OSS communities value open information, open standards, equality, sustainability, the public good, and social responsibility. These values may seem at odds with the capitalistic, profit-driven principles of corporations, including library vendors. However, the OSS movement has a long history of corporate involvement; in fact, the GNU Project encourages it, as mentioned on its "What Is Free Software?" web essay: "'Free software' does not mean 'noncommercial.' On the contrary, a free program must be available for commercial use, commercial development, and commercial distribution.... [T]o exclude commercial use, commercial development or commercial distribution would hobble the free software community and obstruct its path to success."[5] To the extent that OSS is built on this principle, institutions' acceptance of OSS implies acceptance of that corporation-friendly environment.[§]

Software vendors offer important components for the process of development and maintaining library OSS, some of which the library community and the institutions in which they operate cannot provide as effectively. These services can be grouped into three categories: technical infrastructure, agile-based product development, and system-implementation project management.

Technical Infrastructure

Library vendors' most critical contribution to library OSS communities is providing the technical infrastructure and resources for hosting, compiling, updating, and writing the software itself. Although all three library OSS systems discussed in this chapter can be hosted locally, the vast majority of libraries using them have implementations that are hosted and maintained by vendors. In essence, vendors provide the software-as-a-service (SaaS) component of OSS software, delivering its significant benefits. This includes the physical hardware for hosting the software as well as dedicated IT support staff for deployment, security, data backups, and configuration. In addition, many library OSS developers are paid employees of the involved vendors, similar to other OSS projects as mentioned above. Having a core set of paid contributors provides project growth and stability and allows the community to focus on other aspects of the project, such as feature development and release planning.

Product-Management Expertise and Agile Processes

Library vendors can also lend non-developer resources to a project at scale, such as product managers who specialize in formalized agile development methodologies such as Kanban, Lean, or Scrum. Although agile methods are increasingly being adopted by libraries for internal processes and projects,[**] libraries and their staff typically do not have much experience in this field, due to its underrepresentation in library school curricula as well as libraries' general orientation toward customer service rather than software development. The injection of library OSS software projects with dedicated and professional product managers allows projects to become increasingly organized, further formalizes communication channels between developers and users as stipulated by governance models, and allows projects to scale up and increase development speed. Feature requests generated from the community can be planned out and resourced in advance, leading to the development and execution of time-bound project-release roadmaps. These roadmaps communicate to both current and potential users the direction a project is heading.

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§ The ACRL principles call out "commercial motives" as a negative value, but this is explicitly done in the context of bias in information resources, which is not directly applicable to the provision of information systems.

** See, for example, Linda Ballinger, "Agile-Ish: Bringing Agile and Scrum into Project Management for Digital Collections Metadata," *Library Leadership & Management* 33, no. 4 (September 26, 2019), https://doi.org/10.5860/LLM.V33I4.7365; Eric Prosser, "Navigating Projects in Academic Libraries: A Scoping Review of Project Management Applications," *The Journal of Academic Librarianship* 50, no. 5 (September 1, 2024): 102929, https://doi.org/10.1016/j.acalib.2024.102929; Morgan M. Stoddard, Bill Gillis, and Peter Cohn, "Agile Project Management in Libraries: Creating Collaborative, Resilient, Responsive Organizations," *Journal of Library Administration* 59, no. 5 (July 4, 2019): 492–511, https://doi.org/10.1080/01930826.2019.1616971.

Migration/Implementation Project Management

The final component that vendors provide library communities is project implementation services and best practices. Almost all libraries adopting a library OSS system will need to migrate bibliographic, vendor, and patron data from their old system to the new one. Considering the volume of holdings and user records for even moderately sized libraries, system migration is a daunting task involving data cleanup, extraction, transformation, and loading of thousands if not millions of records. Library OSS vendors can fill the implementation knowledge gap in the community by offering expert implementation project management specific to that ILS or LSP as part of their services. Implementation managers, in addition to their professional roles at their employers, often participate in documentation and community support roles due to their in-depth knowledge of the software and their experience implementing the software at many different kinds of libraries.

Vendor Involvement in Koha, Evergreen, and FOLIO

By the accepted industry definitions, Koha and Evergreen are integrated library systems (ILS) while FOLIO is a library services platform (LSP; definitions outlined in more detail in table 1). Koha is the oldest of the three, first adopted in 2000, followed by Evergreen in 2007 and FOLIO in 2019. Figure 1 shows the timing and rate of each product's growth. With hundreds of libraries using each of these systems, they all play a prominent role in the marketplace. The communities supporting these systems have much in common, but their histories, origins, principles of licensing and governance, and range of contributors also reflect important differences.

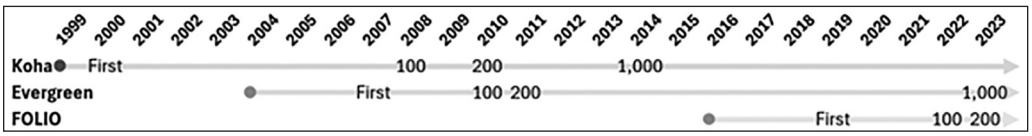


Figure 1. Timeline of adoption milestones. Respective timelines begin with the year each product was announced. Numbers are of adopting institutions, not individual sites (a calculation from the raw numbers shows an average of roughly 1 to 2.5 sites per institution). Based on data from Marshall Breeding, "Libraries.org: Search for Libraries," Library Technology, accessed December 4, 2024, https://librarytechnology.org/libraries/advanced/. After logging in to this URL with a free account, data was compiled by entering each system's name as the "Current Automation System," checking the "Main Site" box, entering one year as the "Year ILS Installed," recording the number returned, and repeating for each year of the system's existence. Because of how the database is structured, this query excludes adopters no longer using the system, and therefore totals for past years may decline in future searches. Dates or definition of adoption given in this source may differ from those in community or institutional announcements.

Koha Community Operations

The Koha OSS ILS project is one of the oldest continuously developed library OSS projects.[6] According to Marshall Breeding's Library Technology Reports website, over 4,200 libraries of all types worldwide use the system as their ILS.[7] Over 600 individual developers have contributed to the code base over the twenty-five-year life of the project. Chris Cormack of Katipo Communications initially created Koha in 1999 to replace an ILS affected by a Y2K bug, and since then, Koha has greatly expanded and is supported by both its community of users and a

core group of vendors that contribute support for development, hosting, and implementation. In many ways, it was the pioneering library OSS system and community.[8]

Koha is licensed under the GNU Public License version 3 (GPLv3),[9] making it a copyleft-oriented project. This orientation is clearly reflected in its governance structure, its development cycle, and its release management. The overall project, and in particular the Koha project logo trademark, is legally overseen by the Horowhenua Library Trust (HLT). This trust, in conjunction with the Koha community, appoints a seven-member committee to make recommendations and represent the interests of the community to the trust. Typically, five or six members of the committee hold certain roles of the project, such as maintainer of the current release, manager of the next release, translation managers, the "Kaitiaki" (Maori term for guardian or steward), the head librarian of the HLT, and a trustee of the HLT.[10] One member is chosen from the wider community for a two-year term. There are also explicit limitations on involvement by corporate interests in Koha; no more than 30 percent of the members of the committee can "have a material financial interest in or [be] employees of the same organisation or whose organisations have a supplier/customer relationship."[11] In addition, members of the committee who fall under the above definition are required to abstain from participation in any votes in which they or their organization may have a financial or material interest.[12]

Koha's ongoing development framework is agile-based, with a set release schedule twice a year in May and November.[13] A general list of project goals is set for each release, but they are not binding, and the release managers reserve the right to set priorities and postpone specific ideas until later. The project uses Bugzilla to track bugs and features being worked on.[14] As these are finished, they are vetted in a quality-assurance process and then committed to the Main Koha branch. Some time before each upcoming feature release date, the main Koha branch is frozen, final testing is done, and it is released as a new major version. Thus, development is fairly decentralized, with individual contributors having the freedom to pursue a variety of features or bugs based on their interests or needs.[15]

The above notwithstanding, Koha's development is primarily done by individuals affiliated with vendors. Using commits per developer as a measure of contribution, and based on data available on openhub.net, the authors determined that of the top ten contributors to Koha over the lifetime of the project, eight either worked for or were associated with known Koha vendors such as ByWater, BibLibre, PTFS Europe, and Equinox Open Library Initiative. The other two were affiliated with library systems using Koha. Together, these contributors account for approximately 56 percent of total commits to Koha.[16] Measuring and comparing total code commits is an imperfect way of gauging the composition of user participation in an OSS project; however, it does illustrate that despite Koha's more egalitarian licensing and governance structure, vendors play a significant and beneficial role in its development.

Evergreen Community Operations

Evergreen is another OSS ILS project; similar to Koha, it has been around for almost two decades. Evergreen is installed and utilized by over 2,000 libraries globally, ranging across most library types. According to the Evergreen website, the project positions itself as "highly-scalable software for libraries that helps library patrons find library materials, and helps libraries manage, catalog, and circulate those materials, no matter how large or complex the libraries."[17] It has a small but dedicated team of eleven core code contributors, supplemented by a much larger group of individuals who regularly contribute patches as well as documentation, error reporting, and "organizational energy."[18] Evergreen was started in 2005 when the Georgia Public Library System needed a new ILS that could meet the complex resource-sharing needs of a 275-member public library consortium. After a failed request for proposal (RFP), the organization, with the backing of the State of Georgia, decided to develop its own OSS ILS. The initial code base completion and launch of Evergreen occurred twenty-eight months later in late 2006. Soon after the launch of Evergreen, the development team split off from the state and founded their own company, Equinox Software, dedicated to the continued development

and support of the new ILS. Today, this company lives on in the non-profit organization Equinox Open Library Initiative.[19]

Similar to Koha, Evergreen is licensed under the GNU Public License version 2 (GPLv2).[20] The nine-member Evergreen Project Board, elected annually by participating members (defined as those who contribute code or documentation or are employed by Evergreen libraries), oversees the legal interests of the project as well as serving as occasional arbiters of strategic or operational decisions.[21] This board is governed by the project's bylaws, which enumerate the responsibilities of the board, authorize its committee structure, and outline procedures for elections. Unlike its counterpart in the Koha community, this board is not subject to limitations based on corporate or organizational membership; however, the bylaws do contain restrictions on voting for board members who might have a material interest in decisions being considered.[22]

Compared to Koha, Evergreen has a moderately centralized decision-making and development structure. Equinox Open Library Initiative, as the organizational successor of the original developers of the project, still carries out most of the project's development.[23] However, development is also driven by participation of member libraries, organized into the Evergreen Community Development Initiative (ECDI). The ECDI consists of several library consortia that utilize Evergreen, and it serves as a means to focus development priorities based on community needs "to drive useful and in-demand development by focusing the available funding and staff support on community targeted projects."[24] The ECDI holds annual votes among its membership to determine the Evergreen project priorities that it will fund.[25] Thus, a predetermined feature roadmap mostly drives Evergreen's development and release cycle.[26]

FOLIO Community Operations

When work on FOLIO began in earnest in 2016, three organizations (sometimes referred to as the "tripartite"[27]) supplied funding and leadership: EBSCO, Index Data, and the Open Library Environment (OLE).[28] As development progressed, thirteen organizations, representing more than 227 libraries[††] in a broad array of sizes, types, and locations, committed to adopting a "market viable platform" by summer 2020.[29] Chalmers University of Technology in Sweden adopted FOLIO for daily library operations in the fall of 2019,[30] followed less than a year later by the first FOLIO institution in the United States, Missouri State University.[31] Upon Chalmers's adoption of FOLIO (as of the "Daisy" release),[‡‡] the LSP featured basic functions in electronic-resource management (ERM), acquisitions, cataloging, and circulation; many more features were yet to come, including data import and export, user management, a circulation log, bulk editing, order templates, ERM dashboards, and many usability enhancements before FOLIO could be considered fully featured.[32] As a testament to the LSP's growth since that time, just under a decade after its announcement, FOLIO is used for day-to-day workflows at hundreds of libraries, including libraries of all sizes and types as well as shared consortial catalogs. The project and product are international in reach, with libraries live on FOLIO that are based in North and South America, Europe, Asia, Australia, and Africa.[33] Prominent international examples include: the Shanghai Library,[34] the largest public library in China; member libraries in Germany's GBV Common Library Network;[35] the National Library of Australia;[36] and, of course, the Library of Congress,[37] by many measures the world's largest library.

Various principles of licensing, architecture, and governance come together to make the FOLIO OSS community work. FOLIO is licensed under the Apache 2.0 License,[38] which, as described earlier, is permissive, allowing even commercial reuse and extension. Architecturally, FOLIO is built around "microservices," which is "a flexible approach where modules or

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[††] Along with German service provider hbz and 9 one-member institutions, GBV is a network of over 200 libraries in Germany, the Fenway Library Organization has 12 member libraries, the Five College Consortium has 5.

[‡‡] FOLIO's major release versions are named after pollinating flowers. See "Flower Release Names," FOLIO Wiki, October 17, 2024, https://folio-org.atlassian.net/wiki/spaces/REL/pages/5210505/Flower+Release+Names. Where FOLIO's flower releases increment alphabetically, Daisy was FOLIO's fourth release version.

apps related to any given area of functionality can be inserted or replaced."[39] "As long as the FOLIO RESTful API contracts are met,"[§§] any interested party can build new features that implementing libraries can optionally use instead of or alongside those adopted by the community at large. This allows local customization as well as organic evolution over time. One example of code contributed under contract with a specific organization is Knowledge Integration's work on behalf of GBV to develop and maintain FOLIO apps specific to ERM.[40]

Representatives from libraries, vendors, and the Open Library Foundation (OLF)[***] began work on a revised, long-term governance model in 2019, and FOLIO stakeholders approved the resulting document in January 2021.[41] By that point, the FOLIO community included twenty-five member organizations.[42] Solidifying a mature model was significant because it "substantially turned FOLIO to an open and inclusive community that is managed and built by all the member institutions, whether libraries or companies, and regardless of their sizes, locations and budgets."[43] The 2021 model, with minor subsequent revisions, remains in effect today. Under it, a tri-council system governs FOLIO's development. The Community Council, Product Council, and Technical Council all have important roles in moving the work forward:

- the Community Council "ensure[s] the community is strong and healthy," including maintaining its character as "collaborative and inclusive";
- the Product Council "act[s] on behalf of the FOLIO Community to express and champion priorities in the development of the FOLIO Library Service Platform"; and
- the Technical Council "provides technical guidance to the FOLIO project."[44]

Members and chairs of each council are elected to staggered terms from among member organizations, with each organization eligible to hold only one elected position per council at a time.[45] As of June 2024, both EBSCO and Index Data were represented on all three of the governing councils.[46] These councils average twelve members in size, so the "maximum of one member" rule wisely limits any organization's influence and "foster[s] diversity and involvement by a wide group of members."[47] Whereas many libraries cannot dedicate the time and energy to this type of OSS community engagement, vendors' representation is commensurate with their high level of development contribution, without giving their voice disproportionate weight. Diverse libraries and consortia are represented on the councils, including Lehigh University, Texas A&M University, Cornell University, and the Five Colleges Consortium.

FOLIO is developed under an agile methodology,[48] where product owners (POs) coordinate the efforts of development teams in consultation with subject-matter experts in special interest groups (SIGs). The work of POs is funded by the sponsoring organizations for which they work, but the community wiki specifies that they are to "support the entire community," "[l]iste[n] to all perspectives … to design common solutions," and "[v]erif[y] with the broader community before making changes requested by a single institution."[49] As one article describes, FOLIO's "[p]roduct development is based on the actual needs of library staff and their patrons. Users' requirements are well expressed, considered and incorporated in the design, development and testing processes through close communications and interactions with library staff."[50] POs step through this process primarily through SIGs, which operate under the direction of the governing councils. Because anyone can attend SIG meetings, it is difficult to capture their "membership" accurately, though meeting attendance is recorded in their respective public

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§§ Don Watkins, "Why Libraries Need to Get with Apps and APIs," Opensource.com, June 27, 2018, https://opensource.com/article/18/6/folio. An API contract "defines rules, specifications, and expectations for data exchange, functionality, and communication" among software and services (see "API Contract: What It Is and How to Use It (Examples)," Adobe Acrobat, 2024, https://www.adobe.com/acrobat/business/hub/what-s-included-in-an-api-contract.html). RESTful APIs are defined as featuring a client-server architecture that sends "stateless" calls over hypertext transfer protocol (HTTP), meaning it does not open a database-connection session but instead displays or modifies specific resources in response to individual requests (see "What Is a REST API?," RedHat.com, May 8, 2020, https://www.redhat.com/en/topics/api/what-is-a-rest-api).

*** The OLF was formed in 2016 to manage the FOLIO project's governance (see Breeding, "EBSCO Supports New Open Source Project"), manage the open licensing and availability of project code, and encourage contributions from the library community (see FOLIO.org, "Open Library Foundation Established"). In June 2024, the Koha Project Foundation also voted to join the OLF (see "Koha Community Newsletter: July 2024," Koha Library Software, July 31, 2024, https://koha-community.org/koha-community-newsletter-july-2024/).

wiki pages. Along with a large number of libraries, vendors such as EBSCO, Index Data, and ByWater Solutions send participants to these important groups as well as provide some of the SIGs' "conveners," who schedule meetings, call for agenda items, and facilitate discussion.

Feature requirements, developed through POs' work, iteratively inform the programming work done by software development teams, guided by the Product Council's roadmap.††† By one estimate from 2023, EBSCO is responsible for about three-quarters of FOLIO's development manpower.[51] Anecdotally, most libraries are unable to dedicate software developer positions to open-source projects such as FOLIO, and even when (generally large) libraries do so, it is contingent on that work directly benefiting the sponsoring institution. This leads logically to development being one of the strongest areas of EBSCO's contribution to the FOLIO project, the company stepping up to fill the needs least likely to be met by member libraries alone. (To diversify the base of code contributors in the face of the many technical barriers to getting started,[52] and particularly to include individual software developers and those at smaller institutions, the FOLIO community created a "developer advocate" role in 2024.)[53] Closely related to development proper, POs also play an important role in writing test cases and coordinating follow-up via the FOLIO testing community, which brings together subject-matter experts, POs, and vendor organizations to ensure the software remains robust with each release.[54]

Summary of Vendor Roles in Each Community

Table 3 highlights how vendors participate in the Koha, Evergreen, and FOLIO OSS communities' respective governing bodies and practices of product management and software development.

	Governance	Product Management	Software Development
Koha	Via HLT Koha Committee. Limits are placed on vendor representation, and explicit rules handle conflicts of interest in decision-making.	Decentralized project contributions made sporadically are merged into code base on a regular schedule.	Majority of coding is sponsored, funded, and carried out by ByWater, PTFS Europe, BibLibre, and Catalyst, but nearly half of code contributions come from developers not affiliated with vendors.
Evergreen	Via Project Board. Membership is restricted to current development contributors; no strict limitations on vendor representation.	Driven by ECDI prioritization of community needs.	Majority of coding is sponsored and funded by large public library consortia, carried out by Equinox.
FOLIO	Via OLF Board, Product Council, Technical Council, and Community Council. Elected membership, limits are placed on vendor representation.	Defined standards for project and feature development are embodied in product roadmap.	Majority of coding is sponsored, funded, and carried out by EBSCO, Index Data, and Knowledge Integration.

Table 3. Comparison of vendor roles in OSS ILS/LSP communities.

††† An example of an 18-month FOLIO roadmap can be found at Jesse Koennecke, "FOLIO Roadmap," FOLIO Wiki, September 24, 2024, https://folio-org.atlassian.net/wiki/spaces/PC/pages/4883619/FOLIO+Roadmap.

Conclusion

Absent from this chapter so far is an opposing viewpoint, that vendors are perhaps unwelcome in library OSS communities or a detriment to the ones in which they participate. Little in the scholarly literature directly identifies such a tension in the ILS/LSP environment. Still, parallels in other sectors of the Open Movement can give clues about where discontent might arise.

Red Hat, Inc., has had a "revolutionary" role in developing and spreading Linux[55] and has been a driving force in "how open source ate software" more broadly.[56] Through its distribution of Linux, Red Hat "became the largest [company] and [the] only publicly traded company whose business model was built entirely around free software."[57] Despite this—or perhaps because of it—some claim that its "appropriation of the copyright to the collective work of the Linux distribution from the community is not carried out in the interests of the community, but in the interest of its shareholders."[58] Non-scholarly sources express similar frustration, with titles such as "I'm done with Red Hat,"[59] "The Linux Community Is Circumventing Red Hat's Controversial New Strategy,"[60] and "So if corporate Linux is so evil, why are we still defending it?"[61] As the last title suggests, many are supportive of Red Hat's substantial contributions to the Linux community, and more to the point, OSS at large endorses making a profit. The influential "GNU Manifesto" defends corporate involvement in OSS, saying, "If people would rather pay for [free software] plus service than get [free software] free without service, a company [that provides] service to people who have obtained [the software] free ought to be profitable."[62] This suggests an underlying philosophy of letting market forces regulate corporate activity.

More directly in the library market, the practices of some publishers in the era of open access provide another example. One *Scholarly Kitchen* author accused a "new breed of technology-driven and copyright-disdaining publishers" of benefiting financially from open content licensing without delivering comparable value back to the research community. Those same parties then allegedly tip the scales to influence copyright law and its interpretation toward further self-interested openness.[63] This might be seen as contradicting the spirit of open access while complying with its legal requirements—a loophole that open licensors only sometimes choose to close.

Based on plenty of real-world examples of corporate greed, an extreme view might hold that "business profits are either, at best, amoral or, at worst, immoral."[64] If one holds that "there is no moral justification for business profits,"[65] then the corresponding moral imperative would likely prioritize reforming industries and products other than libraries and library software. A pragmatic view might more reasonably find, "A world without profit is … a world with a lot less innovation" and posit that library vendors can realistically offer libraries the benefits described in this chapter only to the extent that they sustain profits.[66]

Fittingly, the OSS space offers a mechanism for mitigating such concerns in the case of library OSS. Recognizing that the strengths of OSS "are threatened when a single organization can exert exclusive control over the future evolution of an OSS project,"[67] an individual OSS community serves as a "boundary organization." In this framework, the community is "created to negotiate and establish boundaries between two [or more] parties who may have both shared and disparate interests."[68] Naturally, just as library staff desire "agency in the development of the system that is so critical to functions across [their] library,"[69] representatives of a corporation also want autonomy in the work they do and the freedom to make meaningful connections between the ILS or LSP and the company's other offerings. The community's governing structures and documents become a way for all parties to externalize and negotiate their concerns into a concrete social contract. If the reality of life under a governance model proves unbalanced or unstable, parties can come and go, and new models can evolve. The communities described here have so far proven resilient and hold promise for the future.

As discussed above in the "Values and Services" section and as embodied in the operations of these three OSS communities, the libraries, vendors, and others that maintain Koha, Evergreen, and FOLIO hold many shared values. Each party works on these library systems to

provide equitable access to information resources and related services. They understand the importance of patron privacy and develop tools and procedures for ensuring it. They value what libraries offer the world, and they build their staffing, budget structures, and innovation cycles around helping libraries do what they do. They increasingly do business with a view toward sustainable, long-term value. Such values are shared in common among all those involved in these communities. As this chapter has illustrated, library OSS is a complex undertaking, and it succeeds only to the degree that each party contributes its unique strengths toward rich, supportive community engagement.

Conflict-of-Interest Statement

The authors wish to disclose that they are employed by EBSCO and participate actively in the FOLIO community, and they recognize the potential for perceived bias or conflicts of interest. Where possible, they have made every effort to research and document the facts and positions presented, using reliable sources to supplement their own experience. In producing this chapter, the authors represent themselves and not their employer and take full responsibility for its content. They also disclose that they have received no special compensation (beyond their regular salaries) for writing this chapter or for approaching it in a certain way.

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CHAPTER 12

Organizing for Open: Envisioning Next-Generation Open Services and Organizational Frameworks at Oregon State University Libraries and Press

Diana Castillo, Cara Key, Taylor Ralph, and Michael Boock

Introduction

Libraries have traditionally been organized around a central mission of reducing barriers to access for our communities. As the scholarly publishing landscape becomes increasingly open, the scale of this shift poses a significant challenge to academic libraries, requiring a widespread reevaluation of how we conceptualize and fulfill our mission. In 2023, the library at Oregon State University began a self-reflective process to consider possibilities for real-locating resources and reorienting services to align with the trend toward open. The dean of libraries commissioned a group of library faculty to produce a white paper exploring the topic of open, which provided a local framework for strategic thinking about organizational change. This chapter discusses what followed and offers a model for other academic libraries engaging with the effects of the movement toward open on their work and, potentially, on the library as a whole.

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Background

Oregon State University (OSU) is an R1 public research university. With an enrollment of over 36,000 students across its main campus, satellite campuses, and eCampus program, OSU is the largest university in the state.[1] In the most recent fiscal year, OSU received $480 million in research funding.[2] Its designation as a land grant as well as sea, sun, and space grant university means OSU must serve the public through its research—a theme that is central to the expressed mission and values of the institution.[3]

Oregon State University Libraries and Press (OSULP)* has a legacy of contributing to this commitment as a champion of open access (OA). Like many university libraries, OSULP directly supports broad dissemination of the scholarship and digital resources of its community through repository and hosting services, including the institutional repository, a digital collections repository, a scholarly journal hosting platform, and DOI minting services. Oregon State University has been a leader in OA policy development: the library faculty passed an OA policy in 2009,[4] and the OSU Faculty Senate adopted a campus-wide policy in 2013 that directed faculty authors to submit accepted manuscript versions of their scholarly articles to the library for deposit to the institutional repository.[5] Library faculty and staff have engaged in numerous initiatives to facilitate and automate OSU faculty article deposits to the institutional repository.[6] Furthermore, OSULP advocates on behalf of OSU authors' open publishing rights, for instance, by providing direct support for amending publisher agreements to comply with the campus policy where necessary.[7]

In addition, OSULP also works with OSU faculty and students to comply with the new OSTP (Office of Science and Technology Policy) public access requirements. The OSTP's 2022 policy, often referred to as the "Nelson Memo," mandates that all federally funded research publications and their supporting data be made immediately available to the public without embargo.[8] To support the university community, OSULP offers guidance on deposit requirements, repository use, and strategies for making data and articles publicly available. The library plays a key role in ensuring that OSU researchers understand and meet these obligations, providing training, consultations, and tools to streamline compliance while promoting open scholarship practices.

The Green OA model, in which a copy of a publication is made publicly available in an institutional repository or similar online archive, has thus been a focus of OSULP initiatives, but it is only a small portion of the changing landscape. As of 2022, Basson et al. found that nearly half of all new scholarly articles were published as open access.[9] A study by Piwowar, Priem, and Orr in 2019 projected that by 2025, 44 percent of all journal articles published since 1950 would be available as OA.[10] At OSULP, the library's role in unlocking access to resources for our researchers via library subscriptions is becoming less clear as the scholarly publishing market moves in this direction. During our 2023 renewal negotiations with Elsevier, we discovered that during the calendar year 2023, nearly a quarter of the items accessed from the OSULP library discovery system were Gold OA content from the top five publishers: Elsevier, Springer Nature, Wiley, Taylor & Francis, and Sage. Gold OA refers to publishers making content freely and immediately available to readers under newer revenue models that shift the costs for publishing onto authors through article processing charges and transformative agreements, or onto third parties. Traditional library subscription dollars do not provide access to those resources, yet the library still pays continuously rising subscription costs.[11] Too often, university libraries are paying publishers for subscription packages while the same publishers charge university researchers to make their work open access.

* OSU Press, which publishes monographs relevant to the community and region, is administratively situated within the Libraries organization with its director reporting to the dean of libraries. However, the Press's day-to-day operations are largely independent from those of the Library. The Press was not significantly involved in the work described here, in spite of the usage of the formal "OSULP" acronym when referring to the Library. Publishing for scholarly communications, including the IR and hosted OA journals, does fall within the scope of OSU Library operations

Relative to the university's research expenditures, OSULP's materials budget is low, at between $5–$6 million per year. Its staffing is also modest compared to peer libraries at doctoral universities, with approximately eighty full-time equivalent employees, twenty-five of whom are faculty librarians.[12] So, it is critical that the library makes carefully considered, sustainable decisions about resource spending. In 2019, the Open & Sustainable Scholarly Communication Advisory Committee (OSSC) was established by the university, producing a set of principles to guide negotiations with journal vendors.[13] These principles served as a foundation for negotiations between OSU, its partner institutions University of Oregon and Portland State University, and Elsevier in 2022 that led to the suspension of our Elsevier contract for over a year.[14] In 2024, OSULP signed a new contract after Elsevier agreed to match some of these principles and allowed the contract to be openly shared. This established a significant precedent, as the library, with the support of the University, held firm to its principles and explored alternate strategies for resource allocation and content delivery.

To summarize, OSU and OSULP have been active and engaged in conversations and policy-making decisions to support OA. The timeline is as follows, and is illustrated in figure 1:

- **2009:** The library creates and passes an OA policy that encourages materials deposit into OSULP's institutional repository.
- **2013:** The OSU faculty senate adopts a campus-wide OA policy that directs faculty members to deposit scholarly materials into the institutional repository.
- **2019:** The OSSC is established by the university.
- **2023:** OSU loses current access to Elsevier content while negotiations are paused. The dean of libraries commissions the "Organizing for Open" whitepaper.
- **2024:** The dean of libraries meets with each library department to discuss the whitepaper and the impact OA might have on their work.
- **2024 and beyond:** OSULP will continue to have conversations about OA and workflow implications. Open scholarship initiatives are outlined in the new OSULP strategic plan, and specific actions, including APC payment services in place of selected subscriptions, are currently under discussion.

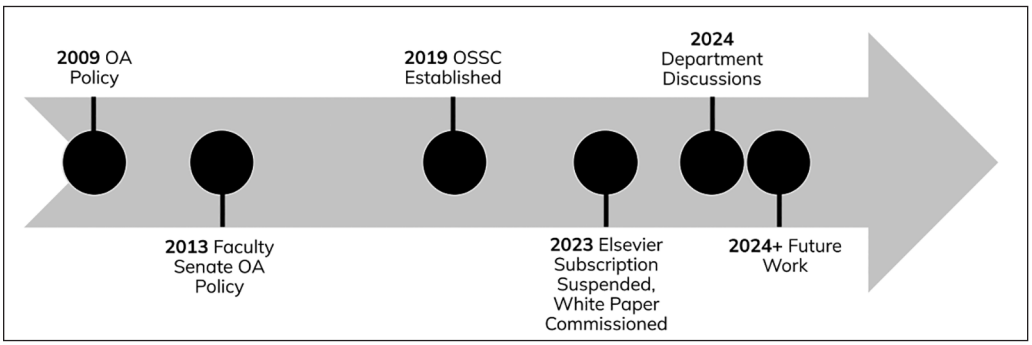


Figure 1. A timeline illustration of OSU's commitments to open access.

White Paper Creation

Several organizational factors were converging at the end of 2022, including leadership changes, anticipated strategic planning efforts at both the library and campus levels, and key open positions. Recognizing the opportunity to consider the future of open scholarship and its direct impact on library services and personnel, the dean of libraries charged a library faculty task force with drafting a white paper that explored repositioning the library in a world of open scholarship. The goal of the paper aimed to answer two questions: "Why is it important to organize around open?" and "What kinds of changes are involved when we talk about around open?"

The writing group included four faculty from OSULP's Research and Learning, Library Information Technology, and Resource Acquisition and Sharing departments. The librarian positions covered outreach and instruction, digital repository management, scholarly communications, and collection assessment. The representatives were asked to rely on expertise from their respective positions and experiences, while also seeking broader feedback from library experts at other institutions and internal stakeholders. Comprehensive feedback from the library as a whole would later be solicited through the dean's planned tour to each library department following the paper's completion. The smaller group had logistical advantages, such as easier scheduling, increased accountability, and focused feedback.

Initially, the dean met individually with each librarian involved to introduce the project, including anticipated outcomes, expectations, timelines, and contextual information. The group had a timeline of two to three months to complete the paper, with an estimated length of eight pages. During initial discussions, the group formulated an outline focused on the following elements, not all of which were included in the final paper:

- a literature review in our introduction;
- an analysis of OSULP's past efforts concerning open scholarship;
- interviews with faculty from peer institutions;
- interviews with internal employees;
- synthesized responses from the interviews;
- suggestions for new positions and/or departmental organization;
- an exploration of how shifts to open could impact each library department; and
- a list of idea-generative questions for different areas of librarianship (collections; discovery, metadata, and cataloging; instruction and outreach; and scholarly communication).

After outlining the paper, we identified external and internal interview candidates. We selected peer institutions with positions, departments, or action statements related to open scholarship, and we conducted seven virtual interviews with librarians in positions related to scholarly communications at those institutions. The predetermined questions focused on how the library and university support open, and whether their libraries have implemented subscription or structural changes as a result of that support. We also held internal interviews with six OSULP employees, identified as having specialized knowledge or being particularly impacted by the movement toward open based on position descriptions. For the internal interviews, we used questions tailored to individual positions. The external interviews supported the white paper as we brainstormed new positions and library organizational structures; the internal interviews helped determine the extent of work currently supporting open scholarship and gauge capacity for added responsibilities.

The white paper authors held weekly meetings, during which we reviewed our progress, wrote, and planned action items and goals for the next meeting. The research, writing, and additional workload were distributed evenly and divided by areas of expertise or interest. Halfway through the project timeline, the group shared a full rough draft with the dean, then met with her to review our progress and discuss the current direction. We revisited the initial charge and the paper's ultimate goal, which was to act as an idea-generative resource for library discussion. This well-timed conversation allowed us to resolve outstanding questions and resulted in the removal of direct proposals in favor of a tone more conducive to further brainstorming within departments. Specifically, we cut a section of the paper that proposed new organizational structures, positions and interdepartmental groups, and reconfigured departments. The dean expressed concern that providing concrete restructuring proposals would limit creativity and conversation in the planned department meetings. The subsequent work on the paper focused on generating discussion prompts around anticipated changes in the academic landscape, potential uses of current OSULP resources, and brainstorming larger changes OSULP could make to support open long-term. The goals of these prompts were to spark thinking on behalf of the dean and provide conversation-starters for the different departments.

White Paper Dissemination and Internal Conversations

We completed the white paper in June 2023, within the expected two- to three-month timeline, and deposited it in OSU's institutional repository.[15] We then presented it to the dean, who did not request any additional changes. With the completion of the white paper, the dean began holding departmental conversations around the prospect of organizing the library around open, as well as using our library's annual in-service day to solicit additional input. She used two framing questions to start the conversations; a third question came up in all discussions, although it was not formally asked.

The first framing question, "Do you buy the premise [of the white paper]?" also included the subquestions, "What do we risk if we don't do this?" and "What do we gain if we do?" when presented at the in-service event. Overall, departments responded favorably that they did buy in to the premise. In-service participants noted that the risks of not having at least some reorganization around open included limiting access to research and risking irrelevance. In contrast, the benefits included promoting more equitable access, increasing affordability for students, and increasing discoverability of open materials. OSULP employees expressed that benefits align with observations in the literature of how libraries have centered themselves within the discussion around open as well as affordability in higher education.[16] In an article looking at trends within academic libraries, open is described as an evolving concept that should be a key feature moving forward,[17] and OSULP staff demonstrated that they recognized its importance.

The second framing question focused on what the change might mean for a person's work. During the in-service event, this framing question included the narrower questions: "What existing strengths could OSULP build on?" and "What gaps do we need to fill?" Participants cited strengths, including the alignment of organizing around open with existing library values, particularly our value of access, as well as building on our existing open services and infrastructure, such as digital repositories. Some gaps that were identified included existing library infrastructure not being designed for open scholarship, needing to change how instruction librarians teach identifying reliable resources, and accessibility of resources.

The third informally asked question pertained to what the library needed to figure out in terms of resources and partnerships to organize around open. Staff and faculty discussed how the library would allocate resources to meet this goal—essentially, how the costs would be paid, since open doesn't necessarily equal free. Given that the university was engaged in strategic planning at the time of the white paper creation and the library was looking ahead to its own related strategic planning effort, the discussions also surfaced the question of how this potential reorganization around open might be incorporated into a new strategic plan. Finally, when thinking about how this might affect how the library focuses on partnerships, participants expressed concerns about capacity and about those we collaborate with not sharing our library's commitment to open.

Discussion

As a result of the white paper project and the library-wide conversations that followed, OSULP identified areas that could be reorganized or reimagined to better support open initiatives. Following this project, open topics resurfaced frequently during library-wide strategic planning work. The new strategic action plan includes three actions explicitly tied to support for open scholarship:

- **Open up library collections**: Increase the percentage of open content and improve the accessibility and discoverability of open resources.
- **Supplement services**: Enhance on-demand delivery of journal articles and open access content discovery with targeted acquisitions supporting OSU's 2024–2030 strategic plan, Prosperity Widely Shared.

- **Remove barriers**: Eliminate obstacles preventing OSU students and researchers from making their creative work broadly accessible.

Library management is still determining specific, achievable tactics for completing these actions. We expect some of those tactics will reflect the recommendations and conversations from the white paper. One possible new direction is dedicating an endowed, named library position, the Gray Family Chair, to advancing open access initiatives. Currently vacant, the Gray Family Chair has been responsible in the past for "expand[ing] innovative means of accessing and delivering information to students, faculty, staff, and other constituents of the university, with the end goal of further strengthening OSU's distinction as an information provider."[18] Previously, the position was filled by experts in open-source computing, digital preservation, and publishing, but the endowment had accumulated without use over the past few years. If the position was filled with an eye to a future reorganization around open, the librarian would be responsible for negotiating terms with publishers that meet the university's principles for openness as stated in the *Principles Guiding Negotiations with Journal Vendors at Oregon State University*.[19] This includes engaging in discussions with publishers to ensure that the terms of agreements support open access initiatives and adhere to the values of the institution. Additionally, the Gray Family Chair may be involved in advocating for open publishing rights for OSU authors and coordinating efforts to promote open scholarship within the university community. Overall, the position would play a crucial role in advancing OSULP's commitment to open access and supporting the dissemination of scholarly research if the dean decided to use it this way. Final decisions about the chair have yet to be made, as the library finishes its strategic planning.

More broadly, we found that our organization is receptive to the idea that we need to change. Overall, the ideas expressed during the feedback portion mirrored the SWOT analysis John Cox published in *The New Review of Academic Librarianship* in 2023. Of the unique strengths, weaknesses, opportunities, and threats identified in the pair of articles, "The Position and Prospects of Academic Libraries: Weaknesses, Threats and Proposed Strategic Directions"[20] and "The Position and Prospects of Academic Libraries: Strengths and Opportunities,"[21] there was at least one of each that surfaced in OSULP stakeholder discussions. In particular, according to Cox, two of the current weaknesses in academic libraries include a failure of libraries to quickly adapt to new situations, leaving behind old methods that no longer reflect the current situation, as well as being trapped in a financial situation fueled by rising subscription costs and stagnant budgets.[22] This is the exact situation that OSULP is trying to proactively address through the exercise of reorganizing around open. We want to better meet stakeholder needs, while finding alternatives to the current subscription model that is not working well for anyone except the publishers. Cox also notes that a major threat to academic libraries is a loss of identity in the changing digital and scholarly landscape.[23] As mentioned earlier, the rise of open is changing the library's role to provide access to scholarly works. The white paper was written to help our library consider making strategic updates to our identity so that we can continue to serve our community.

While there are threats and weaknesses to be addressed within academic libraries, the initial white paper and the following discussions also mirrored the strengths and opportunities Cox found. In particular, the premise of organizing for open aligns with an opportunity for digital and open scholarship to significantly reshape how academic libraries function,[24] especially given an overall positive reception to the idea. One staff member, when thinking about the premise, called the process of organizing around open to be "proactive future building," where we think about our present, the future we want to see, and what steps we need to take to achieve it. This also ties into one of the key strengths of academic libraries Cox discusses, namely the fact that libraries can reinvent themselves to accommodate changing digital landscapes and remain relevant to future generations of students and researchers.[25] As OSULP navigates the changing scholarly communication landscape, a substantial question will be how we balance our traditional gatekeeping capacity with wanting to make information more accessible and remain relevant within the university community.

The approach that we took in creating and disseminating the white paper, while centered on OSULP's experiences and current situation, could be replicated at other institutions. From the start, one important aspect of the project was the inclusion of faculty and staff from all departments of the library, not only those who work on scholarly communication or collections. This was reflected in the writing group's makeup and the internal experts we consulted as part of the process. Based on library and staff feedback, we tailored our proposals to accommodate recommendations and highlight potential issues. Once the paper was completed, the rest of the library was brought in to deliberate on the white paper's premise and its conclusions. Even departments less affected by this shift, like Special Collections and Archives or public services, were involved; each brought unique insights to the process.

While OSULP takes independent steps toward open, for the movement to gain effectiveness, work must be done collaboratively. Individual efforts are valuable but have greater impact when pursued more broadly across academic libraries. The more libraries actively invest in open access and take proactive action, rather than waiting on outside factors like federal research sharing requirements, the more new opportunities and investments are created. Existing networks, such as the Orbis-Cascade Alliance in the Pacific Northwest, could facilitate discussions about transitioning from subscriptions to open access and providing guidance to institutions at all levels. Lasting change can only be achieved through internal and external collaboration, combined with a deliberate focus on advancing openness.

Conclusion

University-wide frustration with the current publishing model, alongside an evolving journal publishing landscape and federal mandates for open, indicates an imminent tipping point toward openness and away from the traditional subscription model. A 2022 survey of OSU faculty found overwhelming support for OSULP to find ways to support open availability of OSU's scholarship.[26] This is an opportune moment for OSULP to leverage its previous work and establish a strategic trajectory, one that boosts the capacity of our faculty and students to publish scholarly articles openly or deposit them in open repositories. This effort aids them in fulfilling federal mandates and amplifying the impact and availability of the university's research, all in alignment with the new 2024–2030 university Prosperity Widely Shared strategic mission.[27]

One crucial lesson from this process is the value of broad stakeholder engagement when navigating such a pivotal shift in library services. As we've seen, inviting staff from diverse departments to participate in strategic conversations ensured a richer, more holistic view of how reorienting around open scholarship could affect the entire organization. This inclusive approach enabled us to uncover hidden opportunities and challenges, fostering a collective sense of ownership over the resulting decisions. For libraries at other institutions, adopting a similar strategy of widespread consultation can lead to more sustainable, long-term organizational change, as all voices are considered in shaping a future that embraces openness.

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PART 3

IMPROVING ACCESS POINTS FOR OPEN

CHAPTER 13

Opening Research Communication: Improving Public Access by Changing How We Communicate

Ellen Dubinsky

Introduction

Scholarly communication librarians often balance support for "traditional" practices of how research is conducted, communicated, and preserved with efforts to introduce and promote open access publication, open data, and other open practices that improve access, equity, and research impact. This chapter argues that librarians, especially scholarly communication librarians, should embrace a broader conception of open access and further research communication by prioritizing making research accessible and understandable to the many communities interested in engaging with the research.

Context

I have long been a general interest reader—I love long-form journalism and nonfiction and have always enjoyed learning about research that is not associated with librarianship. I had been considering writing about more effective scholarly communication methods aimed at non-specialists for a few years but never could quite devote the time and energy to put pen to paper. A letter to the editor of the *New York Times Magazine*—in reference to an article about Columbia University's COVID oral history project—made me sit upright and recommit, as the author of the letter succinctly expressed what I had long felt:

> *This is an outstanding piece of writing. The study is impressive in and of itself, but the author of this report has done a masterful job of describing the work of these sociologists in a very accessible manner … that alone makes this research so much more powerful to the lay person. I have often marveled at the gift of making academic research more readable and meaningful to the lay public. This is a gem of an example. Bravo!*[1]

CC BY

As an academic librarian, my outreach to researchers and authors focuses on traditional scholarly communication channels and concerns. I talk about how to choose an appropriate journal for manuscript submissions, journal metrics, the benefits of open access publication and the dangers of predatory publishers, copyright concerns, proper citation styles, peer review, research impact, and building one's scholarly identity. I do that in the context of a scholarly publishing ecosystem that in many ways closely resembles scholarly publishing in the seventeenth century. The initial scientific findings disseminated in the literature were referred to as "letters," authored by members of learned societies. These letters were periodically published in the journals of these societies, organized into distinct issues and volumes. More than three hundred years later, research articles are published by a diverse range of publishers, including commercial entities, nonprofit organizations, scholarly societies, and academic institutions. Despite this evolution in the publishing landscape, contemporary research articles continue to be presented within journals that maintain a structure of issues and volumes, closely resembling the format of publications from centuries past.

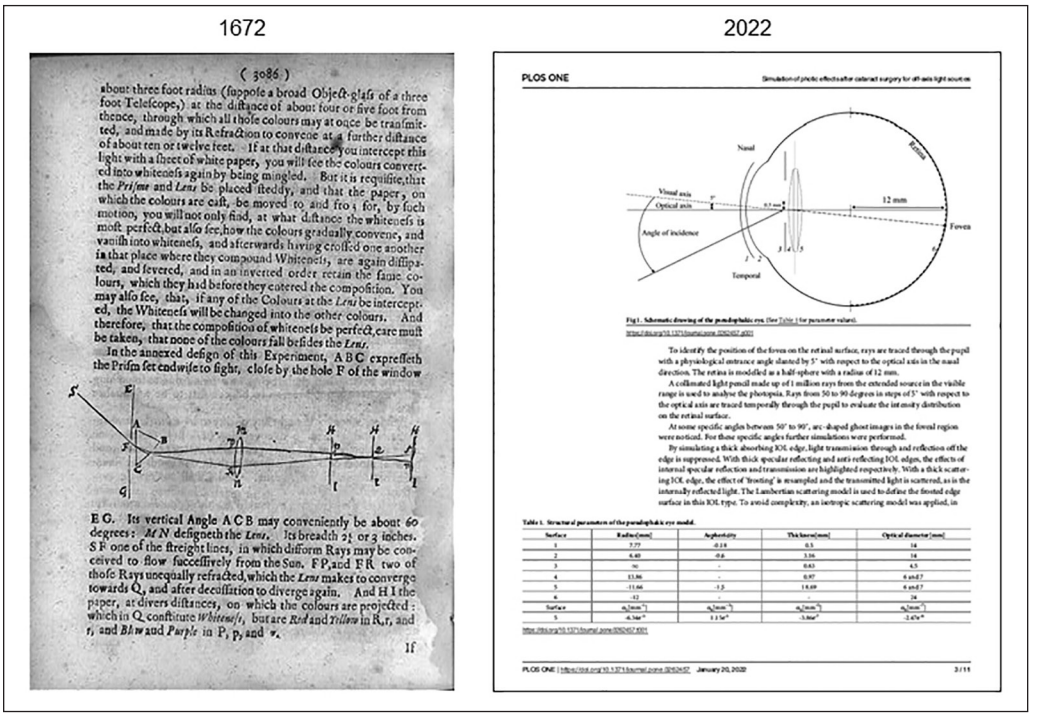


Figure 1. A research article published in the twenty-first century is formatted similarly to a research "letter" published in the seventeenth century. On the left (diagram of experiment with prisms that led Newton to conclude light is a mixture of differently refrangible rays; with surrounding text), published in *Philosophical Transactions*, vol. 6, February 1672. On the right, "Simulation of photic effects after cataract surgery for off-axis light sources" by P. Omidi, A. Cayliss, and A. Langenbucher, published in *PLOS One* 17, no. 1, 2022: e0262457, https://doi.org/10.1371/journal.pone.0262457.

The familiar reasoning recognizes that research publications are almost always published in scholarly journals for a specific audience—namely, other researchers in the same discipline who are familiar with the jargon, norms, and prior research in the field. When in the role of author, researchers know a disciplinary-specific audience is well-versed in the progression of research in that field. This learned audience understands the context of research in this field and how a new report of research relates to what has preceded it and what might succeed it.

The impact or reputation of a researcher is tied to their published research. The traditional academic career trajectory involves promotion, tenure, and retention review decisions that evaluate—and prioritize—metrics such as citation counts and journal impact factors. These metrics only measure interactions with a limited audience composed of researchers in a specialized field; they don't tell a complete story. Evidence of societal impact—changes to policies, practices, and programs that benefit the broader society—reflects much deeper engagement with and the value of scientific research.[2,3]

To realize societal impacts, research outputs need to reach a broader audience. Open access publishing makes research outputs available to the public, but not necessarily accessible in terms of understanding or engagement. Barriers to accessibility include language, jargon, prior technical knowledge, and inexperience in understanding conclusions or possible practical applications. If a psychologist opens an article about economics, can they understand the article and recognize the value of its findings? A less technical version of the article might pique the interest of a non-economist researcher interested in the psychology behind economic decision-making. A brief and jargon-free plain language or lay summary could expand the audience of readers who would better understand the value of this research and how the findings could be relevant to their lives. Translations might allow audiences around the world to more quickly engage with the content and assess the relevance of the findings for different global communities. Similarly, a podcast, video, or infographic might help to convey research findings more effectively to a diverse audience. Would these other forms of less formal communication lead to beneficial social, political, or economic outcomes?

Making Science Accessible

The release of the August 2022 Office of Science and Technology Policy (OSTP) Memo[4] reignited discussions about public access to funded scientific research outputs, though these conversations have been occurring on and off since the early 2000s. The first public access policy in the United States was issued in 2004.[5] Subsequent policies were mandated both in the US and abroad. Other nations, such as the United Kingdom, Ireland, Japan, and Australia, have issued policies calling for or mandating open or public access to research outputs, as have international organizations such as the European Commission and UNESCO. Non-governmental funders, such as the Wellcome Trust (beginning in 2005) and the Howard Hughes Medical Institute (beginning in 2007), require open access to research resulting from their grants.

The 2022 OSTP Memo emphasizes the importance of sharing research results freely and openly with both the scientific community and the public. The access policies developed by each federal funding agency *must* allow all Americans to benefit from the government's investments in research and development. However, the language in the funding agencies' compliance policies refers to research outputs such as "author accepted manuscripts," datasets, and open access "versions of record," which are terms that may be unfamiliar to the public. The federal funding agency compliance plans required by the 2022 OSTP Memo are expected to indicate where text-based outputs and data should be deposited. Existing federal funding agency policies (as well as the policies from some non-governmental funders) designate repositories such as PubMed Central (hosted by the National Institutes of Health),[6] NSF-PAR (hosted by the National Science Foundation in partnership with the Department of Energy),[7] and Europe PubMed Central (hosted by EMBL-EBI [European Molecular Biology Laboratory's European Bioinformatics Institute]).[8] These repositories are known to researchers but not necessarily to the broader public in the US or abroad. People who are not involved in research might not know that there are alternative versions of research articles besides the official versions published in journals. Although the official version of an open access research article can be discoverable by anyone, most articles are published without open licenses and are often behind paywalls. This means they can only be accessed by those who can afford to pay for them or by individuals affiliated with institutions that have paid for access.

Researchers do a disservice to the general public and the research itself by solely making the traditional forms of research outputs available (i.e., formal scholarly articles written for a disciplinary, researcher-centric audience). The potential benefits of research diminish if we focus solely on providing access to research outputs. True accessibility involves utilizing formats and multiple languages beyond our current practice. We are making research available, but not accessible. Rather than focusing on *scholarly* communication, let's focus on broader *science* communication.

What's a Scholarly Communication Librarian To Do?

Scholarly communication librarians are already leading efforts promoting open access and open research practices. This puts us in a position to promote alternative avenues of science communication. We can help the research community benefit from the opportunities made possible by communicating with non-disciplinary audiences. Specifically, let's promote reaching

- researchers from other disciplines, opening up possibilities for cross-disciplinary collaborations;
- members of the communities involved in or impacted by research about those communities, leading to improved societal impacts; and
- the broader general public via formats that are freely accessible and intended to foster direct communication between specialist researchers and non-specialist audiences, leading to more public engagement with science, combating misinformation and disinformation, and possibly increasing trust in scientific results.

Communicating with Researchers in Other Disciplines

The early 2020s have been full of examples of challenges to the health and well-being of people across the world. People across the globe faced the disruption, uncertainties, and difficulties of a global pandemic. Regions throughout the world have experienced the effects of rapidly accelerating climate change. Migration and displaced populations have increased due to political conflict, persecution, and economic insecurity. These problems are often framed as "grand challenges"[9] and are related to energy, health, education, the environment, global development, food, prosperity, water, space, disaster resilience, shelter, and governance. While not framed as grand challenges, the United Nations' Sustainable Development Goals share the same desired outcomes of improving human lives and protecting the environment.[10]

Though these grand challenges seem overwhelming, they can and have been framed as opportunities. Past American Association for the Advancement of Science (AAAS) President Gilbert S. Omenn opined that the "concept and promotion of Grand Challenges can help energize not only the scientific and engineering community, but also students, journalists, the public, and their elected representatives, to develop a sense of the possibilities, an appreciation of the risks, and an urgent commitment to accelerate progress."[11] The Obama presidential administration described grand challenges as "ambitious but achievable" with "the potential to capture the public's imagination."[12,*]

Interdisciplinary research[†] has become an increasingly important component of science policy initiatives and research funding schemes—especially to address these grand

[*] Clearly, grand challenges have captured the interest of researchers. Of the 3,525 articles published between1970 and 2023, with the term "grand challenges" in the title included in the OpenAlex database, 82% were published between 2010 and 2023. (Data pulled late May 2024).

[†] For simplicity's sake I use the term interdisciplinary research to describe research where two or more disciplines work together. Other common terms used for these collaborative research approaches include transdisciplinary, multidisciplinary, post-disciplinary, or cross-disciplinary, each with its own nuanced difference in meaning.

challenges—based on the confidence that interdisciplinarity will lead to both new scientific discoveries and solutions to real-life problems.[13] Scientific teamwork or "team science" is "expected to combine specialized expertise, theoretical approaches, and research methods across disciplinary boundaries, solving these complex problems and producing high-impact science."[14] Interdisciplinary research is credited with its ability to tackle complex, real-world problems that require a combination of skills and expertise from different fields. It allows researchers to work on bigger scientific and societal questions, and it can lead to innovative solutions. It can also foster more collaborations with industry, which is often focused on real-world problems that cross disciplines.[15,16]

Evidence of interdisciplinary work has been noted in the references included in published research articles. Researchers frequently cite works coming from other fields. In the natural sciences and engineering, more than one-third of the references point to research in other disciplines. In the social sciences, almost half of the references cite research in other disciplines.[17]

Researchers are already familiar with writing to an audience of peers and colleagues steeped in their discipline. From the beginning of their advanced training and throughout their careers as researchers or academics, reputation and impact and career trajectory have been closely linked to what and how much has been published in specific, selective scholarly journals. Researchers may hesitate to leave that long-established comfort zone. However, staying within one's comfort zone prevents reaching new audiences. Scholarly communication librarians can encourage researchers to develop research outputs targeted at researchers outside their own specialized discipline. These other researchers will have a baseline understanding of scientific methods and scientific advancement in other fields. They are familiar with and accustomed to the format and formality of traditional scientific research articles. Publication in journals that attract readership across the disciplines or sub-disciplines will likely be discoverable by a wider readership of researchers, some of whom might be interested or intrigued about research in another field. Communication with this different but related audience ultimately could be more impactful if it allows for collaborative research that leads to new insights and societal benefits and helps solve global challenges.

As a bonus, there may be career advantages to engaging in interdisciplinary research. These collaborations can provide a researcher with the opportunity to

- expand their network of collaborators and colleagues;
- widen their perspective toward a research topic;
- gain visibility in a related (or even divergent) field;
- find new funding sources;
- learn new research methods and protocols; and
- engage in mixed methods research.

Communicating with the Communities Impacted or Involved in the Research

Advances in scientific knowledge are dependent not only on communication and collaboration between researchers across disciplines but also on the engagement between researchers and the communities most impacted by the research. Science communication improves with the involvement of the public throughout the scientific process and the use of effective and accessible channels of communication. Direct communication with the communities most impacted by the research or those communities that contributed in some way to the research not only acknowledges the stakeholders' contributions but also demonstrates the intention that research results be transformed into real-world applications. The social impact of research goes beyond openness and transparency and toward community-engaged scholarship and community-engaged research.

The US Centers for Disease Control (CDC) defines community engagement as "the process of working collaboratively with groups of people who are affiliated by geographic proximity, special interests, or similar situations with respect to issues affecting their well-being."[18] An active partnership between researchers and the public is implied in the Academy of Community Engagement Scholarship's (ACES) definition of community engaged scholarship: "The collaborative generation, refinement, conservation, and exchange of mutually beneficial and societally relevant knowledge that is communicated to and validated by peers in academe and the community."[19] The White House Office of Science and Technology Policy (OSTP) defines community-engaged research as research in which "[r]esearchers engage with a community to develop research questions, design a study, and collect data."[20]

The terms "knowledge translation" and "knowledge mobilization" are frequently used to describe the activities that lead to the sharing and dissemination of research results. Knowledge translation, most commonly used in the health sciences, focuses on moving research from "bench to bedside"—from the laboratory into the hands of clinicians, patients, social workers, policy experts, legislators, and organizations who put the research to practical use. The Social Science and Humanities Research Council (SSHRC) in Canada defines the term "knowledge mobilization" as "the reciprocal and complementary flow and uptake of research knowledge between researchers, knowledge brokers and knowledge users," resulting in positive impacts that have the potential to "enhance the profile, reach and impact of social sciences and humanities research."[21]

For research in medicine and health sciences, the community most affected by the research consists of patients. Including patients early and often in the research enterprise can lead to asking better research questions and conducting higher-quality research.[22] Patient involvement in the research process builds trust between researcher and patient, and it recognizes the broad expertise and value of patients, including and beyond their illness experience. It may allow patients to feel more like collaborators in their treatment rather than simply recipients of that treatment. Patients can be involved in multiple stages of the research process, including research design, data collection, and dissemination of results through presentations to expert and lay audiences, contributions to research reports, sharing research with other patients and caregivers, and development of plain language summaries.[23]

Funders can invite patients, caregivers, and impacted groups into the funding process. The Wellcome Trust uses "lived experience advisors" who are involved in the decision-making processes that determine what proposals are funded. Researchers applying for funding are encouraged to make lived experience central to all stages of their research projects—for example, by including those with lived experience as co-applicants, peer researchers, collaborators, advisors, or workshop participants.[24]

Community engagement can encompass the inclusion of patients, caregivers, and members of the general public in the research publication process. Researchers can include those non-researchers as co-authors of manuscripts. Journal publishers can invite non-researchers to participate as reviewers of submitted manuscripts. The inaugural issue of *Research Involvement and Engagement* made clear that the inclusion of non-researcher stakeholders "encapsulate[s] the vision of 'nothing about me without me.'"[25] The *British Medical Journal* began embedding patient peer review of research papers in 2014.[26] Cochrane incorporates "consumer peer reviews" in its editorial process. These consumers—patients, caregivers, and members of the public interested in health care—provide feedback on Cochrane protocols and reviews prior to publication.[27]

Communicating in Additional Formats and New Ways

Members of the public are willing to and do engage with research and scholarly content. Non-specialist audiences are willing to engage with research, even when the targeted audience

consists of specialist researchers. A SpringerNature white paper noted that about 40 percent of readers of open access content accessed on SpringerNature websites were "non-academic audiences," even though the content was primarily targeted to researchers, not other audience groups. These non-academic consumers were divided into "halo" users—those reading research for professional purposes, but not conducting or publishing research themselves—and "general" users who don't conduct, publish, or cite research but who engage with research out of personal or professional interest.[28]

Recent research explored the composition of the readers who downloaded and used National Academies of Sciences, Engineering, and Medicine (NASEM) consensus reports. These reports, targeted at the federal government agencies that commission them and researchers in the relevant fields, have been free to download since 2011. The study sought to find out if nonscientists who are not steeped in the foundational knowledge required to understand technical publications use this information and, if so, why did they seek it out and what needs did it address? Comments left by readers showed heavy usage for education and research needs, but there was also considerable engagement for information gathering, self-directed learning, or other personal use. The researchers concluded that readers were "motivated to seek out the most credible sources, engage with challenging material, use it to improve the services they provide, and learn more about the world they live in."[29]

The SpringerNature white paper and the public engagement with the NASEM consensus reports demonstrate that a significant percentage of the public is willing to engage with research jargon to gain some insight or information of use to them personally. What if we tried to make engaging with research a little easier? Approaching science communication as storytelling may be an effective route.

Using a narrative structure describes science as a human-driven endeavor, one full of successes, uncertainties, and failures. Communicating science as a story means making science-related information engaging and relevant. In short, it is about making people care. Researchers should go beyond presenting facts and evidence and toward creating emotional connections between scientists and the many communities that have a vested interest in the research.[30]

Some researchers have embraced principles of marketing communication as the framework for telling visual stories about science using short-format videos. In addition to elements such as simplicity (focusing on the core idea) and credibility (delivering a believable message), effective communication includes several elements typical of a story:

- unexpectedness—generating interest and curiosity by deviating from the audience's expectation with an element of novelty;
- emotional—making the audience care by incorporating an emotional appeal, which can be more effective than providing information alone; and
- storytelling—science storytelling needs to have hooks that connect to the audience's curiosity and relevance to their lives.[31]

Articles that use concrete language, examples, common words, and active voice can contribute to better understanding and engagement with readers. Marketing researchers have looked at how the norms of academic writing hinder its potential impact. Academic writing can become unclear when authors fail to recognize that some readers may not have the same level of familiarity with and knowledge of the subject matter as they do. Having a great deal of knowledge can lead to writing that is harder to understand, especially when it is full of abstractions, technical language, and passive writing. When writers fail to ground their ideas around concrete examples, they may confuse readers with the abstractions. Researchers often overestimate the amount of technical language that their readers understand. Using passive voice hides the "actor" who is performing an "action," thus confusing the reader. From a marketing perspective:

> *The only way for scholars to build bridges and to disseminate knowledge to a broad audience is to cast aside the curse of knowledge and write so that readers*

> *outside of their narrow subject area understand what they have to say. By writing clearly, scholars will expand the market for their ideas.*[32]

These same principles easily apply to other forms of non-technical research communication, including visualizations and infographics, blog posts, media releases, and illustrated review articles.‡

There are tools available online to help academic authors, such as Writing Clarity Calculator,[33] that provide an objective measure of the extent to which manuscript text is abstract, technical, or passive. Artificial intelligence (AI) tools such as ChatGPT can be prompted to evaluate one's writing tone, rephrase sections to avoid repetition, assess readability, simplify the writing, or make it sound more professional. Journal publishers and disciplinary repositories such as BioRxiv[34] are developing or piloting AI tools for generating lay summaries. Tools such as SciSummary,[35] Scholarcy,[36] and SciSpace[37] are available, though there is concern about errors in the summaries. These same tools could be used to generate press releases, blog posts, news items, or briefings.[38]

AI can also provide robust language translation capabilities, which would help address the most glaring barrier to effective communication between researchers and stakeholder communities. The predominance of English-language research articles excludes the non-English-speaking populace from learning about and contributing to most of the published scientific literature. According to Ethnologue,[39] there were 380 million native English speakers in the world in 2023, which represents a little less than 5 percent of the world's population. If we add the number of non-native English speakers (those who speak English as a second language), there were 1.5 billion speakers of English, which still represents only 19 percent of the world's population.[40]

In May 2024, a check of the active titles in the Scopus database showed that 79 percent of the 29,000 active indexed journals are published in English only. The Web of Science Core Collection is similarly composed: of the over 23,300 journals indexed, 78 percent are published in English only. The emphasis on and predominance of English-language research outputs create a barrier that blocks the non-English-speaking populace from learning about, engaging with, and contributing to the majority of scientific literature. Journal publishers could allow or even provide duplicate publication in a second language, acknowledging the authors' preferred language or the language of the population related to the research.

An October 2023 Pew Research Center survey of Americans' Trust in Science revealed a fall in trust in scientists since April 2020, at the start of the COVID pandemic. Distrust of scientists grew during that period as well. The survey did not delve into why public trust declined and distrust increased, so there may not be any relationship to how the research was communicated.[41] At a minimum, exploring alternative channels of research communication could improve engagement with the wider public and contribute to a slowing or reversal of the trend noted by Pew.

Conclusion

Clear and accessible communication helps the public comprehend the significance of research and its implications for society. These efforts may encourage public participation and promote a sense of engagement with scientific advancements. When the public is well-informed about scientific research, they can make informed decisions on matters that impact their lives, such as health, environment, and policy issues.

Increased engagement with and understanding of scientific developments can also combat both misinformation and disinformation. Misinformation is usually defined as misleading or inaccurate information, whereas disinformation is the spreading of false information with the intention to deceive. Providing channels for the public to engage with reputable science

‡ Article type that was rolled out by Research and Practice in Thrombosis and Haemostasis in 2018.

helps the public build more familiarity with science topics, providing the skills and confidence to question misleading scientific claims. A better understanding of the research process and exposure to reliable research outlets can provide a basis by which to evaluate fringe news outlets. Engagement with and inclusion of community members impacted by research builds both familiarity with the research process and trust between researchers and the public.

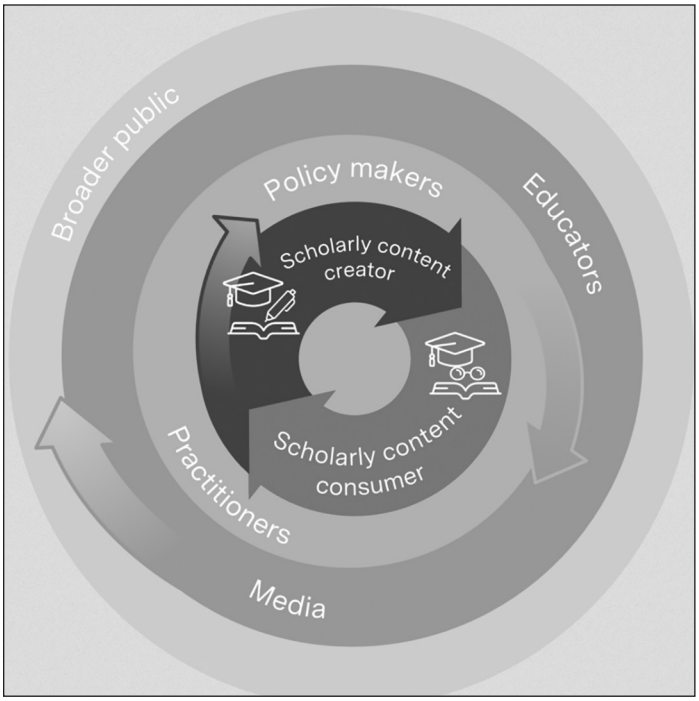


Figure 2. The expanding ring of audience groups interested in engaging with published research. © Charlie Rapple (co-founder of research communication platform Kudos, growkudos.com) 2024.

Traditional scholarly communication is a conversation between scientists that leads to new knowledge and discoveries. Open research communication is a conversation between a researcher and many different audience groups. As scholarly communication librarians, we can advocate within our institutions for broader research communication and encourage more exchange between scientists and non-scientists. At a time when federal governments and funding agencies are emphasizing the need to make research outputs publicly accessible without delay, we have the opportunity to think about more effective ways to make these outputs more engaging and consumable by a broader audience that includes both researchers in other disciplines and the wider public.

As librarians who work with researchers and authors, we can start conversations about creating content for specialized or general science media outlets and general media outlets (newspapers, television, radio). We can recommend resources for developing podcasts or video summaries. We can connect researchers with our university media and marketing units to assist in the wider dissemination of research outputs. We can encourage authors to provide plain language or lay summaries of their research findings. We can suggest effective communication strategies.

As a reader, I want researchers to tell me, in terms that I can understand, what they explored and why it was important to them. I want them to convince me, as a non-expert, why I should be interested in this topic that intrigues them and drives their conviction to explore it more deeply. I want researchers to share their wonder of scientific discovery with me.

I have tried to take my own advice. Using the first-person singular voice and a less formal writing style was my first leap of faith. My early drafts of this chapter ended with the story

about a letter to the editor of the *New York Times Magazine*, but I decided to open the chapter with that story in an attempt to better engage readers. Many of us are already strong and vocal advocates for open access publication and increasing the volume of openly available research outputs. Now it is time to shift more focus explicitly to improving the "access" component. Research is a dialogue. If that conversation is taking place behind a paywall or is inaccessible due to language or necessary technical knowledge expertise, the public cannot fully reap the rewards of the findings. Scholarly communication librarians have a key role to play in opening research communication. We can encourage and assist researchers in broadening their audience, which can lead to increased public engagement with science and to benefits for both the public and the researchers.

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CHAPTER 14

Distorting Our Message, Misrepresenting Ourselves: Siloing in Open Movements

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Introduction

As with all specialized disciplines, the language that we use as open advocates can create barriers between ourselves and others who are less familiar with our movements' norms. These barriers are expected in any interdisciplinary discussion; however, they are less expected—and may be more disruptive—when they occur across open movements. One might expect us to have a shared understanding of Open, but the messaging projected from any single movement might not reflect the values of other open movements or even of individuals within the same movement. When communication barriers create the impression that one open movement has no interconnected tissue with others, those barriers can become siloes.

Heery and Noon define a silo mentality as

> *an approach to working, whereby a manager concentrates on their own function or area without thinking about how they might integrate with others. Whilst this has often occurred across different business functions, such as between sales and manufacturing, it can also be seen within business functions.*[1]

In other words, siloing is a common occurrence in businesses when a staff member or department focuses their labor entirely on their own projects or a single aspect of their own work. Is this always negative? Not necessarily. But it can become detrimental to an organization when the silo mentality of a single manager or group infringes on their team's ability to make connections and build sustainable processes.

Like the business definition, we define siloing as the process whereby an individual or group builds an inflexible boundary around a topic in which they are invested. This boundary, in turn, limits the group's scope and obscures its purpose from external parties. For those whose labor is directly related to open knowledge, this barrier is more like the siloing present in the business world, as the situation may affect their working relationships with peers.

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The Siloing Process

The process of siloing is not always a selfish or even a conscious endeavor. It might start as a fact-finding process, through the development of definitions or guidelines. Delineations are not bad by themselves; they become siloes when we can no longer recognize the common ground between our work and those whose work is adjacent to ours. Think of it like this: defining your role, focus, and area of influence is like setting up a simple fence around a property line. You can see through the fence, and others can see into it. When that wire fence is replaced with an eight-foot, solid wood fence, then you can no longer see out of it, and you are blocked from others. This is where siloing occurs. You aren't only obscuring *your* view. You are also limiting the ability of others to learn about you.

Siloing may occur when we assume that those who want to be part of our community are either already invested in our work or able to find and engage with us easily. In these cases, it can perpetuate marginalization within our communities, both for historically underrepresented groups and for those encountering Open for the first time. This situation may be compared to someone engaging with an open-source project with an established user base. A newcomer might enter with enthusiasm to learn about the project, merely to find that there is limited or no documentation available on the main project website, and other members of this community have joined a closed forum that is not easy to locate. Existing members of the community have a choice: they can either make their documentation more transparent or they can respond, "This is really easy to pick up; just ask us if you have a question." Requiring that extra level of self-advocacy for the newest member of a group could be inviting for some, but it can be a daunting task to come up with questions to ask while you are still integrating into a new community and lacking background knowledge that can inform your questions. This is also a concern for the long-term health and sustainability of a project.[2]

Siloing may also occur when we go out of our way to make our work digestible for an external audience. We might try to "spin" our work in a way that makes it more approachable or exciting to a specific group and lose some of the clarity around why Open is valuable in the process. We might even hone in on one aspect of our movement, defining it so narrowly that from the outside, it's unclear how any other open movement relates.

Open movements are not always siloed by advocates directly, as they might not have the agency to choose how they talk about Open. There are systemically driven differences in who has access to different amounts of agency and why. For example, aligning with institutional expectations, there is a growing interest in academia to limit strategic initiatives to projects with a high return on investment (ROI).[3] This focus creates a situation wherein an institution or initiative has to set a metric they can track for their work, limiting the scope of the projects they pursue and inflicting a narrow focus on how Open is practiced. Does the need to align one's initiatives with easy-to-track metrics like download statistics or citations limit the ways we can invest in Open? Perhaps.

Among open access (OA) advocates, a focus on metrics has advanced the often-maligned use of journal impact factor and citation counts as markers of success for research.[4] This process affects how others perceive us and moves us away from our goals as we jockey to align with institutional or organizational expectations. Furthermore, this process of leaning into marketing tactics entrenches inequitable systems rather than countering them by highlighting the real value of Open: broadening access to knowledge beyond those with the means to pay for it. There are other goals we can pursue, but if we cannot operationalize them in a way that is consistent with metric-gathering and analytics to the benefit of our organizations, then many of us will not be allowed to pursue those goals at all. One might even ask, is the ROI model incompatible with the values of Open?

As we dig deeper into the practices that signpost siloing behaviors throughout the rest of this chapter, we want you to consider how these actions reflect the values of our communities, and, if they do not, how we might go about changing them.

Standards Can Become Siloes

In this section, we explore how expectations for what an open movement *should* be affect our ability to build new structures between our work and the work of advocates in other open movements. The forces imposing these expectations on us can come in several forms, from larger collectives within our own movements to institutions, governing bodies, and even commercial actors operating within the Open space.

Declarations: Defining a Movement

Open movements are often described in similar ways.[5] Individual declarations, mission statements, government mandates, and organizational initiatives offer frameworks that can be useful for communicating the value of our work. These standards can also help practitioners across communities come together, find common ground, and coordinate efforts toward larger, interconnected goals. However, declarations can also create siloes when they are followed prescriptively.

Declarations are at their best when they are used as general guidelines for a community to rally around. For example, three declarations for OA from the early twenty-first century[6] outline clear action steps for advancing OA: promoting self-archiving of scholarly works in open repositories, launching new OA journals, and advocating for promotion and tenure evaluation changes to affirm the value of open access publishing. The declarations for other open movements offer complementary values, centering expansive visions for the global public good that can be achieved through Open. The Barcelona Declaration on Open Research Information, for example, argues, "To advance responsible research assessment and open science and to promote unbiased high-quality decision making, there is an urgent need to make research information openly available through open scholarly infrastructures."[7] Similarly, the Cape Town Open Education Declaration states that access to open resources "nourish(es) the kind of participatory culture of learning, creating, sharing and cooperation that rapidly changing knowledge societies need."[8] When we look at these declarations side by side, it is clear that our movements have shared goals. By connecting with our peers, we can work toward these shared goals together.

While developing declarations helps to situate open movements—and their associated geographic, social, political, and economic context—defining ourselves strictly can also constrain the way we think about, talk about, and build upon our work in Open. When we depend on declarations or mandates to validate the worthiness of our cause, we risk shifting power from open communities as a collective to authorities that codify certain aspects of our goals. Viewing an open movement as a monolithic ideal rather than a community with evolving priorities can make it harder for us to raise questions about the way things work or propose solutions to challenges. To move forward sustainably, we should view these materials as living documents and toolkits that can be leveraged for future collaboration and development. The declarations themselves tend to encourage a wide range of actions, infrastructure development, and new paradigms. However, when we position declarations in ways that suggest a centralized vision for Open, we risk creating silos and discouraging connections across our communities.

Opening Metadata: When our Systems are Siloes

Siloing can also occur when we lean on systems and dependencies that limit interoperability. Is it possible that a service or object can be defined as open while its associated metadata remains locked behind systems, workflows, schema, or ontologies that are either proprietary or somehow restrictive? For instance, if an object or resource is given the designation "open access" and then housed in a repository that relies on a proprietary metadata schema that either intentionally or unintentionally restricts access or discovery, does that limit its technical openness?

On the surface, open metadata seems like a simple proposition. Open metadata is data about data that allows for access, redistribution, and reuse of the data independent of proprietary platforms or licensing.[9] Metadata development is often a local endeavor that results in records owned and managed by the creating institution. In these cases, the sharing and reuse of descriptive, structural, and administrative metadata is often both open and free, albeit not necessarily with the priority placed on ease of distribution and acquisition. The potential for locally created metadata exists for all types of collections and platforms, including bibliographic metadata typically housed in an integrated library system (ILS).

MARC bibliographic metadata is especially challenging to share or reuse because of its dependency on OCLC, a large-scale vendor for libraries.[10] In an effort to make ILS data openly available, institutions must presumably exclude records received from sources that impose restrictions. Statements to this effect can be seen describing best efforts to make metadata open and free for reuse and analysis: "Toward this end, we release our metadata as openly and widely as possible. We avoid placing restrictions on reuse, except in cases of ethical, contractual or legal obligations, e.g., metadata received on condition that we not share them further, or metadata for which sharing might compromise user privacy."[11] When we limit the sharing of original metadata in all its forms and schemas to paid distribution mechanisms, we silo ourselves off from potential partnerships. Union[12] or shared catalogs are structurally effective mechanisms to share and distribute data, but they are functionally restrictive in their need to charge a fee and in their restrictions on metadata reuse. Within the relationship between fees and restricted reuse lies the barrier to open metadata: a need to protect a product and its operating costs, which necessitates the restriction on free and open distribution of their records.[13]

Looking back at the question we started this section with: is it possible to call something truly open if its metadata is closed? Yes; however, there is an opportunity for us to open up more of our work by looking beyond the content we share and toward the systems that enclose, describe, and support it. Open metadata should not be seen as just added value but also as a standard we can work toward and collaborate on together. Operating as if it is okay for metadata to be commercialized and enclosed, when the content it describes may not be, implies a deprioritization that belittles the work of people in the open metadata space and siloes them from other open communities. Advocating for open metadata should not be seen as a "nice to have" but as an improvement to our infrastructures in support of Open as a whole.

Misconceptions Can Create Siloes

Siloing in open movements does not always occur because of the choices open advocates make. Sometimes, a movement is siloed because of misconceptions about its focus or scope. These misconceptions might come about because of "spin," when marketing is leveraged to simplify Open to reach a broad audience. In other cases, misconceptions arise from disputes about where a movement's focus should be. Whatever causes these misconceptions, they restrict how our communities are perceived, obfuscating our goals and values.

"Open Access is Publishing with Fees"

One of the most pervasive misconceptions in the open access (OA) movement is the idea that OA is synonymous with publishing, and specifically with publishing with fees. However, this reductive interpretation can be harmful to authors hoping to share their work openly. OA is "literature [that] is digital, online, free of charge, and free of most copyright and licensing restrictions."[14] This access may be accomplished through publishing (gold OA) or it may come about through a process known as self-archiving (green OA), wherein a copy of a work is shared in an online repository or archive.[15] Even within the bounds of OA publishing, there are several different business models that do not require author-facing fees or article processing charges (APCs). For example, "diamond" open access models support publishers so that content can be shared openly without charging authors fees or charging readers to access publications.[16]

Although these different types of OA exist, the assertion that OA is a fee-based publishing system is pervasive. One example can be seen in an article by Soyer et al., who assert, "In the Open Access model, the individual researcher pays an article process charge (APC)."[17] This is not true on several levels. As previously discussed, OA may also include self-archiving, and not all OA publishing requires fees. Even among the venues where charges are levied for the publication of OA content, APCs are not necessarily paid by authors alone. As Suber shares in a review of the literature about who pays for OA, multiple studies have found that less than 50 percent of authors pay APCs personally,[18] with that number being as low as 16 percent in a recent study from the Association for Information Science and Technology.[19] Implying that authors of OA scholarship must pay APCs can alienate those who might otherwise share their work openly and normalizes the belief that OA publishing is a predatory, anti-author practice.[20]

In addition to making OA less appealing to authors and potential partners, the normalization of APCs has enabled new business models to emerge that leech off of the OA movement by incentivizing the transfer of funds to publishers and platforms that can simplify the APC payment process, such as "transformative" read and publish agreements.[21] The growth of read and publish agreements indicates interest from commercial providers in OA, which is technically a good sign. It signals that more people want to broaden access to research, and their support is pushing groups who would traditionally oppose OA to instead work with OA advocates. However, these changes have also created room for commercial actors to entrench themselves in the OA movement to profit from its growth, a practice sometimes referred to as openwashing.[22] Furthermore, the growth of read and publish agreements has given more power to scholars in the Global North who are already set up to succeed through their access to closed research, funding, and time.[23] Providing accurate information about the many ways that content can be made open helps demystify OA for readers and authors alike.

The misconception that OA publishing requires APCs does not just limit our ability to connect with authors; it also makes OA opaque for those in other open movements, limiting the potential for collaboration. There are several groups that could learn from the experimental business models that the OA movement has tested over the past decade; however, it is impossible for those outside of the OA space to recognize that opportunity if all that they know about OA is APC-based publishing. Even within the OA movement, this misconception can create siloes for those whose work does not align with the popular depiction of OA. Focusing on "publishing OA articles" disregards the rich history of OA in the humanities and social sciences, which goes beyond journal publishing to include the sharing of nontraditional works and OA books, alongside other scholarly outputs.[24]

So, what can we do? As individuals, we can highlight the many and varied ways that OA can present itself, including both publishing and self-archiving. As organizations, we can do even more, advancing support for non-APC-based OA and highlighting when that work is being done well. Projects like Subscribe 2 Open are one example of large-scale efforts to make more scholarship OA without resorting to author-facing APCs or read and publish agreements, but there are others as well.[25] And for those who offer support for authors pursuing APC-based gold OA, it is worth noting that we *do not* need to stop saying "we offer support for OA fees" to address broad misconceptions about our work. These statements are not communicated to maliciously misrepresent OA but to let our audiences know that support is available for them. It is easier for us to share that message by highlighting the support we have for APCs rather than first defining the range of flavors of open access that might not be relevant for everyone. Moving beyond a siloed view is not something that happens overnight. Supporting our community to engage with the complexity of OA requires us to be aware of when the messaging presented to them might be incorrect. But we should not limit ourselves to only the most "accurate" representation of our movements at all times. Starting by sharing easy-to-digest information and answering questions is necessary before anyone can fully engage in the complexities of OA.

"Open Data is FAIR Data"

Another common misconception is that FAIR data and open data are interchangeable, with the belief that FAIR includes openness and open data is FAIR. This is understandable, as some of their goals overlap, supporting aims such as increased data access, reuse, collaboration, and transparency. Both guidelines are widely considered tenets of good data sharing and open science, and both offer positive value propositions. However, it is important to distinguish between the two to avoid distorting the definition of open data in such a way that could cause siloes within and outside the open data community.

The FAIR Guiding Principles for data outline major foundations to increase data discovery, integration, and innovation.[26] The practices and structures suggested to make data FAIR (findable, accessible, interoperable, and reusable) are distinct from those that make data open. This is understandable, as the FAIR principles were made in response to growing data sharing practices driven by the open data movement.[27] In contrast to FAIR, which focuses on technical discoverability and reuse, open data principles prioritize access, availability, and open licensing.[28] Open data can be freely used, modified, and shared by anyone for any purpose, thanks to its licensing,[29] but it is not necessarily structured or presented in a way that implements the FAIR data principles.* Similarly, the FAIR principles can be applied to data with restrictive licenses and access.

Data can be open, FAIR, both, or neither. These statuses are not prerequisites for each other. When these terms are treated as interchangeable, the differences between open and FAIR data, the utilities of each, and how they interact are lost. This creates confusion both within the open data movement and for those trying to navigate different data sharing practices and expectations. By recognizing the differences between open and FAIR data principles, we can make more thoughtful decisions regarding data management, preservation, and sharing. A common saying is that data should be "as open as possible, as closed as necessary." What is often missing in these discussions is that all data, including data that is not shared, can still benefit from the FAIR principles.

It is equally important to acknowledge that FAIR and open principles are not the only frameworks to consider when evaluating how data should be shared and made usable by others. Indigenous groups have presented principles to use in support of Indigenous data sovereignty, such as CARE (collective benefit, authority to control, responsibility, and ethics).[30] The CARE principles can also help in determining whether it is appropriate for data to be open and which FAIR principles should be applied. As Stephanie Ruso Carroll et al. have noted, "The CARE Principles should not be thought of as only applying to Indigenous Knowledge or Traditional Knowledge but also scientific data. The CARE Principles speak to how scientific data are used in ways that are purposeful and oriented towards enhancing the well-being of people."[31] Just as open data is not automatically FAIR, neither open nor FAIR data automatically follows the CARE principles. By appreciating the ways different sets of data principles layer and interact, we are better able to understand how they work in concert with each other.

Promoting an understanding of these different frameworks and recognizing the utility of their varying foci helps minimize misconceptions that can result from over-generalizations and lead to siloing. While this process can have a steep learning curve, it's an important step for developing interoperable open infrastructure and can ultimately aid citizens, scholars, and newcomers in participating in open movements to the fullest extent possible.

"OER Are Affordable Teaching Materials"

Open educational resources (OER) are often marketed as tools for making education more affordable, with messaging focused on lowering the cost of teaching materials to support students. This is useful framing for institutions investing in open education, as the value of

* Open licensing, too, is not one-size fits all; the Citizen's Guide to Open Data outlines different options and considerations.

lowering costs is easy to communicate. However, there are two problems with this approach: 1) this messaging can be co-opted by commercial providers if they lower costs to increase affordability,[32] and 2) it leaves out the main component of "openness," which is necessary for an OER to be considered open.[33] Like OA advocates before them, OE advocates find themselves in a precipitous position: lean on a successful marketing tactic or clarify the boundaries of our work to reinforce the value of Open?

OER messaging that focuses on affordability can lead to positive outcomes. "Affordability" is understandable to a wide range of audiences and communicates the OE movement's goals of increasing access to education. However, positioning an open movement primarily around affordability also leaves it ripe for exploitation. This is a variable term, one that can be levied in many different ways. If something is "more affordable," what is the alternative price point we are comparing to, and who benefits?

We are currently in the middle of a huge shift in the way that teaching materials are assigned, bought, and used in higher education. Many institutions across North America are seeing a concerted push from textbook and courseware providers to switch to a subscription-based model for accessing their content at a discounted price.[34] These subscription-based course material platforms, as well as freemium educational tools, artificial intelligence "partners," book rentals, and courseware, have all been labeled as affordable over the past decade, despite prices ranging above one hundred dollars each.[35] Leaning into affordability within our own communities allows commercial providers to make the case that their work increases access for students and lowers costs, just like OER!

However, OER are not just "affordable." By definition, they're both free of cost and open. One of the earliest definitions of OER clarified that these materials must be "licensed in a manner that provides everyone with free and perpetual permission to engage in the 5R activities," to retain, revise, remix, reuse, and redistribute open content.[36] This distillation of the 5Rs was built upon larger discussions in the burgeoning OE community about access to learning. An article from 2007 from the Creative Commons ccLearn team argued that, far from being a bonus, the permissions built into OER through licensing are what make them such a powerful tool for education.[37] The permissive licensing for OER allows educators to leverage the materials flexibly in their courses and leaves room for engaging in more open teaching practices, such as open pedagogy. Ignoring those possibilities focuses the OE movement solely on materials rather than people and practices and does a disservice to the educators whose work underpins the movement as a whole.

Focusing our messaging on affordability brings more attention to OE work and allows institutional leaders to acquire easy wins for their administration; however, broadening our definitions can distance the OE movement from other open movements when affordability takes precedence over openness. As Wiley states in a blog post about arguments over definitions:

> *As far as I can tell, the only people actively engaged in a debate about the meaning of the word "open" in the educational context are (1) those who genuinely misunderstand it because they haven't become part of the community yet, and (2) those whose business models would collapse if the public had free access to and open licenses for their products.*[38]

In trying to craft an accessible message, individuals and groups within the OE community have created opportunities for their work to be simplified in a way that ignores the actual meaning of open. Other open movements face similar challenges with the push-and-pull between pragmatic goals and idealistic values. This has been called out in the past, but it remains a pivotal tension point among OE advocates.[39] In order to move past these siloing behaviors, it's important to listen to the many different voices within and outside of the OE community. Those within the OE movement are not just advocates but also educators, support staff, and students who engage with OER in their day-to-day lives. These individuals can speak to the value of open content as flexible and adaptable materials. And listening to those outside of OE is just as important. We can learn from those in the open-source software community

who have championed the ability to not only access but also reuse and adapt code as a core value of their movement. By finding common ground with our partners and peers, we can build stronger messages about our open movements, ones that are true to our core values and which cannot be overtaken easily.

Misinterpreting and Overhyping Open

Well-intentioned discussions about open movements can still result in siloing when the messages we project distort our core values. Two examples of this distortion happening can be seen in practices known as openwashing and openwishing. In 2009, Michelle Thorne defined openwashing as "to spin a product or company as open, although it is not,"[40] similar to the concept of "greenwashing." As we've discussed in the open access publishing section, openwashing is often perpetuated by companies that want to reap the rewards of Open without contributing back to the commons.[41]

Openwashing is not the only way that our messaging can get distorted. Andy Dickinson created the term openwishing to describe the "big claims of social and individual impact"[42] routinely made by open declarations, practitioners, and organizations. Openwishing, he claims, can backfire in overpromising vague public goods and benefits that might be difficult to deliver in a discrete time frame and that are challenging to consistently measure.[43] This leaves open movements susceptible to criticisms about their feasibility and can distance potential partners from the premise of Open entirely. For example, the metaphor "free as in puppies" is often used as a "gotcha" in arguments about the potential benefits of Open by highlighting that to support open infrastructures, advocates need to invest funding, labor, and time, not just positive thought.

Openwashing and openwishing are often non-malicious behaviors. Openwishing can stem from a sincere and founded expression of hope for what Open can achieve, reflecting a belief that change is inherently possible, even for entrenched norms. While openwashing is more likely to come from external forces, it can be seen as an indication that Open has achieved enough progress and buy-in that it has become desirable to imitate. Nonetheless, both of these practices can exacerbate siloing, miscommunication, and general confusion around the idea of Open. Anticipating the criticisms that can be levied against us due to openwishing and the miscommunications created by openwashing from external groups can help us respond effectively. It can also help us minimize direct harm that occurs when openwashing and openwishing are presented without reflection.

How to Recognize Siloing Behaviors

Throughout this chapter, we have given examples of ways in which siloes can occur. These siloes might be built out of necessity, in response to outside forces, or they might be imposed on us by forces beyond our control. Now we want to take a step back and think about how we can avoid siloing behaviors. If we can spot the signs of siloing, we can take measures to move away from them so that the values that we embody across our movements are able to shine through without distortion.

As we've discussed, siloing behaviors occur when:

- We are **reactionary**—to mandates, marketing, and messaging. In these situations, it is common to lose sight of our goals as we scramble to meet expectations that have been placed on us or to fight against a perceived threat.
- We **overgeneralize** our work's purpose and try to meet all the needs in our community, regardless of whether those needs actually align with the specific focus of our work. This can lead to the duplication of efforts underway both within and across our movements. Overstretching the definition of what an individual open movement covers can create situations where other open movements can't distinguish our original purpose or shared goals.

- We are so **prescriptive** in our boundaries that we refuse to support projects that might align with our values, just because they don't 100 percent align with our definitions. This is not to say definitions don't matter, but that there is a balance to strike between overgeneralizing and being prescriptive.

Conclusion: Looking Ahead Together

We initially extended an invitation for you to consider how our actions reflect the values of the open movements we work within. Now, we circle back to this invitation and offer the following as we look ahead to more exciting possibilities away from siloing tendencies.

We need to consider how open movements support the systems that enclose and describe our content. Is there greater potential for us to leverage existing open infrastructures that meet the needs of multiple open movements or to build new ones? If so, we should listen to our peers in other open movements and learn from their expertise. For instance, OA support staff could work with metadata experts to develop standard open metadata practices for OA repositories.

Next, we need to seek out information from other open movements in the places where their work is discussed. In addition to reviewing the declarations that provide an easy entry point for understanding the underpinning values of a movement, we can also learn about the current work of other movements through email listservs, social media platforms, associations, and forums. This is a critical step, as even if we are up-to-date and engaged in our own communities, there is more we can discover from the work of our peers. Once we have built more connections across our movements, we can propose better-informed projects to bridge our work. For example, OA advocates could help those in the OE space recognize that OA books that have an educational component could be leveraged as OER and not seen as simply "other."

Above all, we can acknowledge that across open movements, we all face challenges with communicating our worth and balancing our pragmatic needs with our ideals. The labor tied to Open is often rewarded for siloing, for setting itself apart as a unique outcome that has no peer. So much more can be achieved when we notice the opportunities for collaboration in front of us and work toward a future wherein we can recognize the value of one another's work as both distinct and complementary to our own. But first we must push past the structures that keep us siloed from one another, even if it is difficult or slow in the beginning. We share a common purpose: to expand access to information for everyone in the world. To reach that goal together, we need to keep our eyes and ears open to those around us.

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CHAPTER 15

Open Archives: Framing Archival and Special Collections Materials as Open Educational Resources

Hilary Barlow

Many archives and special collections departments make portions of their collections available online for the use of the general public or at least their own user base. This aligns their work with open movements, including open access, open science, and open education. While open education is well-understood in the library field overall, it is not fully integrated with special collections librarianship in the same way, and it is much less known among archivists. What may seem like an incompatibility is actually an opening to conversation, learning, and innovation. Through case study interviews and an anonymous survey, this research shows that while their current knowledge of open education principles varies, archivists and special collections librarians are interested in incorporating open education practices into their curation of digital materials toward the goal of making them more accessible online.

The Intersection of Archives, Special Collections, and Open Education

The open education (OE) movement has made great strides in the advocacy for openly licensed digital resources. According to the Scholarly Publishing and Academic Resources Coalition (SPARC), OE "encompasses resources, tools and practices that are free of legal, financial and technical barriers and can be fully used, shared and adapted in the digital environment."[1] Among the products of the OE movement, open educational resources (OER) are the most likely to be encountered by archival and library workers. The United Nations Educational, Scientific and Cultural Organization (UNESCO) defines OER as "learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others."[2] The choice to release educational materials as OER

and the practice of sharing archival and special collections materials online have a similar goal: make these resources accessible to the public.

Some university library systems have succeeded in incorporating OER into the curriculum at their institutions, particularly in the United States. The switch to remote learning that took place early in the COVID-19 pandemic further demonstrated the benefits of low- or no-cost, accessible educational resources to instructors, students, and researchers. Archivists, special collections librarians, and paraprofessionals put great effort into making the materials they curate more accessible, including digital collections that were already available online.[3]

Archives are "permanently valuable records—such as letters, reports, accounts, minute books, draft and final manuscripts, and photographs of people, businesses, and government. These records are kept because they have continuing value to the creating agency and to other potential users. They are the documentary evidence of past events. They are the facts we use to interpret and understand history."[4] "Special collections" is an umbrella term that can include materials varied in both content and format. The Society of American Archivists defines special collections as "an institution or an administrative unit of a library responsible for managing materials outside the general library collection, including rare books, archives, manuscripts, maps, oral history interviews, and ephemera."[5] The distinct characteristics of archival and special collections materials necessitate distinct approaches to their care.

One complex consideration is copyright. Even if materials such as letters or journals are unpublished, they may still be copyrighted. In some cases, the copyright owner is unknown or cannot be contacted, creating a situation where the archives cannot ask permission to, for example, feature the materials in a digital display. Models to display "orphan works," or materials where the copyright holder is unknown, have been successfully implemented at Canadian universities.[6] Nonetheless, it can be daunting to consider sharing such items online for many practitioners.[7] Some records contain personal and identifying information, prompting both legal and ethical concerns as to when and how they should be shared. Additionally, materials may be restricted due to donor agreements.*

Archives and special collections have been engaged in digital sharing for a long time, but the way they do so is necessarily different because of the above-mentioned constraints. Principles such as remixing (an expectation for OER) conflict with archival principles such as original order. "Remixing a resource means you can adapt, reorganize, and remake a resource to create something new."[8] In archival terms, original order is "the organization and sequence of records established by the creator of the records,"[9] and it is an important cornerstone of archival practice to ensure researchers grasp the context in which a record was created. For archival materials to become OER, archivists would need to determine what role original order would play in the sharing of archival records in an OE context. Does the principle of original order make archival materials poor candidates for OER? Or could sharing some records as OER give archivists an opening to educate students and the public?

Nonetheless, archives and special collections staff regularly make materials more available on digital platforms when the permissions and resources align to make that possible. While materials benefit from context, in practice, digital items can easily be removed and posted elsewhere without context or credit to the home institution.[10] Many archivists and special collections librarians engage in activities that might be considered open education work, whether or not they use that terminology.

When materials are shared in an online collection, they are typically shared as one of the following: free online resources under all rights reserved copyright, non-open access

* The Society of American Archivists Dictionary of Archives Terminology defines a donor agreement as "a legal agreement that serves as the formal expression of the terms of a gift to a repository, including the terms of the transfer of ownership of records and/or intellectual property rights," stating further that "Donor agreements may specify use restrictions, relevant policies and procedures to be used in the management of the records, or other terms on which the donor and the repository agree." "SAA Dictionary: donor agreement," Society of American Archivists, accessed December 31, 2024, https://dictionary.archivists.org/entry/donor-agreement.html.

materials available only to library users, or open access materials (see figure 1). According to open access advocate Peter Suber, "Open-access (OA) literature is digital, online, free of charge, and free of most copyright and licensing restrictions."[11] It is a commonly used criterion in library circles, particularly in academic institutions, and is another step toward OER status if the item released as open access is modifiable according to its license agreement. Materials in the public domain can be shared and remixed, assuming there aren't other concerns with the material, such as privacy.

Material type	Openly licensed	Freely available	Modifiable
OER	Yes	Yes	Yes
Free online resources under all rights reserved copyright	No	Yes	No
Non-open access materials available through the library	No	Yes (but only to current students and faculty)	No
Open access materials and monographs	Yes	Yes	Maybe (if the Creative Commons license allows derivatives)

Figure 1. Components of OER and free resources. Adapted by Stephanie Quail and Hilary Barlow. This table is a derivative work of Abbey Elder, "Table 1: Components of an OER," in *The OER Starter Kit* (Ames, IA: Iowa State University Digital Press, 2019), https://iastate.pressbooks.pub/oerstarterkit/chapter/introduction/.

I believe the archival profession could benefit from exposure to the open education movement, that open education could benefit from the specialized knowledge of archivists and special collections librarians, and that there are many ways such a connection could expand current practices. While some collections would not be compatible with OE practices due to the above-mentioned constraints, others could be shared on OE platforms. Archivists could incorporate OE considerations into donor agreements to make archival materials that much more accessible to the public, beginning at the acquisition stage. In cases where the archives retain copyright, archives could license materials using open licensing systems like Creative Commons, while works in the public domain could be shared without restriction. Digital archivists could bring their knowledge of digital preservation to OER collections, and both archivists and special collections librarians could bring unique collections to OE audiences.

Special collections and archives in particular can present challenges for novice researchers because search strategies when using archives are different than when using most library collections,[12] and this challenge is still present when querying digital collections.[13] The specialized vocabulary of archival finding aids[†] is often distinct from that used when browsing a typical library catalog[14] and, as a result, using archival or special collections materials requires a different skill set compared to conducting research with materials found in non-special collections. In many, though not all cases, archives and special collections materials are primary sources.

[†] The Society of American Archivists Dictionary of Archives Terminology defines a finding aid as "a description that typically consists of contextual and structural information about an archival resource." The entry also states that "Often a finding aid places archival resources in context by consolidating information about the collection, such as acquisition and processing; provenance, including administrative history or biographical note; scope of the collection, including size, subjects, media; organization and arrangement; and an inventory of the series and the folders. Finding aids could also describe a single level or a single item." "SAA Dictionary: finding aid," Society of American Archivists, accessed December 31, 2024, https://dictionary.archivists.org/entry/finding-aid.html.

While many archivists build connections with schools and universities at all levels, instructors may lack the specialized knowledge about primary sources and archival research,[15] leaving space for archivists and special collections librarians to make in-person or digital connections. This is not a critique of current digital archival practice, which has engaged significantly in the creation of digital collections, but rather a place for expansion and connection between different fields and professional practices.

In 2022, I was hired for a two-year position as the W.P. Scott Chair in Librarianship at York University. This gave me the opportunity to explore these ideas and conduct original research at the intersection of archives, special collections, and open education. The research project was based on the substantial literature about OE and OER and on the challenges (legal, logistical, and financial) of bringing archival and special collections materials to the wider public, as reported by archives and special collections practitioners.[16]

In contrast, there is not a great deal of academic literature about the intersection of archives and open education. Archivists and special collections librarians routinely use digital collections platforms, but the language of the OE movement is consistently associated with non-special library collections and not with archives. Archives and special collections typically appear in OE literature only when they are part of a larger institutional embrace of OE principles. I believe this stems from a hesitancy to confront the very real structural challenges of making archives "open" in the OE sense, which are distinct from the challenges that apply to most library collections.

That being said, while there may be a dearth of literature on the subject, it is by no means nonexistent. In the search for on-campus allies for OER initiatives, archives and special collections have been called "untapped, unique [...] material that can be used as teaching materials."[17] Digital archival materials have been used in open pedagogy contexts to teach history and information literacy.[18] The Open GLAM organization promotes equitable standards and practices for galleries, archives, libraries, and museums ("GLAM" institutions).[19] In its organizational principles, Open GLAM specifically recommends sharing public domain items using the Creative Commons Zero Waiver[20] in line with OE practices.

Archives and special collections engage in the realm of open digital resources, whether the terminology particular to the OE movement is in use or not. These departments make their collections available online when it is legally and logistically possible, contributing to wider open access initiatives with unique resources.[21] Digital archivists have much expertise to contribute toward the long-term preservation of OER.[22] While it may seem like the archival principle of original order stands in opposition to the OE principle of remixing, there are practitioners changing what it means to remix archival materials.[23]

In order to determine if there was a point of collaboration between archives, special collections, and open education, I wanted to test my assumptions. Was it true that OE concepts hadn't been integrated into the archival profession as they had in librarianship? I wrote a series of interview questions to gauge my colleagues' knowledge of OE and OER and their openness to incorporating OE practices into their own work. I decided to begin with case study interviews and then launch a larger, anonymous survey to see if conclusions from my interviews were reflected in a larger sample of practitioners.

Case Study Stage

I conducted twenty-two case study interviews with archivists, librarians, paraprofessionals, and scholars in Canada and the United States. My interviewees were mostly based in Canada—fifteen in Ontario, three in Quebec, and one each in British Columbia, Manitoba, and Nova Scotia—with one interviewee in Pennsylvania to see if an American practitioner would answer differently. Ultimately, the American interviewee did not diverge greatly from their Canadian peers but confirmed overall trends.

In terms of occupation, I talked to eleven archivists, seven librarians, two paraprofessionals, and two teaching faculty who had incorporated OER into their work. I attempted to make the group professionally diverse by including multiple specialties, such as a map librarian, audiovisual archivist, and a rare books librarian. The interviewees worked in a variety of settings, including universities, museums, and public libraries.

I asked twenty-three open-ended questions about their knowledge of OE, OER, and Creative Commons (CC). Additionally, I asked about their general openness to incorporating OE in their work, presenting concrete examples of how OE could be incorporated to gauge their reactions to different practices. I preempted questions about OE and OER with definitions in case an interviewee was unfamiliar with the terminology. My goals were to establish the interviewees' current professional work and relationship to digital collections, gauge familiarity with OE, OER, and Creative Commons, and gauge openness to potentially new practices. I went through several drafts, refining the list of questions with input from my project supervisors and colleagues at York University. In order to make my interviewees more comfortable and encourage them to speak freely, I gave them a great deal of leeway in how they wanted to be quoted: not at all, anonymously, or with their names attached.

After transcribing interviews, I coded responses for easier comparisons. For example, when I asked about participants' familiarity with OE, depending on how they rated themselves, I categorized the response as high familiarity, medium to high, medium, low to medium, and none to low familiarity. I sorted their responses based on keywords in their description of their knowledge. For example, if they said they had never heard of OER or had only heard the term but had no context for it, I rated that as "none to low." Having a vague understanding of a term categorized the response as "low to medium." A clear comprehension of a concept was coded as "medium," with a detailed understanding of how a concept was implemented resulted in a "medium to high." Comprehensive understanding and usage of a concept or practice was categorized as "high."

While twenty-two is a small sample size, I did find some differences between archivists and librarians when it came to OE. I asked all of my interviewees, "What is your level of familiarity with the open education (OE) movement?" While levels of familiarity varied within the archivist and librarian sub-groups, librarians tended to rate themselves with a higher level of familiarity.

The seven librarians rated between "none to low" familiarity and "medium to high," with "none to low" the most common at three responses. At two responses, "medium" familiarity was the highest any of the eleven archivists rated themselves. Five said "low to medium" and the remaining four reported "little to none." I only interviewed two paraprofessionals, one reporting "low-to-medium" familiarity and one "medium to high" (see figure 2). The latter paraprofessional had worked on an OE-heavy digital repository project, and they consistently rated their knowledge highly.

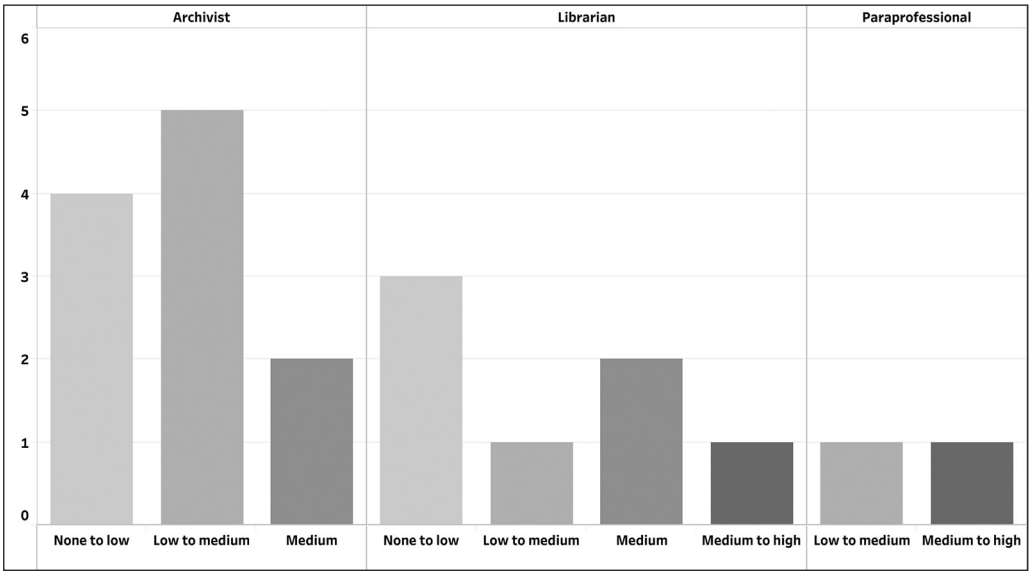


Figure 2. Open education familiarity by profession. Coded responses to the question, "What is your level of familiarity with the open education (OE) movement?" arranged by profession. Created by Hilary Barlow.

I also asked, "What is your level of familiarity with open educational resources (OER)?" Generally, interviewees were more familiar with this well-known product of the movement than the OE movement itself. The seven librarians ranged relatively evenly between "none to low" (three responses), "medium" (two), and "medium to high" (two). Archivists trended lower. "Medium" was their highest rating as a group at four responses, and a majority fell in either the "none to low" (four) "low to medium" (three) category. One paraprofessional rated themselves at "medium" familiarity, and the second, a very OE-proficient paraprofessional, was the only interviewee to describe themselves as "highly familiar" with OER.

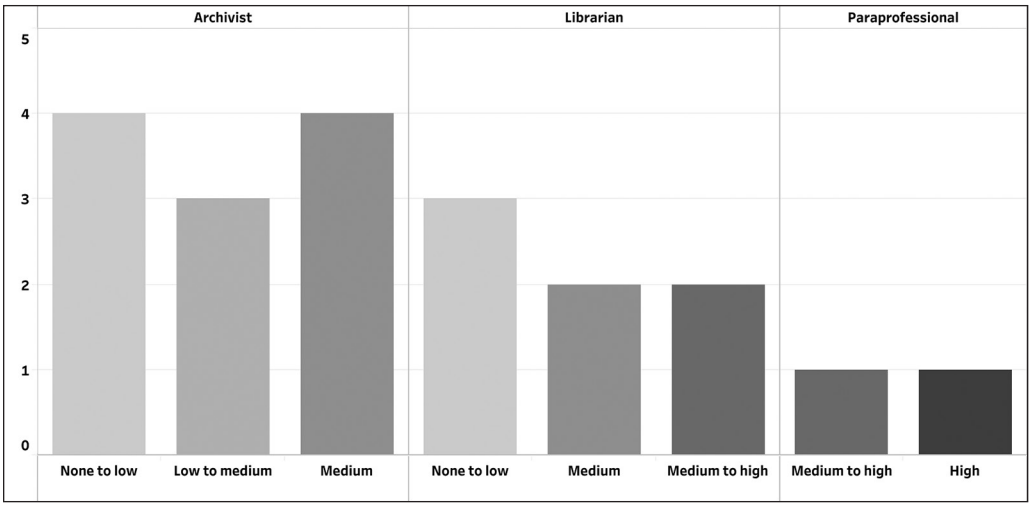


Figure 3. OER familiarity by profession. Coded responses to the question, "What is your level of familiarity with open educational resources (OER)?" arranged by profession. Created by Hilary Barlow.

Open licenses provided by the Creative Commons (CC) organization are a frequently utilized set of permissions in the open education movement and in digital collections generally. Creative Commons licenses "give everyone from individual creators to large institutions a standardized way to grant the public permission to use their creative work under copyright law."[24]

Since CC licenses are so commonly employed in the OE movement, I was also curious to know how practiced my interviewees were with CC generally and open licenses in particular. When it came to CC, there was a more noticeable difference in both knowledge and practice between archivists and librarians. The seven librarians rated themselves as having "medium familiarity" (three respondents) at minimum, with one "medium to high" and three rating themselves "highly familiar." While none of the eleven archivists rated themselves as highly familiar, four were in the "medium to high" range, with five in the "low to medium" range. The paraprofessionals were very knowledgeable about CC, with one "medium to high" and the other "highly familiar" (see figure 4).

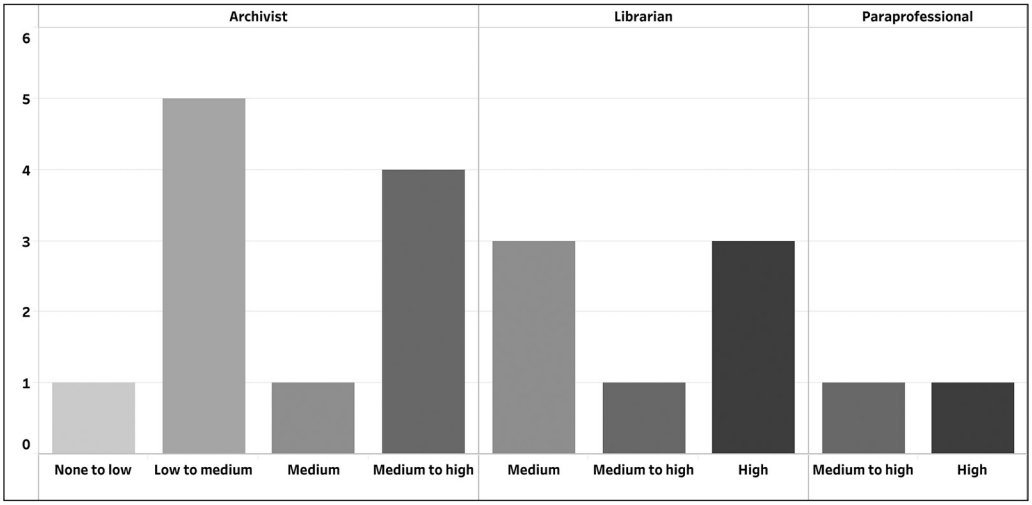


Figure 4. Creative Commons familiarity by profession. Coded responses to the question, "What is your level of familiarity with Creative Commons, including Creative Commons licenses?" arranged by profession. Created by Hilary Barlow.

Use of CC licensing in one's own work was the starkest difference between archivists and librarians. I asked, "Do you ever assign Creative Commons licenses to your work? If not, do you use another licensing system?" In asking about their "own work," I meant publications such as conference presentations and journal articles as opposed to archival or library items submitted to an online collection with a license attached. Only three out of eleven archivists interviewed said they assigned open licenses to their own work, while five of the seven librarians said that they did. Neither of the paraprofessionals assigned licenses to their own work, though they reported assigning them to materials in online collections (see figure 5).

It's worth noting that in Canada, where most interviewees and all of the paraprofessionals reside, employers retain copyright over work completed by employees while on the job in many cases. Archivists and librarians who are members of faculty unions, as many are in university settings, may retain copyright over their work as part of their bargaining agreements,[‡]something that would not commonly be the case for paraprofessionals.[§] This would not necessarily apply to external publications and conference presentations. Paraprofessionals may engage in those types of outside work, but it is unlikely that they would be under the expectation of publishing for the tenure track, as many university librarians and archivists are.

‡ Using my own institution as an example, the York University Faculty Association Collective Agreement states that its members retain copyright over their own work under article 23, section 2: York University Faculty Association, *Collective Agreement 1 May 2021 – 30 April 2024*, York University Faculty Association and York University Board of Governors (CUPE 1281 and Thistle Printing), 140–41.

§ In contrast, the union for paraprofessionals has no such statement as part of its collective agreement: York University Staff Association, *Collective Agreement between York University and York University Staff Association/ Association du Personnel de l'Université York*, Unit 1, 2021–2024.

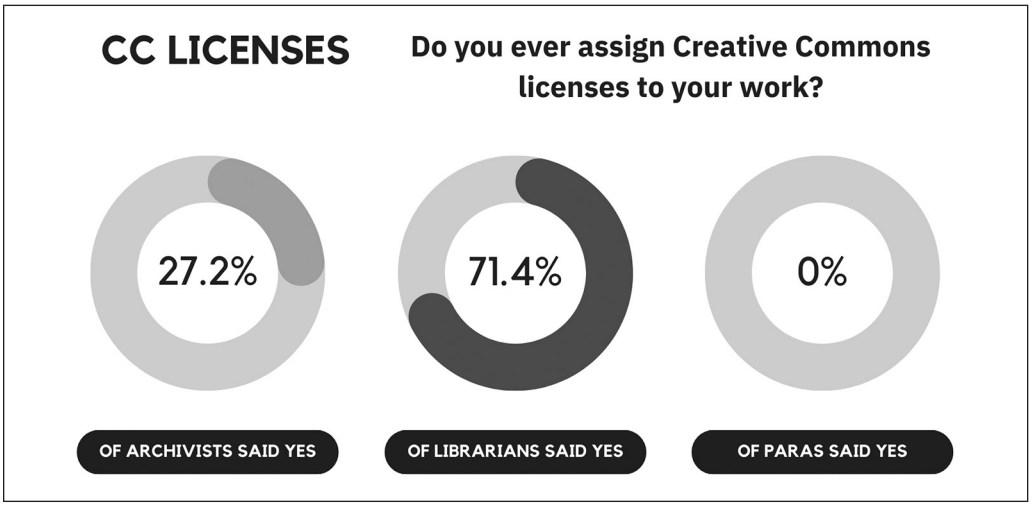


Figure 5. Creative Commons licenses. Interviewee answers to the question, "Do you ever assign Creative Commons licenses to your work? If not, do you use another licensing system?" arranged by profession. Created by Hilary Barlow.

Another big question was whether archivists and special collections librarians would be open to incorporating OE principles into their regular practice. Archival practice and librarianship are both fields where constantly learning and updating practices is normalized, particularly when it comes to technology. Incorporating new best practices is tricky, especially with frequently limited budgets and resources; nonetheless, it's something librarians and archivists do all the time.

Both special collections librarians and archivists tended to be open to the idea of incorporating OE principles into their work, while cognizant of the challenges that would entail. An archivist based in Nova Scotia said, "I do think that the potential is there." A University Archivist at York University said, "There's no point in us preserving them unless the material is discoverable and being used."

Some interviewees challenged me by asking if this was already being done. Perhaps the terminology wasn't the same, but weren't archives and special collections libraries already sharing in a way aligned with the OE movement? "A lot [of objects] already are being shared in that way through the attribution of Creative Commons licenses," argued an archivist based in Manitoba. A paraprofessional based in Ontario said, "I think they certainly can be and I believe they already are [being shared in a way aligned with OE]. I think that they aren't really being promoted that way."

Multiple practitioners were open to incorporating new ideas, while emphasizing that it had to be done right to respect copyright, privacy, donor agreements, and the archives itself. An academic librarian based in Ontario said, "Hell yes, open license it, but let's be attentive to the terms of the license." A paraprofessional in Ontario shared, "I don't have a problem with the concept of people remixing and doing whatever, but I do have a problem with not linking back to where you got it from or the original source material." This paraprofessional's view underscores another common set of concerns among practitioners that materials be shared in a way that acknowledges both the work of their professions and their institutions.[25]

Survey Stage

In preparation for the survey stage, I revised and pared down my original interview questions so they could be answered in an anonymous survey format. As with my case study interviews, I asked some demographic questions, had participants gauge their familiarity with OE

principles, and asked if they felt archival and special collections materials could potentially be shared as OER. In the survey format, these became eighteen questions with either multiple choice or an open text entry box, depending on the question. Having more participants enabled me to hear from practitioners in more places, particularly the United States.

My target survey participants were archivists, special collections librarians, and paraprofessionals who worked with digital materials in some capacity. To find them, I promoted my survey on professional listservs devoted to those fields. At the close of my survey, I had received 165 responses. Around 57 percent came from practitioners based in the United States and 43 percent from Canada. A majority of respondents (61 percent) worked in university settings, with the second highest place of employment being "other private institutions" at 11 percent. Eighty-four percent said they managed digital materials in some way as part of their work.

In terms of their professional role, 60 percent of respondents were archivists (ninety-nine individuals) and 26 percent self-identified as librarians (forty-two). Eleven individuals self-identified as either library or archives paraprofessionals, and the remaining thirteen people selected "Other" and had the option to give a custom description of their role.

As reflected in the case study interviews, librarian survey respondents were more likely to be versed in Creative Commons than archivists. Among librarians, 52 percent described themselves as "very familiar," while 25 percent of archivists did the same. Five percent of archivists selected "not at all familiar," but no librarians did (see figure 6).

While only eleven survey participants identified themselves as paraprofessionals, the responses differed between archives and library paraprofessionals. The four archivist paraprofessionals answered that they were either "somewhat familiar" (two responses) or "not very familiar" (also two responses) with CC. On the other hand, the seven library paraprofessionals said they were either "very familiar" (four) or "somewhat familiar" (three).

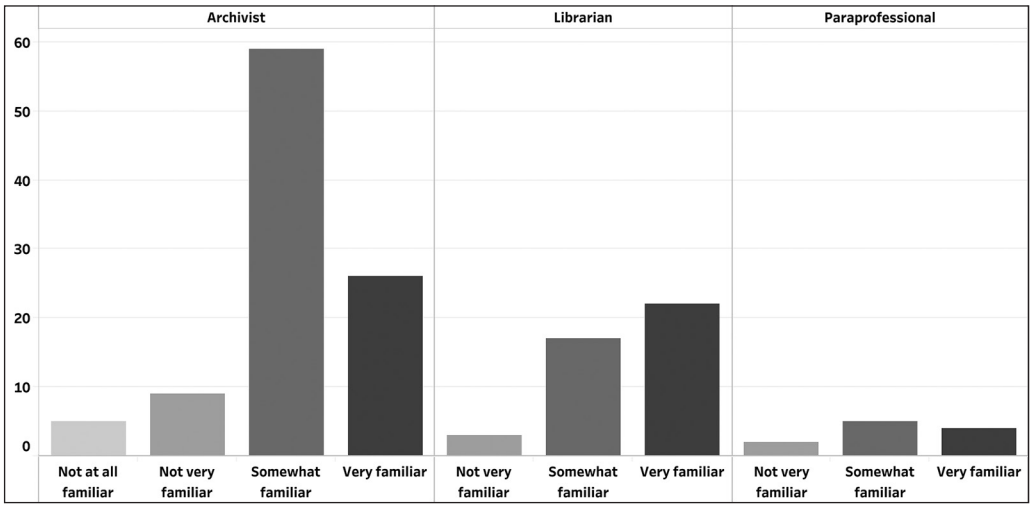


Figure 6. Creative Commons familiarity by profession. Multiple choice survey answers to the question, "What is your level of familiarity with Creative Commons, including Creative Commons licenses?" arranged by profession. Created by Hilary Barlow.

The big question was the same as in the case study interviews: "Do you think digital archival and special collections materials could potentially be shared as OER?" Respondents could answer "Yes, absolutely," "Yes, if certain conditions were met," "I'm not sure," "In exceptional circumstances," "No, not at all," or "Other." They could make one selection, and selecting "other" would give them a chance to input custom text. Again, there was a great deal of openness. "Yes, absolutely" was selected by fifty-six individuals, 34 percent of respondents. A

majority of 58 percent (ninety-four individuals) selected the still encouraging "Yes, if certain conditions were met." Six percent, or nine individuals, selected "I'm not sure." The remaining options of "in exceptional circumstances," "no, not at all," and "other" made up the remaining 2 percent or four individuals (see figure 7).

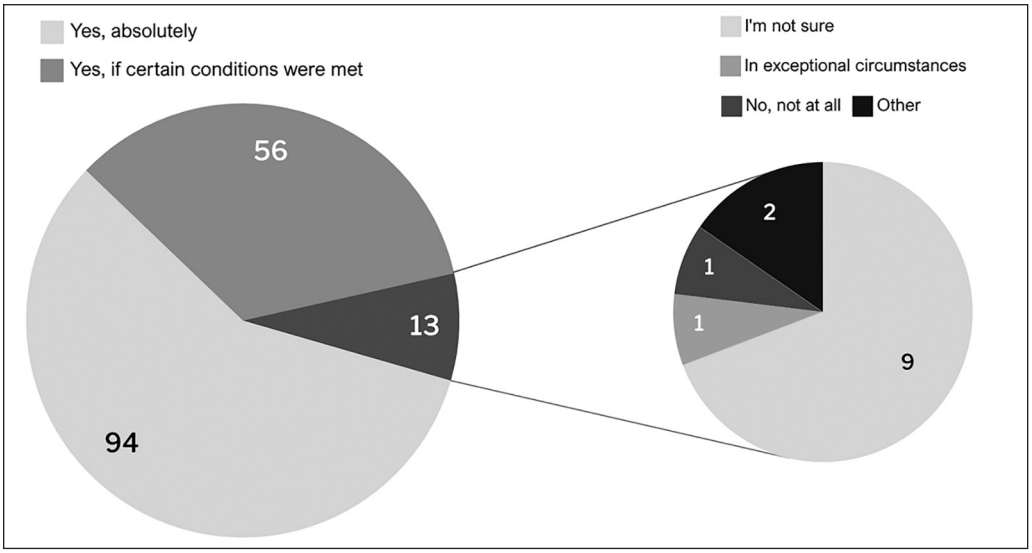


Figure 7. Archives and special collections as OER. Responses to the question, "Do you think digital archival and special collections materials could potentially be shared as OER?" The numbers along the edge of the pie chart are the number of selections for each answer. Respondents could only make one selection.

Cognizant that a theoretical eagerness might dissipate if applied to one's own work environment, I also asked, "Could the collections you work with be shared as OER?" There was more hesitancy in these responses but still a fair amount of openness. Thirty-four percent of participants (fifty-five individuals) selected "yes, many of them could," while 44 percent (seventy-two individuals) said "maybe some of them," with 7 percent (eleven) for "I'm not sure," 9 percent (fifteen) for "very few," and only 5 percent (eight) saying "no, not at all."

In the case study stage, I offered interviewees possible scenarios where they could incorporate OE practices into their work to see how they would react. One theoretical scenario seemed best suited for conversion into a survey question: incorporating OE into conversations with donors and into donor agreements at the acquisition stage.

I asked survey participants, "Do you think it would be feasible for your institution to start including open education principles in donor agreements and/or talk to donors about making their materials freely available online?" They could select "yes," "no," "maybe," "unlikely," "not sure," "my institution already does this," and "other," with the last option providing an opportunity for a custom text response. Again, respondents communicated openness to the possibility. Thirty-three percent said "yes" and just over 32 percent said "maybe." Nine respondents (a little over 5 percent) said their institution already did this.

During my case study interviews, when I asked about barriers to sharing materials as OER, my colleagues gave a range of answers. At the survey stage, one of my questions was, "What do you see as the biggest potential barrier(s) to the collections you work with being shared as OER?" In this multiple-choice question, I provided the most common replies from the case studies: copyright, privacy, donor agreements, donor hesitancy, technological resources, human resources/staff time, financial resources, hesitancy at the administration or board level, lack of staff knowledge or training, and "other." Respondents could select multiple

options and provide a custom response if they chose "other" (see figure 8). The top selections were copyright (20 percent of selections), human resources/staff time (18 percent), and privacy (14 percent).

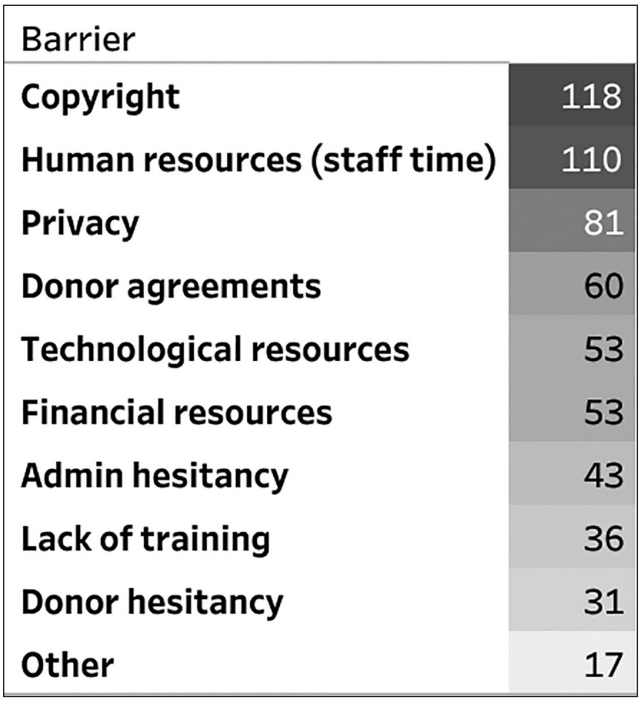


Figure 8. Barriers to sharing materials as OER. Number of selections for answers to the question, "What do you see as the biggest potential barrier(s) to the collections you work with being shared as OER?" Respondents could make multiple selections. Created by Hilary Barlow.

Questions of copyright, privacy, and online sharing can be very specific depending on the materials one manages. Toward the end of the survey, I asked a few open-ended questions for participants to share in their own words if they so chose. I asked, "Do you work with specialized collections that make copyright and online sharing more complex?" A librarian at a university in Canada shared, "Copyright and permissions for media materials are more complex in terms of IP legislation and the number of people and/or corporate bodies involved in a single media work. These complexities play out in donor agreements and in public domain rights research as well."

In response to the same prompt, an archivist at a university in the United States said, "We have many literary 20th century/21st century collections—these authors are unlikely, I think, to make their own work freely available. I would also say based on recent experience that some of our donors who come from marginalized backgrounds would be less trusting and more hesitant to freely share their materials online (and have in fact been, though in one case they did agree to creative commons licensing)."

The last question I asked was, "Is there anything else you would like to share about these topics?" An archivist at a private institution in the United States responded, "[O]ur archives are for a religious community. When it comes to others remixing material from our collection there would be a concern that some would do so in irreverent or offensive ways. I think this would not happen often but I think it is a legitimate concern." Finally, a librarian at a university in the United States shared, "I believe the process to digitize and openly license/publish existing archival materials held in libraries is an opportunity to highlight and center Indigenous knowledge but has to be done thoughtfully and in collaboration with tribal communities. It is a way libraries/archives can counter the continuing marginalization of Indigenous voices in commercial publishing avenues." The custom responses show the complex concerns of

librarians and archivists that vary greatly depending on their institution and the collections they manage. Hesitation may come from governing bodies, donors, or privacy concerns, but the inspiration to innovative potential also came from practitioners' unique experiences and situations.

Conclusion

In most cases, incorporating OE principles into archival or special collections work would not require an overhaul of operations. For many practitioners, a positive first step is learning about the open education movement, open educational resources, and Creative Commons. Information and cultural heritage practitioners already work in a culture that encourages continuing education, and I believe that once exposed to new concepts, many connections to their own work would become evident.

There are many places to start learning, some of which have already been cited in this chapter. *Intersections of Open Educational Resources and Information Literacy* contains multiple chapters about OER in the library context and an excellent chapter about the Open GLAM organization that includes archives. The book is open access and easily available online through the Association of College and Research Libraries.[26] The Open GLAM organization, while not exclusively focused on OER, has published numerous articles on incorporating open access and Creative Commons at GLAM institutions.[27] They also offer a mailing list and regular calls to connect practitioners.[28]

Incorporating OE principles can be as simple as re-examining resources that already exist at one's institution and pushing them a little farther. Perhaps there are online resources that could be shared with a CC license or with a less restrictive license than they currently have. Maybe there are colleagues or faculty at one's institution interested in or already engaged in OER projects. Donor agreement templates could be re-examined to see if language could be included to make a donation an OER or open access digital object. Practitioners already put a great deal of thought and effort into making collections available, and OE is but another avenue to bring those efforts further.

The special collections librarianship and archival fields are communities of innovation, continuing education, and embracing technology. They are also restricted by resources, decreasing budgets, and the important considerations of copyright and privacy. While open education and open educational resources have not fully penetrated these fields, there is space for growth and even more accessible digital collections. Archivists and special collections librarians are well-versed in both the challenges and best practices to make the materials they manage work for the communities they serve.

Appendix A: Survey Questions

The type of question and the type of answers permitted are noted in parentheses. When "Other" was an option in a multiple-choice question, selecting it would produce a custom text area where respondents could provide a custom answer.

1. Which most closely describes your professional role? (Multiple choice, one selection permitted)
    a. Archivist
    b. Librarian
    c. Archives paraprofessional
    d. Library paraprofessional
    e. Scholar
    f. Other

2. How would you describe your gender? Select all that apply. (Multiple choice, multiple selections permitted)
    a. Male,
    b. Female
    c. Non-binary
    d. Genderfluid
    e. Other
    f. Prefer not to say
3. How would you describe your race/ethnicity? Select all that apply. (Multiple choice, multiple selections permitted)
    a. White
    b. Black
    c. South Asian
    d. East Asian
    e. Southeast Asian
    f. Latin American
    g. Middle Eastern
    h. Indigenous
    i. Other
    j. Prefer not to say
4. In what region do you primarily work? (Multiple choice, one selection permitted, choosing a region other than Canada or the United States would prompt the respondent with a text entry box)
    a. Canada
    b. United States
    c. Latin America (please enter country name)
    d. Europe (please enter country name)
    e. Asia (please enter country name)
    f. Africa (please enter country name)
    g. Oceania (please enter country name)
    h. Other (please enter country name)
5. What kind of institution do you work in? (Multiple choice, one selection permitted)
    a. University
    b. Private company
    c. Government agency
    d. Public library or archive
    e. Other private institution
    f. None/freelance or consultant
    g. Not currently employed
    h. Other
6. Do you manage digital resources (digitized or born-digital) as part of your work? (Multiple choice, one selection permitted)
    a. Yes
    b. Not in my current position, but I have in the past
    c. No, I never have
    d. Unsure
    e. Other

7. "Open Education encompasses resources, tools and practices that are free of legal, financial and technical barriers and can be fully used, shared and adapted in the digital environment. Open Education maximizes the power of the Internet to make education more affordable, accessible and effective." Source: Scholarly Publishing and Academic Resources Coalition (SPARC). Before reading the above quote, what was your level of familiarity with Open Education (OE)? (Multiple choice, one selection permitted)
    a. Very familiar
    b. Somewhat familiar
    c. Not very familiar
    d. Not at all familiar
8. "Open Educational Resources (OER) are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others." Source: United Nations Educational, Scientific and Cultural Organization (UNESCO) Before reading the above quote, what was your level of familiarity with Open Educational Resources (OER)? (Multiple choice, one selection permitted)
    a. Very familiar
    b. Somewhat familiar
    c. Not very familiar
    d. Not at all familiar
9. Do you engage with Open Educational Resources (OER) in some way in your work? (Multiple choice, one selection permitted)
    a. Yes, regularly
    b. Sometimes
    c. Some of my colleagues do but not me
    d. Rarely
    e. Not at all
10. The Open Education (OE) movement encourages making digital resources freely available to all online. It also promotes the use of open licensing such as Creative Commons and encourages creators to allow others to remix or change aspects of their creations with credit. These creations are called Open Educational Resources (OER). Do you think digital archival and special collections materials could potentially be shared as OER? (Multiple choice, one selection permitted)
    a. Yes, absolutely
    b. Yes, if certain conditions were met
    c. I'm not sure
    d. In exceptional circumstances
    e. No, not at all
    f. Other
11. Could the collections you work with be shared as OER? (Multiple choice, one selection permitted)
    a. Yes, many of them could
    b. Maybe some of them
    c. I'm not sure
    d. Very few
    e. No, not at all
    f. Other

12. What do you see as the biggest potential barrier(s) to the collections you work with being shared as OER? Select all that apply. (Multiple choice, multiple selections permitted)
    a. Copyright
    b. Privacy
    c. Donor agreements
    d. Donor hesitancy
    e. Technological resources
    f. Human resources (e.g., staff time)
    g. Financial resources
    h. Hesitancy at the administration or board level
    i. Lack of staff knowledge/training
    j. Other
13. What is your level of familiarity with Creative Commons, including Creative Commons licenses? (Multiple choice, one selection permitted)
    a. Very familiar
    b. Somewhat familiar
    c. Not very familiar
    d. Not at all familiar
14. Have you assigned a Creative Commons license (or other type of open license) to an item or collection as part of your work? (Multiple choice, one selection permitted)
    a. Yes
    b. No
    c. Unsure
15. Have you assigned a Creative Commons license (or other type of open license) to your own work (e.g., blog post, journal article, presentation)? (Multiple choice, one selection permitted)
    a. Yes
    b. No
    c. Unsure
16. Do you think it would be feasible for your institution to start including Open Education principles in donor agreements and/or talk to donors about making their materials freely available online? (Multiple choice, one selection permitted)
    a. Yes
    b. Maybe
    c. Not sure
    d. Unlikely
    e. No
    f. My institution already does this
    g. Other
17. Do you work with specialized collections that make copyright and online sharing more complex? For example, is copyright more complicated because of the format of your collections? Is your institution less inclined to share online because of the nature of the material, etc.? If so, please share below: (Open text box)
18. Is there anything else you would like to share about these topics? (Open text box)

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CHAPTER 16

Inaccessible Open Access? Why Inclusive Design is Essential to Open Movements

Stewart Baker

Introduction

In the decades since the publication of the Budapest Open Access Initiative (BOAI) Declaration in 2001 first defined the term open access (OA), global academic communities have made great strides in using the internet to provide scholarly articles that users can "read, download, copy, distribute, print, search, or link to" at no additional cost.[1] A major large-scale study in 2018 showed that 28 percent of all journal articles ever published were freely available online, with the percentage rising to as much as 47 percent for articles published between 2015 and 2017.[2] Although more recent data are not available, the increasing adoption of OA publishing licenses by publishers,[3] as well as major government and funder initiatives like Plan S, which require the open publication of research outputs,[4] suggests that OA's success will continue, making more and more research freely available online.

Another concept that has become more mainstream is that of web accessibility—the idea that those with disabilities, slow or unstable internet connections, and other usability needs should be able to find and use content online without barriers.[5] Although web accessibility has long been considered "esoteric"[6] and difficult, it has increasingly become a recognized international standard, due in part to the efforts of the World Wide Web Consortium's Web Content Accessibility Guidelines (WCAG) and the adoption of accessibility laws in many nations. Just as open access seems poised to continue its successes, web accessibility seems certain to become more relevant and important in our always-connected world.

In this chapter, I look at the context of the two movements to determine how they intersect and why proponents of open access do not typically include web accessibility in their discussions. I will also introduce readers to the concept of universal design and discuss how creating accessible materials improves them for *all* users, not just those with additional needs. Although my focus is primarily on scholarly articles and the open access movement, the principles I discuss can be applied equally to all kinds of open content.

Access and Accessibility

OA statements and texts typically use the word "accessible" to mean "available," usually under an open license or some other method of removing barriers. However, in this chapter, I will be using the word in a more technical way, specifically as it applies to the process of making content usable by those with disabilities or other needs. To avoid confusion, I will use "accessibility" or "accessible" to refer to web accessibility and "open availability" or "openly available" to refer to whether something meets the BOAI definition of open access (see figure 1).

Accessibility / Accessible	Open Availability / Openly Available
Content can be accessed by users with disabilities, either natively or through the use of assistive technology.	Content can be accessed without "financial, legal, or technical barriers other than those inseparable from gaining access to the internet itself."

Figure 1. Accessibility vs. availability.

To better understand the difference between these concepts, we will need to take a quick trip back in time to the early 2000s, when both the open access and web accessibility movements were relatively new, and review the context to which both movements are responding.

In addition to the BOAI (mentioned above), the two other primary OA statements are the Berlin Declaration and the Bethesda Statement. All three statements grew out of a conversation around the "economic realities of distributing published scientific knowledge" and the potential of the internet to fundamentally change those realities.[7] These realities included the serials crisis, which saw the cost of library subscriptions to academic journals skyrocket since at least the early 1980s, as well as with general scholarly discontent with the business model of scholarly publishers like Elsevier, arguing that publishers prioritize shareholder profits over genuine scientific discovery.[8]

Because of this context, OA tends to focus primarily on the costs of access to materials. In the words of BOAI, the goal of the open access movement is to remove "access barriers" to scientific knowledge, especially those that are "price barriers."[9] Peter Suber, in his foundational book *Open Access*, focuses even more specifically on costs, arguing that "OA isn't universal access" and isn't intended to remove "handicap access barriers," "connectivity barriers," or barriers related to censorship and language use.[10] To paraphrase Suber using a common aphorism, the perfect should not become the enemy of the good.

Web accessibility, like OA, is also concerned with barriers. In this context, however, the purpose of the movement is to remove "accessibility barriers" or design practices and development tools that make website[s] and other digital content "difficult or impossible for some people to use."[11] Although the Department of Justice ruled that the Americans with Disabilities Act (ADA) applied to websites as early as 1994,[12] the first version of the Web Content Accessibility Guidelines (WCAG)—the major global standard for ensuring that websites and documents are usable by those with disabilities—was only released in 1999.[13] In this context, it is perhaps unsurprising that OA advocates either decided to set web accessibility to one side or did not know enough about it to address it at all.

Today, there are robust tools and established best practices for creating content in a manner that meets the needs of users. Despite this, OA journals still struggle when it comes to meeting web accessibility standards.[14] As Johnson and Abumeeiz argue, by using "accessible" with the narrower meaning of freely available and by focusing primarily on paywalls and publisher behavior, the OA movement has ultimately set web accessibility aside as irrelevant and out of scope, and as a result has "reproduced and enshrined paradigms of inaccessibility."[15]

This may sound harsh, or like it's intended to blame early advocates of OA for things that were outside of their area of expertise. We may find ourselves inclined to agree with Suber

that web accessibility is rightly outside the scope of OA or other open movements, and that it is quite reasonable to leave it to web accessibility experts instead.

I would like to argue that we should ask ourselves a different question instead: How can we do better?

Why Web Accessibility Matters: Three Arguments for Accessible Open

There are many common arguments about why web accessibility is unnecessary or too difficult to bother with. Content developers and designers may think it's too expensive, that it benefits too few people to be worthwhile, or that it can be handled by the user with assistive technologies.[16] To counter these perceptions, here are three arguments for building accessibility into open movements.

First of all, accessibility of web and other content is a legal requirement in many places. In the United States, Section 508 of the Rehabilitation Act of 1973 requires that federal and state agencies, as well as some other publicly funded organizations such as public universities, ensure that web content is accessible to users with disabilities,[17] while Title II and Title III of the Americans with Disabilities Act of 1999 prohibit public and private schools (respectively) from discriminating against students with disabilities, including in the provision of course materials, textbooks, and web content.[18] State and local statutes may cover web accessibility more broadly,[19] and similar laws exist in Europe[20] as well as South Korea, Canada, the United Kingdom, and many other places around the world.[21]

A second practical argument is that creating accessible websites, documents, and other content improves everybody's experience—not just the experience of specific user groups. This isn't just empty rhetoric. Studies have shown that following web accessibility guidelines like WCAG creates significant improvements to a site's usability for non-disabled users.[22] In other words, by designing materials that follow web accessibility and universal design principles, you can make your materials better for everyone and potentially improve the success of your open initiatives.

Finally, there is a strong ethical argument to be made about implementing Open with web accessibility principles in mind. Web accessibility is not just a legal or practical issue. Designers and content creators sometimes feel they have an ethical responsibility to provide equal access to web content to all users, not just those without a disability.[23] In OA, some argue that the use of public funding for research creates an "obligatio[n] of reciprocity" to provide research outputs to the general public who ultimately provided those funds through tax revenues and other contributions to their governments.[24] Some open movements even make explicit connections between inclusion and ethics: the UNESCO Recommendation on Open Science, for example, situates "access to scientific knowledge" as a fundamental human right,[25] while Creative Commons argues for "the right to universal access to information" as important for the health and safety of people around the world.[26]

Smits and Pells, in their book *Plan S for Shock,* describe the stories of patients who are unable to access published articles that may describe life-saving medical treatments, strongly implying that open access could save lives or at least improve them.[27] If this is the case, surely we must also argue for a need for content that is accessible to all users regardless of ability or other needs—not simply content that does not require a purchase from a publisher. From an ethical perspective, saying that accessibility is too hard to implement is tantamount to saying that these users' needs do not matter. At best, this is a position that unknowingly disenfranchises those with disabilities; at worst, it amounts to ableism—a form of discrimination.

Even when people agree with the ethical, legal, and practical considerations, they may have concerns about the time and expertise required to create content under the principle of web accessibility. While there is something to these concerns, web accessibility does not need to be a burden, nor does it necessarily require highly specialized technical knowledge. In fact, the

most important steps to take when creating content accessibly do not involve technology at all: thinking about how and why you are organizing and creating your content is equally important.

By learning and applying a few high-level concepts and by being intentional with how learning objects, texts, images, audio-visual presentations, and other types of content are created and presented, *anyone* can make broadly accessible content. Even better, content created using this mindset tends to be easier and more intuitive for everybody—not just those with a disability.

Universal Design and Universal Design for Learning

The idea that we can better serve *all* people by thinking inclusively is not new. The term "universal design" was introduced by architect Ron Mace, who defines it as a way of designing things that makes them "both attractive and functional for all people, disabled or not."[28] Initially, the term was used primarily to refer to architecture and building design—universal design was a way to ensure that physical spaces and objects were accessible to all rather than a more general design philosophy.

In 1997, Mace and nine other universal design advocates working out of the North Carolina State University (NCSU) released *The Principles of Universal Design*, a document laying out seven principles for how to actually create things that meet Mace's definition (see figure 2). Despite its initial focus on physical objects or environments, many universal design principles can be applied to analog and digital documents, presentations, and other, less tangible forms of transmitting information, making them usable by and useful to educators, writers, content creators, and other open movement advocates.

Principle	Short Definition
Equitable use	The design is useful and marketable to people with diverse abilities.
Flexibility in use	The design accommodates a wide range of individual preferences and abilities.
Simple and intuitive use	Use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level.
Perceptible information	The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.
Tolerance for error	The design minimizes hazards and the adverse consequences of accidental or unintended actions.
Low physical effort	The design can be used efficiently, comfortably, and with a minimum of fatigue.
Size and space for approach and use	Appropriate size and space are provided for approach, reach, manipulation, and use regardless of the user's body size, posture, or mobility.

Figure 2. Principles of universal design.[29]

Although the names for these principles are not always clear, the short definitions show how they are broadly applicable to nearly anything. The "undo" button—an essential feature on many software programs—is a perfect example of the tolerance for error principle. By

providing a way to get rid of mistakes, the undo button allows users of all ability levels to "minimize … the adverse consequences of accidental or unintended actions."[30]

Everyone who has ever used a word processor or graphics program has likely used an "undo" button to fix a typo, and this feature is so ubiquitous that it may not seem like a web accessibility element. That, though, is the central concept of universal design: when implemented with thoughtfulness and intentionality, universal design benefits *everybody*, not just users with a specific disability or other need. The same basic tenet can also be applied to educational materials and practices.

While universal design was being refined, CAST* adapted its principles to improve educational content and the way classes were taught. In 1998, CAST's work was addressed in a federal topical brief that argued for the importance of a curriculum that gave "meaningful" access, allowing "each student, regardless of disability, difference, or diversity" to "use his or her strengths."[31] By the early 2000s, universal design for learning (UDL) was an established pedagogical method, and in 2009, CAST released the first iteration of its UDL Guidelines, establishing a formal, evidence-based framework that educators and others could use to ensure that their learning materials and classroom practices met the needs of as many learners as possible.[32]

Much like the universal design principles document published by the Center for Universal Design, the UDL Guidelines breaks UDL best practices into discrete conceptual areas. However, while universal design is a list of principles, UDL concepts are typically presented as a matrix. The UDL Guidelines also present these conceptual areas as varying things educators *do* rather than things that a design needs to have done to it. These differences make UDL much easier to put into practice for educators and others involved in learning and make it a great fit for anyone who wants to develop broadly accessible open initiatives.

The UDL Guidelines are organized around the core "principles" of engagement, representation, and action and expression (see figure 3), each with multiple "guidelines" that home in on a specific aspect of the principle. These guidelines are further subdivided into checkpoints: concrete approaches to implementation that present users of UDL with examples and actions to take to make learning more broadly accessible.

UDL Principle	Brief Summary
Engagement	Provide learners with multiple ways of engaging with learning material in recognition of different needs and preferences. This principle includes guidelines for getting learners interested in material, for building up their persistence and adjusting effort levels, and for helping them develop self-regulation skills.
Representation	Provide multiple representations of learning materials so that as many learners as possible can perceive and comprehend them. This principle includes guidelines for ensuring that content is presented in multiple media (e.g., subtitles, alternative text, text-to-speech), clarifying the language and symbols used in learning materials, and ensuring that learners can construct meaning from the materials.
Action and expression	Provide multiple means of action and expression, in recognition of the fact that learners may need to approach learning tasks differently. This principle includes guidelines for ensuring that learners can interact physically with materials and tools, express ideas and communicate them appropriately, and set goals and develop strategies to reach them.

Figure 3. Principles of universal design for learning (UDL).[33]

* CAST is a nonprofit organization which focuses on promoting equitable, inclusive learning. The name originally stood for the Center for Applied Special Technology, an acronym which the organization no longer uses as its focus has broadened.

Just like universal design, the UDL Guidelines don't *just* benefit those with disabilities or specific learning needs. Instead, as the guidelines note, their goal is to create learners who are "purposeful and motivated, resourceful and knowledgeable, and strategic and goal directed."[34] This holds true regardless of ability level, native language, or technology availability. All students and researchers, and every other person who will interact with your materials (or attend your symposium, etc.), will benefit from your use of UDL principles throughout your project.

For those who have a legal obligation to build accessibility into their content, it is worth noting that UDL is not a legal standard. That means its use alone may not guarantee that content conforms to the Americans with Disabilities Act, the European Accessibility Act, or even to standards-based accessibility guidelines like WCAG. However, following the UDL checkpoints typically makes it easier to meet official legal standards, which share some guiding principles with the UDL Guidelines.

In short, with its focus on practical steps that educators and content creators can take and its simple, plain-language explanations, UDL is an effective tool for creating open content that can be easily used by as many people as possible and is a great starting point for anyone who cares about bringing accessibility to Open.

Building Web Accessibility into Open with the UDL Guidelines

Of course, it's one thing to decide that you're going to make your open initiative or its content easier to use and understand or that you're going to use UDL principles when creating learning materials. Putting these promises into action can be a struggle, especially if you are new to thinking about web accessibility and user design.

As an example, let's examine a chapter in a published book using the UDL guideline of perception. This is the first of three guidelines in the "engagement" principle and is designed to "reduce barriers in learning" by ensuring that "key information is equally perceptible to all learners" through the use of multiple modes of representation and making sure that information is provided in a customizable manner.[35] The guideline includes three checkpoints: "offer ways of customizing information," "offer alternatives for auditory information," and "offer alternatives for visual information [including written text]," each of which provides a detailed rationale and suggestions for meeting the guideline.[36] Figure 4 offers possible ways a written artifact like a book chapter could meet these checkpoints.

Checkpoint	Possible Strategies
Offer ways of customizing information	• Provide access to an ebook version where users can enlarge or reduce text and change font if desired. • Provide text equivalent descriptions for complex data (e.g., charts, tables). • Make sure textual and data figures in electronic versions are text-based and not image files. • Allow for multiple ways of accessing the text (e.g., print book, ebook, audiobook).
Offer alternatives for auditory information	• Make sure audio versions include a transcript or link to the text version. • Make sure audio versions include clear descriptions of charts and tables, as well as access to text versions.
Offer alternatives for visual information	• Follow accessibility standards when creating electronic texts so that users can use assistive technology such as a screen reader if needed. • Provide audio versions of text-only content.

Figure 4. Strategies for ensuring a book chapter meets the UDL perception guideline.

Although UDL is designed for learning materials, it can be applied to other types of content—or even to in-person events. Conferences can ensure that spoken presentations include a sign language interpreter and that attendees with wheelchairs or mobility needs can easily move around the venue; virtual events can make sure websites are screen reader-friendly and that video conferencing technologies like Zoom include subtitles and a transcript. In other words, it's important for open initiatives to think about inclusion and equity no matter what they're doing—not just limiting these preparations to websites or digital documents. While such strategies will ensure that attendees with specific needs don't encounter barriers to access, they will also likely improve the experience for *all* attendees.

Note that the strategies above do not provide specific technological means to achieve them. Rather, they are broad ideas that anyone creating content or an event should think about ahead of time. This may seem overwhelming, but there are many technological tools and existing projects that can be adopted or adapted to better meet the needs of users. If you are still struggling to think of ways to apply UDL to your open initiative and its content, consider creating user personas—hypothetical example users who have specific needs and goals when interacting with your content—to help guide your process by allowing you to think about specific problems people may encounter and how to solve them.[37]

Of course, it goes without saying that all of these things take time and may increase costs. Regardless of the approach you take when implementing UDL, be kind to yourself and stay focused on the goal of broader accessibility, however you can accomplish that. Your open initiative or movement can only benefit from any time you take to think about and implement web accessibility and UDL principles.

Conclusion

Forty years have passed since Ronald Mace first outlined universal design, and nearly twenty-five years since the foundational open access statements like BOAI and Bethesda. Despite that, there is still work to be done in ensuring that open movements are truly removing barriers to access rather than simply removing financial barriers for a limited set of users. Open will be stronger and more equitable when it enables access and supports the needs of people of all ability levels, all ages, and all backgrounds. Although this may seem daunting, it doesn't need to be. It's only when inclusion is an afterthought that things get difficult. The concepts of universal design and the UDL Guidelines present a roadmap for movements to follow and concrete steps to take long before content is created or events are planned.

If you take only one suggestion with you regarding accessibility as you embark on a journey to Open, let it be this: web accessibility is not a specialized, esoteric practice that requires significant technical knowledge. Instead, like universal design, it simply requires you to acknowledge that "the essence of good design [is] that which is practical and can be used by all."[38] By keeping universal design and web accessibility in mind, in addition to access to content, you will not only improve open content for groups of individuals with specific needs but also for everyone who finds and uses it.

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CHAPTER 17

We're Open! (Sort of): The Challenges of Promoting Open Education Across IU's Regional Campuses

Beth South, Martha Barker, Scott Hudnall, Julie Feighery, and Mitchell Scott

Introduction

In recent years, open education has become an increasingly popular movement across higher education. As with other open initiatives, much of the work to implement open education initiatives falls on university libraries or other similar departments. The labor and logistics involved in getting these types of initiatives established can be extra challenging when more than one campus library across a university system is tasked with promoting open educational practices on their campuses. Indiana University (IU) has five regional campuses across the state: IU Northwest in Gary, IU South Bend, IU Kokomo, IU East in Richmond, and IU Southeast in New Albany.

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Campus	Location	Enrollment	Full-Time Instructors	Open Initiatives Currently Active
Indiana University East	Richmond (Borders Ohio)	3,203	102	CMT Fellowship; OER Chats with Faculty
Indiana University Kokomo	Kokomo (Central Indiana)	2,921	124	Textbook Affordability Working Group
Indiana University Northwest	Gary (Borders Illinois)	3,041	119	Free4All Resource Guide
Indiana University South Bend	South Bend (Borders Michigan)	4,631	198	Zero Textbook Cost Adoption Program
Indiana University Southeast	New Albany (Borders Kentucky)	3,736	160	Zero Textbook Cost 3-Tiered program

Table 1. Campus snapshot, fall 2024.

Despite sharing the same library systems and having some oversight from the Herman B Wells Library at the main campus in Bloomington, the IU regional libraries have all operated as individual entities, with varying budgets, services, staff sizes, and regional cultural differences. These differences led each regional library to pursue different avenues of open education, such as providing open textbook review workshops, incentive programs for adopting or creating open or zero cost materials; or use library funding to purchase assigned textbooks that meet the requirements of being electronic, providing unlimited user access, being DRM-free,[1] and being accessible system-wide through IU's online library catalog, IUCAT, ensuring all IU students can have access to the book through the library (see table 1).

It can be intimidating to begin open education initiatives at one campus, let alone five. This chapter highlights how the IU regional libraries support open education at the individual campuses, including the shared challenges and the collaborative efforts to overcome some of these challenges. By sharing our experience, we hope readers in a similar situation as presented here can feel encouraged and discover new ideas to move forward at their own institutions.

IU OER Advisory Council

In 2019, Indiana University established the IU OER (open educational resources) Advisory Council, a group charged by IU's Academic Leadership Council with supporting and facilitating an IU strategy for developing and using OER, including policy, best practices, and procedures surrounding OER. The council is mostly composed of regional library deans or directors, or an appointed library representative, and for the first five years was convened by the assistant vice president for University Academic Policy. At this time, all the regional librarians involved with open education initiatives are members of the council. For its first task, the council led the entire IU system in joining the Open Education Network (OEN) in spring 2020.[2] However, the COVID-19 pandemic, along with librarian and administrative turnover, drastically slowed the council's efforts in the last four years.

Beginning in spring 2024, two librarians took on the role of co-chairing this committee, allowing those most involved with open initiatives to take the lead. The council decided to

utilize a tool we all have access to, the OEN's Data Dashboard, to collect open education and textbook affordability program data from the last several years across all IU campuses. The collected data will be used to support a proposal in 2025 requesting university-wide funding for open education initiatives across all campuses. This will provide a consistent source of funding to support OER adoptions, reviews, and other open educational efforts. The council will also begin meeting quarterly starting in the 2024–25 academic year to maintain growth in these efforts. Previously, the council met biannually, and many in the group felt this contributed to the group's slow progress.

The OER Advisory Council has also played a pivotal role in connecting the regional campus librarians involved with open education together, and there is growing optimism that the work the council is currently doing will help the regional libraries scale up their open education efforts. However, even with this general support at the university level, numerous challenges remain in promoting and supporting open education at the individual campuses.

Challenges/Collaborations

The regional librarians have identified five main challenges that are shared by all campuses:

1. a cohesive messaging or promotion of OER
2. a sustainable budget to support open education initiatives
3. consistent staffing
4. competition with e-Text and faculty perceptions
5. faculty engagement

These challenges can slow or stall many collaborative efforts, but we've also reached a point where we now must reach out system-wide for support and advice. When it comes to possible solutions, the Indiana University regional campuses can share specific resources and tools for open education initiatives.

Messaging: OER Versus Affordability

One formal collaboration or partnership within the IU system has been the IU Advisory Council's negotiation and support of a system-wide membership to the Open Education Network (OEN). The IU system negotiated a cost-sharing membership for all IU campuses, and this membership allows for access to OEN training materials, the OEN review platform, and the OEN Data Dashboard. Outside of some formal and informal collaborations that occur within the IU OER Advisory Council or among OEN members, most IU regional campuses operate independently in terms of OER programming and support. This decentralized approach means each campus in the Indiana University system tailors its OER initiatives according to its own staff, budget, and institutional priorities.

While IU Kokomo and IU Northwest are in the preliminary stages of exploring library support for OER, IU East, IU Southeast, and IU South Bend have expanded their programming beyond the basics. These campuses have routinely hosted the annual Open Education Network Introduction to OER workshops, held OER office hours, and distributed faculty surveys on OER awareness and students' textbook costs. While these efforts effectively raised campus awareness of OER and encouraged adoption, regionals offering these programs and services have found that, despite these efforts, OER adoption has been minimal or leveling off. Recent research suggests that since 2019–2020, higher education has seen a widespread plateau in OER adoptions, as increased awareness has not equated to increased adoptions.[3] Therefore, for those that can support new initiatives, IU Regionals have created incentive programs attempting to move the needle on OER adoption.

Where collaboration and messaging have coalesced is in the adoption of "no-cost" or "affordable content" programs within the IU System. For example, IU East drew inspiration from IU Bloomington's discussion of its Course Material Transformation Fellowship Program at an IU OER Advisory Council meeting.[4] Similarly, IU South Bend and IU Southeast based their

programs on other IU initiatives and models from across the US. No-cost incentive programs are distinct from traditional OER incentive programs in that no-cost incentive programs encourage faculty to transition to OER, use library-licensed content (primarily unlimited-access ebooks), or other freely available materials rather than just mandating OER as the lone path to affordability.

While never discussed as a formal IU System messaging direction, the labeling of these initiatives as "no-cost" or "affordable course content" rather than OER alone has been intentional and deliberate. After participating in the 2023–2024 American Association of Colleges and Universities (AAC&U) OER Institute, consulting with an AAC&U faculty member, and inspired by similar programs in the IU System, IU South Bend adopted the "no-cost" label for its incentive program. Although OER adoptions are prioritized within these programs, this approach provides faculty with a broader selection of content for creating no-cost courses and for expanding affordability efforts into areas where traditional OER may have limited resources and application. Research has shown that no-cost incentive programs can increase adoptions, as they raise faculty awareness and willingness to transition to OER.[5]

Another affordability initiative within the IU System, also largely decentralized, is support for ebook-based, library-supplied affordability. Although licensed ebook content is part of each IU campus's no-cost incentive program, IU regional campuses are in various stages of implementation and support of formalized and centralized library-led ebook affordability initiatives. These initiatives typically involve obtaining data on required materials—often from campus bookstores—and determining which materials the library can provide as unlimited-access ebooks and then notifying instructors and students that a free option is available.[6] For example, while IU Kokomo had a bookstore partnership in the past, it was lost due to library staff changes in the position who managed this program. IU East, IU South Bend, and IU Northwest, despite requests, have not been granted access to required material data through their bookstores. However, IU Southeast has an active program with a current bookstore partnership.

Since IU students can take online courses across the IU system, IU Southeast's list of required materials included courses from other campuses with enrolled IU Southeast students. To support these students, IU Southeast purchased ebooks through a vendor that permits system-wide access and contacted other IU campuses to share information about the courses and library-provided ebooks. Although the IU Southeast librarian managing this program has since left, this program fostered a brief period of system-wide collaboration for ebook purchases that benefited students at multiple campuses. Recently, IU Kokomo reestablished a bookstore connection, and IU Bloomington launched initiatives to leverage library-licensed ebooks in affordability efforts.

While these individual approaches can be effective, their independent nature also makes them susceptible to local institutional changes in staffing, budget, or other environmental variables. Instead of funding these locally, the IU System would benefit from a centralized fund to expand these no-cost incentive programs to all IU regionals and potentially expand the budgets for existing programs. While several no-cost programs do incentivize OER creation, this incentive level was omitted from IU Southeast's program due to the elevated cost of incentivizing creation and the lack of available funds to support creation in addition to adoption. A centralized IU fund could also potentially address this disparity in programs, allow for expanded incentives at the regional level, and result in more OER creation within the IU System. With various IU System ebook-based affordability programs also in place, and others being denied access from the same bookstore franchise, the IU System would also benefit from negotiating access to the materials list for all regionals and potentially pooling funds to support these purchases that would benefit students enrolled in online courses across the IU System. Both recommendations are tasks that the IU OER Advisory Council plans to take on in 2025.

Library Budget and Open Access Support

While the larger IU campus libraries at IU Bloomington and IU Indianapolis have regular funding to support zero-textbook-cost programming or open access publishing for their faculty, funding for these initiatives at the regional libraries ranges from inconsistent to unsustainable to non-existent.

The overall budget situation for each regional library directly impacts efforts toward open education. As of fall 2024, most regional libraries have operated under a flat budget for a decade or more, with IU Kokomo tasked with cutting its budget. Flat and decreasing budgets are also constrained by the inflationary costs of electronic resources, most notably databases and journal subscriptions. In this academic landscape, the regional libraries must rely on solutions such as canceling less-used subscriptions to fall within budget, leaving little room for dedicated open access initiatives as funded on the individual campus level. Any incentives provided by regional libraries on a regular basis require the carving out of funds in the budget while balancing the resource needs of various academic programs.

Specific regional campus libraries have found success through collaboration with other departments. In fall 2022, IU East library partnered with its Center for Faculty Development (CFD) to establish a Course Material Transformation (CMT) Fellowship. This fellowship provides a $2,000 stipend to faculty who modify or create courses to fit the definition of a zero-textbook-cost (ZTC) course, which helps students identify which courses have no textbook fee. Through this partnership, CFD promotes faculty professional development and funds online course development, and the library provides technical support and knowledge of OER and affordable options throughout the fellowship.[7] As of fall 2024, the fellowship has saved IU East students an estimated $18,800 in textbook costs. Initially using one-time funds to pilot an incentive program, IU South Bend formed a partnership with its University Center for Excellence in Teaching (UCET), attending the AAC&U OER Institute together. After that experience, they have partnered with the library to provide funds for the Zero Textbook Cost Adoption project to continue.[8] IU Southeast's Zero Textbook Cost incentive program receives $3,000 annually from the office of the Executive Vice Chancellor for Academic Affairs (EVCAA).[9] IU Kokomo library created the Textbook Affordability Working Group (TAWG) in 2020. Through TAWG, each academic school and the campus bookstore assigns a member to participate in exploring strategies to lower textbook costs and adopt OER. These three campus libraries serve as models for their regional peers to argue in favor of open education and launch similar initiatives.

Not all the regional libraries can fully lean into OER-focused initiatives at this time, but the broader IU Libraries system does provide a framework for open access discovery, which is a step in the right direction in helping faculty and students utilize open access materials. The IU Libraries adopted the LibKey Discovery suite as its core article-linking solution in spring 2023.[10] The regional libraries would not have been able to afford this service separately but are able to participate in a group subscription, with the pricing scaled to each regional library's budget. The service brings with it a wider array of open access resources for faculty and student use. Through LibKey, each library has a set of tools providing direct links to both subscribed databases and open access resources. For example, the LibKey Nomad browser extension reduces costs for students by ensuring they have a one-click button for access, even if they encounter paywall barriers.[11] The risk of students mistakenly purchasing a paywalled article despite no-cost options increases when they perform research off campus. LibKey Nomad minimizes this risk by meeting students where they are while conducting research. Additionally, all IU libraries coordinate to "subscribe" to the same open access book and database collections in SerialsSolutions, an e-resources access and management service, to increase the number of open access resources in our library catalog system, IUCAT.

Staffing and Support

Open education can be an important avenue with the growth of online programs within Indiana University, and all the regional libraries agree that efforts to promote open education play an important role in the library. Each regional librarian has duties that justify their pursuit of open education: reference, instruction, and collection management. Access serves as the key factor in each role. However, librarian turnover proves a major challenge to implementing a university-wide open education initiative. IU South Bend, IU Kokomo, IU Southeast, and IU Northwest all had librarians who started open initiatives on their campuses, but those librarians have now left. With declining enrollment and library budget, most regionals are unable to fill these vacancies, leading to librarians in other areas taking on open education with little or no experience of open education practices.

At IU East, the assistant librarian of Access Services was assigned the task of overseeing open education initiatives when appointed tenure-track but now continues to take on other responsibilities with the retirement of IU East's library director. The associate librarian of Research and Outreach Services at IU South Bend and the associate librarian of Reference Services and Emerging Technologies at Northwest both took on open education initiatives as additional duties due to librarian turnover. The assistant librarian of Library Collections at IU Kokomo is new to their role and is just beginning to explore open education practices, and IU Southeast's assistant librarian of Collections and Online Resources left in November 2023, and the management of the zero-cost incentive has transitioned to the coordinator of Access Services.

The addition of new responsibilities, or stepping into a new role entirely, is challenging, but the OER Advisory Council has helped keep some consistency by keeping all librarians involved with open education across the system connected. This connection has allowed those who have had the opportunity to accomplish more with open initiatives to assist those new to the role. While the campuses don't currently share funding, the expertise and open education program material created by librarians across the IU system is shared readily through Zoom calls, email, and Microsoft Teams.

Sharing open education outreach strategies and recommendations for professional development opportunities, like the Creative Commons Certificate program or OEN's Open Education Librarianship Certificate, provides an educational support system for those newly tasked with open education efforts and helps build their own networks within open movements. The sharing of OER program material also allows the regional campuses to start some programs on a smaller scale. For example, the Course Material Transformation Fellowship at the IU East Campus was a program first piloted at IU's main campus in 2020. The Open Education librarian at IU Bloomington shared all fellowship material, including calls for proposals, survey metrics, and faculty support documentation, allowing IU East to adapt the fellowship to fit the size and scope of their campus needs.

University Technology: E-text Versus Pressbooks

IU first piloted an eText or inclusive access program in 2009, officially rolling it out in 2012.[12] "Inclusive Access is a textbook sales model that adds the cost of digital course content into students' tuition and fees."[13] While the IU eTexts program has some positive features—lower costs, ease of integration into Canvas, etc.—there are some drawbacks. It is difficult for students to opt out of using an ebook in a course; they lose their access to the eText at the end of their time at the university, and while it is a reduced cost, it is not free.

The eText initiative is run by the University Information Technology Services (UITS) and has a lot of marketing support and institutional and faculty buy-in, such as a complete systemwide website and Canvas tutorial. OER is not promoted in the same manner. One would need to look at each campus's websites or subject guides for additional information. Another example of administrative support for eTexts: students and advisors can filter in the schedule of classes

for courses that have eTexts. The IU OER Advisory Council has been trying to get the same filter for ZTC courses, but progress has been slow.

The differences between eTexts and OER are not well understood by faculty or administrators. Libraries are more likely to have faculty engage with library-licensed content or use an IU eText rather than adopting or creating OER. Part of this is due to a lack of awareness of OER but also concern from the faculty about their time as well as their reliance on the ancillary features in eTexts, such as built-in quizzes and homework assignments. While the Physics department was an early adopter of OER at IU South Bend, when talking to other faculty about OER, they have noted that the eText is a reduction in price and easily provides the homework setup that they need. Compensation for creating new quizzes, even if just adopting an existing OER, has been noted as a needed incentive to explore OER or zero-cost resources.

IU is part of the Unizin consortium, which, among other things, provides a platform for the hosting of eTexts and OER within Canvas.[14] Through Unizin, everyone at IU has access to Pressbooks, an ebook publishing platform that easily allows for the creation and/or modification of OER. Many faculty are unaware that they have this access. At IU South Bend, there was a faculty workshop on Pressbooks at UCET where faculty explored a recent information literacy text that had been modified to meet campus needs and were able to log in to their own Pressbooks accounts and explore its ease of use. Additional OER-focused workshops at IU South Bend and IU East feature a section on using Pressbooks.

A running theme, of course, is the limited staffing and time to promote tools, like Pressbooks, that encourage faculty to explore creating or adapting OER. This means that there is a need to approach faculty engagement differently and try out new collaborative methods.

Faculty Engagement

Raising faculty awareness of open initiatives has been a challenge. All campuses are experiencing a combination of declining enrollment and administrative changes, such as school or department mergers and turnover in leadership. Lack of consistent funding to incentivize and support faculty to make the switch to OER exacerbates this issue, as well as some faculty not finding OER suitable for their needs. Other issues, such as AI (artificial intelligence) and concerns over academic freedom, such as the newly enacted Indiana law, Senate Enrolled Act 202, which mandates "intellectual diversity" within the classroom and requires faculty to introduce scholarly works from a variety of political or ideological frameworks, also draw attention away from OER.[15]

One way to address the lack of faculty engagement is the regional librarians' effort to collaborate with IU's eText department and UITS to participate in a Tech Fair at the beginning of the fall semesters. Formerly called the Scholars Book Fair, showcasing affordable content options to faculty, the IU Tech Fair is now also open to students and staff to learn about various technologies and cost-saving resources, such as eTexts, OER, Pressbooks, 3D printing, Library Resources, etc.[16] The centralized units, such as IU eTexts and UITS, travel to each regional campus to host this fair, and the libraries participate, usually in promotion of OER and other library services. This is an event where librarians can show off OER as well as direct them to the tools IU offers, like Pressbooks, that can help in creating and adapting OER. These well-publicized events in busy campus areas offer food and giveaways, attracting both students and faculty. Connecting with students at these events and speaking with them about the benefits of OER is another way to indirectly connect with faculty. Faculty will often listen and respond to students' needs, and if more students bring up OER with their faculty, then those faculty members may be more responsive to open initiatives. This partnership helps regional libraries reach a broad audience to promote open initiatives, while easing the workload of the solo librarian responsible for these efforts.

In addition to the IU Tech Fairs, some campuses are also reaching out directly to students to create awareness. As part of Open Education week at IU South Bend in 2024, students were asked to share what they spent on textbooks, what they would have used that money for instead

or what they saved when their professor used zero-cost options, and what they used that money for. The responses were posted to the library's social media. Representatives from UCET and the library presented to the IU South Bend Student Government Association (SGA) in the fall of 2023 and 2024 on zero-cost resources, and in November 2024, South Bend's SGA passed a resolution in favor of OER and zero-cost resources. We are hopeful that visible student support for zero-cost materials may lead to administrative support.

Conclusion

With two librarians co-chairing the IU OER Advisory Council and planning to use the OEN Dashboard to gather data for a central funding proposal supporting open education efforts, IU regional libraries can now better collaborate and advocate through the council. They can share workflows and initiatives related to library-led ebook textbook affordability and rely on the expertise of IU regional colleagues to exchange best practices and local successes. Creating this community of practice within IU regional libraries has fostered a network of support among colleagues who share and understand institutional and system pain points. This network provides a shared voice advocating for system-wide open initiatives that consider the unique situations of regional campuses compared to their R1 and R2 counterparts. While IU regionals still face challenges around open education initiatives, like limited staffing and faculty engagement, there is renewed energy taking place within the IU OER Advisory Council, and the IU regional librarians are hopeful that their small individual successes will strengthen the shared efforts in the future.

Notes

1. DRM stands for Digital Rights Management. DRM-free means that a user can download and access the content without the use of specific software or app, making it more accessible to users.
2. Open Education Network (OEN) is a network that supports the use of open educational resources, providing a reviewed open textbook library, OER workshop materials, advocacy support and training, and a tool for collecting and analyzing data related to open education and textbook affordability.
3. Julia E. Seaman and Jeff Seaman, "Turning Point for Digital Curricula: Educational Resources in U.S. Higher Education, 2022," *Bayview Analytics* (January 2022): 41, https://eric.ed.gov/?id=ED622348.
4. "Course Material Transformation Fellowship Program," Indiana University Bloomington Libraries, accessed November 18, 2024, https://libraries.indiana.edu/cmfp.
5. Tomohiro Nagashima and Susan Harch, "Motivating Factors among University Faculty for Adopting Open Educational Resources: Incentives Matter," *Journal of Interactive Media* 19, no. 1 (2021): 5, https://eric.ed.gov/?id=EJ1342044.
6. Mitchell Scott and Rachel E. Scott, "A Comprehensive Study of Library-Led Textbook Affordability Initiatives in the United States," *portal: Libraries and the Academy* (2025).
7. Elizabeth South, "Textbook Savings and Customized Teaching: The 2023-24 Course Material Transformation Fellowship Award Winners," *IU East Library Blog*, December 18, 2023.
8. "Open Educational Resources," Indiana University South Bend Libraries, last modified September 2024, https://library.iusb.edu/research-support/schpub/oer.html.
9. "Zero Textbook Cost Initiative," Indiana University Southeast Library, accessed November 18, 2024, https://southeast.iu.edu/library/faculty-services/zero-cost-initiative.html.
10. LibKey Discovery is a tool designed by vendor ThirdIron to streamline access to academic articles and other scholarly content.
11. Paywall-requiring payment or a subscription to access content.

12. Mark Lieberman, "How a Digital Textbook Initiative Achieved Liftoff," *Inside Higher Ed* (May 1, 2018), https://www.insidehighered.com/digital-learning/article/2018/05/02/indiana-university-inclusive-access-model-offers-road-map-others#.
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PART 4

NAVIGATING CHOICE AND ENGAGEMENT IN OPEN

CHAPTER 18

Sharing (or Not Sharing) Sensitive Data

Sara Mannheimer and Venice Bayrd

Introduction

Recent years have seen the rise of initiatives to share research data openly to facilitate research transparency and reproducibility and to advance and accelerate scientific discovery. However, not all research data should be made openly available. In this chapter, we examine sensitive data from the perspective of data curators. We provide an overview of sensitive data and discuss strategies that can enable sensitive data sharing or facilitate transparency when data cannot be shared. To illustrate the complexities of sensitive data, we provide an example of a form of data that may be less commonly recognized as sensitive—research data on water quality from private wells in Montana. We conclude the chapter with lessons learned and advocacy for an ethically centered, highly collaborative, interdisciplinary future for open data initiatives.

Open Science and Open Research Data

Open research data is a key element of open science* and helps promote the open science goal of "making research products and processes available to all."[1] These research products can include scholarly publications, research data, and research workflows, such as the code used to generate and analyze data.[2] When we refer to open data and data sharing in this chapter, our intention is to refer to research data as well as other supporting documentation as applicable—e.g., field protocols, metadata, software, code, raw and intermediary data products, and so on.†

In her book *Big Data, Little Data, No Data*, Borgman suggests four key rationales for sharing research data: "...(1) to reproduce research, (2) to make public assets available to the public, (3) to leverage investments in research, and (4) to advance research and innovation."[3] Open research data produces additional transparency and reproducibility for open science by providing unrestricted and open access to the data that was analyzed to produce research results.[4] Open research data can also be reused for future research, thus advancing and accelerating

* The terms open scholarship and/or open research may also be used to more clearly include all disciplines, including humanities, arts, science, and social science. We use the term open science in this chapter, but our intention is to consider all disciplines in our discussion.

† For a fuller discussion of the complexities of defining data in different disciplines, see: Christine L. Borgman, *Big Data, Little Data, No Data: Scholarship in the Networked World* (Cambridge, MA: The MIT Press, 2015), https://doi.org/10.7551/mitpress/9963.001.

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science and potentially reducing financial and labor-related burdens on researchers and participants.[5] Disciplines that were early adopters of open data practices include: physics, which pioneered open access articles with arXiv[6] and has gone on to lead a generalist research data repository effort through Zenodo, a project from CERN;[7] ecology and environmental science, which produced The Joint Data Archiving Policy and launched the generalist Dryad Repository;[8] and genomics, which contributed to the open data movement with the Human Genome Project and Genbank[9] in the 1990s.[10] In social science, political science and psychology have been early adopters of data sharing practices, including scholarly conversations[11] and repository infrastructure for data sharing.[12]

In response to the potential benefits of open research data, policies to encourage data sharing began emerging in the early 2000s in the United States and worldwide.‡ Such policies have continued to grow since then. In 2013, the White House Office of Science and Technology Policy (OSTP) issued a memo regarding a national commitment to data sharing;[13] OSTP updated this guidance in 2022 to call for immediate, free access to federally funded research, including research data.[14] These United States policies have led to a requirement to include data management and sharing plans (DMSPs) in federal grant proposals. In Europe, the Horizon 2020 Programme open access pilot launched in 2014,[15] and the Open Data Directive was established in 2019.[16] Some private funders, such as Wellcome[17] and Bill & Melinda Gates Foundation,[18] require DMSPs. Academic societies and journals have also adopted data sharing guidelines and policies. These policies illustrate the widespread adoption of open data policies; however, most of these policies provide exceptions for various forms of sensitive data.

Many institutions have developed services and resources to facilitate research data management throughout the research lifecycle.[19] Help with data management planning, storage, curation, and sharing may be provided by university research offices, institutional review boards (IRBs), research cyberinfrastructure, academic libraries, and other university offices and services; collaborative partnerships between such entities are emerging to facilitate data-intensive research and research data management.[20] In academic libraries, the roles of data curator and data librarian are growing, providing guidance and training for researchers on data management planning, metadata, data use and reuse, repository selection, and data sharing considerations.[21]

Researchers have also responded to trends toward open data. A trio of articles by the DataONE Usability and Assessment group shows that researcher perspectives on open data have evolved over time—from limited understanding and acceptance of open data practices in 2011 to more widespread support for data management and sharing in 2020.[22] However, these studies found that, as of 2020, there is still "a discrepancy between positive attitudes and actual implementation of those principles by the scientific community."[23] These discrepancies stem from a variety of concerns from researchers, including the need to publish before sharing data, insufficient time or funding, lack of rights to make the data public, institutional support, and the potential risks of data sharing.[24]

Sensitive Data

As noted above, despite prevailing rationales and trends, not all research data may be appropriate to share openly. This section begins by addressing some of the challenges specific to sharing sensitive data, including legal, cultural, ethical, and administrative concerns. The

‡ A note about the scope of this chapter: we come at this from a United States perspective because that is where we live and work. We cite some US-specific policies in this chapter, and we have also tried to include some international policy examples. Regardless of policies cited and examples provided, most of the ideas in this chapter will be transferable to many different contexts, including internationally. For more information on global open science efforts starting in the early 2000s, see the "Background Information" section of: OECD, "Recommendation of the Council concerning Access to Research Data from Public Funding," OECD/LEGAL/0347.

section closes with a discussion of several strategies that researchers and curators can employ to make data "as open as possible, as closed as necessary."[25]

Challenges to Sharing Sensitive Data

The rise of data sharing has brought conversations about sensitive data to the fore. As open data becomes more common, it is all the more important to consider what makes data sensitive, including unexpected sensitivities that may arise. Responsible data sharing requires awareness of potential risks and potential harms to research stakeholders, and situations may arise in which the risks outweigh the rationales for sharing data openly.[26] Data sensitivity can arise in multiple scenarios. These situations may be related to legal constraints (e.g., proprietary data, intellectual property concerns), cultural considerations (e.g., Indigenous data sovereignty), human subjects considerations (e.g., privacy protections, consent, vulnerable behaviors or identities), and other types of confidential information. Sensitive data can arise in unexpected places as well, as illustrated in the subsequent "Water Quality Data from Private Wells" example.

Some examples of sensitive data include[27]

- human subjects data from research participants from vulnerable populations—e.g., children, people involved in illegal activities, political dissidents, religious minority communities, or respondents from marginalized and minoritized communities such as Black, Indigenous, LGBTQIA+, or disabled communities;
- human subjects data from research on sensitive topics—e.g., drug or alcohol use, details of sexual activity, relationship/domestic violence, criminal behavior, health and mental health-related questions;
- human subjects data relating to Indigenous cultures, communities, and knowledge;
- archaeological site information with geolocation data;§
- endangered species data;**
- patentable inventions or other proprietary data; and
- human subjects data that contains personally identifiable information that cannot be deidentified.

Internationally, various frameworks provide a spectrum of sensitive data guidance ranging from laws to non-binding recommendations to global initiatives that shape policy. The mismatch between international data privacy laws is one of the challenges to sharing sensitive data across borders,[28] adding an important layer of complexity to international research collaborations. In the United States, treatment of many categories of sensitive data is encompassed by technical implementations of Controlled Unclassified Information (CUI) protection in academic and other non-governmental systems.†† The Australian Privacy Act of 1988[29] and the Australian Research Data Commons provide legal guidance and resources, respectively, on sensitive data;[30] and the European General Data Protection Regulation (GDPR) and related directives[31] provide stringent protection of individual privacy rights. Global initiatives to address sensitive data and data privacy include the United Nations Global Pulse initiative,[32] the United Nations Declaration on the Rights of Indigenous Peoples,[33] and UN initiatives connecting big data and sustainable development goals.[34] The 2021 version of the Organisation for Economic Co-operation and Development (OECD) *Recommendation of the Council concerning Access to Research Data from Public Funding,*[35] also acknowledged that legal protections and cultural principles affect whether data may be openly shared.

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§ To explain further, keeping archaeological site geolocation data private protects the location of sensitive cultural heritage sites and prevents looting and vandalism.

** To explain further, keeping endangered species data private prevents poaching and species removal.

†† CUI standards in the US are addressed by Ron Ross and Victoria Pillitteri, "Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations" (Gaithersburg, MD: National Institute of Standards and Technology, 2024).

The landscape of guidance addressing ethical sensitive data reuse is rapidly shifting. Sharing sensitive data has the potential to benefit science and society—as in the case of sharing clinical trial data to support global health.[36] But sharing sensitive data also has the potential to do harm to communities—as in the numerous documented instances of harms to Indigenous peoples as a result of data sharing.[37] Integrating culturally appropriate methods into curation practices is critical as open data initiatives advance. Methods can involve principles, including: findable, accessible, interoperable, reusable (FAIR)[38] and collective benefit, authority to control, responsibility, and ethics (CARE);[39] standards such as the draft IEEE P2890 Recommended Practice on Provenance of Indigenous Peoples' Data;[40] and distinctions-based approaches[41] early in the research process. Examples of operationalizing ethical practices to decolonize the curatorial process are emerging.[42] Thielen, et al. have begun to outline what a more thoughtful curatorial pace might resemble in practice.[43] Critical engagement with culturally appropriate practices will require relationship-building and time.

On the administrative side, challenges to handling, storing, and facilitating access to sensitive data can provide collaboration opportunities for librarians, information security staff, research IT professionals, and others.[44] Institution-level data governance policies may define various colloquial classes of sensitive data and suggest local storage guidance. However, the focus of such policies tends to be enterprise data (i.e., personnel, student, financial, and administrative data) rather than research data. As one example, at Montana State University, where we work, we have nuanced guidance for enterprise data, but less so for research data. Different types of enterprise data have known and predictable sensitivity/protection status, and therefore their treatment is defined within data governance policy. Research data, in contrast, are highly heterogeneous and unpredictable by nature, a fact that frequently (though certainly not always) renders sensitivity classification an ad hoc activity. Research data management is also often interwoven into IRB human subjects data decisions and/or informed consent agreements, and research compliance offices may be involved to provide guidance on research data protected by law. Determining storage solutions for sensitive data can thus become a collaborative, case-by-case process. This remains a topic to watch in the near future, as awareness of the enterprise versus research data governance gap is beginning to take shape in the library, information science, and IT communities (e.g., Nexus Day 2024 *Data Governance and Working with Data* workshop,[45] Research Data Alliance Sensitive Data Interest Group,[46] Sensitive Data Expert Group of the Digital Research Alliance of Canada).[47]

Enabling Sensitive Data Sharing

As outlined above, sensitive data may not always be appropriate to share. However, there are strategies and considerations that can help enable responsible sharing of sensitive data. We suggest some of these strategies below.

Data Deidentification

Data deidentification procedures aim to mask participant identities or other sensitive information by removing direct and indirect identifiers. Direct identifiers are pieces of information that can specifically identify an individual person or place, such as name, social security number, or address. Indirect identifiers are pieces of information that, on their own, do not identify a specific person but can be linked to other information to potentially reveal individual identities. Some examples of indirect identifiers are date of birth, zip code, gender, or geolocation.[48] However, deidentified data still holds a risk of deductive disclosure or re-identification,[49] and some data may be too sensitive to take on this risk.[50]

When sharing a dataset that includes deidentified data, data curators can help researchers include details about data status in the metadata, and they can work with researchers to help with deidentification. Some deidentification methods can introduce uncertainties or affect replicability and thus must be weighed carefully when evaluating data sharing options. If deidentification reduces the value of the shared data, sharing the data with access restrictions could be a better option.

Restricted Access

If researchers wish to provide access to data that are too sensitive to be shared openly, some repositories provide resources and infrastructure to support various levels of restricted access to sensitive data,[51] including, in some cases, providing data enclaves in which data can only be accessed for data analysis, not download.[52]

Data Licensing and Data Use Agreements

Regardless of how restricted or open sensitive data access may be, a license or public domain waiver can help make it clear to future users how data may be used and reused.[53] Data repositories can also provide data transfer and data use agreements that facilitate the responsible transfer and use of sensitive data.

Informed Consent

Another important practice to enable sensitive data sharing is to appropriately obtain consent from research participants to share research data. Informed consent goes beyond simple consent to participation in research. To obtain informed consent, researchers must: 1) provide the necessary information for research participants to make an informed decision; 2) ensure that research participants fully understand this information; and 3) ensure that the decision to participate in the research is voluntary.[54] Informed consent practices have an impact on data sharing. In order to share data, even with restricted access, informed consent documents must include details about any potential future data sharing so that participants are aware of how data will be shared.[55] Researchers can also offer tiered consent.[‡‡] Tiered consent structures provide various consent options for participants—for example, opting into the study but opting out of data sharing, consenting to restricted data sharing, or allowing a subsection of data to be shared.[56] IRBs that oversee ethical practice in human subjects research,[57] funding agencies,[58] data curation professional groups,[59] and data repositories[60] may provide template language to facilitate participant consent to data sharing.

Transparency Strategies When Sharing is Not Possible

For especially sensitive data that truly cannot be shared (or in the case that researchers have not obtained proper consent for data sharing), other strategies can support transparency. For example, sharing a subset of data or sharing aggregated data can mask participant identities and protect sensitive material while still providing some data access. Publishing a metadata record, sharing code books, or sharing an overview of the data can also enable discovery and enhance knowledge of the research without providing direct data access.

For sensitive data that cannot be shared, researchers may invoke exceptions to data sharing mandates from funding agencies. For example, in the United States, the National Institute of on Alcohol Abuse and Alcoholism provides guidance on exceptions,[61] the National Science Foundation (NSF) states that exceptions to its data-sharing requirements may be made "based on legal, privacy, ethical, intellectual property and national security considerations,"[62] and National Institutes of Health (NIH) includes "justifiable reasons for limiting sharing of data" in its Data Sharing FAQs.[63] Journal data sharing policies also allow researchers to explain any data sharing exceptions in the data availability statement.[§§] Sometimes data status may change during the course of a research project, as illustrated in the following "Water Quality Data from Private Wells" example. In the event of data status changes in grant-funded research,

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‡‡ Tiered consent may also be referred to as flexible consent, line-item consent, or multilayered consent.

§§ For example, see: "Reporting Standards and Availability of Data, Materials, Code and Protocols," Nature Editorial Policies, Nature, 2024; "Understanding Our Data Sharing Policies," Taylor & Francis, 2024, https://authorservices.taylorandfrancis.com/data-sharing-policies/; "Wiley's Data Sharing Policies," Wiley, 2024, https://authorservices.wiley.com/author-resources/Journal-Authors/open-access/data-sharing-citation/data-sharing-policy.html.

proactive DMSP changes made in consultation with funding agency program officers are both acceptable and expected. DMSPs are living documents—that is, DMSP updates are best seen as normative rather than exceptional or undesirable. Updating a DMSP may be necessary if the data status changes from shareable to non-shareable. For research that is not federally funded, DMSPs can be updated according to specific practices outlined by funders, research teams, or institutions.

An Illustration of Sensitive Data

The strategies described in the previous section for facilitating sensitive data sharing can introduce uncertainties. For instance, masking or eliding spatial data fields is one option for geotagged data, but methods such as location offsets cannot be described in public metadata (due to potential for re-identification) and may affect replicability and reusability. The following illustration, drawn from our data curation work with researchers, is an example of geotagged data status shifting through the life of a project. This example illustrates the complexities of real-world sensitive data curation.

In addition to methodological uncertainties, this case study also demonstrates how multi-layered trust relationships within a research project can surface complex data management scenarios. We chose this example because it represents a less commonly considered situation relating to sensitive data and thus raises a variety of questions. We also chose it because we were engaged only at the end of the project, at which time we determined that the data could not responsibly be shared. Given the emergent and shifting nature of the issues described, we invite readers to consider, as we have done: what would have been a responsible course of curatorial action in this scenario had curators been involved in the project beginning in the research planning phase? There is no presumption here that data curators have all of the answers; rather, there is an earnest interest to examine the layers of complexity that can arise in the course of research, and to reflect on the resulting ramifications for our professional curatorial practice.

Water Quality Data from Private Wells

Under the aegis of the Consortium for Research on Environmental Water Systems (CREWS) project (NSF-OIA award #1757351), researchers in Montana investigated water quality issues specific to intensive agriculture, hard rock mining, and energy extraction. As part of their investigations, CREWS researchers collected water quality data from wells on private farmland. One facet of privately held lands is economic: they are a real estate commodity. Well data that reveal the presence of contaminants can therefore become sensitive data because contaminant data could harm land valuation.

If sharing well data could jeopardize the researcher-partner relationship and/or permission to do the research on privately held land, researchers might choose to anonymize or exclude those data from a shared research dataset. For example, one CREWS researcher noted:

> *We did work to protect locations of some of our data collection on private land in [region] to protect the privacy of cooperating farmers. In my experience it is common that by the end of a project, farmers are fine with sharing data, but during the relationship building stage, it can be reassuring to them that we will not share the data with locations.*[64]

This scenario illustrates how sharing data from research conducted on private lands may depend upon the trust relationship built between researchers and private landowners. Research data resulting from a trust relationship may necessitate additional care to create a balanced approach to curation, considering trust relationships alongside privacy and research integrity protections and exemptions to sharing.

It can be difficult to foresee all potential data sensitivities ahead of time. Sometimes, as in this case, data curators are not consulted in advance, so whether approaches such as informed

consent are considered during the research planning phase becomes dependent on researchers' awareness of available strategies; this awareness, it bears mentioning, differs among disciplines. When data curators are involved early in the research lifecycle, it becomes possible to collaborate with researchers to continually reevaluate data stewardship responsibilities, especially prioritizing the relationship between the researchers and the research partners or research participants, and continually assessing the potential risks and harms that could be associated with sharing research data. As noted previously in the discussion of ethical data sharing, prioritizing ethical curation practices will require not only relationship-building but will also require taking the time to carefully consider data sharing from the outset of a project.

Conclusion and Future Directions

It may not always be appropriate to openly share research data. In this chapter, we have discussed how data curators can facilitate sharing for sensitive research data, when appropriate. This chapter suggests that data curation professionals can assist with data deidentification, restricted access, data licensing and data use agreements, informed consent, and transparency strategies when sharing is not possible.

In an illustration of a real-world research scenario, we explored the complexities of handling sensitive data and the challenges of relationship-building among stakeholders when sharing sensitive data. In the course of our work as data curators, we have observed that trusted relationships between curators and researchers can make the difference between researchers either co-developing solutions for complex data sharing scenarios in collaboration with curators or not involving curators at all. As described by Farrell in 2023,[65] building collaborative partnerships between data curators and researchers is a critical pillar of future open data initiatives. As we have personally seen, building such trusted, collaborative partnerships is particularly important for projects involving sensitive data. Sensitive data sharing is context-specific and requires case-by-case consideration. With the proper relationship-building and collaborative structures, data curators can facilitate either sharing or not sharing sensitive data, depending on the specific context of each research project.

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CHAPTER 19

When to Open, When to Close? The Benefits and Risks of Open Scholarship

Chelsee Dickson and Sabrina Davis

Introduction

The world of Open is constantly expanding, changing, and evolving. Open scholarship, a term used to describe a movement that "seeks to make knowledge of all kinds openly shared, transparent, rigorously researched, and inclusive,"[1] encompasses open access (OA), open education, open pedagogy, open data, open source, and open science. Additionally, open peer review is gaining recognition as an innovative, avant-garde practice moving scholarly publishing toward greater transparency in reviewing manuscripts for publication. Each movement nestled under the umbrella of open scholarship seeks to further the exchange of knowledge worldwide. Though the benefits of Open may seem abundant, there remain barriers and risks to participating in such movements. Unethical, discriminatory, and predatory practices are unfortunately problems that plague scholarship as a whole, which can distance open scholarship practices from universal acceptance.

The authors of this chapter are two librarians working at different institutions in the United States. One holds the title of assistant director of Academic Engagement & Instruction, Scholarly Publishing and Research (formerly the scholarly communications librarian) at a Carnegie class* R2 institution in Georgia. The other is an open educational resources (OER) librarian at a Carnegie class R1 institution in Texas. While their paths have ultimately led them to a similar facet of librarianship, institutional cultures and job responsibilities have affected how they advocate for open movements. Their similar yet different paths allow them to address the concerns of faculty regarding scholarly publishing and contributions to open scholarship with a unique approach, all of which are discussed in the chapter.

Throughout the chapter, the authors present a path forward for advocates of Open that addresses the barriers individuals face in publishing and creating open scholarship.[2] It is imperative that faculty, students, and the academic community are aware of multiple perspectives so they may make an informed choice when it comes to open scholarship practices. The chapter reviews scholarship alongside real scenarios encountered by the authors regarding

* More information about Carnegie classifications can be found at https://carnegieclassifications.acenet.edu/.

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faculty concerns about scooping—with an emphasis on Black, Indigenous, and People of Color (BIPOC) faculty members—as well as funding, copyright, and promotion and tenure (P&T) requirements as they relate to open practices. Woven throughout the chapter are examples of how the authors have addressed these concerns and helped open scholarship creators continue to uphold and celebrate open principles. The authors encourage readers to test these solutions at their own institutions and consider unique, future directions.

BIPOC Scooping Concerns

Scooping is a common concern in the Open community, and preprints are thought to be the most vulnerable since this type of work is shared before or during formal peer review.[3] However, early release of research may also provide a solid and tangible record of creation—including a date and the author's name—which can combat scooping and foster open dialogue. Generally, the act of appropriating someone else's research or ideas through scooping is considered plagiarism or misappropriation (which could ruin the robber's reputation if discovered) and can "undermine years' worth of work if a line of research gets abandoned due to scooping."[4] As a result, wary researchers may stick to publishing in subscription-based, closed journals, hiding their work behind a digital lock and key. Laine states that "[t]he risk of scooping is often used as a counterargument for open science, especially open data,"[5] since researchers can be distrustful of competitors in their field and "open by default" mandates. To share work openly, however, creators must have faith in the ethics and honesty of humanity and scientific integrity, though this can be challenging.

Concerns about scooping are all the more significant to BIPOC faculty members. Academic publishing sits at the intersection of three predominantly white professions: academia, publishing, and librarianship.[6] Inefuku[7] notes that the overwhelming whiteness of these professions has shaped the scholarly record and the norms of academia that traditionally exclude or create bias against BIPOC scholars. The main tenets of open data are accessibility, research scrutiny, and transparency,[8] but once a researcher—particularly a BIPOC faculty member—has completed their data collection and is ready to publish the results, they are sometimes hesitant to share their data and datasets openly. They may worry that openly sharing would allow others to reuse or scoop the data without providing the original author credit, as well as restricting the original author from reusing the data themselves. When BIPOC faculty members fear their intellectual property will be scooped and stolen by peers or colleagues, compounded by bias against BIPOC academics, it prevents said faculty from sharing their work as openly as possible.

How Do I Share My Data Without the Fear of Getting Scooped?

A BIPOC faculty member at Kennesaw State University (KSU) brought scooping concerns to the scholarly communications librarian and institutional repository staff in the fall of 2022. The faculty member received a grant from the National Science Foundation, completed their research in the field of geography and anthropology, and, after writing up their results, was ready to publish in a scholarly journal. Editors of said journal requested that the faculty member include in the publication a dataset that reflected data collected from the faculty member's study. The journal provided the faculty member with a choice: submit the dataset to the journal and allow them to publish it or host the dataset in an open repository so others may view the data via a permalink. The faculty member was wary; they were concerned their data could end up in the wrong hands if displayed alongside their narrative.

After numerous exchanges with KSU libraries' staff, the faculty member decided to deposit a copy of their dataset into the Digital Commons @ Kennesaw State University, KSU's institutional repository. The institutional repository houses a growing collection of datasets, and the institutional repository staff were keen to grow its holdings. The scholarly communications

librarian and institutional repository supervisor were more than willing to help the faculty member with a mediated deposit of their dataset. The faculty member's decision came with a caveat in the form of a very strict copyright statement. In no uncertain terms, the faculty member delineated restricted usage of the data with an all-rights-reserved statement that prohibited reproduction and distribution without written permission.

The scholarly communications librarian and the institutional repository supervisor were in a moral quandary: did this go against the entire ethos of the institutional repository—namely, its emphasis on Green Open Access? In short, the scholarly communications librarian and the institutional repository supervisor felt the answer was a resounding yes; the restrictive copyright statement went against the values and tenets of the open movements—values and tenets the institutional repository was meant to uphold. Both made a concerted effort to educate the faculty member on copyright and the implications of such a strict statement, explained the benefits of open access and the global sharing of knowledge, and provided examples of the high research impact through the inclusion of the dataset in the repository. Still, the faculty member was adamant: they did not want their dataset to be built upon or contribute to a global knowledge base.

The faculty member worried that if others were to build upon their work, it would damage their credibility in the future—that is, if a downstream user were to use the data and stumbled upon a ground-breaking conclusion, the original author would not retain their right to attribution. The scholarly communications librarian and the institutional repository supervisor attempted to assuage the faculty member's fears by describing the Creative Commons Licenses, which require users to give the author credit, as well as the fact that the dataset would now have an attached digital object identifier (DOI) generated by the institutional repository, meaning the faculty member's name would be attached to the dataset in perpetuity. It was to no avail. Currently, the strict language and copyright statement remain on the dataset's record in the institutional repository. The dataset itself has since been removed by the author and is unable to be viewed by anyone as of the time of writing in 2024. The author requested that the institutional repository supervisor remove the dataset after only one month of its deposit. To maintain ethical practices, the scholarly communications librarian and the institutional repository supervisor agreed to uphold the author's choice in light of their desire to contribute to open data.

Tested and Hypothetical Solutions for BIPOC Scooping Concerns

Lack of copyright education and open access advocacy are the crux of this specific scooping concern. Similar concerns can be mitigated through the following: consistent educational copyright workshops, institutional repository upload days during which faculty are provided support from library employees while uploading and depositing their work into the institutional repository, and encouragement from an institution's administration through open access policies and guidelines. Faculty may benefit from support in the form of official institutional open access and open education policies. These policies are usually created interdepartmentally with input from the library, an office of research, or the institution's faculty senate. Furthermore, some institutions[†] have turned to creating a specific office or drafting departmental policies, missions, and strategic plans about how the library will work with other areas on campus to lead cultural change at the institution regarding open practices. The Texas Tech University (TTU) OER Office has created such a mission and vision statement and is currently working on a strategic plan. Both consistent educational opportunities and enforcement of open access policies help to provide a broader base of well-informed faculty scholars.

[†] Examples of institutions include Emory Libraries' Scholarly Communications Office, Stanford University's Office of Scholarly Communications, Harvard Library's Office for Scholarly Communication, and Texas A&M University Libraries' Office of Scholarly Communication.

Webinars and conferences that focus on open movements can also be effective ways to educate the academic community and beyond. The KSU Scholarly Publishing and Research team hosts annual, virtual, free events for Open Access Week, an international event that recognizes the benefits of the open access movement and framework.[9] In 2023, the KSU team developed All Things Open Week, a bespoke week of events at KSU that bring together all open movements.[10] The events are in the same vein as Open Access Week—free and virtual—and the team is working to make them as accessible as possible. KSU and TTU recently formed a partnership, and both university libraries celebrated All Things Open Week together in 2024. With a low barrier to entry, events have garnered national and international attention and serve to educate the community about the benefits and successes of all open movements.

Funding

Throughout the years, the scholarly communications librarian receives a steady trickle of faculty inquiring about funding for both open and traditional publishing projects. Many have recently received acceptance from a publisher and are ready to take the next step in sharing their work with the world. The scholarly communications librarian actively promotes all OA routes since OA strives to make "scholarly, peer-reviewed journal articles freely available to anyone, anywhere" through a myriad of publishing models.[11] After a researcher, author, or creator makes the decision to publish their work openly, they must decide which publishing model—be it gold, green, diamond, or platinum, to name a few—best meets their personal and academic criteria. However, not all routes to OA are equally viable for authors. If the researcher decides to publish via the gold OA route, next they must find a way to pay the article processing charges (APC) that many gold OA journals require.

Some APCs can be in excess of $5,000, and authors and creators may opt to pay out of their own pocket.[12] Researchers without financial support may choose to forgo Open altogether, pivoting to for-profit publishers that do not charge to publish or decide against publishing at all.[13] Many funded researchers are at rich universities in developed countries in the Global North,[14] leaving underfunded and less privileged researchers in the Global South to fend for themselves, meaning their knowledge is not shared widely—or even at all.

Shaw and Elger[15] identify three unethical aspects of for-profit, gold OA publishing: discrimination against researchers, editorial conflicts of interest, and funding diversions from conducting research. They argue that, though researchers can include payment for OA fees in grant proposal budget breakdowns, this only shifts the problem of financing OA journal articles and may serve to "diminish the amount of money going toward actually doing relevant research."[16] This shift can lead to discrimination in research topics, as higher competition for funds means less established areas of research are passed over by funders in favor of mainstream projects.

Another concern is invisible labor. Faculty who create and adopt open educational resources (OER) strive to engage students more fully in their coursework, though Jordan[17] acknowledges that such creation often involves invisible labor—particularly for People of Color. Jordan[18] argues that higher education institutions must provide OER creators with course releases and a clear OER funding policy to ensure faculty have time and space to expend on writing, researching, developing, and sharing OER. Without these safeguards, the open education movement will continue to favor privileged academics while turning a blind eye to marginalized and underrepresented creators, leading to cultural taxation and burnout, both unseen and unacknowledged.

How Do I Fund My Open Scholarship Project When I Have No Money?

One faculty member at KSU reached out to the scholarly communications librarian in the hopes of creating an OER[19] and utilizing it in their classroom. The faculty member was unaware of the monetary compensation and support available for this type of work. Another hoped to

publish a book and wondered if the library could assist in paying the production costs. The scholarly communications librarian directed them to federal and state grants and KSU's Open Access Publishing Fund. Monetary incentives, like grants and open access publishing funds, can serve as solutions to the aforementioned funding problems found in the open movements. In the State of Georgia, an initiative known as Affordable Learning Georgia encourages the spread of low- and no-cost course materials. They do this by awarding educators with Affordable Materials Grants to transform their course materials from a traditional textbook into an open resource or use open and original material to create OER.[20] The grants help offset the cost of a faculty member's time spent creating an OER.

Additionally, OA publishing funds, like the Open Access Publishing Fund at KSU, can incentivize faculty to publish OA.[21] Funds like this are dedicated by an institution to financing the cost of APCs, alleviating stress for authors and creators. The scholarly communications librarian regularly directs faculty to this fund, but an increase in journal fee waivers from publishers and additional funding for underserved institutions are sorely needed. Though diamond and platinum open access journals, which do not charge any fees for authors nor readers, are gaining traction and are the ultimate solution to monetary concerns for authors, grants and funds that specifically support the advancement of open movements, as well as initiatives such as Subscribe to Open (S20),[22] are the metaphorical Band-Aids for the time being.

Tested and Hypothetical Solutions for Funding

Higher education institutions should consider offering grants in-house to motivate and compensate faculty for their human capital. Many libraries have found ways to provide financial incentives to their faculty for the creation of OER through awards and authors' receptions. At TTU, the libraries are fortunate to be able to provide financial support to faculty as well as other types of recognition. While these avenues may not be as highly regarded as federal grants, institutional funding can cover certain publishing fees and usually counts in a P&T review since these programs typically require an application process. Other institutions may not have the same level of funding as TTU; therefore, it should be noted that the green OA route allows authors without funding to deposit their work in a repository free of charge.

Copyright

Today, there are many scholarly communication avenues that faculty can choose from when determining how best to disseminate their scholarship. One concern faculty have when navigating this landscape revolves around copyright. If a faculty member chooses to forego traditional publishing in place of Open, there are several things to consider. For instance, some are concerned that publishing openly could potentially lead to someone stealing information or plagiarizing their copyrighted work.[23] As a copyright-related process, open licensing can be extremely confusing and keep faculty from fully understanding their protections.[24]

Can I Rescind My Open License After the Fact?

Copyright and its protections are typically thought to provide creators with peace of mind. All works in a tangible medium have automatic copyright protection, and a copyright statement is not necessarily needed to protect a work, unless the creator plans to pursue litigation.[25] In certain situations, however, copyright can be stifling to both creators and users. With the advent of copyleft[26] and the Creative Commons licenses,[27] authors and creators can decide on the rights granted to downstream users by picking one of six licenses. Creators may choose from the most open license, CC BY, which only requires that users provide attribution to the original author, to the most strict, CC BY-NC-ND, which prohibits adaptations, commercial use, and the creation of derivatives.[28] These licenses work as a safety valve for the unbending law of copyright, allowing downstream users to share, utilize, and incorporate copyrighted works into their own work, teaching, and research without fear of infringement.

Creative Commons licenses are particularly useful for faculty, especially when creating an OER.[29] Though faculty spend many hours of labor and human capital and often go uncompensated, recouping the cost from that labor is difficult, and changing the status of a work from open to closed can prove even more daunting. One faculty member at KSU in the field of computer science and information technology requested such a status change for an open textbook they created for undergraduate students. While enrolled in the faculty members' course, students receive a free, electronic PDF copy of the textbook through the learning management system, Desire2Learn (D2L). If students prefer a hard copy, they have the option to purchase a print-on-demand version from Amazon, Google Play, and Apple Books; the faculty member receives compensation when students select this route.

Though the KSU computer science and information technology textbook has copyright protection, this free version is essentially under a CC BY license, and students can share, download, and print the book as they see fit. Now, the faculty member hopes to recoup monetary costs—as a result of their time, effort, and production—for the free version of the textbook. A consultation with the scholarly communications librarian resulted in the following solutions: the faculty member could lean into the open movement and link the digital copy to the institutional repository's faculty bookshelf, thereby ensuring the faculty member receives a dashboard with impact factor metrics from PlumX as well as download counts; the faculty member could remove the free version from D2L and replace it with a link to the paid, closed version on retail websites; or the faculty member could publish an updated version of the textbook with a more restrictive Creative Commons license.

There was, however, no solution that would allow the faculty member to rescind their chosen open license for the current version of the textbook. As Creative Commons[30] states, "CC licenses are not revocable once granted unless there has been a breach…." After all, once the free version has been provided to students, they are able to duplicate it, share it with others, and print it out. Copyright law and its protections still stand, and unless another version is produced with a new license, the old license remains valid. At the time of writing, the faculty member has allowed the textbook to be placed on the digital faculty bookshelf, and they are currently contemplating the creation of a new, updated, and more restrictive version of the textbook. This dilemma blurs the line between supporting open education altruistically yet wanting to be paid for the work and human capital required to create an OER. Ultimately, the best solution largely depends on the faculty members' intrinsic (and sometimes extrinsic) motivations and the institutional systems in place that allow them to create OER with support.

Can Someone Sell My Open Work for a Profit?

A faculty member in the family and consumer sciences education department at TTU had been working with the OER librarian on the creation of an open textbook for nearly a year. One day, the librarian received an email from the faculty member asking whether users could capitalize on their openness and sell the faculty members' work for profit. In the email, the faculty member explained they had a colleague who chose to release an OER under a CC BY license and later realized their OER was being sold by a third party on Amazon. The TTU faculty member was concerned that the same thing could happen to their current in-progress open textbook.

The concern that this TTU faculty member expressed is not unique nor a surprise to those championing open movements on college and university campuses, especially at TTU. While many who choose to publish openly are not aiming to profit from their scholarly material, it is understandable that they are not comfortable with others profiting from their work. To ease some of these concerns, the TTU OER librarian met with the faculty member to explain Creative Commons licensing and answer any questions. While these types of meetings are helpful, the question of whether someone can capitalize on openness has led nearly all open textbook authors at TTU to use the most restrictive license, CC BY-NC-ND.

The family and consumer sciences education faculty member opted for that license, so the OER librarian was able to address their concern by reminding them of the terms of the license, along with another reminder that Creative Commons licenses work within copyright, meaning the faculty member has protections. While this faculty member's work was not sold for profit elsewhere (and without their knowledge), this concern exists because it is the reality for some open textbook authors who select a less restrictive license or are not as familiar with the different protections provided under each of the licenses.

Tested and Hypothetical Solutions for Copyright

Copyright education is one solution to fears of inappropriate Creative Commons license usage and general confusion over the rights afforded to a creator via copyright law. It is imperative that copyright education be available to researchers as they first embark upon the research cycle to avoid mistakes throughout the process of publication. For instance, at KSU, the scholarly publishing and research team organized a week-long celebration of copyright exceptions in 2021 titled Fair Use Week, in honor of the events founded by librarian Kenneth Crews.[31] Events highlighted the bundle of rights that make up copyright but also explored the exceptions to copyright, like the fair use doctrine, Creative Commons licenses, and other ways to safely use and share copyrighted work without infringement. Scholarly communications librarians, or other library employees involved in this work, may consider implementing a workflow that includes copyright education early in a faculty member's publication process to address copyright or authors' rights questions and dispel any misconceptions.

Within the TTU OER Office, the OER librarian provides copyright training for all faculty creators who wish to produce an OER. This training takes place before the faculty member's project begins and includes an in-depth overview of the Creative Commons licenses and how these licenses work in tandem with copyright. During the training, the OER librarian will ask faculty members, "With what level of openness are you comfortable, and will you allow others to adapt your work?" Toward the end of the training session, the TTU OER librarian will help the faculty member pick a license that best fits their work. Training is one-on-one, individualized, and customized to the faculty member's needs and focuses on the faculty member's specific project.

Promotion and Tenure

While many faculty appear to recognize the benefits of open practices on their scholarship and their students, one of the biggest barriers continues to be a lack of recognition within the P&T process.[32] Moreover, some faculty believe that publishing their article OA or in an OA journal could negatively impact their P&T reviews.[33] As more institutions begin to create OA policies and evaluate how to incorporate OA into P&T guidelines,[34] some may find that both a top-down and bottom-up approach that focuses on policy change and education efforts is needed to see a culture shift.[35] The creation of OER and use of open educational practices are usually not a consideration in P&T portfolios, though these practices could fit into any of the three areas in which faculty are assessed—research, teaching, and service.[36]

Will I Receive Credit for Open Scholarship Work in My Promotion and Tenure Portfolio?

Many institutions, particularly those categorized within the Carnegie classification system as R1, have P&T guidelines that emphasize research. In addition, existing initiatives attempt to address challenges with the current research assessment process, including the Declaration on Research Assessment (DORA),[37] Tools to Advance Research Assessment (Project TARA),[38] and

the Coalition for Advancing Research Assessment (CoARA).[39] When tenure-track faculty wish to achieve coveted tenure status, they realize that published research will have more impact than the creation of OER.[40] Therefore, concerns surrounding P&T guidelines look a bit different for the TTU OER librarian based on whether a concerned faculty member is interested in publishing an article OA versus creating an educational resource.

A tenure-track faculty member in the TTU engineering department scheduled a consultation with the OER librarian to learn more about OER. During this consultation, the faculty member asked several questions about Creative Commons licensing and its relationship to OER creation, and they voiced concerns regarding P&T. The OER librarian asked the faculty member if they had ever published in an OA academic journal. The faculty member excitedly answered "yes" because they believed in the importance of globally accessible scholarly work. While consulting with the faculty member, the OER librarian accessed a PDF copy of the faculty member's article and pointed out the Creative Commons license displayed prominently on the first page.

Next, the librarian explained that OA journals use the exact same licenses used for OER. This appeared to clear up confusion about open licensing, but the faculty member reiterated their concerns surrounding OER creation and P&T—namely, they would be happy to use preexisting OER in their courses, but they could not spend time adapting or creating original OER due to P&T guidelines. The faculty member continued to explain that, while they were tenure-track, they needed to focus on conducting and publishing research because that carries more weight in their dossier. This was not the first time (nor would it be the last) that the TTU OER librarian could not overcome the P&T hurdle while trying to expand OER usage and awareness on campus.

Tested and Hypothetical Solutions for Promotion and Tenure

TTU library faculty have worked to include open practices, such as OA publishing and OER creation, in their P&T guidelines. While the library can be a great place to start, there is value in institutional support—external to the library—for open practices. Intellectual contributions are the most common pieces of evidence in a P&T portfolio, but there are other ways faculty can demonstrate impact on their discipline and institution in the P&T process. In addition to funding, TTU provides other types of recognition that faculty can include in P&T dossiers. For example, faculty can apply for the institution's Open Access Week Awards, and winners are provided with certificates and gift baskets of library swag during International Open Access Week.

The TTU Libraries OER Office also worked with the student government association (SGA) to create the SGA Open Educator Award. Students nominate a faculty member, and the winner is awarded during the annual SGA banquet every spring semester. Other institutions should find ways to highlight faculty who participate in open practices. Shoutouts and spotlights can be done through social media posts or newsletters. When faculty ask the institution's OER librarian how they can include OER in their P&T dossier, the TTU OER librarian uses their own annual evaluations and recent third-year dossier as examples that illustrate how to include OER creation and OA publications into the library's P&T guidelines as they work toward tenure. The OER Librarian also provides faculty with guidelines and examples from other institutions, such as the work done by DOERS3 (Driving OER Sustainability for Student Success).[41]

Conclusion

Open scholarship practices have gained traction across the world. Such open movements serve to create a global knowledge base that is transparent, accessible, and available to anyone with an internet connection. Promoters of the open movements include creators, librarians, higher education institutions, and publishers. Though there are many benefits, problems still exist

that serve to undermine Open. Unethical processes, corruption, and discrimination within the scholarly communication and higher education landscapes can prevent researchers, authors, and creators from participating in open scholarship. The authors of this chapter recognize problems in the open movements and have witnessed concerns and fielded questions from faculty about the benefits of making their work open.

The authors' responses to these concerns are based on their belief that open movements promote wide research distribution and increased citation counts. Their tested and hypothetical methods for communicating the purpose and value of open are only the beginning; there is a long way to go in reimagining the current open landscape. We as a global community must work together to ensure society is knowledge-rich through the fair and transparent sharing of research. It is the responsibility of change agents—like the authors of this chapter, librarians, and other open advocates—to communicate the purpose and value of open scholarship while keeping creators and authors interested in participating in open scholarship practices.

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CHAPTER 20

"For Authors": What Publishers are Saying about Open Access

*Nancy Foasberg**

Introduction

Academic publishers play a major role in shaping the perceptions and implementation of open access. Not only do publishers set the conditions under which scholarly works are (or can be made) available, they also enjoy direct access to the authors, institutions, funders, and so on, often at exactly the moment that the question of access is most relevant. Thus, publisher rhetoric around open access has implications for the future of scholarly publishing.

In particular, publishers' definitions of open access are important, as they have the potential to change the public perception of what open access is. The struggle over what "counts" as open access can influence both policy and the assessment of the open access movement's success.† The initiatives that "count" as open access are the ones that will receive attention, funding, and other resources. Many open access advocates are sharply critical of large scholarly publishers, due to their high processing charges, long embargoes, and continued unsustainable subscriptions. However, these same publishers can claim that they are advancing the cause of open access by providing openly available materials. Still, it does not necessarily follow that they support the goals of the open access movement as understood by librarians, scholars, and other open access advocates. Many librarians and scholars are drawn to open access because it holds the promise of a more equitable system of scholarly communication, one which will "share the learning of the rich with the poor and the poor with the rich."[1] More prosaically, many librarians are interested in open access as a way to control the costs of scholarly information.[2] However, many large publishers prefer to fund open access via author-facing charges, which do not advance either of these goals.

The purpose of this chapter is to examine publisher rhetoric around open access, with a focus on publisher definitions of open access as they are presented to potential authors.

* The author composed part of this chapter while employed at CUNY Queens College. She wishes to express gratitude for the PSC-CUNY Professional Reassignment Leave, under which she completed much of this work.

† I have referred throughout this chapter to the open access movement, but like other Open movements, it is fractured among many stakeholders and priorities, as I discuss in the chapter.

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Defining Open Access

The most well-known definition of open access is the BBB (Budapest, Bethesda, Berlin) definition, which has been formally embraced by both advocates and publishers. Peter Suber, an influential leader in the open access movement, succinctly summarizes it as follows: "Open access (OA) literature is digital, online, free of charge, and free of most copyright and licensing restrictions."[3] This definition is an amalgamation of three statements adopted in the early days of the open access movement: the Budapest Open Access Initiative from 2001 and the Bethesda and Berlin statements from 2003. These three statements describe many of the practices that characterize open access today: self-archiving, open journals, open licensing of journal articles, the development of open access infrastructure, and even article processing charges.

Even in these early statements, we can see a debate over goals and priorities, which the combined BBB definition obscures. The Budapest statement, which was the first to be adopted, carefully separates its working definition of open access from the specific strategies it proposes. It calls for scholars to self-archive their work by depositing it in open repositories and to establish (new) open access journals, in that order. In addition to these approaches, it recommends that the community experiment to find additional models. Although nodding to existing journals, the Budapest Declaration suggests an active role for scholars in the development of open access.

The Bethesda and Berlin open access statements, which share a single definition of open access, envision a different open access landscape. They require that open access publications meet two requirements: open licensing and archiving, in that order—the opposite of the order in which they are listed in the Budapest Declaration. Rather than self-archiving, these documents refer to *long-term* archiving, for instance, in PubMed, which could also be done by publishers or funders. Both these differences highlight the role of the publishers in the scholarly communication landscape. These two documents put more emphasis on ensuring that works are made open via publishing rather than archiving. At the same time, their description of open publishing articulates the rights granted to readers in a way that translates easily to specific Creative Commons licenses. The Bethesda and Berlin statements even refer to Article Processing Charges (APCs), pledging to help fund them, where the Budapest declaration notes that there are "many alternative sources of funds for these purposes," only *possibly* including APCs. In 2020, however, the Budapest Open Access Initiative, in view of the growth of commercial open access, was updated with recommendations for the next ten years; their highest recommendations included moving away from article processing charges and read-and-publish agreements.[4] Thus, the BBB definition holds two sets of principles, one of which is friendlier to commercial publishers than the other.

With different goals, there are different ideas of what really "counts" as open access, and definitions are often used inconsistently.[5] Are we seeking rights, use, transparency, participation, or something else?[6] Historically, there are (at least) two lineages to the open access movement: the branch descending from the open source software movement emphasizes open licenses and reuse rights (*libre* open access), whereas the side of the movement that cares more about public access to research prioritizes simply making works available without imposing a cost on the reader (*gratis* open access).[7] The existence of multiple, nuanced definitions, however, carries the risk of co-option; Pomerantz and Peek warn of openwashing, while Moore argues that the push for strictly *libre* open access is linked with a willingness to partner with business and leaves OA open for exploitation by publishers.

These priorities can be roughly mapped to the two major "paths" to open access identified by early advocates: green and gold. "Gold" open access refers to open access via the publisher. Materials in this category are made openly available on the publisher's website. "Green" open access refers to the work of authors to make their publications open access, while following publisher policies. Authors achieve green open access by posting their publications—preprints, postprints, or publisher PDFs—on their website or in institutional or subject repositories. In practice, *libre* open access requires the cooperation of the publisher and is therefore strongly linked with publisher-driven or "gold" open access, while repository-based ("green") open access is very often *gratis* (but not *libre*).

The gold/green open access distinction has also inspired many definitional arguments, which have led to the proliferation of colors or flavors of open access. Some authors have called for abandoning these terms as vague and uninformative, instead proposing that the concept of OA be unbundled into various criteria.[8] However, others have continued to add terms. The term "diamond" open access originated in an attempt to distinguish between APC-funded and non-APC-funded models of publisher-supported open access; Fuchs and Sandoval proposed a distinction between "diamond" and "corporate" open access, but only the "diamond" half of this proposal became popular terminology, while APC models continued to be referred to as "gold."[9] Works made temporarily available by publishers have come to be referred to as "bronze" open access;[10] Costello suggests instead referring to them as "fourrée"—that is, counterfeit open access, provided at the whim of the publisher.[11] Others have argued, provocatively, that works made available via illegal channels could be considered "black open access" (and have pointed out that this may dilute green open access).[12]

The proliferation of types of open access, then, can also be seen as a struggle over the definition of open access and the validity of various open access strategies. At the funding level, the implementation and funding of open access hinge on how it is defined. Plan S, an influential European initiative from 2021, refers explicitly to the Berlin definition, requires open licenses, and calls for funders or institutions (rather than authors) to pay APCs.[13] In the United States, the 2022 OSTP memo requires recipients of federal research grants to make their work immediately available in agency-approved open repositories, a clear signal of support for green open access. However, the memo also refers more circumspectly to "public access," which allows publishers to claim that this is *not* open access, in contrast to the paid services that they offer.

Publishers' Open Access Preferences

Publishers are not neutral participants in the conversation about open access. It is important to understand publishers' goals and perspectives because publishers play a role in communicating about open access. Most publishers make their open access policies publicly available on their website; they need potential authors to know that the publishers will help them comply with funding or institutional requirements, and this is an opportunity to market their services in that respect. However, it is also an opportunity to frame open access in light of their own policies.

Most commercial academic publishers engage in APC-driven open access publishing, under which the author pays for the privilege of placing a Creative Commons license on their work. An article by three Wiley authors argues that APCs actually represent a "rights transaction rather than an article-processing fee"[14]—that is, authors are not covering publishing costs but rather purchasing back the rights to their own work! The nature of this transaction is also reflected in the language of many publisher websites, as we shall see below. Since publishers control how works are licensed, it is in their interest to promote a definition of open access that mandatorily includes open licensing.

Thus, it is unsurprising that large publishers prefer APC-driven models of open access over self-archiving and have even criticized self-archiving mandates, which typically do not rely on fees paid to the publisher. In the article referenced above, Wiley editors described green open access as "government expropriation of assets that have benefited from private investment," along with other criticism.[15] More recently, the aforementioned 2022 OSTP memo, which will require immediate open access via archiving for federally funded research, has attracted three letters of protest from publishers and scholarly societies.[16]

More often, publishers express their skepticism of green open access via restrictive self-archiving policies rather than explicit critique. As publishers have come to offer more APC options, they have also imposed a greater number of restrictions on self-archiving, thus reducing the attractiveness of self-archiving as an alternative to APCs.[17] In some cases, publishers' self-archiving policies frame restrictions on reuse as *benefits* to the author![18]

In the meantime, APCs have become a source of profit for publishers, who encourage authors to pay them. Publishers engage in profit-seeking strategies, such as funneling papers into APC-charging "companion" journals.[19] Meanwhile, hybrid journals' APCs continue to rise alongside their subscription prices.[20] APC-funded open access models do not reduce costs,[21] but they serve publishers well by providing an additional revenue stream.

Authors and Open Access Policies

Publishers' emphasis on APC-driven open access contributes to the prominence of that model and can sometimes lead authors to assume that the term "open access," and particularly "gold open access," refers to APC-supported publishing. Some academic articles outside of librarianship treat them as synonyms,[22] including one arguing that "reader-paid" and "author-paid" are accurate terms equivalent to "subscription-access" and "open-access,"[23] and another suggesting that journal publishers should "develop new approaches for open access publishing free of charge," apparently not realizing that such approaches already exist.[24] Another article states more baldly that "the OA model uses Article Processing Charges."[25] These articles do not recognize the range of open access models that exist. Publishers are not wholly to blame for these misconceptions, but the perception that APC-funded models are the primary or perhaps the only way to make a work open access clearly benefits their business model.

This conflation of open access with APC-supported publishing may drive some authors away from open access. Many authors see these fees as a deterrent to open access publishing[26] and find them detrimental to the progress of science.[27] Authors who practice self-archiving were especially hostile to APCs.[28] At the same time, authors do not necessarily respect (or understand) the restrictions that publishers place on self-archiving.[29]

Methods

To better understand how publisher websites define and frame the concept of open access to authors, I analyzed the author-facing websites of thirty-two major publishers. This analysis focuses on large journal publishers because they control a sizable portion of the scholarly communication infrastructure and are influential in conversations about open access. Furthermore, they are most likely to have consistent policies across their publications and to articulate these policies on their websites. Book publishers were not included in this analysis because many of them still do not make their open access policies public.

Selecting Publisher Websites

Academic publishing is dominated by five major publishers.[30] Due to their reach and influence, the policies and rhetoric of these publishers have the potential to affect what approaches are seen as standard. Any project that seeks to understand academic publishing must take them into account. However, because of their very size and influence, they are not necessarily representative of academic publishing as a whole or even the world of large academic publishers. To present a more complete picture, then, this article also considers several other large publishers.

To get a reasonable list representing most of the largest publishers, I used Sherpa-ROMEO, which compiles scholarly journals' open access policies. While Sherpa-ROMEO is not a comprehensive list of scholarly journal publishers, major publishers and many smaller publishers are represented there. The database captures enough policies to make it a useful tool for repository managers and others who are interested in publisher policies. Additionally, because Sherpa-ROMEO is a compilation of publicly available policies, the publishers listed there are likely to make their policies available on their website. Perhaps a future analysis could gain a more international perspective by using a list consolidated from multiple sources.‡

‡ One example of such a list can be found in Andreas Nishikawa-Pacher, "Who Are the 100 Largest Publishers by Journal Count? A Web-Scraping Approach," *Journal of Documentation* 78, no. 7 (2022): 450–63, https://doi.org/10.1108/JD-04-2022-0083.

Using the Sherpa-ROMEO website, I generated a list of publishers and the number of journals published by each, then sorted it by the number of journals published. Imprints of larger publishers were removed from the list; for example, Routledge was removed because it is an imprint of Taylor and Francis, which is also included in this database. Publishers without English-language websites were also removed from the list. Finally, publishers that do not market their services to authors, such as Philosophy Documentation Center, were removed. Publishers were not added or removed based on the quality of their reputation. The final list of publishers examined is available on Knowledge Commons.[31]

Publisher Definitions of Open Access

Overall Definitions

All the publishers I looked at included information on their websites about open access, such as definitions, policies, and explanations of the benefits of open access. Some, but not all, of the publishers provided explicit definitions of open access or a brief explanation of its benefits. Since publisher websites are large and complex, they might include multiple definitions on different parts of the website; for the purposes of this chapter, I was particularly interested in definitions provided on or linked from "author services" pages. Publishers that did not provide any definition of open access have not been included.

I have collected (and in some cases abridged) these definitions in the second tab of the data. A few of them explicitly cite the BBB definition of open access. Wiley provides a link to the Bethesda statement, also quoting it at length. MDPI references the Budapest, Berlin, and Bethesda declarations when giving its definition of open access, and Scientific Research Publishing quotes both the Budapest and the Berlin definitions. By invoking these external definitions, the publishers acknowledge that open access is part of the larger scholarly communication landscape, assuring readers that they are following appropriate standards.

It is unsurprising, then, that most of them also reference reuse rights, immediate access, or both. These aspects of open access are included in the BBB definition, as discussed above. Springer, Wiley, and DeGruyter (among others) prominently feature reuse rights in their descriptions of open access, though they phrase this in various ways: "the rights to use these outputs fully," "freely available to read, download, and share," and "available to read, share, and reuse." Wiley, Brill, and several others refer to specific licenses here; this clarifies their meaning but also creates an association between open access and the license that the publisher applies for a fee. A few publishers allow for more variety in models; for instance, Cambridge University Press notes that, "typically," open access involves reuse rights. A smaller number of publishers don't mention reuse rights in their explicit definitions; for instance, SAGE emphasizes that open access research is "without paywall barriers." Taylor and Francis's definition describes open access as "the process of making published academic articles freely and permanently available online," but later on the same page, when discussing the articles they make temporarily available, they state that this is "free access" and not open access, because "open access articles will always have a license which allows reuse of the work by third parties."

The emphasis on reuse rights is in line with many of the goals of the open access movement and has many benefits for interoperability, preservation, and so on. However, most large publishers' policies only make these rights available with the payment of processing charges. Few of the publishers went so far as to explicitly reference fees in their definitions; only publishers like MDPI, Pleiades Publishing, and IGI Global did that, and even they were relatively cautious about it—for instance, Pleiades refers to APCs as "the common business model" in open access publishing. Still, most publishers define open access as something that their authors can achieve via payment, and helpfully also include information about charges on the same page of the website.

References to OA Models

Looking elsewhere on the publishers' websites, most referred to both gold and green models of open access. A few publishers explicitly encouraged authors to consider gold open access in preference to green. Elsevier frames this as a way to fulfill requirements, rather than a benefit for the author:

> *If you choose to publish a subscription article, we can still help you meet your open access funding requirements. All of our subscription journals offer a* ***green open access*** *option, meaning you can post a version of your article in a repository after an embargo, so people can access it freely.*[32]

DeGruyter more straightforwardly extols the value of gold open access: "Publishing your article as Gold Open Access will maximise its openness, discoverability and re-usability."[33] IOP and Springer both link gold open access to "sustainability": "IOP Publishing believes that gold open access is the most sustainable route to open access but provides green open access options to those authors wanting or needing to share their work in this way."[34] Springer uses similar language: **"**We believe Gold OA publication offers the simplest most open and most sustainable route to OA and open science and research**.**"[35] The reference to "sustainability" is interesting. It would appear to imply that either these publishers consider green open access a threat to their business model or else that they depend on APCs for their continued profitability. In each of these cases, it is clear that these publishers view APC-supported open access as preferable and hope that authors will agree, even encouraging them to do so.

Several publishers' websites (Taylor and Francis,[36] Springer,[37] and Karger[38]) include tables comparing gold and green open access; Karger's table also includes a hybrid and diamond open access as categories, while Taylor and Francis and Springer do not distinguish between hybrid and gold models.[§] These tables compare embargo periods, fees, and licensing. These tables, while ostensibly informative in nature, provide another opportunity for publishers to argue for their preferred approach to open access. In particular, Springer's table pushes authors toward gold open access, suggesting that green open access is a threat to the integrity of the scholarly record and arguing that a transition to full open access is only possible via payments to publishers, which fund "publishing infrastructure"—that is, the established infrastructure that relies heavily on large commercial publishers such as Springer.

Restrictions on green open access, which commonly include embargos and rules about which version of the article can be shared, are often framed as features inherent to this model of open access, rather than publisher policies. For instance, the comparative tables of both Karger and Taylor & Francis note that embargoes are "sometimes" or "usually" required, that the publisher "generally" holds the copyright, and so on. Similarly, Cambridge University Press notes that green open access "typically" involves the submitted or accepted manuscript.[39] Wiley's definition of self-archiving begins: "self-archiving allows non-final versions to be hosted on a personal or institutional website,"[40] and IOP specifically defines green open access as "the process of archiving the copy of an accepted manuscript."[41] ACS's definition is even more hands-off, noting that green open access "involves archiving a version" of the manuscript.[42] Indeed, many of Springer's concerns regarding the scholarly record stem from the availability of different versions of an article. All this language is notable because both embargoes and the use of submitted or accepted manuscripts are the result of publisher policies. Rather than explaining that their policies do not allow immediate green open access or the sharing of the version of record, several publishers use language suggesting that these restrictions are features of green open access itself, or else they use descriptive language gesturing at the prevalence of these restrictions, minimizing their own role in these decisions.

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§ Hybrid open access is an approach wherein authors pay to make a single article openly available in a journal that is toll-access by default. This approach has been widely criticized on the grounds that it allows the publisher to profit from both APCs and subscriptions, which may be paid by the same institution. However, it is now very common for subscription journals to offer a hybrid OA option.

Other publishers organize their websites in a way that separates green open access from their pages on open access, hosting it instead under "rights and permissions" or in a similar part of their website. For instance, De Gruyter's page on green open access is called "Repository Policy for Journal Publications," and it links to the open access page, but navigating from the open access page to the repository policy page is more complex and relies on a link under the heading of "read-only journals" on the "Open Access Articles" page.[43] Brill has a page titled "Sharing Your Work/Self-Archiving," which covers repository deposit,[44] but it is housed several clicks down from their "Rights and Permissions" page and is not linked from Brill's set of pages on open access. Additionally, not only does it not identify self-archiving as a form of open access, it warns authors that these rules "do not apply for any articles published in open access on Brill.com with a Creative Commons license." This is because the Creative Commons license allows broader reuse, meaning the author could archive the article with no restrictions, but the way this is phrased could easily confuse someone who is not familiar with these licenses. IEEE has an entire website devoted to open access, but green open access does not feature prominently; aside from some news posts and instructions for UK authors specifically, it is only mentioned in the FAQs, along with a link to more information in the Author Center (that is, on their main website, separate from the open access website). Their main page on green open access is the Author Center, under "Publishing Ethics"; the page itself is titled "Post-Publication Policies" and does not refer to self-archiving as a form of open access.[45]

The information provided on these websites is useful and accurate; however, authors who rely on sites like this as a source of information about open access are likely to come away with the impression that open access is a paid service offered by publishers. This is certainly advantageous to publishers, who benefit from APCs and read-and-publish agreements, but it does not move us closer to an equitable scholarly publishing landscape. Rather, APCs exclude many authors on economic grounds, entrench the advantages of large publishers, and favor "traditionally advantaged actors."[46]

Conclusion

While the BBB documents have been deeply important in the development of the open access movement, they have not prevented the development of what Fuchs and Sandoval call "corporate open access."[47] The revision of the Budapest declaration is a welcome step in a different direction; it may be a useful document in arguing for new and more equitable conceptions of open access.

Commercial publishers must act in their own commercial interests, but those interests may not align with the goals that brought librarians and many scholars into the open access movement to begin with, including building a more equitable scholarly communication system and controlling the costs of scholarly information. The APC/read-and-publish models of open access do not solve the pricing crisis, nor do they provide equitable access to scholarly publishing. It is very important that publishers not be allowed to shape the conversation around open access.

Advocates for equitable open access must continue to educate our communities; I believe that librarians, in particular, have a strong role to play in that respect. Those who are not deeply immersed in conversations about scholarly communication have likely heard about open access in the terms in which publishers present it. It is essential to provide an alternate narrative, including the benefits of non-APC open access. We can ensure that equity remains part of the conversations around scholarly communication. We can help authors understand how publishers' policies influence the scholarly communication landscape, even when their descriptions of these policies serve to draw attention away from their role.

In order to play this role, it's important to understand how publishers are talking about open access, and how we want to talk about it. As publishers continue to update their policies and their own outreach to authors, let's make sure that we are prepared to offer a larger and less publisher-centric view.

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CHAPTER 21

Capitalizing on Shared Motivations for Successful Open Initiatives

Jessica Kirschner, Preeti Kamat, and Nina Exner

Introduction

This chapter draws on data from a pilot study designed to explore higher education faculty motivations for engaging with open practices—specifically, open-access article publishing and open educational resource (OER) creation. By delving into these motivations, the research team hoped to begin building a map for how our open education and open access initiatives could more significantly encourage faculty participation.

At Virginia Commonwealth University (VCU) Libraries, where the research team was based, all open initiatives are organized in the same unit. However, as a large R1 university, VCU is lucky enough to have a separate staff member responsible for each open movement. Thus, the initiatives are, in essence, distinct. Our research team included VCU's open education librarian, scholarly communication librarian (who supports open access), and the research data librarian (who supports open data). As such, though our study explored multiple open movements, we opted to do so independently to reflect the separate focuses of each initiative.

Such distinct support for open initiatives is not unique to VCU Libraries. Many other academic libraries in the US, especially those that are well-resourced, have separate staffing for their open initiatives.* These separate initiatives are often aligned with associated faculty motivations and categories of US higher education faculty responsibilities. For example, academic libraries often promote:

- Open access alongside publication support. Many faculty are required to publish scholarship, and open access emerged to increase access to these publications.[1]
- Open data alongside research support. Data underscores many scholarly publications, although it was not often shared. In a push for transparency and reproducibility of research studies, open data emerged.[2]
- Open education alongside teaching support. Teaching faculty focus on delivering a quality education to their students, and open education emerged out of a desire to make education more affordable for students by reducing textbook costs.[3]

* Similar structures occur at large libraries such as University of Houston Libraries, University of North Carolina, Utah State University, and Florida State University, to name just a few.

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These distinctions are often echoed in the US higher education institutions themselves. For example, open access support may be placed within libraries, open data support within offices of research, and open education at centers for teaching and learning.

However, open movements are not as distinct as their initiatives are often treated. All open movements are motivated by an underlying desire to increase free access to information. While each open movement may reflect different sharing, the distinction between each movement isn't always clear due to the overarching label of "open." For example, when a faculty member contacts the library, they just want someone to answer their question without really caring who the answer comes from, and they often start by contacting the person with whom they have an existing relationship. Such confusion extends beyond faculty to others in higher education, such as administration, students, and even some in the library. This confusion likely exists beyond higher education, although that is beyond the scope of this chapter.

This conflation of various types of open creates an opportunity for open practitioners and advocates to grow various open movements together rather than separately. By identifying similarities and differences in our data between higher education faculty motivations for engaging with open access and open education, we can begin to explore avenues of overlap between these two open movements. This preliminary research shows that faculty actually share concerns and motivators across both open movements, suggesting that a shared advocacy approach could be beneficial.

Pilot Study

Our pilot study explored motivations for faculty engagement with open practices. We focused on the active creation and dissemination of resources, specifically the publication of articles in open access journals and the creation or customization of OER. The pilot study was conducted at VCU, a large R1 institution located in Richmond, Virginia.[†]

Methods

Because expectations, culture, and promotion and tenure (P&T) policies vary by discipline,[4] our research team decided the pilot study would focus on one discipline: education. All non-adjunct faculty in VCU's School of Education were invited to participate in both phases of the study. Those who may engage in research, scholarship, or teaching yet are not eligible for P&T, such as adjunct faculty or graduate students, were excluded from the study because P&T's influence on open engagement was of special interest to the research team.

The study explored which factors influence faculty engagement with open practices using the framework from Lwonga and Questier's exploration of open access motivations in Tanzanian health sciences universities.[5] Following Lwonga and Questier's model, we organized potential motivations into five categories:

- Facilitating conditions (FC), or conditions that help or hinder engagement with open practices
- Costs, or any real or anticipated effort, loss, or sacrifice necessary to engage with practices
- Intrinsic factors (IF), or internal motivations
- Extrinsic factors (EF), or external motivations
- Contextual factors (CF), or contextual and environmental influences

A list of factors within each category can be found in our final codebook. [‡]

[†] Further information on the project can be found at Jessica Kirschner, Hillary Miller, et al., "To Open or Not to Open: An Exploration of Faculty Decisions to Publish Open-Access Articles," Journal of Librarianship and Scholarly Communication 12, no. 1 (2024): eP16894, https://doi.org/10.31274/jlsc.16894.

[‡] Data available at Jessica Kirschner, Preeti Kamat, Hillary Miller, Jose Alcaine, Sergio Chaparro, and Nina Exner. "Data for Exploring Factors Influencing Faculty Engagement with Open Practices," OSF, 2023, https://doi.org/10.17605/OSF.IO/AQXT4.

We employed a mixed-methods sequential explanatory design. The first phase, an electronic survey, was modeled after the survey used in Lwonga and Questier's study. It included demographic and general questions, such as respondents' publication practices and views of P&T. Then, for each open movement (open access articles, OER), respondents were asked to answer Likert-type scale questions asking for agreement with statements inspired by Lwoga and Questier's survey instrument.[6] The five-point Likert-type response options were strongly agree, agree, neutral, disagree, and strongly disagree.[§] The quantitative data gathered in the survey would allow us to use social exchange theory (SET)[**] to provide a baseline in gathering how faculty feelings toward aspects of open may relate to their actual practices.

To delve further into the initial findings from the quantitative data, the research team conducted interviews and focus groups with members of the same population. Gathering this qualitative data would enable us to interrogate the results of the survey by providing more contextual information, confirming results, and querying where confusion might have arisen to improve the survey instrument for further iterations. Invitations for participation in this phase were organized based on faculty status:

- tenured faculty were invited to be interviewed to ensure anonymity when discussing P&T committees;
- term faculty were invited to participate in focus groups to allow for a larger participant pool; and
- tenure-track faculty were invited to participate in either option in an attempt to oversample this key population, which was undersampled during phase one.

In this study, "tenure-track" participants are those who are on the tenure track but have not yet received tenure ("tenured"), and "term" is VCU's label for annually contracted faculty. Eight interviews and three focus groups were conducted based on responses to recruitment.[††]

Limitations

While all non-adjunct faculty at the School of Education were invited to participate, this was still a limited convenience population, selected due to the pilot nature of the study and the research team's membership. Due to this small target population (100 faculty), the study's sample size (n=15 for both phases) was also fairly small, which limited nuanced statistical analysis. The team hopes that the research protocol will eventually be expanded on a broader scale and/or conducted at additional institutions to build on our preliminary exploratory findings and increase the generalizability of our results.

Further, the self-selection of participants may have biased respondents to those in the target population who are already aware of open practices. Additionally, tenure-track faculty had low participation rates, limiting analysis in terms of faculty classification.

The survey elicited a high number of neutral responses to many questions. It is unclear if these neutrals indicate a true neutral, a lack of understanding of the topic, or if respondents found the question unclear.

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§ The full survey instrument is also included on our OSF site, included in the references.

** SET is the framework used in Lwonga and Questier (2014). For more on SET, read Kim (2010) or Homans (1958, 1961).

†† Additional information on methodology and an overview of data analysis can be found in Kirschner et al., "To Open or Not to Open."

Results

Fifteen faculty members at VCU's School of Education participated in each phase of the study (table 1). These fifteen participants were not necessarily the same between phases; we did not collect identifying information in the survey, so it is not clear which, if any, survey respondents also participated in the qualitative data phase.

Category	Measure	Quantitative Phase	Qualitative Phase	
		Survey	Interview	Focus Group
Faculty classification	Tenured	7	6	0
	Term	7	0	6
	Tenure-track	1	2	1
Rank	Professor	2	1	1
	Associate professor	6	5	1
	Assistant professor	7	2	5

Table 1. Demographics of fifteen faculty members at VCU's School of Education who participated in each phase of the study.

Survey (Quantitative) Results

Publication Practices

Survey respondents were asked about their publication history and then how many of these publications were open.[‡‡] Overall, more respondents published articles than educational resources, and they did so in a higher volume. Even those respondents who had not published an article had expectations of publishing one in the future. Conversely, many respondents were unsure whether they would publish an open textbook, with a majority saying they have not considered it, and three without any plans to do so. This could be reflective of an institutional culture that prioritizes research and scholarship over teaching.[7]

When delving more closely into overall versus open publications, six respondents indicated that their OER publications were greater in number than their total course material publications. For example, one faculty responded that they had not published any course materials but then reported that they had created/shared five OER. While this may suggest some confusion about the survey question, the misreporting only occurred once for open access articles. Thus, it more strongly suggests some confusion about the definition of OER, which had not yet been provided to faculty in the survey.

‡‡ Questions 1 and 2 of the survey:

1. You may have the opportunity to share the findings of your scholarly research or your teaching materials in a variety of different formats. In the past five years, about how many times have you shared your materials in each of the following ways:
   a. Published an article in a peer-reviewed journal
   b. Published a textbook
   c. Published or shared other course materials
2. Within these methods, you may have had the option to publish your works openly. In the past five years, about how many times have you shared your materials in each of the following ways:
   a. Published an article in a fully open access journal
   b. Published an open access article in a journal with an open access option
   c. Customized an open textbook
   d. Created an open textbook
   e. Created/shared other types of open educational resources

Factors Influencing Engagement with Open

This study had hoped to use social exchange theory (SET) to draw a connection between publishing history and the Likert scale question responses to best pinpoint which factors were the strongest indicators of faculty open engagement in practice. While this analysis was possible for open access publishing,[§§] the small number of respondents publishing course materials and the disconnect between the reported numbers of OER versus course materials published made such analysis difficult for OER. Instead, to compare motivations for engaging with open access and open education, this chapter looks at wider trends in faculty responses, specifically which Likert scale questions had no agreement (strongly agree or agree) or no disagreement (strongly disagree or disagree) (table 2).

Question	Open access articles	OER
[OA or OER][***] will increase the chance to communicate my research findings to those outside of the academy, including community members, practitioners, or policymakers.	No disagreement	1 disagree
[OA or OER] will increase the potential impact of my work.	No disagreement or neutrals	1 disagree
[OA or OER] will increase my visibility within the discipline(s) to which I belong.	No disagreement	2 disagrees
[OA or OER] will help other researchers build on my research findings.	No disagreement or neutrals	No disagreement
[OA or OER] allows others to access [scholarship/teaching and learning materials] that they could not otherwise freely access.	No disagreement or neutrals	No disagreement or neutrals
I support the principle of [OA or OER]	No disagreement	No disagreement
[OA or OER] are beneficial to the scholarly community.	No disagreement	No disagreement
[OA or OER] are beneficial to the common good.	No disagreement or neutrals	No disagreement
I will [OA or OER] if my research funding agency requires me to make research outputs [open access/create customize OER].	No disagreement or neutrals	No disagreement
I will [OA or OER] if my co-authors or collaborators [OA or OER].	No disagreement	No disagreement

Table 2. Likert questions without agreement or disagreement. (*continues*)

§§ See Kirschner et al., "To Open or Not to Open," for in-depth results and discussion of this SET analysis for open access.

*** For the open access section, most questions began with "Publishing open access articles…" while the open education section began with "creating or customizing OER." Questions only veered for structural agreement with the statement (e.g. "I support the principle of open access.") To streamline the table, we are abbreviating here "[OA or OER]".

Question	Open access articles	OER
My school values [OA or OER] as evidence for tenure and/or promotion.	3 agrees	No agreement
[OA or OER] will adversely affect my chances of tenure and/or promotion.	2 agrees	No agreement
Scholars who [OA or OER] have more prestige than those who do not.	No agreement	No agreement
[OA or OER] will improve my professional reputation within the discipline(s) to which I belong.	No agreement	2 agrees
I have the necessary funding to publish my work in journals that require open access article processing charges.	No agreement	*[Not asked in open education section]*
In my field, it is common for researchers to [OA or OER]	1 strongly agree	No agreement
In my school, it is common for faculty to [OA or OER]	No agreement	No agreement

Table 2. Likert questions without agreement or disagreement.

Exploring which questions lack (dis)agreement can show which items elicited strong and clear feelings across all respondents. For example, one question had no disagreement or neutrals across either open movement studies: "[OA or OER] allows others to access [scholarship/teaching and learning materials] that they could not otherwise freely access." From this, we can learn that it is clear to respondents that these open practices increase free access to materials.

Qualitative Results

Interviews and focus groups further explored the quantitative results, asking participants to discuss their decisions to engage with both open movements more in-depth.[†††] To analyze the qualitative data, we coded excerpts according to each category[‡‡‡] and factor (e.g., Contextual Factors (category) - Culture (factor) is coded as CF_Culture), with a third level specifying if the comment referred to a specific open movement ("JOA" for journals open access and "OER" for open education). Occurrences for each code in the qualitative data are presented in table 3.

[†††] Questions can be found on our OSF site (https://doi.org/10.17605/OSF.IO/AQXT4), cited in the references.

[‡‡‡] Facilitating conditions (FC); Costs; Intrinsic factors (IF); Extrinsic factors (EF); Contextual factors (CF).

Code	Broadly	Open Access Journals	OER
CF_Broadly	5	5	3
CF_Culture	19	30	45
CF_Influence	16	30	17
Costs_Broadly	2	3	8
Costs_Attribution	2	0	15
Costs_Efforts	1	4	23
Costs_Monetary	1	22	6
Costs_Time	5	7	32
EF_Broadly	0	8	2
EF_PromTenure	38	50	34
EF_RecogPub	4	41	2
EF_Trust	0	44	10
FC_Broadly	1	0	1
FC_FundSupp	1	6	1
FC_InstSupp	2	4	8
IF_Broadly	1	8	6
IF_Altruism	8	14	35
IF_Knowledge	4	13	18

Table 3. Code occurrences.

Shared Motivations for Engaging with Open

Our study explored publishing open access articles and OER separately to allow for an in-depth exploration of US higher education faculty's unique motivations to engage with each movement. We also hoped that the results would help shape the various open movements the research team supported, including suggesting strategies for what would make our work most impactful. VCU's R1 status and large student population have translated to a large university library staff, enabling simultaneous yet separate support for open access, open education, and open data. Each initiative has its own staffing, supports, priorities, and alignment with institutional structures, such as other departments (e.g., Office of the Vice President of Research partners with the library for open data support).

Yet, each open initiative does not exist independently. Even though VCU Libraries has dedicated scholarly communication (open access), OER (open education), and research data (open data) librarians, others within the library, such as subject liaisons, have to be able to assist faculty, students, and staff with aspects of all open initiatives, even if that just means referring the patron to the appropriate specialist.

Further, the distinction between open initiatives is not always clear to the patrons that academic libraries serve. In fact, these patrons often conflate the various types of open.[§§§] For example, a faculty previously awarded an OER grant might email the OER librarian with questions about APCs, which then have to be redirected to the scholarly communications librarian. Our data also emphasizes this confluence, with confusion between open access and open education noted at least seven times in the qualitative data. We suspect that this count does not capture all instances of conflation, as the confusion may not always have been evident to the interviewer or coder. This intermixing of open initiatives in faculty's minds suggests that shared advocacy strategies and outreach may increase the impact of all open initiatives. Drawing from the pilot study data, we will explore which aspects of open movements can cross-pollinate for maximum uptake of open initiatives.

External Pressures and Contexts

Promotion and Tenure (P&T)

Since P&T policies and guidelines provide a scaffolding for US higher education faculty advancement, retention, and recruitment, it is no surprise that these faculty often make decisions based on P&T expectations.[8] For example, the main author of this chapter was told multiple times that faculty will consider OER only after tenure is received because, before then, it is seen as a distraction. Her experience is not unique, and many groups have identified and begun to combat this perception by showcasing how to highlight different aspects of OER for inclusion in P&T documentation.[****] The importance of P&T as an influence on faculty decisions also applies to other open movements—for example, motivators for green open access sharing,[9] calls for revisions to promotion valuing systems,[10] and drivers behind data linkage and citation projects like Make Data Count.[11]

The influence of P&T on faculty decisions is also reflected in our data, with P&T occurring first (OA), fifth (general), and seventh (OER) in qualitative code occurrences. P&T is also tied to other high-occurring factors, like trust (e.g., editorial board members) and recognition and publicity (e.g., number of citations), which are perceived to be evidence of the quality of a publication for many P&T committees.[12]

Despite P&T seemingly being the driving force behind faculty decisions, our data suggests it isn't always the central focus. While a large number of respondents agreed that they shape their choices for both article publishing and course material creation around P&T guidelines, there were also a small number who disagreed, and about a third or more who provided a neutral response (figure 1).

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§§§ The authors had difficulty finding literature supporting this dilemma, perhaps because the initiatives are often studied separately, reinforcing the need for a volume like this one! However, the existence of many webpages, often from academic libraries, highlights that this confusion does exist. For example: https://www.library.rochester.edu/services/open/are-oer-and-oa-same-thing and https://libanswers.depauw.edu/faq/380179.

**** For example, the DOERS3 matrix (https://www.doers3.org/tenure-and-promotion.html) has been adapted for other contexts, such as Iowa and Oregon. Also, books like Andrew McKinney's edited volume *Valuing OER in the Tenure, Promotion, and Reappointment Process* (Montreal: Pressbooks, 2024), https://pressbooks.cuny.edu/tenureandpromotioncasestudies/.

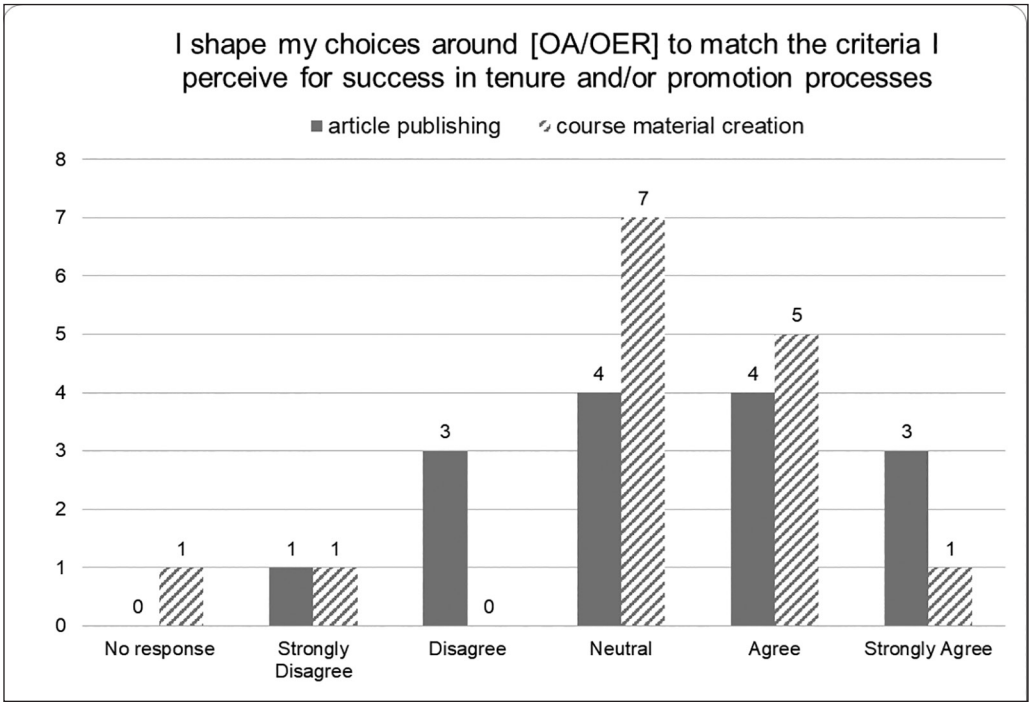


Figure 1. Responses to "Do faculty shape publishing decisions around P&T?"

These results are echoed in other P&T-related questions (e.g., Would OA/OER be valued or accepted as evidence?), for which there were a number of neutral responses. The neutrals of these latter questions suggest that faculty may not shape their publication decisions around P&T because they're unsure how P&T committees will receive their open publications. The literature has reported similarly complex pressures in other forms of resource sharing, such as open science,[13] where institutional policies and individual uncertainty make a complex interrelationship. While clearly articulated support for open resource sharing in P&T guidelines would pave the path for many who have been hesitant to participate in these open movements, this complicated interrelationship suggests the need for more complex solutions.

Culture and Influence

As a part of these complex solutions, we can identify which other factors have a strong influence on faculty decisions. Our data highlights culture (i.e., departmental, school, or disciplinary norms and expectations) and influence (i.e., what others in your field, department, or school are doing). In fact, culture was the most common theme for OER in the qualitative data, suggesting that unofficial guidance from departmental colleagues has a stronger impact on decisions to publish OER than what is articulated in P&T policies. Influence and culture both appeared thirty times in the qualitative data for open access articles, suggesting both have strong influences on faculty decision-making.

The survey results are murkier. Respondents were not unified in whether they would be influenced by what others in their department or leading researchers in their field do, with a slight trend toward agreement. Simultaneously, respondents were unsure about what their colleagues were doing, with the four questions asking about the activities of others receiving some of the highest number of neutral responses in the survey. The non-neutral responses about others' activities almost universally indicated that it was not common for others to publish articles or course materials openly. This trend may be reflective of the discipline rather than universal, as open engagement, including data sharing, is known to be highly

disciplinary.[14] However, one would think the survey would show many faculty do publish OER, given that education as a field has a culture of sharing teaching resources. The authors can also attest that faculty in the School of Education do engage with open education, open access, and open data. Thus, there seems to be an opportunity to better publicize partnerships and existing work within the open space to increase engagement.

A need for publicity of existing engagement is reinforced by respondents agreeing that they would follow their co-author's decisions to engage with open. Since co-authors have close working relationships, it is understandable that one author can influence another's decision, especially given the conversations that likely occur prior to a decision being reached. This deep communication between authors suggests another opportunity to increase engagement by increasing communication about open more generally. If more colleagues share how they engage with open and how they have benefited from this decision, it may pave the path for departmental and disciplinary culture to change. Our data suggests that a culture where open engagement is commonplace may lead to increased participation in open. Such a change in culture may also lead to changes in P&T policies. This path has had some success for open access, with some faculty senates introducing open access statements and some eventually leading to inclusion of open in P&T documentation.[15]

Requirements

Requiring publishing openly can also be a motivator for faculty to participate in open access and open education. All survey respondents said they would publish an open-access article if required by a funder, with all but three respondents (neutrals) saying they would do so for course materials. Perhaps this is not surprising, as open access article and open data mandates for federally funded works are becoming common in the US.[16] Actual engagement with data sharing has had mixed results,[17] although similar requirements have advanced the visibility of other open research initiatives, such as the sharing of scientific protocols and analytical code.[18]

Faculty were not as easily swayed toward open if their school required them to publish openly, with more neutrals and one "disagree" for both open movements. This was echoed in the qualitative results, with at least one participant noting that top-down mandates would not be welcome. Requirements from funders may be more palatable because funders are seen as external entities and the requirements for open a byproduct of funding for larger projects. Our data suggests OER may have similar luck motivating faculty through funder requirements as a byproduct of grants—for example, teaching material created as a part of an NSF grant, which could also increase the "broader impacts" score during application review.[19]

Balancing Costs

The factors in the "costs" category occurred often in the qualitative data, but not consistently across the two open movements. Thus, it is a bit more difficult to identify shared strategies to advance advocacy across open movements. Participants were concerned about the amount of time and effort required to create or customize OER[20] and the funding needed to publish open access articles.[21] These concerns echo barriers discussed in the open science community, with publishing being discussed as a financial and reputational barrier, while data sharing encounters cultural and infrastructural barriers.[22] These discrepancies make sense. Course material creation is often considered a part of teaching and, while a significant lift, authoring a textbook is often not considered scholarship and may be seen as a detraction from required research and article publications.[23] Meanwhile, open access article publication is often tied to article processing charges, while there typically is not a charge associated with publishing OER. Unless these two types of publications are treated the same financially, culturally, and within P&T (e.g., given the same weight), it may be hard to find strategies to cross-pollinate movements to overcome these cost concerns.

Attribution and Plagiarism

Respondents expressed worry about the lack of attribution and possible plagiarism when sharing educational materials, although this concern was basically absent when considering open access articles. Concerns about OER attribution appeared fifteen times in the qualitative data, but, interestingly, zero times for open access articles. In the survey, six respondents agreed that their work may be plagiarized for OER, but only one agreed for open access articles.

We wonder if this concern about attribution is somewhat due to the use of open licensing, such as Creative Commons, for OER and open data.[24] Many faculty do not clearly understand open licensing.[25] The preemptive release of certain rights under open licenses, such as the ability to share and edit, may also make faculty uncomfortable. Faculty are worried about their work being taken out of context or adapted to make it factually inaccurate or misrepresent them or their original ideas, a worry that appeared multiple times across our interviews and focus groups.

Since open licensing is not a universal requirement for open access articles, it is not clear how the associated lack of worry around attribution can be carried over to benefit or strengthen other open movements. As the difference in attitude toward attribution was so striking between the two movements, we suggest it as an avenue for further exploration; reducing this worry has the potential for significant impact on engagement with open education.

Beliefs and Values as Motivators

Faculty beliefs and values also have a significant impact on their choices. Five out of seven questions with no disagreement from both open movements were categorized under intrinsic motivations. These questions emphasized faculty's belief in these open concepts, the benefit to the community, and the ability to allow others to access and build on your materials. In fact, the only question with no neutrals or disagreements for either open movement was about enabling access to materials. Faculty clearly understand that openness means increasing access and almost all agree it is beneficial to the community, a potential strong motivator for engaging in these practices.

The qualitative data further emphasized altruism as a motivator for engaging in open education. Those using or creating OER are often motivated by a desire to help their students and improve learning outcomes, which has an immediate and visible impact.[††††] Those publishing open access articles are motivated by altruism to a lesser extent, perhaps because increasing access to their articles for others in the scholarly community is more removed. And while the principle of access has been much-stated in public access policy for article and data sharing (e.g., the OSTP memo),[26] researcher implementation still lags. Thus, there may be opportunities for other open movements to draw from open education's strong altruistic influence by focusing on tangible impacts.

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†††† Cost as motivator, e.g.: Lisa Petrides, Cynthia Jimes, Clare Middleton-Detzner, Julie Walling, and Shenandoah Weiss, "Open Textbook Adoption and Use: Implications for Teachers and Learners," *Open Learning: The Journal of Open, Distance and e-Learning* 26, no. 1 (February 2011): 39–49, https://doi.org/10.1080/02680513.2011.538563; Ozgur Ozdemir and Christina Hendricks, "Instructor and Student Experiences with Open Textbooks, from the California Open Online Library for Education (Cool4Ed)," *Journal of Computing in Higher Education* 29, no. 1 (April 2017): 98–113, https://doi.org/10.1007/s12528-017-9138-0. Increased learning outcomes, e.g.: Virginia Clinton and Shafiq Khan, "Efficacy of Open Textbook Adoption on Learning Performance and Course Withdrawal Rates: A Meta-Analysis," *AERA Open* 5, no. 3 (July 2019): 2332858419872212, https://doi.org/10.1177/2332858419872212; Nicholas B. Colvard, C. Edward Watson, and Hyojin Park, "The Impact of Open Educational Resources on Various Student Success Metrics," *International Journal of Teaching and Learning in Higher Education* 30, no. 2 (2018): 262–76, https://eric.ed.gov/?id=EJ1184998.

Conclusion: Main Motivators and Potential Solutions

Our research indicated that P&T, departmental and disciplinary culture, funder requirements, altruism, and attribution have a strong influence on higher education faculty's engagement with open education and open access. However, not all of these factors have an equal impact on faculty publication decisions in practice.

We began our research with the hypothesis that P&T would be the main motivating factor in faculty decisions about whether to engage with open. This hypothesis was partially confirmed by the data: in the SET analysis of the open access portion of this study, our team found that P&T strongly influenced faculty's decisions to *not* publish an article in an open access journal, with many hesitant due to fear of negative impact on their P&T reviews.

On the other hand, those who *do* publish open access did so despite these P&T worries, instead leaning into their beliefs in sharing resources and benefiting the community.[27] We have not duplicated the SET analysis for OER due to the lack of foundational data.[§§§§] However, the similarity of responses to the Likert-type scale questions, especially those about shaping decisions around P&T and intrinsic motivations, suggests similar results in a potential SET analysis of the open education portion. This potential for similar results is only emphasized by the intermixing of these open movements in many faculty's minds; if faculty do not clearly distinguish the two initiatives, it follows that their feelings toward them will also be intermixed.

The fact that other factors, such as altruism, can override the impulse to steer away from publishing open access or creating OER suggests that open movement advocates' efforts would have the most impact by emphasizing these other factors. Specifically, our data suggests the following strategies would have the most impact:

- **Finding faculty advocates to begin culture change.** If conversations with others can influence faculty decisions to publish open access or create OER, advocates need to work to ensure that the stories and opinions shared in these conversations have a wide reach. While some of these conversations occur naturally, there would be further impact if those engaging in an open movement would share their experiences widely. Some templates for this type of program occur, especially in open education: programs titled "Textbook Heroes" or something similar celebrate the efforts of higher education faculty in open education.[*****] If these types of recognition and opportunities to share experiences and successes become more commonplace, the work they celebrate can integrate itself into institutional and departmental culture to normalize engagement with the associated open movement.
- **Emphasizing altruistic impacts.** Since altruism and other internal motivators can overcome the fear of P&T repercussions, emphasizing these motivators in advocacy has the potential for a significant impact on levels of engagement with open. Some of this already occurs, such as when academic libraries talk about the increased readership or citations for open access publications or student savings for open educational resources in their advocacy efforts. Putting these and other intrinsic benefits, such as other ways to benefit student learning experiences when using OER, at the forefront of reasons to engage with open can only reinforce these underpinning motivations.

§§§§ As noted in the Results section of this chapter, the small number of respondents publishing course materials and the disconnect between the reported numbers of course materials published and how many of those were open made us hesitant to perform an SET-type analysis for lack of validity of the results.

***** For example: https://lib.ku.edu/textbook-heroes; https://libguides.mnsu.edu/textbook-affordability/textbook-heroes; https://sites.psu.edu/altoonaoaertextbookheroes/2024-textbook-heroes/.

††††† The DOERS3 matrix, https://www.doers3.org/tenure-and-promotion.html.

- **Quelling cost worries.** Assuaging concerns about P&T reception of open work, such as the work being done through DOERS3 and others for open education,[†††††] will help pave the way for participation in open, especially for those for whom P&T is the biggest barrier. However, our data suggests that focusing on the worries in the cost category, such as concerns about attribution for OER, could also have a significant impact on faculty engagement. For example, additional copyright education and webinars about remixing content could help calm concerns about reuse, attribution, and plagiarism for OER.

We also want to emphasize that the intermixing of these initiatives in many minds suggests that there exists an opportunity to take what is successful in one open movement and apply it to another. For example, building on the success of Textbook Heroes programs to create similar programs highlighting successful open access publications, whether articles or books. There are also opportunities to combine successful programs across open movements. For example, copyright education, for which a variety of programs already exist on many campuses, could expand from focusing on focusing on open access publication or ETD sharing to open licensing in general and include examples from all open movements.

Last, while this research focused on open access articles and open education, the same principles can be applied to other open initiatives, such as open data, open science, or open-source software. By highlighting existing work, promoting intrinsic motivations, and quelling cost concerns for any open initiative, advocates can work to increase engagement with any open movement. And by working on these initiatives in tandem and building on successes across open initiatives, we can only further capitalize on "openness" to increase engagement.

Acknowledgments

The data in this article were originally collected as part of a pilot study partially funded by a 2021 VCU Quest (formerly PeRQ) award. Members of the original project team include all three authors as well as Hillary Miller, Sergio Chaparro, and Jose Alcaine.

Deidentified, disaggregated data from the research project are available for public download via OSF at https://doi.org/10.17605/OSF.IO/AQXT4. Identifiable data cannot be shared, based on the consent forms signed by the participants.

Data Set

Kirschner, Jessica, Preeti Kamat, Hillary Miller, Jose Alcaine, Sergio Chaparro, and Nina Exner N. (2023). Data for "Exploring Factors Influencing Faculty Engagement with Open Practices." https://doi.org/10.17605/OSF.IO/AQXT4.

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CHAPTER 22

From Open Metadata Practice to Open Metadata Policy: Developing An Open Metadata Sharing Framework

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Introduction

Open metadata (also referred to as metadata sharing or open metadata sharing)* involves ensuring that library metadata is openly available and reusable to the widest extent possible, often achieved through applying an open license(s), either in a statement or to datasets of metadata or catalogue records, or both. Open metadata sharing offers immense possibilities that encourage collaboration, innovation, and research and formalizes the longstanding sharing and reuse practices of the library metadata community. This chapter describes how open metadata sharing connects to, supports, and is supported by other open movements and discusses the necessity and importance of open metadata sharing within the library community. Further, this chapter offers a glimpse into the history of the open metadata movement in Canada and the United States specifically, though the implications of open metadata sharing are broader, as well as insight into the challenges that library institutions may encounter in their effort to participate in fulsome open metadata sharing. Lastly, we share a case study of the University of Alberta Library (UAL)'s process in approaching and creating an open metadata sharing framework.

* For consistency of use in this chapter, we use "open metadata sharing."

CC BY NC

Open Metadata and Open Movements

According to the Open Data Handbook,[1] open movements "seek to work towards solutions of many of the world's most pressing problems in a spirit of transparency, collaboration, re-use and free access." These movements include open data, open science, open government, open scholarship, open publishing, open peer review, and others. These open movements are built on the principles of openness, participation, and collaboration, and on a belief that the common good is not only served by and enriched by open and transparent sharing of information in all its forms.[2] Open metadata also falls under the umbrella of open movements, but is perhaps uniquely situated as being both supported by and supporting the other open initiatives.

Open movements involve making content and data that "can be freely used, modified, and shared by anyone for any purpose"[3] freely available. A critical piece of this puzzle is metadata. Metadata is what makes that data findable and accessible; it is what allows humans and machines to know what is out there that they may want or need to use. Metadata also provides the necessary information to understand what data or content are and whether or not they are suitable for any given need or interest. Importantly, metadata is what outlines how that data or content can or cannot be used. The services and infrastructure that underlie the various open initiatives are heavily reliant on metadata, and so it is critical for metadata to exist and for that metadata itself to be open.

The open metadata movement, in turn, can be supported and bolstered by the other open movements. Metadata has long been understood as "data about data"[4] and as critical to describing information objects for findability, accessibility, and usability, as described in the previous paragraph. However, whether as individual records or instances or in aggregate form, metadata is increasingly seen as having inherent value in and of itself, as a knowledge product that can be "created and gathered, shared, manipulated and transformed, interrogated, and visualized."[5] Metadata creators, including "libraries, publishers, universities, scholars or social reference management communities have an important role to play in supporting the advance of humanity's knowledge."[6] The metadata that libraries create or curate is critical to their public and educational missions, not only enabling them to carry out their core functions of helping communities create, share, and use information but also allowing others to understand those functions and their impacts. Thinking of metadata as knowledge that is a part of the common good makes the application of the principles of the open movement to metadata seem both obvious and logical.

Moreover, there is a democratic role for open data where "[a]ccess to the entirety of the intellectual output of a society is an important condition of the democratic debate and a citizen's informed decision-making. This access is, of course, not possible if said output is not described with the appropriate metadata."[7] Just as within library collection development something is not truly a part of the collection until it is described and made available, similarly, open scholarship would be troubled if no one can find it, and for this, open metadata is needed.

History and Importance of Open Metadata Sharing

Resource description—the practice of creating library metadata or catalogue records—is complex work. Unfortunately, the complexity of the work and particularly the reliance on open sharing of bibliographic data to support library workflows is not often well understood. As indicated by Flynn in *Open Access Metadata, Catalogers, and Vendors: The Future of Cataloging Records*:

> *OA [Open access] metadata enables sharing cataloging records in order to improve cataloging efficiency and quality as well as promoting timelier, more consistent access to information. It ensures that any library can obtain needed records in a timely manner to better their library and users, while also allowing libraries to contribute records themselves.*[8]

However, while open metadata policy and collaboration on it are more emergent, open metadata has been part of resource description practice since the shift from catalogue cards. From the outset of the move from cards to online catalogues, an encoding method was needed for sharing bibliographic metadata from one system to another, leading to the development of the MARC (Machine Readable Cataloging) standard in the 1960s by Henriette Avram, and "[b]y 1971, the MARC record had become the defacto standard for sharing bibliographic cataloging in the United States and within a few years it had been adopted worldwide."[9]

Along with the technical infrastructure for data sharing, policy and practice developed early as well through universal bibliographic control (UBC). A fundamental concept for UBC is that in order to retain all published heritage worldwide, it must also be made accessible through full description. As such, to support this and share the resource burden of accomplishing said objective, "[e]ach National Bibliographic Agency (NBA) is responsible for producing authoritative bibliographic and authority data for resources published in its own territory and making this data available internationally to libraries and other NBAs."[10]

UBC has not simply been a theoretical notion either. Indeed, "[d]uring the 1970's IFLA [International Federation of Library Associations and Institutions] established an office for Universal Bibliographic Control following several years of discussion about the importance of international sharing of bibliographic data to help reduce costs and to encourage greater cooperation worldwide. That office became known as UBCIM (Universal Bibliographic Control and International MARC)."[11] Moreover, while much emphasis for UBC initially centred on NBAs, more generally, if perhaps less formally, this concept extends to other libraries (not strictly NBAs) worldwide, sharing the burden of resource description and contributing to the overall description of published heritage.

This is also closely related to the International Cataloguing Principles, where 2.10 on interoperability states that "[a]ll efforts should be made to ensure the sharing and reuse of bibliographic and authority data within and outside the library community. For the exchange of data and discovery tools, the use of vocabularies facilitating automatic translation and disambiguation is highly recommended." Additionally, 2.11 on Openness states that "[r]estrictions on data should be minimal in order to foster transparency and conform to Open Access principles, as declared also in the IFLA Statement on Open Access. Any restriction on data access should be fully stated."[12] This is crucial as it sets the stage for open library metadata, given that since libraries started moving from card catalogues to computers, libraries have created standards and practices with open sharing not just in mind but also as a requirement to function.

Since the very formation of online catalogues, resource description standards and practice have been built on supporting interchange and open sharing of data. As captured by Koskas, referring to UBC, "In a way, the principles that presided over its conception and development have become such a fundamental part of bibliographic activities in libraries as to become perhaps largely implicit to librarians."[13]

Moreover, the development of MARC and other library standards to support the open interchange of data rapidly saw efficiency in workflow. The case study by Andrews and Kelley for the adoption of OCLC at Texas Tech is a good example of these, demonstrating the adoption (through OCLC in that case) of metadata re-use supporting efficiency.[14] The success of copy cataloguing, the use of Z39.50, "a client/server based service and protocol for Information Retrieval"[15] commonly used for the exchange of MARC data, and data pools like OCLC, rapidly led to not just an assumed usage of open metadata but also a reliance on it to achieve a certain brutal efficiency in resource description workflows.

This implicit or even assumed availability of open metadata has brought us to this point where renewed focus on open metadata policy is required in the face of efforts to monetize and close down such practice. As outlined by Courtney et al. in "No One 'Owns' That: Metadata, Copyright, and the Problems with [Library] Vendor Agreements," "While there are many players in this space, Online Computer Library Center (OCLC), the organization that most prominently coordinates metadata sharing, has recently increased their claims to having a

proprietary interest in library metadata."[16] As a result, while the same principles are important to consider as libraries navigate this landscape with other vendors, publishers, and institutions, OCLC is also a focus of some of the discussion below.

One key example of efforts that infringe on the previously typical standard of open metadata sharing has been OCLC and the controversial application of the OCLC WorldCat Rights and Responsibilities (WCRR) in 2010,[17] which replaced the "Guidelines for the Use and Transfer of OCLC-Derived Records" that had been in place since 1987.[18] The initial proposed changes to the 1987 guidelines were met with strong concern from the library community regarding the new restrictions on record use and transfer and concepts of singular ownership over metadata, where there had previously not been such restrictions or conceptions in the policy.[19] Concerns relating to "unresolved issues regarding bibliographic metadata use and reuse" remain, particularly as library technology continues to evolve.[20]

The case of WCRR is also a prime example of agreements being a key consideration for the sharing of bibliographic metadata. As described by Courtney et al., the case of OCLC v. Clarivate highlights this, as the lawsuit was not based on copyright infringement but instead based on tort, with "[o]ne claim for tortious interference with a contract, and the other claim for tortious interference with business relationships."[21] While ultimately this case did not go to trial, there was a clear implication that libraries themselves were in breach of contract with respect to the WCRR. Courtney et al. raise concerns as, given the legal action taken, the implications are both that the libraries were in breach of contract and that the WCRR policy is viewed as legally enforceable by OCLC leadership, which they demonstrate is not certain at this point.[22]

As outlined earlier, libraries operate in an environment where the open sharing of bibliographic metadata is essential for operations. While awareness and adherence to policies and agreements are important, the spectre of facing legal action against libraries for the sharing of bibliographic metadata is gravely concerning. Moreover, increasingly, such concerns add further pressure for libraries to move from practice to clearly outline open data policies on the one hand but also to carefully consider agreements that curtail the open sharing of bibliographic metadata and weigh the benefits of any associated agreement in terms of products and services against the potential risks, both legal and operational in terms of limits on reuse. To put it another way, with libraries operating in ecosystems dependent on open interchange, any dependencies on bibliographic metadata with constraints on re-use can increasingly be seen as risks that need to be managed.

In the context of the OCLC v. Clarivate case, the International Coalition of Library Consortia (ICOLC) and American Library Association (ALA) Joint Statement on the Metadata Rights of Libraries was issued:

> *Given how important it is for libraries to use, re-use, and share such metadata freely, current industry trends are troubling. Limited choice in affordable metadata service options, along with restrictions imposed on metadata use, re-use, and sharing across collaborations, platforms, system types, and providers, are impeding libraries' rights and ability to innovate and collaborate to serve their users effectively.*[23]

As argued above, the ability to leverage shared bibliographic data is a key support for core operations of all libraries, not just for underfunded or smaller ones, and any efforts by organizations looking to limit this threaten to destabilize core operations everywhere.

Finally, looking towards the future, other technological innovations place further pressure on the situation, given an ever-increasing need for open bibliographic data. Using one example, with libraries worldwide working on linked data implementations, we can expect an ongoing shift from the use of a record covered by a given policy to entity-based description supported by Resource Description Framework (RDF), where re-use will need to be assessed for separate statements about a resource. Here, without leveraging linked open data for web-scale interchange, the technical challenges of entity-based attribution for data used across communities will be difficult to manage.

The Growth of the Open Metadata Movement

The movement of open metadata sharing centres on the imperative for libraries to clearly set and communicate their own standards and vision for data sharing, data interoperability, intellectual property, and innovation.[24] The movement of sharing public-facing policies or frameworks, often using open licenses, to communicate openness around metadata sharing began in the early 2010s and has since seen slow but steady growth.

While sharing library metadata has been common practice within the metadata community for decades, as discussed above, in the early 2010s, university libraries, national libraries, and other knowledge organizations began to develop theory and best practices around more formalized metadata sharing.[25] The University of Michigan was one of the earliest adopters of open metadata sharing, dedicating a large amount of its bibliographic data to the public domain through a CC0 public domain dedication in 2010.[26] Shortly after, in January 2011, guidelines to help set a standard for open metadata sharing were published, recommending the use of explicit open licenses for all published metadata, and, particularly in cases where data is publicly funded, to use a license that permits the broadest reuse and repurposing of data, such as a public domain dedication waiver.[27] Since then, several American university libraries have released public statements, or "open metadata policies," including Harvard University Library, Columbia University Library, University of Pennsylvania Library (Penn Libraries), Princeton University Library, University of California Library – Special Collections, University of Florida Library, Washington University Library, and Yale University Library. [28]

Apart from bibliographic metadata, guidance for sharing metadata that supports open access publications and the scholarly communications ecosystem has also been developed and shared in the form of the Principles for Open Scholarly Infrastructure (POSI).[29] These principles understand metadata to be foundational and critical infrastructure for open scholarly communications, and they also call for metadata to be shared as openly as possible within privacy limits, preferably through CC0 public domain dedication.[30]

Despite over a decade passing since the beginning of this movement, widespread uptake of open metadata or metadata sharing is still in its early stages. In Canada, for example, the University of Alberta Library is, at the time of writing, the first university library institution in the country to create an open metadata sharing framework or policy.[31] Challenges that the open metadata movement must grapple with are the murky waters caused by 1) lack of a legal definition to determine whether or not metadata (or certain parts of metadata) may or may not be protected by copyright, 2) variation in terms of use in contracts pertaining to outsourced metadata, and 3) the fact that there are many different types of metadata within libraries (whether bibliographic, archival, or metadata supporting digital collections or digital repositories), and each have differing sources and differing considerations to address before proceeding with open sharing, which are explored below.

Navigating Copyright, Contracts, and Open Licenses

Selecting and applying an appropriate open license to library metadata is a tangible way to clearly communicate usage guidelines for sharing metadata; however, open licenses can only be applied to content over which the assumed copyright holder is certain to retain copyright. Whether library metadata is protected by copyright in the first place remains a somewhat grey area that differs based on legal jurisdictions and their interpretation of originality. In Canada, facts are not original and cannot be protected by copyright; insofar as metadata relates only facts, perhaps it would be deemed public domain by Canadian courts.[32] However, there are aspects of metadata that require a level of subjectivity and analysis, such as assigning subject headings and call numbers, which could be deemed to go beyond facts alone.[33] Based on the landmark Supreme Court of Canada case, *CCH Canadian Ltd. v. Law Society of Upper Canada*,

the threshold of originality rests on whether the creation of a work, or "expression of an idea," required an intellectual exercise of both skill and judgment.[34] Viewed in this frame, it may be possible that, in Canada, certain parts of library metadata could be found to hold copyright. However, copyright protection must balance the interests of both the public and the copyright holder,[35] as set out in the Supreme Court decision in *Théberge v. Galerie d'Art du Petit Champlain inc.*;[36] with the consideration that library metadata is typically created through public funding and is intended to support public information retrieval and discovery, perhaps library metadata would not be found to hold copyright protection in Canada. Nonetheless, without a definitive stance upheld by law, one way for libraries to proceed with open licensing of metadata is to first undertake a risk assessment, determine the sources of their metadata (whether in-house or outsourced), assess if there are any usage guidelines attached, and determine if any contractual terms must be considered before openly sharing their metadata.

Comparatively, the threshold of originality in the United States (US), as set out in *Feist Publications, Inc., v. Rural Telephone Service Co.*,[37] is determined based on the amount of creativity involved in a work's creation.[38] In the case of library metadata, it could be argued that it may not be protected by copyright in the US, as even subject analysis and assignment of call numbers are carried out based on rules and guidelines and decidedly not creativity.[39] Courtney et al. offer a detailed and comprehensive analysis of the copyrightability of metadata, with reference to US copyright law and surrounding case law.[40] Nonetheless, as with Canada, barring a legal decision on whether library metadata may or may not be protected by copyright, libraries wanting to pursue open metadata sharing can utilize open licenses to waive certain or all copyright protections that they may or may not hold over their metadata.[41]

The standard best practice for open data licensing is to use the most open and least restrictive license possible, such as the Creative Commons Zero (CC0) public domain dedication; to ensure the maximum reuse and interoperability, CC0 is frequently cited as a best practice for licensing open metadata.[42] Institutions may opt to communicate the open availability of their metadata, whether with a CC0 public domain dedication or through a statement with similar intent, and include a request to attribute the source according to scholarly norms, without making attribution legally binding.[43] Examples of institutions or organizations that have chosen this route include the Canadian Research Knowledge Network (CRKN), New York Public Library, Harvard University Library, Princeton University Library, Columbia University Library, University of Washington Library, and the University of California Library Special Collection.[44] Other open licenses can include restrictions that make attribution a legal requirement, such as the Creative Commons Attribution license (CC-BY) or the Open Data Commons Attribution license (ODC-BY).[45] Creative Commons also offers an open license that restricts any commercial uses, in addition to requiring attribution (CC-BY-NC).[46] While attribution is understood to be good scholarly practice, legal requirements for attribution are cited as barriers to use that could lead to the creation of impractical files of metadata to describe metadata (also referred to as attribution stacking), particularly in cases where users are mixing metadata from multiple sources—a standard practice in resource description/metadata creation.[47] Ultimately, in selecting an open license for open metadata, libraries must consider their jurisdiction's copyright law, related jurisprudence, their institution's goals for wider metadata use, such as interoperability, as well as other factors that may influence their choice in open license, including terms of use from contractual agreements related to their metadata.

Case Study: University of Alberta Library (UAL) Process of Creating an "Open Metadata Sharing Framework"

Discussions on approaching the sharing of UAL metadata in a more systematic way began in 2015. The conversation was initially focused on how best to share metadata record sets that

UAL had created or curated. In particular, there was keen interest in posting these record sets in a place that would be accessible to anyone interested in them, which was a shift from our previous practice of sharing these in the OCLC community space, where non-OCLC members could not access them, and providing guidance on how we believed they could or should be appropriately repurposed. Based on a very cursory scan of what other libraries and institutions were doing at the time, as well as brief conversations with staff from the University's Copyright Office, we planned a page within our website that would include various metadata records sets and a statement along the following lines: "The University of Alberta Library believes that the vast majority of metadata is not copyrightable. However, in the case of discrete record sets, a Creative Commons Attribution-Non-Commercial ShareAlike 4.0 License will apply." In early 2016, the first few record sets were made available under this license through the library's open data portal.[48] These record sets described both physical and digital collections and were either original to UAL or extensively enhanced and enriched records originally from elsewhere.

Over the next several years, additional record sets were added to the UAL open data portal using this same approach. Fast forward to 2022, and several initiatives that UAL was (and is) engaged with, including the Program for Cooperative Cataloging (PCC) and SHARE-VDE: (SVDE), were actively developing open metadata policies. This was an opportunity for the library to consider our own interests and needs, both individually and as a member of such collaborative initiatives, and to provide input on policies that would align with our values and commitment to open access and open data. And it raised questions for us in terms of our own metadata and whether it was once again time to revisit the development and articulation of our own open metadata framework. Recognizing that doing so would require focused attention, we proposed that an academic library resident position be devoted to gaining an understanding of the current landscape, consulting with interested teams and units within the library, and developing a framework or policy for metadata sharing. The project began in July 2023.

While the intention has been, and still is, to include all UAL metadata—that is, catalogue metadata, repository and digital collection metadata, and archival finding aids—within the framework, we recognize that each category comes with its own opportunities and challenges that need to be addressed. For example, privacy or ethical issues with openly licensing and/or sharing repository metadata in the form of open datasets may need to be explored, as it is likely to include names of theses and dissertations authors and, if openly licensed, has the potential of being used to train generative AI large language models. Understanding the current metadata sharing approaches and carrying out the necessary consultations within any of these areas would be a large project in and of itself, and so we agreed that we should begin with one type of metadata. Given the rapidly developing work within the cataloguing community around metadata sharing, we agreed that catalogue records (i.e., MARC records) would perhaps be the most logical place to begin.

To start the project, we set out to gain a broad understanding of trends in the open metadata sharing ecosystem by conducting an environmental scan, which included a literature review, a "strengths, weaknesses, opportunities, and threats/risks" (SWOT) analysis, and an examination of as many institutions' public facing open metadata policies/guidelines as could be found at the time of writing. What resulted was a detailed analysis identifying trends, such as using separate open licenses based on different sources or different types of metadata held within academic libraries, the prevalence of encouragement to attribute paired with CC0 public domain dedications among academic libraries, and CC0 alone being utilized more frequently by national libraries or other types of knowledge organizations. Carrying out the environmental scan, sparked the necessity to delve into Canadian-specific research around whether metadata may or may not be protected by copyright according to the Canadian Copyright Act, and into any light that related jurisprudence could shed on the subject as well. From there, we settled into a process wherein, for each metadata dataset under consideration, we would first research trends and practices of other institutions regarding that type of metadata, followed by carrying out a risk assessment that would take into consideration copyright concerns and any terms of contracts associated with the particular metadata. This process

will be iterative and ongoing as more metadata datasets are considered and shared. Further steps, which are intended to be carried out, include technical decisions on how to share our metadata datasets (whether MARC formats, BIBFRAME formats, or others) and through which platform (common options tend to be Dataverse, an HTML website, or GitHub).

The Open Metadata Framework was endorsed by the UAL senior leadership team in June 2024, and the framework has been shared via the library's website.[49]

Conclusion

Open metadata sharing and the open metadata movement seek to formalize and ensure the continuance of longstanding practices of metadata record sharing, re-use, re-mixing, and openness that support not only the practices and workflows of metadata creation and resource description but also the core functions of libraries as a whole. Although challenges remain in navigating contractual terms associated with certain metadata and copyright considerations across different jurisdictions, utilizing open licenses for metadata or publicly sharing policies or frameworks asserting the intended openness and re-usability of library metadata can help to ensure that practices of library metadata sharing continue into the future, embodying the library ethos of open access and bolstering the work of other open movements in turn.

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PART 5

REFLECTING ON OUR PAST TO BUILD AN OPEN FUTURE

CHAPTER 23

Understanding Realities and Assembling Futures: Open-Source Hardware as a Concept, Movement, and Community

Anastasia Ackom Antwi, Antonio Anaya, Samuel Duah Boadu, Nano Castro, Brianna Johns, Sarah Hutton, Jessica Nguema, and Lee Wilkins

Introduction

Democratization of knowledge helps overcome economic and social barriers to information access, allowing more people to leverage quality information to educate themselves, solve problems, or further research. Accessible research and data have supported an increase in collaboration among scholars, scientists, and citizens worldwide, which drives new discoveries and innovative applications of this knowledge.[1]

A rapidly accelerating area of this innovative application of knowledge is in open-source hardware (OSH). Hardware is considered anything with atoms rather than bits; *open-source* hardware relies on the open sharing of design files and other documentation, making designs publicly available to study, modify, distribute, make, and sell the design or hardware based on that design.[2] Providing the capability to organize hardware into shareable pieces of knowledge brings OSH into alignment with the freedom of information access, an area for which libraries have long been strong supporters and advocates.

Peer production (PP) is a means to leverage access to OSH designs and software, furthering the democratization of access through a distributed production model.[3] Hippel discusses this democratization of innovation by user communities as a powerful force in advancing a decentralized pathway forward—it is a *social* movement that provides a new ownership model

CC BY

for commercial production.[4] With any social movement, groups and communities bound by shared goals often struggle to support the sustained community engagement required to impact systemic change.

In this chapter, a group of OSH community advocates shares their experiences in sustaining their own communities and navigating cross-sector collaboration at the global scale as part of the OSH movement. They share their perspective on the hopes and challenges surrounding the creation of a global "network of networks" that seeks to solve some of the most complex challenges humanity faces today.

What is Open-Source Hardware?

Open-source hardware, known as OSH in global open communities, refers to physical devices, machines, or systems whose design specifications, schematics, and source files are made publicly available. The definition from the Open Source Hardware Association[5] is derived from a variety of sources and requires that certified open-source hardware be freely available to anyone, while acknowledging that there may be a plethora of systemic barriers to achieving this ideal. The concept of OSH builds on open-source software (OSS), where the source code is freely accessible for viewing, modifying, and distributing.[6] The three main principles of OSH center around open design, licensing, and community collaboration. While interpretations of these principles can vary across communities and organizations, core benefits of OSH are widely recognized across the global ecosystem, including transparency, customization, cost savings, education, and learning opportunities.[7]

While open-source hardware has gained traction in various fields, it is important to note that physical components and materials involved in any OSH project may still be subject to traditional supply chain and manufacturing processes, which can involve proprietary elements or patents.[8] To gain a better understanding of these subtleties, it is important to further unpack differences in the interpretation of the definition of OSH and how it is applied.

All open-source hardware should be released under a license, which dictates how and under what restrictions a piece of hardware might be used and distributed.[9] A key reason for using licenses in OSH is so that one can fully understand the legality of iterating on a particular design downstream, allowing the creator to not worry about accidental infringement.[10] A challenge for OSH designs and licenses, as opposed to OSS, is that there are design, manufacturing, and output elements, each of which may be the subject of the license individually. Differences in licenses can include copyleft, which is a designation that requires future iterations to be released under the same license, or permissive licenses, which may require credit to the original but don't need to be released under an OSH or OSS license.[11] While a "true" open-source software license requires it to be able to be commercially available, there are very popular Creative Commons licenses that allow people to freely reproduce things so long as they are not for "creative" use. Some feel that any licenses that place restrictions, such as the Climate Strike license that prevents companies engaging in environmental destruction from using their software,[12] are not "true" open source, but there is a continuum of belief on the practice of this.

Definitions that are a bit stricter and specific to the challenges of hardware require the provision of source files in addition to production files. An example of this is a 3D-printed STL file; it does not actually show the source file, but the design file can be opened and modified by 3D modeling software, allowing the user to remix and iterate on the object. Many open-source creators advocate for a requirement that all OSH should be produced with OSS[13] so that using proprietary software does not become entirely ineffective. This alludes to another challenge of OSH, which is that there is no set requirement for how to document the making of a physical object; building every individual thing can be different, and there is no standard agreed-upon approach for documenting this process. OSH is not only computer hardware but can also be any physical matter, from textiles to print media.

It is challenging to set a universal standard on how open-source hardware projects should be documented. It is for this reason that organizations like the Open Source Hardware Association (OSHWA) have a certification program[14] that looks at each project's documentation to understand whether it is, in fact, reproducible through a review by members of the organization. Organizations like the Internet of Production are working on standardizing an approach to documenting the production of items by coordinating community efforts to create open standards and protocols[15] for distributed manufacturing; while some organizations have adopted components of these standards and protocols, there is still a long way to go before uniformity across global OSH use cases can be achieved.

OSH in Action: Case Studies

Open-source hardware is a great way to share knowledge and facilitate the development of new products; initially, the definition of OSH was written with electronics and mechanical designs in mind, but more fields are adopting it and broadening the term "open-source hardware" to mean anything physical that has public source files.[16] Open-source hardware has been applied to fashion, furniture, musical instruments, farm machinery, bioengineering, and much more. The following use cases provide an overview of projects crossing the humanitarian, education, and public sectors.

Humanitarian/Disaster Relief: Tolocar

The word *toloka* stands for a traditional form of solidarity support in Ukraine. With that in mind, the Tolocar project uses mobile makerspaces equipped with tools, open-source hardware machines, and consumables to support humanitarian activities in Ukraine (figure 1). German and Ukrainian partners, with support from the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ),[17] co-created the Tolocar project just a few weeks after the start of the full-scale Russian invasion of Ukraine.[18]



Figure 1. A Tolocar is a converted van that offers the possibilities of a makerspace or "fablab" on wheels. Photo: Nika Popova.

Education/Clinical: OpenFlexure Microscope

The OpenFlexure project uses 3D printers and off-the-shelf components to build open-source, lab-grade microscopes for a fraction of traditional prices. The OpenFlexure Microscope was designed to enable low-volume manufacturing and maintenance by local makers, and over 100 microscopes have been produced in Tanzania and Kenya for educational, scientific, and clinical applications (figure 2).[19]



Figure 2. An OpenFlexure microscope, built with 3D printers and off-the-shelf components.

Public/Sustainability: Appropedia

Appropedia supports the development and sharing of collaborative solutions in sustainability, poverty reduction, and international development through the use of sound principles and appropriate technology, original research, and project information. Appropedia is a wiki that provides a way to contribute and curate open-source hardware project files. Project areas include photovoltaic systems, pedal power generators, water testing systems, and more (figure 3).[20,21]



Figure 3. A UV-based water filter created from Appropedia open-source hardware files. Photo: Vigyan Ashram, CC BY-SA 4.0.

Broader Societal Impact of OSH: Sustainable Fashion in Focus

The textile industry faces many challenges today, sustainability being one of the most pressing issues; it is one of the most polluting industries, responsible for the equivalent of over 4 billion metric tons of carbon dioxide per year, more than the aviation and maritime industries combined.[22] In response to an outcry from fashion industry community members,[23] there are programs, centers, and new technologies that are being developed at accelerating rates to address this growing issue.

In the fashion industry, open-source hardware represents solutions to reduce the waste impact, which is essential to mitigate risks.[24] Open-source hardware machines and tools are bringing new innovative ways of transforming production and manufacturing waste. By going open-source, fashion brands can hold greater control over their production via collaborations for product design, marketing, and fabric innovation.[25] Organizations such as the TextileLab of Amsterdam[26] offer interdisciplinary teams of engineers, designers, and artists to collectively research and educate makers in the fashion industry on the reduction of waste in production.

Researchers at the ATLAS Institute at the University of Colorado Boulder[27] have developed an open-source machine that spins textile fibers made of materials like sustainably sourced gelatin. The gelatin used in this work is waste produced by the meat industry that does not meet requirements for cosmetics or food products; instead of being thrown away, this waste material is being upcycled into a wearable product. This technology can, in turn, be used to dissolve any textile made of gelatin (when you don't want to wear them anymore) and recycle it to make more fibers.[28]

OpenKnit,[29] now kniterate,[30] is an open-source digital fabrication tool that can create ready-to-wear clothes from digital files, like knitting a jumper in under an hour. Anyone can access the instructions on how to build the machine via GitHub,[31] reproduce it, and help to improve it. This project is revolutionizing the fashion industry by democratizing clothing manufacturing and empowering the potential of locally made fashion.

Massachusetts Institute of Technology's (MIT) Fashion Innovation Hub[32] brings together researchers, students, and leaders of the industry to develop ideas and technologies to reduce industry carbon emissions. They are developing technologies for fibers, fabrics, and related applications across the textile supply chain. The goal is to promote research, education, and innovation in support of the textile industry. They are organizing competitions, such as the MIT Climate Grand Challenges, for projects on decarbonizing the industry and developing new smart textiles and wearables.[33]

An example of an educational program is Fabricademy's From Fibers to Fabrics,[34] which is dedicated to viable, sustainable, and fair alternative systems to change the textile and fashion industry. Fabricademy's Textile and Technology Academy is a multidisciplinary program at the intersection of digital fabrication, textiles, and biology (figure 4).[35] In this program, people are learning to update obsolete machines for producing textiles and take advantage of digital fabrication and open-source technology to come up with open-source machines for knitting and weaving and to invent new techniques. Fabricademy's mission is to reshape and redefine the implications and applications of technology in the textile and clothing industry to benefit the future of the industry with themes covering personal fabrication, distributed manufacturing, Industry 4.0, wearable technology, biofabrication, assistive technologies, and sustainability.[36]



Figure 4. Fabricademy uses digital fabrication and open-source technology to build open-source machines for knitting and weaving. Photo: Harshada, 2019, CC BY 4.0.

With these examples of open-source models and technologies, we can envision a fashion market where brands would disclose how and by whom their products are made and when textiles and garments have been created to be innovative, recycled and repaired, on-demand, locally produced, and better for the environment. However, the adoption of open-source models in the fashion and textile industry is relatively slow. The dominance of fast fashion based on mass production is still very much favored despite the terrible impact it has on labour and on the environment.[37] To rapidly create a change, open-source technology needs to be brought into the mainstream of production and manufacturing.[38]

Who is Working on OSH? Communities in Focus

Much like open-source software, OSH is driven by communities of enthusiasts, engineers, and makers who collaborate to create accessible and customizable technology. OSH communities are united by principles of transparency, accessibility, and collaborative innovation.[39] These communities strive to democratize technology by reducing barriers to entry and fostering a culture of shared knowledge.[40] This approach has led to rapid advancements in fields such as prosthetics,[41] environmental monitoring,[42] and low-cost scientific equipment.

These communities are diverse, spanning various fields such as electronics, 3D printing, robotics, and scientific instrumentation. They often organize around online platforms[43], maker spaces[44], and events[45] where members share designs, offer support, and collaborate

on projects. Notable examples include the Arduino community,[46] which focuses on microcontroller boards, and the RepRap project,[47] which aims to create self-replicating 3D printers.

It is well known within the global OSH community that the ecosystem is very large and vastly under-documented. Some of the prominent maker initiatives and communities working in this space, such as Appropedia, the Internet of Production, the Open Source Hardware Association (OSHWA), and Global Open Science Hardware (GOSH), have collaborated to map these sprawling ecosystems (figure 5).

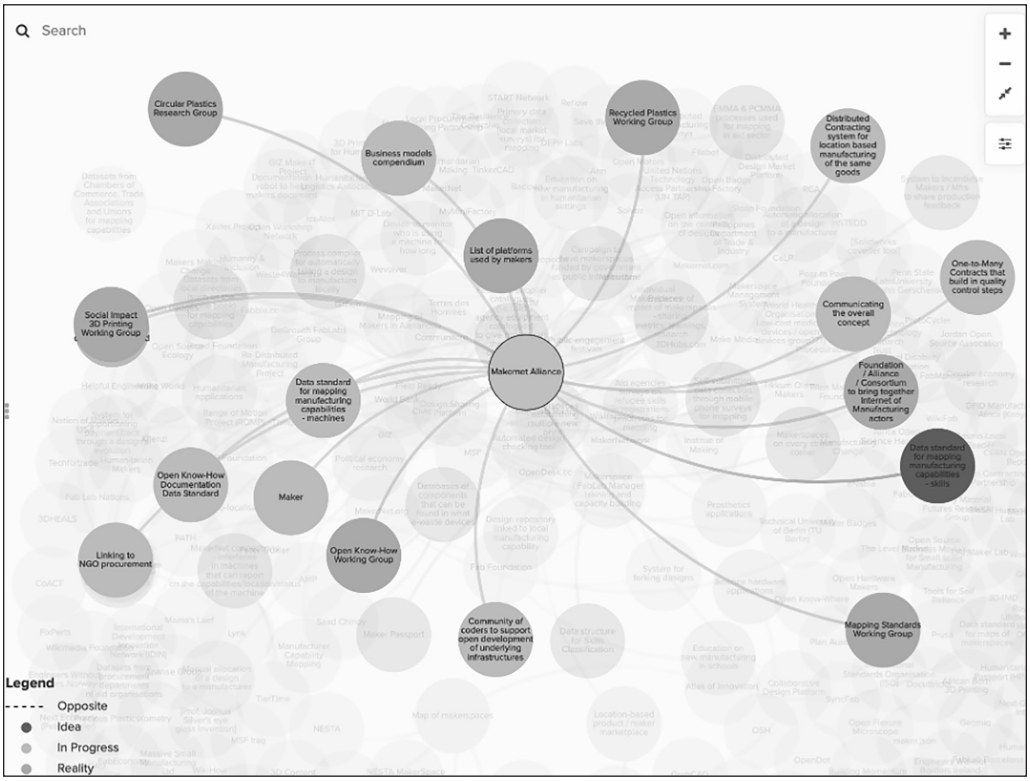


Figure 5. This map of decentralized manufacturing nodes established in 2019 by a group of global community stakeholders has 334 points of interest with overlaps indicated by types of events, specific sector(s), dataset types, and more. A. Lowe, A. Lamb, 2019, CC BY 4.0.

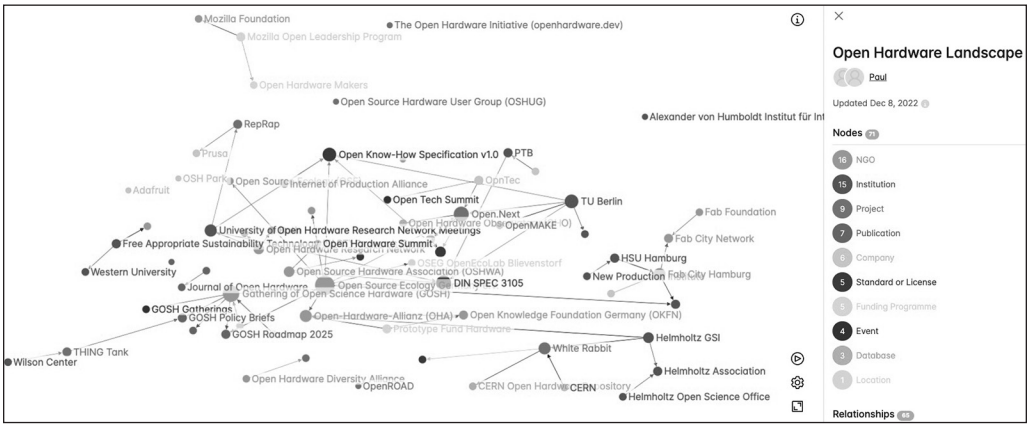


Figure 6. The open hardware landscape created in 2019 illustrating nodes by institution, project, company, and more. Graph Commons.

The visualization in figure 6 of the open hardware landscape was created in 2019 and captures nodes categorically by institution, project, company, and more. While it is recognized that the two inventories of global OSH nodes (figures 5 and 6) should be combined, just as it is challenging to establish a standard by which OSH product files are shared, a standard ontology for how this information should be collected and represented has not been established either. This lack of consensus at the global ecosystem level can make governance, consensus-building, and decision-making across these myriad OSH communities challenging. However, there are many examples of organizations of OSH communities coming together to establish powerful networks, which are moving the larger global conversation in the right direction.

Sharing Our Experiences: Africa in Focus

Africa Open Science and Hardware (AfricaOSH) is a pan-African community of makers, researchers, and educators dedicated to advancing open science and hardware across the African continent.[48] The community was established in 2016 as a response to the need for more inclusive and accessible approaches to science and technology in Africa and currently has over 256 members.

The AfricaOSH community fosters open and collaborative scientific research and innovation methods, focusing on democratizing access to tools and technologies that address local challenges. The community also focuses on capacity-building in open science and hardware, supporting local entrepreneurship, and fostering international collaborations. AfricaOSH's vision is an Africa where hardware designs and blueprints are accessible to everyone to create an ecosystem of shared knowledge, ideas, and resources. AfricaOSH's goal is to create a set of actions and conversations around open science hardware (OSH) among African actors and the international community to adopt OSH principles and practices appropriate to our African context.

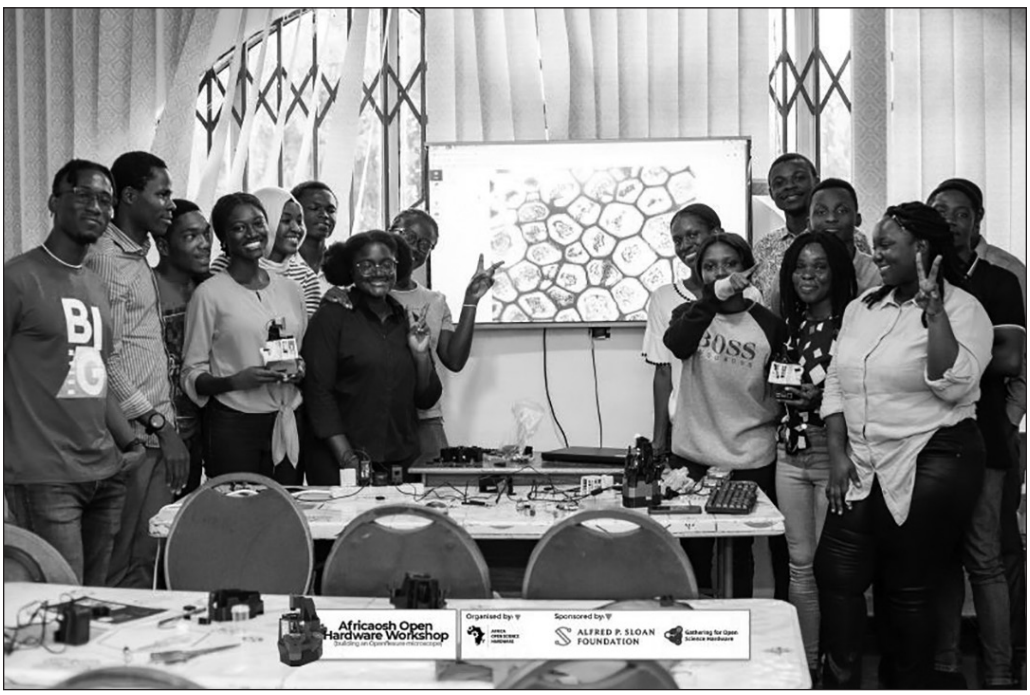


Figure 7. An open hardware workshop in Ghana, June 16, 2023. AfricaOSH, CC BY 4.0.

To achieve these goals, AfricaOSH organizes workshops, training sessions, and hackathons to bring together individuals from diverse backgrounds and disciplines to share knowledge and skills (figure 7). The community provides online resources, including a directory of open science and hardware initiatives in Africa, a forum for discussion and collaboration, and an open-source repository of projects and designs. Members are engaged through a webinar series where experts are invited to discuss and share knowledge and ideas on new trends in open science.



Figure 8. Poster for AfricaOSH, an annual event at which interested parties discuss open science opportunities in Africa. AfricaOSH, 2024, CC BY 4.0.

The largest event is the annual AfricaOSH Summit. The summit is a regional gathering where researchers, technologists, hacker hobbyists, educators, government officials, start-up innovators, and partners from Africa and around the world come together to discuss the various opportunities open science offers to the African continent (figure 8). The Africa OSH gathering aims to achieve an ecosystem for research and innovation that is locally adaptable, culturally relevant, technologically feasible, economically viable, and environmentally sustainable.

Africa faces a myriad of issues, from a lack of teaching and learning materials for STEM students to shortages of essential medical equipment. The absence of policies makes it difficult to promote open science hardware. Through collaboration with community members and other global partners, this community seeks to overcome these barriers and make OSH ubiquitous by the year 2025, in line with the vision of the Global Open Science Hardware (GOSH) vision.[49]

Sharing Our Experiences: Latin America in Focus

Global Open Science and Hardware for Latin America, or Rede Americana de Tecnologias Livres (reGOSH), is a network that connects people, projects, communities, and institutions in Latin America that are working on open-source technologies for science and education.[50] The purpose of this community is to democratize the access, building, and sharing of scientific knowledge through the promotion, adoption, and development of OSH technologies. The origin of reGOSH is linked to an ongoing discussion inside the GOSH community, which began

during the 2017 gathering, regarding the possibility of creating residencies (this is where the "re" in the name came from), where people could meet and continue working within a space and share resources for longer collaborations on specific projects.[51]

Over the course of 2019–2023, with the support of Programa Iberoamericano de Ciencia y Tecnología para el Desarrollo,[52] the reGOSH network planned and organized three residencies with themes linked to common social and environmental issues, including natural resources extractivism, environmental pollution, and the severe difficulties of accessing health and education services. The first residence took place in 2019 at Porto Alegre, hosted by the Centro Tecnologia Academica from UFRGS, during which twenty-six participants from six countries gathered for fifteen days to learn about open-source technologies and work on several educational projects.[53] In 2022, the second residence was organized around tools for agroecology and environmental health in Mendoza[54] because of the ongoing collaboration of network groups with local communities already engaged in environmental monitoring, such as CoSensores[55] and Preservamos.[56] A transition toward agroecology was also supported by the Open Agroecology Lab.[57] This three-week residence was followed by an open gathering with more than 100 participants sharing workshops and talks about the tools; attendees were also able to broaden the conversation around the role of open scientific tools and communities in a larger regional discussion related to environmental health and agroecology movements. As a result of this residence program, a community GitLab repository was created to host the documentation of tools and projects.[58] This residence allowed for the consolidation of existing networks and the expansion of project reach and connection between regional initiatives in the global OSH scene.

As a continuation of this residence, in mid-2023, there was another small gathering called reGOSHito in Buenos Aires,[59] which centered around scientific tools for participative environmental monitoring. This event was a cross-community effort, funded by GOSH regional event funding and hosted by CoSensores. By the end of 2023, the last of the initially planned residencies took place at the Estación Patagonia UC in the Aysén Region in the southern part of Chile. This residence was organized and hosted by the Laboratorio de Tecnología Libre[60] and focused on the development and use of an environmental monitoring system. This residence allowed intensive interactions and the opportunity to test methodologies and devices in extreme conditions.

In between residencies, the network developed several other activities as the "Circuito reGOSH," a series of online talks and webinars organized during pandemics by CTA, and the collaboration of the Laboratorio de Tecnología Libre with the Interspecifics collective in Santiago de Chile[61] for the Speculative Communications[62] audiovisual presentation that was created from patterns and behaviors in bacteria, extracted with computer vision and machine learning techniques from microscopic observations with open source hardware devices. There were also many workshops around the educational use of OSH for educational purposes (e.g., for molecular biology) and some larger initiatives to develop hardware with a commercial profile, like the Pipettin Robot funded by the GOSH collaborative development program.[63]

Since its inception, ReGOSH, as a network, evolved from an institutionalized form of organization by country and a central coordination enforced by CYTED funding to a much more inclusive model. ReGOSH is now a cohesive community that collaborates on different initiatives and activities organized by community members, groups, or projects, but that share the strong imprint of working with communities outside academia to address real regional problems and needs. As it continues to evolve, this network serves as a model to open new possibilities for the broader global OSH ecosystem.

Solving Complex Challenges: What's Next for OSH?

People are still suffering and dying *unnecessarily* during periods of crisis, and millions of people around the world still do not have access to the barest of essentials for human survival on a daily basis. Engineering has already solved problems that prevent people from accessing the essentials to survive and have a good life; the tools and technology to enable geographically dispersed production on a small scale by community members to address urgent, localized demands exist—they are not on some distant horizon. And yet, even with these tools and technologies available, healthcare workers are not receiving life-saving vaccines and personal protective equipment in time to prevent the spread of disease. Refugees displaced by natural disaster or political discord do not have housing, access to an adequate food supply, or clean water. Small business owners whose livelihood depends on operable machinery spend *months* waiting (if they receive them at all) for replacement parts for equipment, without which they cannot plow fields, produce clothing, and make other essential goods for their communities.

By openly sharing designs, specifications, and manuals at different scales (city, regional, national, or global), communities can produce their hardware locally, using local resources and skills. Local production of OSH designs has the potential to reduce transportation and logistics costs, lower the carbon footprint of production, and support the development of more sustainable local economies. Communities created around open-source hardware projects are helping global communities to adapt and overcome issues by providing access to technologies that are otherwise enclosed by initiatives that are too big and cost-intensive to implement at the local level. By putting the tools to design, iterate, and prototype products into the hands of community members, rapid collaboration, new revenue streams, and sustainable business models become achievable for entrepreneurial stakeholders in a global manufacturing ecosystem. Increased access also means that products being made are manufactured with fresh community knowledge and skill, which means better products for our society as well as increasingly diverse engineering talent that will appeal to industry and larger manufacturing companies. *So with the benefits of OSH made so clear—what's the problem? Why isn't everyone doing it?*

Policy Change and Incentivization are Key

OSH is critical to producing new knowledge; it grants researchers more autonomy over the devices they use to collect data and draw scientific conclusions. However, the inclusion of open hardware in open science agendas is often overlooked. While important documents, such as the UNESCO Recommendation on Open Science,[64] include open-source hardware, many open science programs focus their efforts on the final outputs of the research process—namely, data and publications.[65] To foster the inclusion of open hardware and further innovation, there are several measures that can be taken within an academic context.

Incentivizing researchers to publish their hardware designs with an open license is key; a 2023 report reviews strategies and concrete steps that academic institutions can take to better support OSH.[66] Research institutions and funders should also support open science hardware. This would include legitimizing the role of OSH within the broader open science paradigm, creating and enforcing documentation standards, and supporting relevant business models for OSH. In the public sector, working in the OSS and OSH communities is not a career that is well-paid. Those involved do not get the same benefits and support provided by the academe; open source organizations need to strive through difficulties and be sustained, but they often start up then disappear, such as the aforementioned OpenKnit project, or become corporatized; professional support, funding, and stability can be difficult to come by for OSH and have long been a topic of conversation in open-source software development and sustainability.[67]

Conclusion: Recommendations for Next Steps: OSH Cannot Wait

While increased cross-sector collaborations between the academic and public sectors may move us in the right direction, there are no easy answers. Over the course of our combined years of work in the OSH community, we have identified that there needs to be a stronger focus on governance, communications, and community development to help a global ecosystem thrive across sectors. A great deal of our work in OSH is subsidized by grants and temporary funding (or "soft money"), and, unfortunately, many of the grants for innovation in OSH fall within the discipline of engineering or technology and focus on the development of a product or completion of a project, as opposed to the human networks and ecosystem necessary to sustain the work and help the OSH community to thrive.[68] This approach of solving a community-based problem with techno-solutionism will not solve complex global challenges in the longer term.

To achieve transformational change, we need to approach the global adoption of OSH with a focus on not replicating old systems. While there is much to do in support of broadening and sustaining the global OSH ecosystem, we recommend a focus on three primary areas: 1) increased focus on organizational governance; 2) higher education and community partnerships; and 3) provision of economic sovereignty.

While it is recognized that governance frameworks and inter-organizational cooperations serve as predictors for performance outcomes in multiple sectors,[69] additional work is needed to establish effective governance mechanisms necessary for adoption, particularly for quality assurance of open devices produced,[70] for which new tracking systems are being developed.[71] This governance can take the form of community advisory councils, like that which GOSH has assembled,[72] or by establishing product certification programs like OSHWA's.[73] These types of systems for decision-making processes take a larger amount of time investment and support from community members, and, as noted earlier in the chapter, the human element of OSH networks is still grossly underfunded. It is encouraging, however, that organizations such as Code for Science and Society and the National Science Foundation (NSF) are beginning to recognize the necessity to support human relationship development in open ecosystems, such as that supported by Digital Infrastructure Incubator[74] and the Pathways to Enable Open-Source Ecosystems (POSE) program.[75]

Adoption of OSH in academia is on the rise,[76] and this trend is encouraging with products and programs that are gaining recognition, such as the OpenFlexure microscope, which got its start at the University of Bath and University of Cambridge.[77] The benefits of OSH and academic collaborations include greater access to lab space for product testing,[78] the opportunity for increased support in product development through course partnerships,[79] and, hopefully, more opportunities for funding.[80]

However, funding by way of partnership with an academic institution, foundation, or government can also come at a high price; overhead in administration of grants, intellectual property assertions, and exertion of control over program or product direction can all become issues that have the potential to depreciate OSH initiatives.[81] For the open-source hardware community to have full latitude and the agency to maintain community-centered development, new economic models like Grassroots Economics' Commitment Pooling provide an opportunity to compensate community work in OSH outside of volatile currency markets.[82] Once again, it comes back to compensation for community and social engagement—and who is willing to take on the cost?

Developments and increased adoption of OSH cannot wait. We need to match pace with the rapidly increasing threats to humanity we are experiencing as a global community due to climate change, expanding wealth gaps, and political tumult. These issues disproportionately impact the most vulnerable global communities, and the answer is in the hands of our collective network to democratize access for the benefit of humanity.

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CHAPTER 24

A License is Not a Commons: The Creative Commons, Common Pool Resources, and Commoning as a Set of Practices

Jennie Rose Halperin

A few years ago, copyright scholars Kevin Smith and Sara Benson debated whether "U.S. copyright law no longer meets its fundamental purpose to serve the public good" at the Charleston Conference's annual Hyde Park Debate. "Copyright has now outlived its usefulness because in the age of digital technology, creation is less expensive and distribution is essentially free," stated Smith. "Copyright has failed everyone and has thrown up needless obstacles to creators; it cannot be saved by tinkering with it."[1] The Creative Commons licenses, now more than twenty years old, were originally conceived of as a legal "fix" to these digital copyright issues. With billions of works under the licenses, it is worth considering what they actually do and whether they contribute to the development of a global, community-centered commons or whether the licenses are part of a copyright regime that is, in Smith's words, "creating an insurmountable obstacle for users."

From 2016 to 2019, I worked as the senior communications manager at Creative Commons. While there, I engaged with a number of communities that relied on Creative Commons licenses or saw themselves as part of a commons community. I was exposed to scholars like David Bollier, activist artists like Howlround Theater Commons and Double Edge Theater, and scholarly communications communities like Knowledge Commons. But while all of these artists and activists used Creative Commons in their work, the licenses themselves often felt immaterial to the work of community organizing inherent to commons projects, as well as the definition of "commons" in both the economic sense and the social sense.

An oft-repeated maxim in copyright scholarship is that copyright does only one thing: it provides an exclusive legal right for distribution by rightsholders. Trying to wedge other issues (social justice, equity, solidarity) into copyright is a sure road to failure. While the internet has

CC BY

blurred some of the bundle of rights protected by copyright, most disillusionment with open copyright on the internet is largely economic in nature rather than an issue with copyright itself, or "Why pay for something you can get for free?" The founders of Creative Commons understood the distribution issues inherent in digital creation and promoted Creative Commons as a way for creators to maintain "flexibility and control" of their work in order to adapt to new technology. But the internet has changed over the past twenty-five years, and as the web platformed, the flexibility and control of Creative Commons began to feel more like "free" rather than the imagined convivial content commons that would inevitably spring up around the licenses as hypothesized by the Creative Commons founders. Early Creative Commons adopter Molly Crabapple sums up the prevailing mood among many artists in 2023: "I wish I never gave some of my work on Creative Commons. I wish I could take it back. God, I despise them … [a] friend once told me … the intellectual problem is [that] they think that the good guys still have the internet. But the good guys lost it."[2]

Lawyers, Economists, Copyright, and Common-Pool

Creative Commons, a content license managed by a North American private nonprofit, intends to provide an alternative to a restrictive copyright regime. Founded in 2001 by Lawrence Lessig, Hal Abelson, and Eric Eldred, the intention of the licenses is to augment traditional copyright's bundle of "all rights reserved" with a more flexible "some rights reserved" license. Under Creative Commons, the creator keeps their copyright but can layer personal choice onto the content, including the right to remix or reuse, with attribution. Creative Commons licenses draw from contract law and are a lawyerly attempt to update copyright law for the digital age of reproduction and remix. Since the beginning, Creative Commons has seen its role as protecting the digital commons, even though the licenses themselves and the work created under the licenses do not strictly meet the criteria of commons in either definition or management.

The modern study of the commons is due in large part to the work of the pioneering Swedish economist Elinor Ostrom, who won the Nobel Prize in 2008. Ostrom defines common-pool resources as resources that have a high level of *subtractability*, meaning my enjoyment of a resource detracts from another's, and a low level of *excludability*, meaning the man-made or natural resource is sufficiently large to make it costly to exclude others. However, as Johannes Euler writes, Ostrom intentionally did not distinguish between the terms common-pool resources and commons, and there has been an ongoing debate as to whether or not the costliness of exclusion is based on social relations rather than inherent attributes of the good itself. Euler goes on to write that commons are "some tangible and/or intangible matter (resource/product) [and] some sort of social infrastructure, some specific institutional arrangement."[3] In the case of Creative Commons, the licensed works do not meet the criteria of subtractability, but they do meet the criteria of non-excludability, in line with all knowledge commons, as explained below.

To provide an example, a public library is a common-pool resource but not necessarily a commons. While anyone can enter and use a library, if I borrow a printed book from the library and use it, that book becomes unusable by anyone else. In her work on "common-pool resources," such as fisheries, which are subtractive (meaning that fishing depletes the fishery) but non-excludable (everyone can ostensibly fish the waters with the right equipment), Elinor Ostrom identifies three major "grim" social dilemmas within these kinds of goods: free riding, competition for use, and over-harvesting.[4] These dilemmas are social rather than economic in nature and therefore not inherent to the good itself. All of these social dilemmas exist within the Creative Commons community, with free-riding being the most pronounced.

To further expand on the issue of social dilemmas and to illustrate why a public library is not a commons but is a common-pool resource, Ostrom outlines eight rules for managing the commons, including boundaries, adaptation to local contexts, participatory decision-making, monitoring, sanctions for abusers, conflict resolution, legal status and the right to organize,

and the ability to be embedded into larger networks. None of these principles addresses size, but nearly all examples of commons take place in small, intimate, hyper-connected communities. While the resource pool must be large, the management of its community must be intimate enough to encourage social relations. Within the Creative Commons, the primary rule of management appears to be in its ability to be embedded into larger networks. None of the other management rules are addressed by either the licenses or the organization itself, though, as will be later illustrated, some communities choose to build these management rules on top of the licenses.

Ostrom's reconceptualization of common resources was innovative largely due to its focus on *management of goods* rather than *type of goods.* Since its founding, Creative Commons has focused on the growth of the licenses rather than the organization of communities around them, and the internet has changed around the licenses. The majority of digital work now lives on platforms, which the organization initially embraced as a way to quickly grow the licenses. But a collection of free resources on a corporate platform (like the CC BY collection on YouTube or Elsevier's open access journals) does not create a commons. Simply labeling a work as Creative Commons or "open" on a corporate platform does not take into account the social relations that are necessary in building a commons.

An example of this is a 2019 scandal regarding the "Diversity in Faces" dataset. This dataset was compiled by IBM Research using images from Flickr in the YFCC-100M Creative Commons dataset. The YFCC-100M dataset is the largest publicly available multimedia collection, with metadata from 99.2 million photos and .8 million videos, all licensed under Creative Commons.[5] When the dataset of faces was released, IBM framed it in the language of social justice. "Most people generally agree that a fairer world is a better world, and our AI researchers couldn't agree more. That's why we are harnessing the power of science to create AI systems that are more fair and accurate."[6] Needless to say, the Creative Commons photographers did not see this as a social justice issue, other than one that violated their right to privacy. In an article for NBC News, Olivia Solon quoted Meredith Whittaker of the Signal Foundation as saying, "People gave their consent to sharing their photos in a different internet ecosystem.... Now they are being unwillingly or unknowingly cast in the training of systems that could potentially be used in oppressive ways against their communities."[7] In an article for CNN, librarian and technologist Jessamyn West expressed her displeasure at her inclusion in the set and said, "I think if anybody had asked at all, I would have had a different feeling."[8] At the time, CEO Ryan Merkley wrote, "Copyright is not a good tool to protect individual privacy, to address research ethics in AI development, or to regulate the use of surveillance tools employed online."[9]

While this story can be seen as an example of corporate free riding on a public good, it is also an example of the ways in which a lack of governance structure or commoning practice left a community of Creative Commons users with little recourse to address harm. The photos were initially scraped because they did not require licensing fees for use, but when their use became public and monetized, the organization and users were left without any tools for conflict resolution, sanctions for abusers, or monitoring. West's comment that she hoped "someone would ask" is illustrative—if the photos were in an actual commons, IBM would have had to abide by community as well as legal rules.

Attempting to retrofit copyright into a commons framework is well-intentioned, but more than twenty years in, it remains unclear if the licenses work as intended. For example, in his 2007 law review article, Michael Carroll outlines how the licenses are created to be "conversational" due to their "some rights reserved" framework. He writes, "Widespread use of [Creative Commons licenses] would create a commons, a pool of resources open to all to use in conversation and self-expression."[10] In an early video called "A Shared Culture," Creative Commons founders discuss repeatedly how the licenses will increase "sharing."[11] The sentiments make sense conceptually, but as the licenses have become more ubiquitous, particularly in scholarly communication, the imagined commons of twenty years ago is certainly not a conversation, and it is unclear if it is the function of the licenses or the social structures around them that have increased sharing.

The Special Case of Knowledge Commons

While common-pool resources have historically been characterized by their excludability and rivalrousness, knowledge commons present a special case that was theorized by Ostrom and her collaborators, leading to a sea change in the way that knowledge is produced and consumed on the web. Beginning in the mid-1990s, the classification of knowledge as either a public good (non-excludable and non-rivalrous) or as a commons began to take shape, due in large part to the rise of the internet as neither a purely public nor purely private resource and internet content as non-rivalrous due to its reproducibility. Scholars, including Ostrom, Joseph Stiglitz, and Yochai Benkler, were instrumental in considering how both digital goods (like the internet and open-source software) and digital content (like digital art or media) could be reconsidered as commonly managed goods rather than falling under the jurisdiction of the market.[12] In *Understanding Knowledge as a Commons: From Theory to Practice,* Charlotte Hess and Elinor Ostrom write that the primary threats to knowledge commons are commodification or enclosure, pollution or degradation, and non-sustainability.[13] But the issue of subtractability remains at the core of defining knowledge as a commons; most common-pool resources (like fisheries) are highly subtractive, while knowledge commons present the opposite paradigm. The more people use the knowledge commons, the more educated the people, and the more educated the world, the better the outcomes! The issue of non-subtractability presents a new set of issues within knowledge commons and significantly changes the management structure that leads to their success.

The Creative Commons licenses were born from this academic milieu. Advocates assert that rather than creating a commons themselves, the Creative Commons licenses provide the framework upon which to build a knowledge commons.[14] This assertion has often been taken for granted in open knowledge communities and has formed the basis for open access content, particularly access to scientific literature. As Frischman et al. write in *Governing the Knowledge Commons*, "writing about the commons approach is often conceptual or political, using the idea of commons as a rhetorical device to oppose the expansion of intellectual property protection."[15]

Creative Commons as Commons

As communications manager, I wrote the Creative Commons organization's annual State of the Commons report for several years. This now-retired annual report estimated the total licenses at 1.4 billion works in 2017, up from 140 million in 2006.[16] The organization's current project, Quantifying the Commons, estimates the number at 2.7 billion in 2024.[17] But bigger is not necessarily better when it comes to common-pool resources.

There are many examples of knowledge commons built on the Creative Commons licenses: Wikipedia, Humanities Commons, OER Commons, and Arxiv, to name only a few. While the Creative Commons license ostensibly enforces the norms of attribution, share-alike, non-commercial, and other prosocial behaviors, it is unclear whether it is the license that enabled these commons or if individual community norms around scholarly communication and fact sharing evolved alongside but not due to these licenses. The phrase "global digital commons," which crops up in much of the early literature, feels vintage, like "World Wide Web" or "free as in beer."[18] As an illustration of the license's reach, in a 2018 study, it was estimated that 72 percent of all scholarly journal articles remain closed. The paper does not specify license type (Creative Commons or otherwise) in evaluating the 28 percent of open access articles, instead adopting the 2002 Budapest Open Access Initiative definition.[19] In the Directory of Open Access Journals, which indexes more than 21,000 journals, nearly all are under some type of CC license.[20]

For those 28 percent of articles, it is worth considering whether the prosocial behavior—which is gated under "green," "gold," "bronze," etc.—is enabled by the Creative Commons

license, or rather the human imperative to create solutions for sharing knowledge outside of the market or state as supported by the internet. This human urge, as typified by sites like SciHub, could be called "pirate commons," where user-submitted content exists in the gray (or black) zone of legality, to push the limits of sharing and collective resistance in light of unjust market regimes. As the widely circulated "Introduction to Pirate Care" states, "There is a need to revisit piracy and its philosophical implications—such as sharing, openness, decentralization, free access to knowledge and tools … in the light of transformations in access to social goods brought about by digital networks."[21] Further, to quote Creative Commons co-founder Aaron Swartz's seminal "Guerilla Open Access Manifesto," "It's time to come into the light and, in the grand tradition of civil disobedience, declare our opposition to this private theft of public culture.… With enough of us, around the world, we'll not just send a strong message opposing the privatization of knowledge—we'll make it a thing of the past."[22]

What the licenses do effectively is take these knowledge commons out of the legal black or gray and into the legal, mainstream middle, effectively removing risk from a copyright regime that has become increasingly restrictive. This protective impulse is highly positive for sharing online—enormous amounts of online content have been made legally accessible through the Creative Commons licenses. But the internet still is, in many ways, a pirate utopia. The sharing of copyrighted work rather than the widespread adoption of CC licenses is still the norm in most media twenty years later. While Creative Commons has allowed large collections of content to flourish without legal challenge, this legal use remains its primary utility due to its entrenchment in copyright law and contracts. In another universe, global copyright regimes could have been fundamentally changed so that infringement was no longer a threat to sharing on the web. But as that seemed generally out of reach, the Creative Commons provides necessary legal cover.

The book *Made with Creative Commons,* released in 2017, examines twenty Creative Commons case studies. In the book, the authors utilize the commons as a framework, but it is unclear what the licenses do for each of these organizations other than act as a copyright license. "Their use of Creative Commons involves them in the social practice of commoning, managing resources in a collective manner with a community of users."[23] Though the authors' interviews are illustrative of a variety of different business models, it remains unclear what it is about the *license* that enables this social practice, other than shielding the user from the threat of copyright litigation, which is no small contribution.

As Ostrom and Hess wrote in 2006, "Commons analysts have often found it necessary to differentiate between a commons as a resource or resource system and a commons as a property-rights regime."[24] While correlation does not equal causation, the definitional and discursive collapse came about at approximately the same time and were likely influenced by each other as knowledge commons and the internet became more prominent. In the view of early knowledge commons theory, knowledge commons are not independent of markets based on exclusive rights, but they are also not subservient to that market. In knowledge commons, this conceptual approach develops a definition of property rights that is based on "access, democratic guardianship, and convivial abundance."[25]

To draw from *Made with Creative Commons,* the authors write about Arduino, "The CC BY-SA license means anyone can produce copies of their boards, redesign them, and even sell boards that copy the design."[26] This is in line with Ostrom's assertion; rather than a resource system, the CC BY licenses are a property-rights regime that enables duplication and remix. The CC licensed content did create a resource system built on an alternative property-rights regime, which has resulted in a large online community of users. Like with many knowledge commons, the license clarified the legality of duplication and remix. But beyond this clarification, it is unclear whether the use of CC licenses promoted prosocial behavior or if the pro-social behavior would have taken place with or without the license, as long as the threat of litigation was mitigated.[27] To provide another example, *The Conversation,* a news site, uses a CC BY-ND license in order to promote sharing of their content. But the license itself does less heavy lifting than the organizational imperative to build a community. On the internet, copyrighted articles

are often freely shared and republished as long as the copyright owner does not threaten litigation.[28] The value is in the management of the good rather than its individual license. This hairsplitting is not meant to undermine the usefulness of Creative Commons licenses, nor to discredit the claims of the authors of *Made with Creative Commons.* The licenses serve as a useful signal and clarification in many contexts, but it is worth considering the value of the license as independent from its social context.

The dominant utility of CC licenses as signaling was understood by scholars and advocates early in their genesis. In *Understanding Knowledge as a Commons,* multiple authors assert that the value of digital knowledge commons resides in their use of signals and discourse in order to establish a more open property rights regime that sits between the market and the state rather than in their actual transformation of property rights regimes as a whole. While many of the examples in *Made with Creative Commons* provide examples of how small-scale digital knowledge commons have been nurtured and guarded over the organization's history, it is clear that the knowledge economy has unfortunately not shifted into the world of convivially abundant property rights based on commoning principles.

While Creative Commons has absolutely changed the landscape of content on the web, its legibility and legality may have worked to its detriment. Commons flourish when focused on people, not products, and while a content license can provide a legal platform, copyright's focus on individual sovereignty and ownership means that the socialization of the resource is ultimately up to the rightsholder or creator. Adopting a more commons-based approach to the sharing of content on the web that pushed the boundaries of copyright law and focused on the management of the resource rather than its type and expansion may have ultimately created a better web, though it would have come with the inherent stickiness of community organizing, which necessarily means engaging with people and resource management.

Toward an Interdisciplinary and More Creative Commons

While there are exceptions, lawyers do not generally need to make a living from their creative work. The lawyers who created the Creative Commons licenses attempted to make the legal structure more equitable for creators by overlaying a level of control, but because they were based in copyright rather than social organizing or economic realities, the licenses ultimately did not build an adequate set of social practices that would support a "commons" structure.

Building a true "creative commons" would require an interdisciplinary approach that supports the flourishing of human creativity with a focus on resource and human management. It would address Ostrom's "grim social dilemmas" and borrow from modern scholarship on commons and co-creation. A commons of creation would mean allowing one thousand creative commons to grow, each with its own set of shared values and organizing principles based around a rich set of social relations. It would address economic realities and be both intersectional and interdisciplinary in nature. In many ways, the organization has already enabled this vision: hundreds of individual projects self-manage open resources while utilizing Creative Commons licenses, even though the organizing and self-management are not necessarily inherent to the license. While Creative Commons may not offer a comprehensive solution or vision for commons, this foundation is an enormous contribution to content sharing and community organizing on the internet.

Twenty-three years after the conception of the Creative Commons, it is difficult to know what would have happened if the organization and movement had pursued a commons strategy in addition to its legal approach. At its founding, the organization could have further focused on the maintenance and management of sustainable, human-centered commons by engaging with professionals in a variety of fields beyond the law. They could have focused on community and curation rather than the growth and adoption of the licenses as a tool or standard. Rather than attempting to eliminate the legal gray zone of the early web's copyright,

they could have created a completely new paradigm that was more decentralized, messier, and perhaps riskier than the path they followed. Elinor Ostrom writes in *Governing the Commons*, "Organizing is a process; an organization is the result of that process."[29] Organizing the commons necessitates a movement focused on how we manage ourselves, our resources, our work, and our communities. With the puzzle pieces of a global commons framework in place, we can focus our energy on building a commons that envisions another world, built on a license, a community, or an entirely other framework.

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CHAPTER 25

Shining Bright: Diamond Open Access Publishing in UK Libraries

Rebecca Wojturska

Introduction

Diamond open access (OA) is a publishing model that has been utilised for decades under various names, including "platinum." However, the term "diamond" appears to have been coined by Marie Farge in an email to Jean-Pierre Bourguignon in 2012[1] and the definition built upon by Christian Fuchs and Marisol Sandoval in 2013 to "make a conceptual differentiation" from "corporate open access."[2] Fuchs and Sandoval define diamond OA as where "not-for-profit, non-commercial organizations, associations or networks publish material that is made available online in digital format, is free of charge for readers and authors and does not allow commercial and for-profit re-use"[3] but point out that under this model publishers are still able to charge for print formats on a non-profit basis. They clarify that "neither authors nor individual readers nor institutions such as libraries have to pay for obtaining access to the literature [and] authors or their institutions do not have to pay publication fees."[4] However, the definition of diamond OA varies, even within the UK. UKRI (UK Research and Innovation)—the UK's biggest research funding body—defines diamond OA as "publishing models that facilitate open access to the version of record through community funding or subscribe to open models. They do not incur fees to publish and help spread the cost across multiple institutions."[5] This definition appears to exclude publishing initiatives funded directly by a singular library. It does, however, demonstrate some fluidity between diamond and gold OA definitions in the UK landscape, and implies that the funding source can make the difference between diamond OA and gold OA.

Commonly, in the UK, diamond OA is funded directly by the institution responsible for publishing the content. For example, the University of Edinburgh Library funds its own publishing service, Edinburgh Diamond,[6] which publishes books and journals created by staff and students of their institution. Other examples are Liverpool John Moore's Open Journal Service,[7] the University of St Andrew's Journal Hosting Service,[8] and Glasgow Open Journals.[9] These models tend to be known as library publishing services or library hosting services. Sarah Kalikman Lippincott says, "Many library publishing initiatives may best be described as publishing services in that they put less emphasis on acquiring, managing, and owning a coherent portfolio of work and more on providing the necessary technologies and support to

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facilitate content creation and dissemination of all kinds,"[10] which shows another difference between a library publishing service and a commercial press. Interestingly, many library publishing initiatives refer to themselves as hosting services, as opposed to a provider of publishing services, as they provide a hosting platform and the required technical support, but may not extend their services beyond this, or they may not wish to take on the legal role of publisher and so avoid that lingo. Another reason is that the host institution of the service might already have a formal press, which the library may not want to be seen as competing with.

The other facet of diamond OA, if we are to accept UKRI's definition, is collective funding models, such as Subscribe to Open (S2O),[11] which relies on existing library subscription models to flip closed content to open, or community-led funding models. Looking more closely at S2O, the model works by allowing the following:

> *S2O allows publishers to convert journals from subscriptions to OA, one year at a time. Using S2O, a publisher offers a journal's current subscribers continued access. If all current subscribers participate in the S2O offer (simply by not opting out) the publisher opens the content covered by that year's subscription. If participation is not sufficient—for example, if some subscribers delay renewing in the expectation that they can gain access without participating—then that year's content remains gated.*[12]

S2O has a variety of participants, from university presses like Edinburgh University Press[13] and Liverpool University Press[14] to society publishers like the Biochemical Society (via Portland Press)[15] and IWA (International Water Association) Publishing.[16] Writing about their experience of transitioning to an S2O model, IWA Publishing felt "the advantages are clear; S2O relies on existing subscription infrastructure and procurement processes eliminating the need for costly article processing charges (APCs), management software and the associated administrative burden of approving APCs."[17] They go on to state the benefits of the cost remaining with the institution instead of moving to the author, and the infrastructure already being in place, proving particularly beneficial for small society publishers. However, as open access for models such as S2O is conditional on the sale of membership subscriptions, this way of funding can be seen as a precarious way of providing OA.[18] Many presses and services utilising such models are not-for-profit and/or are affiliated with an institution but don't necessarily limit their scope to only publishing the research coming out from their institution. Between the library publishing services and the institutional presses, diamond OA has an increasing presence in the UK, and the main models—institutionally funded, collectively funded, or a mix of both—although different, share common goals: to utilise the power of OA, maintain high-quality publishing evaluations and processes, put the ownership of research back into the hands of the researchers, and open research to all.

Gold OA has different funding options, including author-facing article processing charges (APCs) and book processing charges (BPCs), and they are used by both for-profit and not-for-profit presses. Processing charges are set by publishers and "require authors, instead of readers, to bear the costs of publication."[19] They generally cover the costs of producing the book or article but may also include company overhead, such as staffing and technical infrastructure. Various terms are used to differentiate processing charge models and collective models, such as "Gold OA without APC," "Gold no-APC OA," "Full Gold OA," "Diamond OA," and "sponsor-pays."[20] Many library presses use a mixed funding model, where institutional funding enables them to discount their APC/BPC rates, making their costs more competitive with those of commercial publishers or large university presses. For example, Scottish Universities Press, a small collaborative press consisting of nineteen academic libraries across Scotland, charges between £3,500 and £5,500 to publish a book (cost depending on length and amount of images), whereas the charge from commercial publishers can range from £8,000 to over £15,000.[21]

As various open movements progress, libraries across the United Kingdom are launching their own diamond open access (OA) initiatives through presses, publishing services, and

hosting services. But where do these initiatives sit within the wider cache of various open movements, and what can academic libraries offer? This chapter discusses the current landscape of diamond OA across the UK, the role of the library and librarian, what the library publishing model contributes, and areas requiring development. This chapter also reviews collective strategies for advancing open access within this framework, including the Open Hosting Shared Service in Scotland. Through showcasing collaborative models, this chapter demonstrates the importance of partnership in advancing the diamond open access movement in UK libraries. Overall, this chapter reviews the components needed to ensure that these publishing models are not only sustainable but also that they thrive, shining brightly like the diamonds from which they take their name.

The Landscape of Diamond Open Access in the UK

In 2013, Fuchs and Sandoval claimed that diamond OA was "the dominant reality of open access."[22] A quick look at the Directory of Open Access Journals (DOAJ) in 2025 shows that the database indexes 22,135 journals, 63 percent of which are defined as journals without fees,[23] demonstrating the popularity of the diamond OA publishing model. However, in the UK, this statistic decreases significantly, with 22 percent of the 2,223 journals indexed not charging fees to authors. Also, a study found that "8–9% of the global scholarly output is published in diamond OA journals."[24] It would appear then that diamond OA publishing, albeit a quickly growing area, is still lagging behind its gold OA counterparts in the UK. But why would this be the case?

There have been some major changes from funders regarding open access in the UK, including diamond, gold, and green routes. UKRI mandated that from 1 January 2024, their established policy for peer-reviewed research articles would apply to monographs, book chapters, and edited collections. The policy states that all UKRI-funded research must be made open access, which Judith Fathallah says "constitutes a significant move towards OA as an academic standard."[25] Furthermore, Plan S—an OA initiative launched in 2018—states that "all scholarly articles that result from research funded by members of cOAlition S must be openly available immediately upon publication without any embargo period."[26] UKRI is a cOAlition S[27] member, along with other UK and European funding bodies. Furthermore, the Research Excellence Framework (REF)[28]—the UK's system for assessing the quality of research in UK higher education institutions—has been gathering feedback on a new open access policy. The new policy further pushes towards open access submissions, with a preference for Creative Commons licence to be applied.[29] The rise of OA legislation across these various organisations is promising, but in comparison to other models, there is little in regard to diamond OA library publishing models. This could be because libraries tend to support researchers in adhering to policies but aren't necessarily implementers of these policies themselves. However, the rise of library publishing means librarians are taking on a more active role in the publishing landscape, and library publishing initiatives need to adhere to policies in order to meet funding and policy criteria. If policies from these large organisations develop to put diamond open access on equal footing with other open access routes, it would help the model to be seen as a more viable option for researchers and help cement the role of the library in publishing.

Although there isn't yet substantial direct support for the diamond OA library publishing model from UK funders and organisations, there has been a rise of interest in institutional publishing and in collaborative models. The DIAMAS[30] (Developing Institutional Open Access Publishing Models to Advance Scholarly Communication) project, funded by the European Union, was launched in 2022 to review the institutional OA publishing landscape across Europe (including the UK), with a focus on providing support to diamond OA publishers and encouraging diamond OA policy adoption. 2023 saw the launch of the Open Institutional Publishing Association (OIPA) in the UK, founded to "connect and encourage a diverse range of small to

medium-sized university presses and university-affiliated publishing operations striving for open access."[31] There are currently twenty members, eleven of whom are fully diamond OA. Members include library publishing services, such as Edinburgh Diamond, collective publishers, such as Scottish Universities Press and White Rose University Press, and traditional OA publishers, such as Aberdeen University Press. Furthermore, Copim[32] (Community-led Open Publication Infrastructures for Monographs) is an initiative that supports OA book publishing through collaborative and community-owned funding models, including support for diamond OA.

Another collaborative initiative is the Open Hosting Shared Service in Scotland.[33] Governed by the Scottish Confederation of University & Research Libraries (SCURL) and managed by the University of Edinburgh, the shared service provides hosting, using open-source software, to member institutions around Scotland that wish to launch their own open access initiatives. SCURL is a membership body that supports service development and improvement across Scotland's university and research libraries, and the key areas of work are coordination, collaboration, and advocacy. The Open Hosting Shared Service was launched in 2018, with an aim to equip member institutions with a hosting solution to fulfil their open access publishing activities. Development time is charged back to the University of Edinburgh, with an annual fee of £1,800 1 VAT (value added tax: "a tax added to most products and services sold by VAT-registered businesses"[34] for the UK government) per partner. The service operates on a not-for-profit basis, and all fees are reinvested, covering technical and staffing costs. The service was launched with three members and has now grown to include twelve. Eleven members publish exclusively open access, ten members publish exclusively under the diamond OA model, and ten members are library-led initiatives. The members are:

1. Glasgow Caledonian University
2. Heriot-Watt University
3. Queen Margaret University
4. Robert Gordon University
5. Royal Botanic Garden Edinburgh
6. Scottish Universities Press
7. Society of Antiquaries Scotland
8. St Andrews University
9. University of Dundee
10. University of Edinburgh
11. University of Glasgow
12. The University of the Highlands and Islands

As of 2025, the service includes

- a unique installation of Open Journals System (OJS) and/or Open Monograph Press (OMP);
- training and guidance on using OJS and OMP;
- test and live site setup;
- ongoing technical support, including migrations, upgrades, bug fixes, and general day-to-day maintenance; and
- publishing advice on topics including archiving, copyright, indexing, policies, permissions, publication ethics, and reporting.

The service does not include

- allocation of ISSNs, ISBNs, and DOIs;
- copyediting or typesetting services; and
- workflow management for the content they publish.

Partners are expected to source, fund, and manage the elements that aren't included, although guidance is given in these areas. A steering group, consisting of one or two people from each partner institution, meets four times a year to discuss the direction and growth

of the shared service. This ensures that every partner has a platform to speak and is heard, and means there is a larger diversity of thought and opinion from which to shape the service. Also, most of these publishing initiatives are run by one or two people or a very small team. Working together means having more people to bounce ideas off of and growing confidence in the management of your own initiative. Finally, collaborating means being able to pool together knowledge and resources, getting discounts on third-party services that partners couldn't afford alone, getting answers to tricky questions quickly, saving time by minimising duplication of effort, and helping with establishing best practices. The service continues to grow and flourish, drawing more partners each year, demonstrating the power of unity as more partners become confident in launching their own diamond OA library publishing initiative, knowing they will be supported and among friends.

These collaborative initiatives—including OIPA, Copim, and the SCURL Open Hosting shared service—demonstrate a strong developing investment in diamond OA with a growing potential to affect funding legislation and policies. But where do libraries come into it?

The Role of the Library and Librarian

There is a strong ethical argument to be made for open access—specifically, diamond OA. Thinking of libraries' involvement in particular, Reggie Raju says, "The 'library as a publisher' service brings to the fore and consolidates the social justice imperative of open access"[35] and that "open access empowers those who experience poverty and injustice through lack of access to information as it is a redistribution of information resources to the end user and results in justice by providing 'information fairness' or equal access to information."[36] Libraries and librarians are very much in the business of information, so it makes sense for them to take a more active interest in academic publishing, and diamond OA also presents an opportunity to highlight and showcase work from within their own institutions while removing access barriers globally. Diamond OA, in particular, is more equitable as it doesn't charge fees to authors or readers, allows researchers to take ownership of their work, allows funding bodies to implement policies around publication costs, is more sustainable than gold OA fee models, and enhances diversity and multilingualism opportunities.[37] Libraries are also more likely to get involved as these ethical arguments are often in line with their overall mission.

The 2024 Library Publishing Directory[38] lists thirteen library publishers in the UK, with eight appearing to be diamond OA. However, the UK has many more library publishing initiatives than listed and, as mentioned previously, there is growing support for library involvement in publishing. The DIAMAS project states that "academic libraries play important roles in the Diamond Open Access ecosystem. Libraries provide infrastructure and services, as well as training and expert support to institutional publishers, but they can also be publishers in their own right."[39] Furthermore, Jisc (Joint Information Systems Committee)[40]—the not-for-profit UK digital, data and technology agency focused on tertiary education, research and innovation—published a "new university press toolkit"[41] that provides guidance for setting up and managing a library-led publishing venture or university press. This toolkit also links the two core facets of library publishing: the library publishing services, institutionally-funded and providing a free service to their own staff and students, such as Edinburgh Diamond and Liverpool John Moore's Open Journals Service, and the library presses, funded either institutionally and/or by collective models and/or by APC/BPCs and who publish authors both within and outside their institution, such as UCL Press and White Rose University Press. The library publishing services tend to be exclusively diamond OA, and the library presses usually employ a mixed diamond and gold OA approach.

Although library publishing services have been available in one form or another for years, they are now becoming more robust services that function much as a publisher would. Edinburgh Diamond—the University of Edinburgh's library publishing service—began as a journal hosting service in 2009, before rebranding in 2021, changing its name, growing to include books, and reestablishing itself as a library publishing service (see table 1). The service is

provided free of charge to academics, staff, and students at the University of Edinburgh and exists to support the publication of their research under a diamond OA model with Creative Commons licenses. Proposals are submitted by those who wish to use the service, and a Decisions Group meets monthly to select content based on quality and suitability criteria. As of 2025, there are twenty-three live journals and 87 books—a mix of monographs, textbooks, edited collections, reports, and miscellaneous research projects—published on the Edinburgh Diamond site.

Service Inclusions	Description
Hosting	A hosting platform for publishing content, using open-source software Open Journal System (OJS) for journals and Open Monograph Press (OMP) for books
Ongoing technical support	Including migrations, upgrades, bug fixes, and general day-to-day maintenance
Permanent identifiers	Issuing ISSNs, ISBNs, and DOIs through the British Library, Nielsen, and Crossref
Digital preservation	In-house, with the PKP Preservation Network, and through memberships with LOCKSS and CLOCKSS
Indexing	Including submitting to databases and helping with preparing a book or journal to ensure it meets eligibility criteria
Annual reports	Detailing article downloads, website traffic, journal- and article-level metrics, citations, and other information
Publishing policies	Providing appropriate industry-standard policies, including a publication ethics statement and a digital preservation policy
Publishing advice	On topics including copyright, permissions, publishing workflow, publishing best practice, and any query pertaining to academic publishing
Training & guidance on using OJS	Journal editors only. Full training on use of the workflow system on OJS.

Table 1. Services offered by the University of Edinburgh's "Edinburgh Diamond" library publishing service.

The service currently doesn't include copyediting and typesetting services, due to a lack of resources, or workflow management, as users are responsible for facilitating submissions, peer review, and production for the content they publish.

The Edinburgh Diamond service continues to grow and has attracted the attention of other libraries that have then gone on to launch their own library publishing services, including from libraries that have subsequently signed up to the SCURL Open Hosting Shared Service.

The trajectory is clear: UK libraries are getting involved in academic publishing, either by forming presses or offering robust publishing services, and there is growing support to enable their services. But with the landscape still developing and growing, and with many challenges to face, the question remains: why would academic libraries get involved?

To start, there have been ongoing issues with subscription and acquisition costs charged by publishers to libraries. Ángel Borrego argues that open access developed as a result of "continuous increases in journal subscription prices above the rate of inflation and, as a consequence, the inability of research institutions to buy access to all the scholarly content required

by their researchers,"[42] and that APCs in particular were developed by commercial publishers (namely BioMed Central in 2002) as an income source. Although APCs and BPCs were created as an alternative model to subscriptions and acquisitions, Fathallah points out "they penalise those researchers and institutions least able to bear the burden,"[43] which perpetuates inequitable systems. Yoon, Ku, and Chung also note that the processing charge system is prone to misuse and exploitation "in the publishing of predatory journals for profit,"[44] and librarians increasingly find themselves advising their academics and students on identifying appropriate publishing outlets and avoiding predatory journals. Such advice may include spotting a journal's lack of transparency around publishing charges or even heavily discounted charges, which make them more attractive for students and early career researchers who are already under immense pressure to publish to establish themselves in their field. Serhat Kurt defines four themes explaining why authors publish in predatory journals: social identity threat, unawareness, high pressure, and lack of research proficiency. Kurt's research showed that scholars from less-developed nations and those under pressure to publish for their careers are more likely to publish in predatory journals. He argues, "Predatory publishers have become another way in which people in the developing world are victimized."[45] Predatory publishing, along with high APCs and BPCs, disproportionately affects those least able to bear the burden. Libraries are in a position not only to advise researchers on publishing but also to facilitate equitable publishing through a diamond OA library publishing model. Colliseter et al. point out that "libraries also want to be engaged in the transformation of the current unsustainable commercial subscription pricing system to a more viable system, and to take direct action to support OA."[46] One way is to redirect some of the library funding allocated for the APC/BPC model, and indeed, cOAlition S is already looking at viable ways to facilitate this.[47]

Furthermore, Lippincott says that the growing involvement of libraries in open access publishing is "in response to profound shifts in technology, policy, scholarly practices, and the marketplace."[48] She argues that "publishing has emerged as a natural outgrowth of and corollary to libraries' investment in scholarly communication, digital scholarship, and data management services, among others"[49] and "libraries have brought new models to the table—models infected with the values and principles of academic librarianship, models designed to fill gaps in the publishing landscape, models designed to leverage the unique skills and positioning of libraries."[50] Libraries are well-placed and well-suited to offer publishing services as a natural progression of their existing activities. They already support academics and students in their publishing endeavours, whether through green OA repositories, allocating block grant funds for gold OA, or giving general publishing advice, to name a few. Collister et al. further argue that "publishing is at the core of faculty activities, and libraries should be involved."[51]

Library publishing services are often funded directly from the libraries' core budgets. Lippincott says this allows libraries the "freedom to experiment and to take on publications that are unwieldy or unprofitable" and also "frees them from worrying about cost recovery."[52] Furthermore, Yoon, Ku, and Chung say that diamond OA offers a space for "research that might not fit into traditional disciplinary boundaries or might be considered too risky for commercial publishers."[53] For example, the work may not be considered profitable or may be written for a market considered too small or niche. Library publishers do not need to take profitability into account when choosing books and journals to publish; they can cover a wider range of subject areas, ensure that quality research doesn't fall through the cracks, and introduce a more equitable system of publishing.

Collister et al. summarise, "Library publishing and participation in publishing programs are a natural progression for libraries and a good fit for librarians, who are in an important position to foster innovative publishing initiatives."[54] Librarians contend with high charges from publishers, already give advice on academic publishing to academics and students, and have the ability to take a risk on the content they publish, as the content doesn't need to be profitable. The opportunity for innovation and the ability to create more equitable publishing models is striking—why wouldn't libraries want to be involved?

Challenges for Libraries in Supporting Diamond OA

The library publishing diamond OA model offers an equitable publishing route that resolves librarians' frustrations with commercial publishers and helps put the power of publishing back into the hands of the researchers. However, it is not without its issues and challenges. The DIAMAS project has already identified that "diamond publishers often face similar fragmentation issues, varying service quality, visibility, and sustainability."[55] DIAMAS seeks to address the current gap in guidance for assuring diamond OA quality, which is especially important for a landscape that prioritises impact factors and metrics tied up in for-profit publishing when it comes to quality assessment. Indeed, Bosman, Frantsvåg, Kramer, Langlais, and Proudman, in a report commissioned by cOAlition S, identified the journal impact factor as one of the core issues affecting the sustainability of diamond OA models, along with finance and long-term feasibility.[56] Libraries are entrenched in the wider academic publishing landscape, through subscriptions and APCs and BPCs and indexing metrics, which can cause them to struggle to advance their own publishing activities. Fuchs and Sandoval pushed for evaluation metrics to center on DOAJ instead of Web of Science databases,[57] and indeed many journals published through libraries, including those on the Edinburgh Diamond platform, are indexed in the DOAJ. This shift, however, needs to happen systemically for it to shake up the current academic publishing landscape.

While advocating for this, library publishers also need to think about how to attract editors and authors to publish through their service. De Groote and Case advise that it's "important to think about your primary audience and the services that will attract them to publish their journal through your library."[58] A huge factor in this will be the ability to provide a well-rounded service that can offer everything a traditional publisher would, which in turn removes the risk of putting the role of publisher onto the academic. However, this isn't always possible due to a lack of resources. In a recent study, Dufour, Pontille, and Torny submitted questions to over 1,000 diamond OA journal teams and found that "in a world free of financial constraints, the acts for which journals would be most willing to pay are copy editing (60%), language editing (59%), and format production (56%)."[59] This comes as no surprise, as these are the three most costly aspects of publishing, and are often ones not included in library publishing services (although they are often included in library presses). If asked, it's likely that librarians would also pick these aspects, particularly copyediting and format production. Unfortunately, as discussed, there is a current lack of consistent funding models available to diamond OA library publishing services in the UK, and librarians are in the position of having to pick the most fundamental components needed to run a publishing service. The fundamental components can often run with minimal staffing and resource costs, making them more appealing when requesting internal funding, but this also means that library publishing services aren't often as robust as traditional publishers. Although the basic operational costs of running a publishing service are frequently met by the library's core budget, general budget restrictions and library spending on external publishers' APCs and BPCs mean it is difficult and competitive to secure extra internal funding, which could then be used to offer additional components in a publishing service.

While it is evident that libraries make a good home for academic publishing—and librarians are good leaders of such initiatives—it is important to acknowledge the value that publishers add to scholarship. Copyediting, typesetting, marketing, promotion, archiving, indexing, and the creation of fully accessible formats, to name a few, are skill sets that shouldn't be transferred from the publisher to the researcher. Of course, journal teams within some disciplines may be able to perform publishing tasks more easily than others, but academics are already stretched thin with responsibilities, and adding publishing roles on top of that seems unsustainable. Librarians have the opportunity to evolve and upskill themselves in publishing or to collaborate with publishing professionals to mitigate as much risk as possible and ensure

they have the best chance of meeting and overcoming the challenges of running a library publishing service or library press.

Conclusion

Diamond OA is a more equitable model of publishing that needs to be more heavily invested in to ensure long-term sustainability. The importance of diamond OA cannot be understated. Reggie Raju says that diamond OA "confronts and disrupts the profit-driven monopoly of publishers and the commercialisation of access to information."[60] It's no surprise that not-for-profit academic libraries are becoming involved due to their established involvement in supporting publishing activities and so they can further invest in open research more generally. It is clear that libraries and librarians are well-suited to lead the charge on the growth of diamond OA publishing within the UK. Although already well-suited and well-placed, Raju warns that "the adoption of diamond open access demands a shift in the responsibilities of the librarian."[61] Clearly, the role of the librarian will need to evolve so that important publishing tasks, such as typesetting, are not abandoned or offloaded onto authors, who will then need to learn a new skillset or find the expertise elsewhere. Fuchs and Sandoval even say that diamond OA models entail "the possibility of fostering job security by creating public service publishing jobs,"[62] which would certainly shift power from commercial publishers to librarians and the academics they provide services for.

Wojciechowska and Ramaré argue that "a joint plan of action [...] for library professionals is strongly missing and national networks of libraries are to be involved in projects for Diamond Open Access. Otherwise, librarians not ready to radically alter their job are destined to fade away like a memory of a time gone by."[63] Initiatives like OIPA, Copim, and the SCURL Open Hosting Shared Service demonstrate the importance of collaboration to advance OA and ensure long-term sustainability by pooling resources, infrastructure, guidance, and support. Diamond OA, in particular, thrives under collaboration among libraries and between libraries and academics, as they have an interest in scholarly knowledge and research becoming a common good. Working together means a stronger chance of changing who gets to publish (based on quality and not profit), who gets access to research (everyone), and how quality is measured and perceived (fairly). Librarians have the power to shake up academic publishing, and diamond OA library publishing services and library presses, with help from collaborative initiatives, might just be the way to finally see lasting change.

Notes

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CHAPTER 26

Designing a Curriculum for Openness in LIS Education

Will Cross, Maria Bonn, and Josh Bolick

Introduction

Open movements, including submovements, dedicated to open access, open software, open data, open education and pedagogy, open science and infrastructure, and others have seen explosive growth over the course of the last three decades. Open access (OA), a term that describes scholarly literature that is digital, online, and free of most copyright and reuse restrictions,[1] arose in response to journal price increases, the necessity of access to research, and the opportunity presented by the internet. Though preceded by open-source code, OA is perhaps the most direct antecedent to other open initiatives and movements that have developed in parallel and often in consultation with each other. As open initiatives have proliferated and evolved, it has become clear that openness isn't a goal in itself; it's a means toward better ends.[2] In addition to increasing access to knowledge and resisting exploitation, open movements aim to increase fairness, transparency, reproducibility, ensure authors' rights, and speed innovation, while reducing barriers to knowledge. But how do people learn about open movements and their concepts, given the speed of change and the unique needs of each area of focus?

While the growth of open movements is a remarkable revolution, it has also been intensely contested, at points resisted, and has come with significant growing pains and maybe even a few scars. For example: researchers have had to develop greater savvy at identifying and avoiding scam "predatory" publishers who exploit the existence of publication charges and the highly competitive "publish or perish" research culture;[3] there are nuances to sharing data that must be carefully considered, and labor invested in preparing data and related documentation to support that sharing;[4] there are passionate debates about how open initiatives should be funded, supported, and sustained;[5] and among funders, institutions, and publishers, who sets policy and monitors compliance, or whose policy prevails?[6] Further, there is considerable confusion about terminology (for example, green versus gold versus diamond; gratis versus libre; conflation of concepts like open access and open education), how open licenses work, and numerous additional challenges. Deeply entrenched practices and evaluation norms, such as various methods of impact measurement and processes related to promotion and tenure, intersect with the challenges outlined above and exacerbate these challenges. This is a complex environment, and there are numerous points of tension and ambiguity, which have perhaps contributed to a relative dearth of training in graduate education. That gap, in turn, is a barrier to overcoming challenges and realizing the full potential of openness to increase

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equity and better fulfill institutional missions. As a result, open advocates struggle to find consensus and harmonize efforts within and across stakeholder groups.

Ideally, education about the principles and practices of open movements will be pervasive throughout higher education, starting with undergraduates developing basic skills as scholars and ranging to expert faculty. Our primary intent here is to discuss the role of educating librarians. Libraries and librarians have been essential to the growth of open movements in a variety of capacities. For example, we host and manage institutional repositories of open content, advise students and faculty navigating the open landscape, support the adoption and implementation of institutional open access policies, support open education and open data initiatives, publish journals and open educational resources (OER), and negotiate vendor contracts that may enable or limit open publishing and sharing opportunities for the researcher communities we participate in and serve. There is an increasing sense that all librarians are facilitators of scholarly communication and all in a position to contribute to greater openness, but we lack a shared foundation to build on. As core supporters, advocates, publishers, and partners, and given the explosion and acceleration described above, it's important that there are accessible professional development opportunities for librarians in practice and updates to curricula to adequately support a shifting labor market. It's particularly important to train librarians because we are a nexus in the research landscape, serving all disciplines and interacting with multiple stakeholders. Secondarily, we see increasing knowledge among librarians as a means to expand knowledge of open movements and practices to other disciplines and thus the broader research endeavor. To that end, where are we now, and where should we be headed?

Current State of Education

Educational attention to openness is fragmented, scattered throughout the curricula of different fields and professional development offerings. Scholars at all levels, from school children to senior faculty conducting research and teaching courses, can benefit from and contribute to open culture. To realize those benefits, people need to understand the policies and practices that shape that culture. Although individual educators may be aware of open resources, policies, and opportunities, these topics are largely incidental rather than an integral part of the curriculum. This is true throughout all levels of education. The lack of focused attention is particularly notable in the field of library and information science (LIS). By talking and listening to LIS professionals, and in our own practice, we have found that training for many open practitioners, both those that are active and those that might be, is often absent or incomplete. This is apparent both within specific disciplinary cultures and across higher education. When institutions of higher education seek expertise about OA, they often turn to their librarians; those librarians may lack educational experience of their own with relevant topics and then need to develop it outside of the classroom. For those librarians to be effective and impactful educators, they need education and training. Attention to openness is, at best, nascent in LIS education.

In "Open Access, Scholarly Communication, and Open Science in Psychology: An Overview for Researchers," Laura Bowering Mullen argues:

> *Across the disciplines, scholarly publishing and other aspects of scholarly communication are in a time of disruption and transition. The ongoing trend toward Open Access (OA) to the results of research continues to engage authors, publishers, librarians, research offices and others that seek to maximize impact of scholarly research. However, many treatments of these topics are very broad, and there are very few overviews that focus on how OA is playing out in the disciplines. Each discipline (and even subfield), due to disciplinary differences in scholarly communication practices must be considered separately. Overviews of this new open landscape for each discipline would be useful to researchers and allow comparisons and targeted studies that would help develop best practices that work for the discipline.*[7]

It frequently falls to librarians to pull together these disparate threads and to provide expertise on how and why to open. Consequently, it is important for LIS education to attend to issues and practices of open culture. Many academic librarians, particularly those tasked with scholarly communication services and supporting open education, report that during their education, they were not formally engaged in any discussion of open access, open data, open science, or open education, or given any training in how to support others in understanding and realizing the possibilities of open initiatives. A 2022 study on "Building capacities in Open Knowledge for the LIS (Library and Information Science) sector" found that "while LIS schools are not providing formal training to gain skills and competencies in Openness, their libraries are offering different kinds of training in this respect."[8]

Periodic reviews of the courses offered at the major accredited LIS programs in North America have found few courses addressing open movements as a primary topic. A handful of programs offer courses explicitly about scholarly communication (for instance, at the University at Buffalo, University of Michigan, University of Illinois Urbana-Champaign, and UNC Chapel Hill), but even in those programs, the courses are not taught regularly. These courses almost certainly discuss some aspects of open scholarship and other open movements. There may well be topical attention to issues such as open access policies, open data curation, and open licensing scattered throughout courses on academic librarianship, data management, and copyright, but it is difficult to judge the extent to which these topics are addressed without a granular review of syllabi. In 2018, we conducted a survey of information professionals with scholarly communication responsibilities, asking, among other things, if their LIS education had prepared them for those responsibilities.[9] Few reported curricular attention to scholarly communication. Since open access is a core component of scholarly communication work, it follows that it was also not a subject of study. This is a troubling gap, given the fact that open access is mentioned by name in section 1G of ALA's core competencies for librarianship.[10]

Many people have questions about the implications of open movements for their work and their lives, and service-oriented librarians need to be ready to help answer these questions. Lacking formal education, many librarians turn to professional development opportunities. The library field, and adjacent and allied organizations, have been active in providing those opportunities. For example, the Scholarly Publishing and Access to Resources Coalition (SPARC) has long provided programming to help librarians develop expertise in open practice. The Open Education Network (OEN) offers a Certificate in Open Education Librarianship.[11] There are myriad copyright education programs, such as Harvard's CopyrightX[12] and Copyright for Educators and Librarians.[13] While the content of these programs is typically openly licensed and available, the formal programs often entail cost, which can be substantial, if participants want or need to get a certificate or full experience, which can include formal assessment and a cohort of peers, and which can carry greater weight as a credential.

It is part of professional library culture that the field responds quickly to specific issues and developments—such as new federal policies or changes in licensing models for publications—by organizing and offering educational opportunities in venues like webinars and conference workshops. This collective attention to professional development around open movements is notable in the world of *academic* libraries but not limited to that sector of the profession. For example, as this chapter is being written, American Libraries Live is offering a webcast on Public Libraries and the Open Access Movement,[14] designed to explore

- how the egalitarian roots of the OA movement mirror the public library;
- how OA and open educational resources (OER) can be tools to combat misinformation;
- ways that OA helps counter income inequality, leveling the playing field;
- how OA can help during a time of shrinking library budgets; and
- ways to leverage CloudSource OA to help cultivate lifetime library patrons.

It is an admirable hallmark of librarianship that the profession prioritizes this sort of attention to keep pace with new developments in the information ecosystem and organizes itself to educate and establish practice in response to those developments. That responsiveness does not relieve formal educational programs from doing the same. In the same way that the library

community has created a community of practice around advocating for and supporting open movements, there is opportunity for LIS programs to create an educational community that prepares new professionals to participate fully in that movement.

Where Do We Go From Here?

Openness presents profound opportunities for librarians, scholars, and the global community to remove paywalls, broaden access, and create more inclusive and impactful scholarly practices. LIS education has made great strides in training for specific open practices around access, reusable code, less expensive educational materials, and supporting reproducible data. We are still in the very early stages of developing a coherent Open movement that can be taught across disciplines. As discussed above, we believe that librarianship offers a promising starting point for this work. In order to bring together the various strands of openness, we need a shared understanding and a way to communicate that understanding. Doing that requires a coherent vision that also recognizes the distinct contexts of opening different materials and practices.

Develop a Shared Vision and Values for Openness

Some of the earliest modern conversations about openness come from the free and open-source software (FOSS) community, which recognizes the ambiguity of the English term "free," which can mean both "at no monetary cost" (gratis) and "with little or no restriction" (libre). As the open movement has continued to evolve, this distinction between "working towards public availability with little or no cost" and "working to transform the terms for creation, control, and ownership of knowledge" remains in significant tension. Many of the most common methods for gratis access come bundled with—or at least have resulted in—tighter control by established, often for-profit players. Public access mandates in North America focus primarily on the former, and on those terms, open access has been an unbridled success, with open access publications overtaking subscription journals in the early 2020s.[15] At the same time, as Cambridge Libraries Scholarly Communication Specialist Sam Moore notes, "Although open access has won the day, it is worth considering why this victory also feels like something of a defeat. Publishing is still largely controlled by a handful of profiteering companies who are rapidly expanding into areas beyond research articles, such as research data, user data and other elements in the knowledge production workflow."[16] Is this success, coupled with what Kember and Brand call "the corporate capture of OA,"[17] cause for celebration or alarm? Is "open access" winning? Or losing? Or some of both? (See figure 1.)

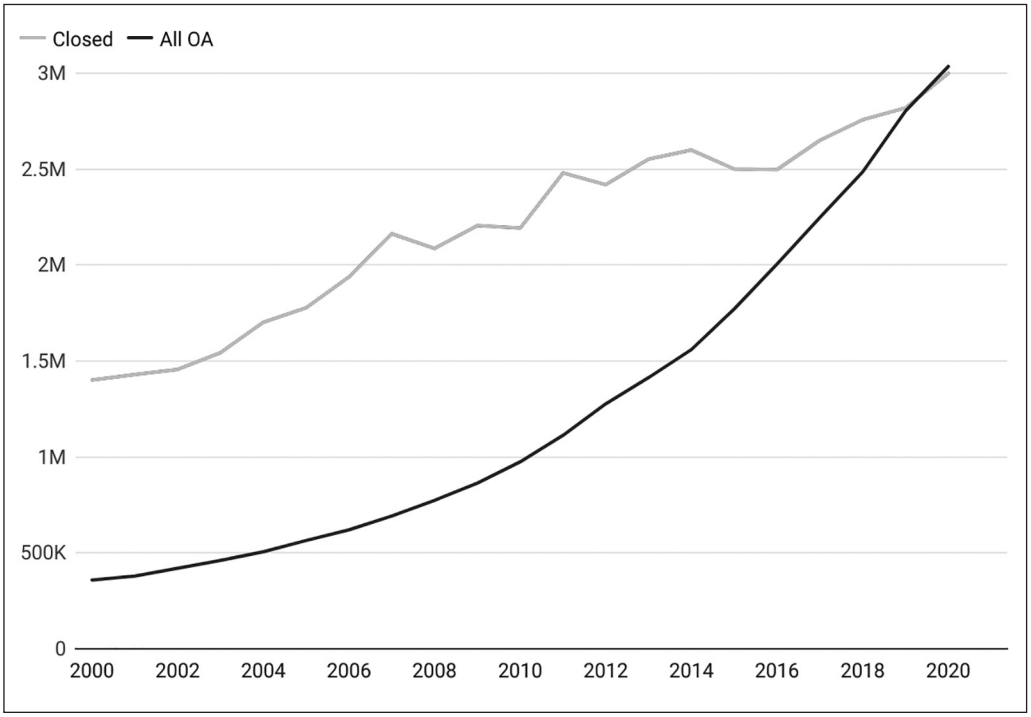


Figure 1. The victory(?) of open access. Open access surpasses subscription publication globally for the first time. Chart used under CC-BY 4.0 license, available at: https://www.dimensions.ai/wp-content/uploads/2021/02/63NYl-oa-versus-closed-access-1.png.

Open educators face a similar dilemma around for-profit participants in the academic ecosystem from commercial publishers offering automatic billing models for textbooks (sometimes framed as "inclusive" or "equitable" access). On the surface, these models promise lower costs for textbooks—the initial driver for many open education advocates. Librarians who remember the journal-focused "Big Deals" of the 2000s wonder if locking students into a commercial bundle will turn out any better than the contracts that led to the "journal pricing crisis" two decades ago.[18] Are for-profit actors (subject perhaps to genuine competition from competing open alternatives) a viable tool for providing less expensive services that do, in the end, require time, labor, expertise, and resources that have a cost that must be paid for in some way? Or are for-profit organizations the serpent in the garden, inexorably driven by quarterly fiduciary duty to extract ever-increasing revenue? Is open a model for reform, or revolution?

Why Open: Begin with First Principles

The first question a curriculum for openness in LIS needs to address is what "Open" is for. Openness, after all, as framed by guiding documents like the Budapest Open Access Initiative 20th Anniversary Recommendations, "is not an end in itself, but a means to further ends."[19] If we truly hope to be part of a unified and coherent movement, the communities working around openness must engage in transparent and intentional discussion about what those ends are. Is Open something like Google Search and Google Drive (where this chapter was written)—totally free for anyone, with some … complicated compromises around corporate capture and data privacy on the back end? Is Open more like a bougie farmer's market—full of local, ethically-sourced goodies at a price that not everyone can afford? Of course, this isn't a binary choice. Whether Open is more like Hardees or Hermès, SpaceX or Soylent Green (as our friend Micah Vandegrift suggested in 2014, "ScholComm is People!"),[20] the library community needs to be clear about our core mission versus what are nice-if-we-can-get-them side benefits.

Developing a curriculum for openness will also require that we build from—and perhaps beyond—the current focus on the "how" of open. Indeed, open advocates like Lauren Collister argue that describing open is the first step in moving toward shared understanding of openness. Introducing a new series called (appropriately) "Describing Open," she concludes that "We are not interested in being the final arbiters of what is or isn't open, but in understanding what is important about openness and what we can do to foster that outcome in the world."[21]

As the authors of this chapter, we believe that openness will thrive when we not only describe what open is but also work to build a shared understanding of what we hope it will be. Of course, technical training on how to set up and stock a repository, negotiate for authors' rights, and so forth is critical today. Happily, this type of training is already thick on the ground. Without a clear understanding of what all this work is for, however, open practitioners may continue to talk past one another with unspoken disagreement undermining the potential for coordinated work and sparking internecine conflicts. A lack of shared definition and mission also leaves the open movement vulnerable to the sort of co-option that Danny Kingsley laments in a recent blog post, where she notes, "Words matter. It is very hard to have meaningful conversations and make informed decisions when we cannot agree what we are talking about."[22]

Curricula in LIS should start with discussion about first principles, even if there is no current consensus on what those principles are. In a perfect world, of course, open communities would quickly come to a rough agreement about the core purpose of openness that educators could use as a cornerstone of the curriculum. To the extent that no such consensus seems imminent, however, LIS educators should present the core issues and speak directly about the tradeoffs involved in pragmatic, values-driven open work. LIS educators should also lead a discussion about how to make openness a tool that works across a spectrum for different ends. A curriculum for openness should clearly surface these questions, prepare practitioners to intentionally work toward clear ends, and, where possible, move toward field-level alignment.

Describing Open: Aligning Language, Tools, and Practices for Openness

In addition to building a shared vision of openness, we also need aligned language, tools, and practices. Terms like "open educational resources" and "open data" are generally clear on their face or with a clear implied parenthetical like "open access (to scholarship)" or "open source (code)." The larger movement has not yet found unified language or often even coordinated vocabulary about the "opens" without awkward quotation marks. This makes internal coordination more complicated than it needs to be and poses significant barriers for education about openness as a field or movement. Likewise, there is work to be done to clearly align the various tools and practices that run throughout open work in various contexts.

The closest to an umbrella term to date may be "open science," which has taken on particular currency in Europe and the EU (figure 2). However, many humanists have suggested that the term feels exclusionary of their work, just as the term "digital humanities" might leave some scientists feeling left out. Some practitioners have advocated for "open knowledge" as an alternative, which has gained some traction in recent years. Others have suggested "open scholarship" as an alternative that may be inclusive, especially where "scholarship" is defined broadly to include knowledge sharing through research outputs, educational resources, and other forms of public engagement.

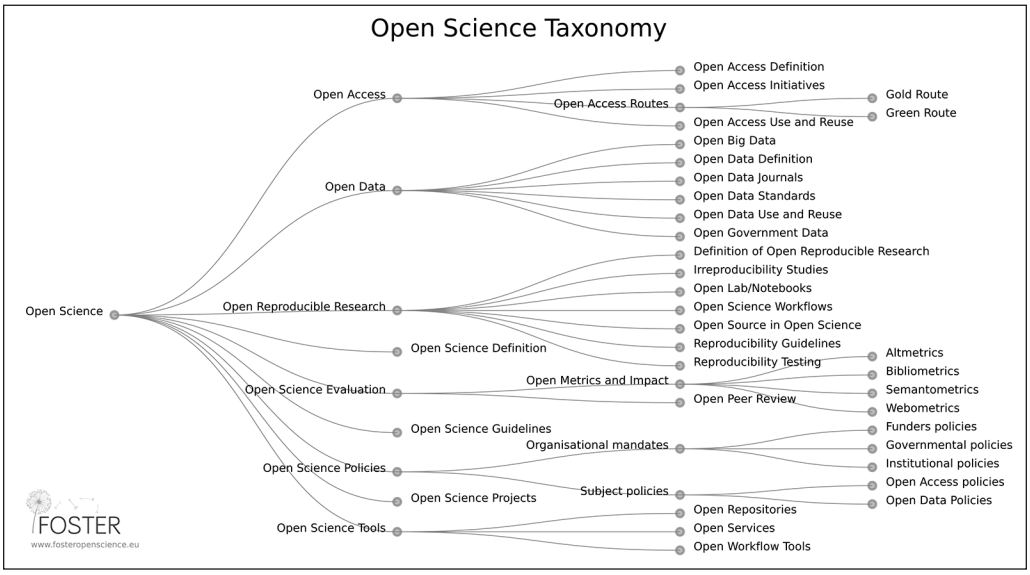


Figure 2. The EU's FOSTER Open Science Taxonomy is an essential tool for visualizing the connections between these areas and referencing basic definitions for the related terms. Infographic used under CC-BY 4.0 license, available at https://figshare.com/articles/figure/Open_Science_Taxonomy/1508606.

A clear term for this work is critical as a banner that open practitioners can rally around and, more importantly, explain intelligibly to administrators, funders, and the public. Clear language can also help move us beyond perceptions that one or another open is the "real" star of the show, with others as friendly fellow-travelers or curious offshoots. If it was ever reasonable to think of open as "OA and the Seven Dwarfs" (with one leading area of practice and several diminutive practices trailing behind), the recent commitment from funders to data sharing and broader impacts should persuade us that those days are past.

Fortunately, this high-level uncertainty about what we call "open" is not always reflected in tools and practice on the ground. While individual open practices naturally incorporate specific terms of art, learning about open source in terms of "push code" or a "commit" or about "green" versus "gold" open access is a natural part of learning about the practices themselves. Further, the practical language often follows a general "make it, license it, share it" framework that can be easily aligned, whether works are shared in DRYAD, PubMedCentral, or GitHub. Whether the open movement ultimately settles on "open science," "open knowledge," "open scholarship," or something totally different, the key work for explaining openness will rest with the alignment of practice-based terminology, as we develop and update educational and training resources across LIS.

Unlike the thornier values-based questions above, this question about language offers more than the opportunity to teach the controversy. Regardless of the term we use for "open [whatever]," curricula should begin with a clear statement about open practices, such as licensing and deposit, and then describe those practices as applied in specific settings. The work for curricula should focus on clear synthesis and alignment so that open practitioners and supporters can contextualize this work across specific practice areas.

(Re)Designing a Curriculum for Openness

The specific practices of openness are and should remain a moving target. In this chapter, we have laid out some of the key aspects of a curriculum for openness in LIS. We would like to close with a final caution and one nascent model for this work. Any resource "is necessarily reflective of the finite set of contributors who created it and specific moment when it was written and released."[23] A curriculum for openness in LIS should begin with a recognition that opening scholarship is an ongoing process where our understanding must be regularly refreshed. Curricular materials should themselves be open so that they can be regularly updated and localized for different communities and contexts.

As one potential case study for this open curriculum, we would like to offer the recently published open textbook, *Scholarly Communication Librarianship and Open Knowledge*.[24] We hope this offers a promising model for the topics, framing, tools, and practices that make up openness, at least from the perspective of a library today. An educator, student, or professional looking to develop curricula in openness could do worse than a foundation in the core issues, deep dive into the current practices, and overview of complementary and adjacent practice on the ground (figure 3).

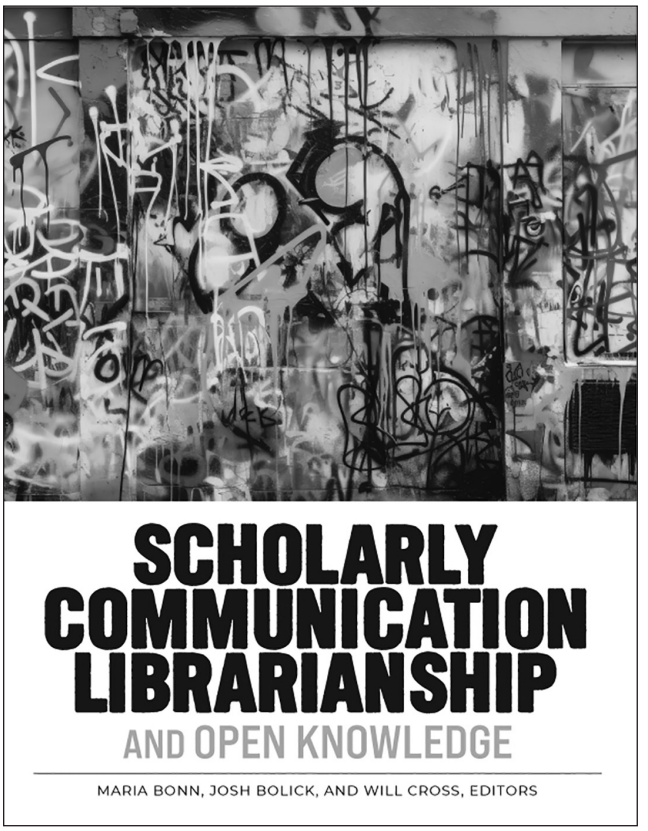


Figure 3. The cover of *Scholarly Communication Librarianship and Open Knowledge*, edited by Bonn, Bolick, and Cross, ACRL, 2023, is licensed CC-BY-NC.

This open resource explicitly centers diverse perspectives and bridges theory and practice. Although we served as the editors, we also invited more than seventy contributors to add their voices and perspectives and linked the text to an open repository for new materials, the Scholarly Communication Notebook.[25] All these resources are living documents, and we hope that you will add your experience and practices as well. We offer guidance on uploading new materials[26] and invite you to help write the next chapter in our evolving curriculum for openness in LIS.

A curriculum for openness should include diverse voices in design, instruction, and assessment. It should also create intentional connections between education and the field by soliciting new materials and guest speakers and by offering students a way to share their own work back out to the field. By using open materials and open pedagogies to connect education and practice and create inclusive educational environments, an open curriculum can demonstrate and benefit from the core values of openness.

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CHAPTER 27

Open Access and Open Education: Missed Connections Between Parallel Movements

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Introduction

Open access (OA) and open education (OE) are deeply related yet distinct movements that advocate for public access to knowledge through different means.[1] In addition to their focus on free access to information, both movements have explored how their values intersect with equity, participation in the creation of knowledge, and responding to commercial pressures.[2] Despite these similarities, there are fundamental differences between the OA and OE movements that have been heightened by rifts between their respective communities.

There have been recent discussions about how to build connections between the OA and OE movements, both explicitly through strategic planning[3] and through connections drawn out by larger groups, like the open science community in the European Union.[4] This isn't a particularly new discussion. In a 2011 interview, Nick Shockey shared his enthusiasm advocating for OE alongside OA: "When we hit roadblocks in one area," said Shockey, "there are often opportunities in others, and advancing one of these pieces (be it OA, OER, open data, open video, etc.) opens the door for further progress in other areas."[5] Despite continued interest in connecting these movements, shared infrastructures between OA and OE remain limited and could be further explored for the benefit of both groups. In this chapter, we discuss some of the ideological and practical differences between the OA and OE movements and highlight areas where collaboration could create positive change.

Background

OA content is typically described as "digital, online, free of charge, and free of most copyright and licensing restrictions."[6] This definition focuses on access first, while also listing reuse as a core part of what makes something open. However, the extent to which OA content is permissively licensed has been debated in the past.[7] Eventually, two varieties of OA were proposed: libre OA, which removes most permission barriers to allow easy redistribution and reuse of content, and gratis OA, which provides only free access without extra copyright

CC BY SA

permissions.[8] While these terms might not be familiar to some newcomers in the OA space, they are an integral part of the OA movement's history. The flexible definition of OA—wherein free access is a necessity and permissive licensing is optional—distinguishes it from how OER is typically defined.

Although the larger umbrella term "open education" is still evolving, the definition for OER has been relatively consistent for over a decade. UNESCO defines OER as

> *learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others. Open license refers to a license that respects the intellectual property rights of the copyright owner and provides permissions granting the public the rights to access, re-use, re-purpose, adapt and redistribute educational materials.*[9]

This definition's inclusion of reuse rights as a fundamental piece of OER was originally codified in the 4R permissions proposed by David Wiley in 2007—the right to reuse, revise, remix, and redistribute open content—before they were adapted to include a fifth R, "the right to retain," in 2014.[10] Through the 5Rs, permissive licensing for OER was made a vital piece of the OE movement in the same way that libre and gratis made room for flexibility in the definition of OA.

There are reasons why OA and OE differ in their approach to defining "open." Sharing scholarship and research outcomes with the intention of expanding access to vetted information does not necessarily require open licensing or downstream reuse. In contrast, it is necessary for OER to be adaptable so these materials can align with the flexible and changing expectations that come with educational accreditation requirements. Neither community is "correct" or "wrong" about their definition of open. Instead, they approach the work with different goals. The differences in definitions across our communities can make it difficult to communicate with one another. For example, new OER advocates who learn their definition of open from OE guides without understanding the foundational histories of OA or open-source movements might be confused when discussing the idea of openness outside of the OE context. One might ask if it is necessary for our definitions to change before a cross-movement collaboration can be viable. This is not the case; however, we do need to set aside our expectations for what Open can be if we want to pursue the many collaborations that are possible with peers working across different open movements.

Lines Drawn in Pencil, Not Ink

At first glance, one would assume that the main difference between OA and OE is their purpose, with OA supporting research needs and OE supporting pedagogical needs. This distinction is validated in a comparison between the two movements by Anita Walz, who describes OA materials as "scholarly works primarily created to document new knowledge and advance the scholarly conversation,"[11] whereas open educational resources (OER) are typically "codified knowledge useful for teaching, learning and stimulating discourse."[12] However, the actual boundary between OA and OE is much less exact, with OER actually falling into the category of OA content in most cases, both from a permissions and cost perspective (figure 1).

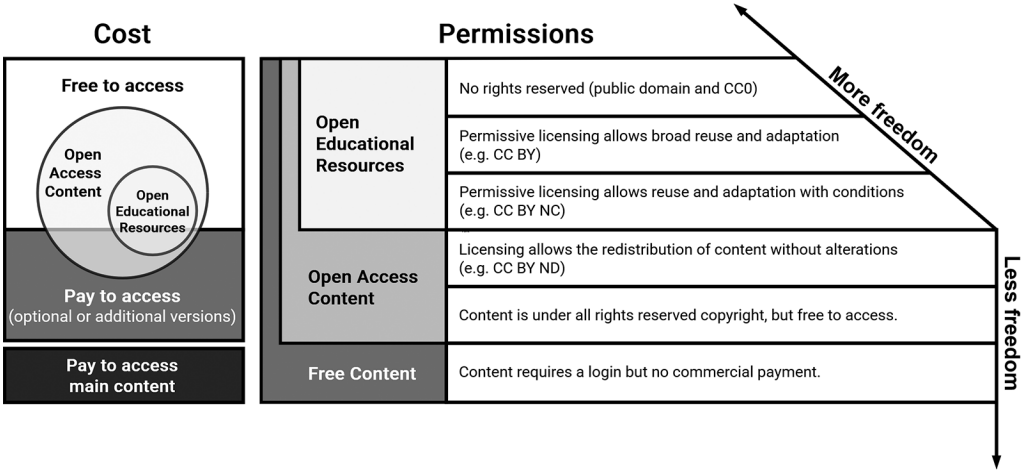


Figure 1. Cost and permissions of OA and OE content. Adapted by Elder (2024) from Walz (2019).

It is easy to overlook that OE can be born of research and OA can be used for education. Even outside of the similarities between our movements' goals and values, the literal content being supported by them can be the same. Within our definitions, there are no clear-cut lines between OA materials and OE materials (or OER). An OA article or book chapter might be considered an OER if it is educational in nature and openly licensed to allow repurposing.[13] Similarly, an OER might be published by a university press under the same processes and practices as an OA monograph, leaving very little distinction between the two besides their original purpose.[14]

There are reasons why we draw distinctions between our movements. Having the space to talk explicitly about learning outcomes in the OE realm is necessary for educators and support staff alike. Similarly, it is good for OA advocates to have the time and space to discuss funder mandates and research reproducibility when that is their strategic focus. As Open Future's "Fields of Open" report phrases it, "Each field has a distinct 'open stack': a set of tools, mechanisms, methods, protocols, norms, and infrastructures that define how the general open norm is expressed in a given field."[15] Still, we can miss the little connections, the pieces of our work where we can learn from one another, when we assume we have nothing in common besides the word "open." As we look at the lines drawn between OA and OE throughout this chapter, we address potential areas where we might learn from or connect with one another for positive change.

Missed Connection: How We Fund Our Work

In recent years, there has been an increase in official policies supporting open movements through requirements, regulations, or grants.[16] This has affected OA and OE in different ways, reflecting the scope and focus of the movements.

Funding that supports open access to research often comes from national research grants, organizations, or institutions that want to support research and innovation, regardless of whether they support the values of OA. The expectation of OA has been "baked into" the system for funding research rather than being the focus of separate OA grants. For example, in the US, grants from the National Science Foundation, Bill & Melinda Gates Foundation, and other funders come with expectations that the outcomes developed through their funding will be made freely available online, and often explicitly open.[17] The funding may support authors in making their work open, or it may only fund the research itself, with authors making their

work open through pre-prints, repositories, or by publishing in diamond OA journals, which incur no fees.[18]

Additional support for OA is provided through institutions and governing bodies that sustain OA infrastructures.[19] One example of this can be seen in the Open Library of Humanities, which prides itself on an equitable business model working with library partners to help pay for the costs of publishing to enable open access to scholarship without author-facing fees.[20] Another example is Subscribe 2 Open, which offers a community of practice that believes in "a pragmatic approach for converting subscription journals to open access … without reliance on either article processing charges (APCs) or altruism."[21] This is in contrast to gold OA, which might require additional fees to enable the publication of an OA work.[22] Supporting open infrastructures is a vital endeavor for maintaining and further developing capacity for publishing and sharing OA content.

Funding for OE has a similar yet less established base. While there are general education funds for the development of open-source software or openly licensed educational materials, these may not be described as "open education" grants,[23] and many individuals in the OE community overlook their relevance. This is not to say that there is no explicit funding available for OER development. In addition to those grants for the creation of instructional materials more broadly, there are grants available from the US Department of Education to support large-scale OER projects,[24] although the amount of funding available at this scale is limited. It is more common to see indirect funding for OE infrastructure in the United States through groups like the Hewlett Foundation[25] or through legislative support for coordinating groups, committees, or grant management. For example, the state of Connecticut created the Connecticut Open Educational Resource Coordinating Council with funding from its legislature, allowing for more in-depth assessment of OE initiatives across the state and the acceleration of OER adoption.[26] Having access to this kind of funding does not guarantee that an OE initiative can support publication, production, or compensation for OER development. Instead, practitioners within the OE space might be aware of smaller, institutional grants[27] available to support their work, which are typically awarded in the hundreds or thousands of dollars rather than the tens of thousands that one might see in research or industry.

There is a dramatic difference between the expectations that authors bring to the development of OER and OA content, with the former often relying on small, local, and contextual funding support and the latter providing broader and more explicit funding opportunities. This has impacted the perception of each movement, which is visible both in the areas of funding and in labor and support structures.

Labor and Support Structures

The history and development of each movement affect the work that happens within it, and the infrastructures that are built to support it. OA saw its big boom following the latter half of the twentieth century as inflating prices, bundling packages, and consolidation among publishers and vendors led to rising access costs for academic research articles, later called the serials crisis.[28] The OA movement hit many milestones in the 1990s and 2000s, with 2003 alone bringing the launch of the Directory of Open Access Journals, the creation of the Public Library of Science, and the release of the Bethesda Statement on Open Access Publishing, among other outcomes.[29] Although grounded in a long history of pedagogical theory, the OE movement did not see much growth until the successes of the OA movement had been realized, with early wins like the release of MIT OpenCourseWare in 2001[30] as an outlier. OE did not gain solid traction until the 2010s with the release of the 2012 Paris UNESCO declaration, after which there was rapid growth, with many new businesses,[31] infrastructures,[32] and support positions created.[33]

The funding models previously discussed are influenced by the expectations of the people participating in the system and the processes they are accustomed to interfacing with. Depending on their discipline, assistance for authors of OA works may be readily available

and integrated into typical workflows through publishers or institutional offices for research. In contrast, the support that creators of OER expect to receive is much more variable, based on the capacities and priorities within their local context. Staffing for OE initiatives is often either tacked onto existing positions (e.g., instruction librarians or scholarly communication librarians), often demanding a wide range of support within a small role, without taking into account the time required to assist faculty with finding and developing OER.[34] More distributed efforts are emerging as instructional designers and other educational support staff are invited into the OE movement, but there is room to learn from peers in the OA movement when it comes to cross-unit and cross-institutional collaborations that support OE practitioners. Because of the differences in our histories, structures for those working in OA are more grounded. OA is central to the wider field of scholarly communication, an area with long-established norms on both the institutional and national scale.[35] The inclusion of OA in national policies and the availability of large-scale funding to support OA publication demonstrate the relevance of OA support for institutions. Since OE is not as integrated in policy and practice, it is often more difficult to advocate for staffing.

Despite the fact that there is typically less staffing for OE initiatives, those developing OER or integrating open educational practices into their work often require more hands-on support than those doing research that is made OA. This is partially due to the lack of professional incentives for participating in OE. Whereas scholarly publishing can easily be linked to tenure and promotion rewards and is seen as an expectation for academics, educational practices are typically undervalued, underserved, or seen as "easy" to implement. When OE work is recognized as relevant for traditional reward structures, it is often lumped into the "teaching" bucket, though work has been done to advocate for its explicit inclusion elsewhere.[36] OA is often undervalued in tenure and promotion processes as well; however, more work has been completed to remedy that over the past decade through groups like DORA.[37] There is more we could do to support the practitioners in the OE space by evaluating the strides the OA movement has made.

Potential Connection Points

Both of our movements could benefit from taking down our blinders and holistically reviewing how we might **utilize shared infrastructure, open calls for innovative projects, or alternative open-source tools.** Consortial models and community-led efforts are particularly powerful options for distributing funding across our movements to achieve widespread benefits.

Because OE advocates often look for funding that specifically meets the needs of "open education," they may ignore opportunities that could support their work. In the same vein, those looking to secure funding for OA work tend to focus on the costs of publishing and meeting the demands of gold OA publishers when they could support broader goals such as building competencies for those learning about OA's relevance for their scholarship. We need to **recognize that funding comes in different packages and that we can consider diverse paths forward.**

It would be too easy to say "we should stop duplicating efforts." Different projects developed across our movements have unique areas of focus. However, it would benefit us all to **survey the projects that intersect with our goals to find inspiration, establish connections, and leverage the work that has already occurred.** The DOERS3 OER Matrix,[38] the various promotion and tenure reform documents developed by DORA,[39] and the Review, Promotion, and Tenure Project from Schol Comm Lab[40] are all excellent examples of community-driven projects to officially acknowledge different forms of scholarship.[41] Understanding the work that each of these groups has done is not just beneficial for us as advocates; it can also help us identify new and different ways we might approach the same problems. By working together on the same issues, we can create shared advocacy networks and educational tools that might serve all of our communities more effectively.

Missed Connection: What We Build and How

How our movements develop and publish open content has some unique differences colored by our histories. One frequent concern among OE practitioners is how we can provide long-term access to OER while meeting the needs of authors developing new materials. Here, OE could learn from the example of diamond OA publishing, which boasts diverse pathways and funding models that sustain publishing practices while remaining free of cost for both readers and authors of OA materials.[42] Diamond OA is not just a useful example because of its variety. It also provides practical examples of how to handle paid physical editions of publications, a frequent concern for OER authors who want to provide a print option for their students.[43]

While OA as a concept has more rooted foundations than OE, some OA practices are more established in the field than others. For example, library publishing is still a relatively young field, similar to OE. Library publishing programs' newness has given them a unique birds-eye perspective of the wider publishing landscape they are situated within, leading to an emphasis on sustainable publishing practices that do not sacrifice values for profits. This approach would make library publishing programs a great partner for OER production.[44] However, because OER and library publishing are both developing fields, their standards are still being built. This "in progress" state makes it difficult to stop and assess our peers or even to acknowledge that they are our peers. There is excellent work happening through groups like the Library Publishing Coalition[45] and the Open Education Network's Pub101 Curriculum[46] to develop more standards for the publication of OA and OE materials. Connecting the efforts and shared expertise of groups like these could be a useful step forward.

The standards we aspire to when developing and publishing content are often difficult to define. For those looking to support OER as a burgeoning area of interest at their institutions, developing sustainable processes can be "somewhat painful."[47] As staff try to meet the growing demands of their institutions[48] and educators alike, strategic planning may become a low-priority goal. It could take several years before OE leads are able to "add guardrails" or make a plan for the longer-term goals of their program.[49] This is not an absolute dichotomy: there are well-established OER publishing services in place today, just as there are commercial publishers with lower-quality outputs and practices.[50] Nevertheless, there is a need within the OE movement to develop more guidance for the production, review, and sharing of publications like textbooks and lab books.

Still, attempting to reproduce systems used by traditional commercial publishers can cause our movements more harm than good. OpenStax, an established OER publisher, has adopted similar production processes to commercial publishers,[51] but conforming to commercial standards can limit our work's flexibility and adaptability. A case study exploring how integrating OER into a library publishing program could affect OER innovation emphasized that there are differences in the capabilities needed to host and share OER in comparison to other types of publications.[52] As the authors assert, "Use of existing publishing platforms may be convenient, but academic libraries considering doing so for publication of OER may find that they are producing work more akin to the work already published rather than implementing an innovative practice or idea."[53] Because of the way that the OE movement emphasizes reuse, the use of paywalled or otherwise less-open platforms and formats can be controversial.

The way that the OA and OE movements approach software and technical infrastructures differs as well. The use of more closed platforms in education is, ironically, more common than in OA. Even in the case study just discussed, the OER developed were typically shared via Canvas Commons, a learning management system that requires an account with the platform to reuse materials.[54] This reflects a long history of platformization happening across the education sphere, which is affecting OE now more than ever.[55] It is worth reflecting on this trend because, as Tel Amiel states,

> *while the field of education has largely embraced the internet and the open web, public educational systems and institutions at all levels have greatly increased*

> *their adoption of proprietary platforms and services from large technology companies. These platforms and services are now indivisible from the day-to-day activities of educational actors.*[56]

The reliance on specific educational platforms creates a unique challenge for OE work. OA is typically a more tool-agnostic field, due to its long-running history and the fact that many platforms have risen and fallen over the past thirty-plus years.

While this section has largely discussed publishing practices, we should not conflate OER with open books, or OER production with traditional publishing, as OER include a diverse range of content types that go beyond the typical "book." In OE, it is common for the community to seek out or share what has been called "little OER,"[57] or materials that are individually produced and created for wide-ranging purposes (e.g., handouts, infographics, and slides). There is an inherent flexibility to these types of materials, and little OER is often where OE's focus on adaptability shines—filling pedagogical needs with ancillary options and being produced in a way that doesn't necessitate a complex workflow. While OA may not be known for demonstrating flexibility in this type of production model, one can draw a similarity between little OER and the sharing of pre-prints and poster presentations in the OA movement. In both cases, the content may not have the explicit goals of a textbook or journal article, but they showcase the breadth of creativity among practitioners.

Production and Priorities: DEI

Diversity, equity, and inclusion (DEI) are difficult to address in open movements, as the application of these concepts varies depending on the context. Arguments that enabling free access to content makes our movements equitable are common in both OA and OE spaces, even when there is no intentional action being taken to make our practices or content equitable.[58] This idea, that "free access creates equality," is not the focus we will take in this section. Instead, we discuss the relevance of DEI as a principle that is embodied by the open content created and shared by our movements.

Because support for OER production is often more hands-on than for OA, there is room for staff to provide additional training on DEI concepts for OER authors.[59] This might be exhibited during the development of an OER or in consultation with OE support staff when an instructor is developing a course utilizing open pedagogy.[60]

Open pedagogy is one of the more prominent areas where DEI is emphasized in the OE movement. Dr. Maultsaid, a professor at Kwantlen Polytechnic University, wrote an essay reflecting on her experience implementing open pedagogy in the classroom, emphasizing its focus on uplifting student voices, building more inclusive classrooms, and making space for representation beyond the readings assigned over the course of a semester.[61] From a theoretical angle, Bali, Cronin, and Jhangiani developed a typology of open educational practices that could be implemented in the classroom to engage students with a social justice frame of reference for assessing practices.[62] However, we can see support for DEI training across the OE movement, through examples like those collected in "Open Voices, Just Choices: OER for Social Justice Faculty Handbook,"[63] guidebooks like "Pulling Together: A Guide for Indigenization of Post-secondary Institutions,"[64] and professional development programs like the Open for Antiracism Program.[65]

While DEI might be structured as part of a training program for OER authors, the application of DEI in research is more likely to come about as a personal decision on the part of the researcher. The majority of OA work is folded into traditional practices around developing, publishing, and sharing scholarship. This leaves little room to support the intentional inclusion of diverse voices, viewpoints, or even theory, unless that is the focus of the research itself or the author chooses to integrate that aspect deliberately.

In addition to authors not being given direct support to incorporate DEI concepts into their research, OA also suffers from a lack of representation due to the proliferation of publishing fees within gold OA journals. This has affected scholars in lower-income countries, making

them less able to participate in this portion of the OA ecosystem.[66] This is not unique to OA, as the OE movement also suffers from a lack of diversity due to the inequitable support available to instructors at smaller institutions who lack the time and funding to invest in exploring OE compared to tenured faculty at well-resourced institutions. As Ma, Buggle, and O'Neill assert, the diversity of voices represented in scholarship is a problem that can be addressed by developing more services that seek out and uplift marginalized voices, like those offered by library publishers or small presses.[67]

Although we have spoken favorably about the OE movement's support for DEI, we acknowledge that integrating DEI concepts into training and development processes for OER is not a panacea for the injustices in our society. Training white authors rooted in Western perspectives to implement DEI in their scholarship does not create tangible representation or diversify participation in open movements. Furthermore, the capacity to support this type of training differs greatly across institutions. While OE training might enable more inclusive practices within the content development process, that support is often limited to institutions with financial means and staffing. We need wider community norms that center diverse voices, not one-off supports provided for those already positioned to accept them. OA and OE could both learn from the open science movement's discussions around broad participation in scholarship and citizen science[68] rather than focusing on specific academic, internal, and institutional communities that we want to appeal to in our work.

Potential Connection Points

We should make concerted efforts to learn from the standards and practices of each other's movements. For example, we can look to library publishers and university presses for excellent standards that inform digital preservation practices and advocate for clearer expectations between authors and publishing services.[69] We can also utilize existing metadata standards used for OA publications and the new ones being developed for OER to create more robust and interoperable standards across our movements.[70] Finally, we can learn from the long history of consortial and collaborative groups that enable access to open platforms to develop more platform-agnostic practices.[71]

The debate about what counts as a "real" publication has a long and troubled history. This is seen in differing views about pre-prints, syllabi, presentations, and other, often unacknowledged scholarship, such as zines, which heavily affect scholars in the arts and humanities.[72] **We need to fight back against the norms that limit our imagination for what our work can be**, effectively keeping power in the hands of commercial entities and those who conform to their expectations.

As with promotion and tenure, we should acknowledge those who are left behind by our movements. This is true not just within our community but also on a global scale. We should strive to connect with organizations and institutions outside of our typical practices and give them space to take a leadership role in our structures or share what we have to help them build their own spaces for their communities. We are already seeing some of this happen through the inclusion of more instructional designers in the OE movement and the focus on building citizen science opportunities into OA programming; however, more could be done to make room for advocates in other open movements and those who are new to the conversation.

Missed Connection: How and Where We Share

While we cannot possibly cover them all, it's important to recognize that there are various ways to make a work open and share it widely outside of the traditional publishing landscape, regardless of your open movement of choice.[73] Like labor and support structures, sharing is influenced by historical norms. When it comes to sharing teaching materials, some sites, like the "for-profit, and largely unvetted, online educational marketplace" Teachers Pay Teachers, leave a negative impression of what "sharing" means.[74] Likewise, in the realm of academic research, concerns about research outcomes being "scooped" and fears about publications

not being accepted if they are already shared elsewhere have led to hesitancy in sharing pre-prints.[75] Concerns about whether or not to make one's work openly available are not unfounded, but they do often overlook the concrete benefits of sharing—namely, making work open helps accelerate the development of new technologies, expand the reach and impact of a work, and build connections with peers across a discipline.[76] Discussions about the benefits of OA have been widely shared, helping to counteract misconceptions about what it means to share one's work.[77] While similar discussions in the OE space are not as well-established, there are examples we can point to that help demystify why sharing one's teaching materials as OER might be a valuable endeavor.[78]

Once we get past the barrier of "why should I care?", we can consider how and where members of the OA and OE movements share their work. OER are often distributed in specialized collections developed for educational resources. For example, OER Commons,[79] a popular tool for finding OER, has filters on its site to search for materials aligned to specific Common Core standards used in primary and secondary US education. Not all OER are shared this way; some authors place their works in Google Drive folders or on personal websites and simply link to them in OER referatories. The referatory is a type of repository that links or refers to materials that are not housed or stored on the search tool itself. Tools like the Open Textbook Library[80] and MERLOT[81] are popular examples in the OE space, alongside the aforementioned OER Commons.

Alternatively, OA content is typically found in subject-specific repositories like CrimRxiv[82] or multi-use platforms like Zenodo.[83] There are some similarities between how content is shared in the OE and OA communities. Scholars often share copies of their works on personal websites and on academic social media sites, just as OER authors might share their work through personal avenues. There are also referatories used in the OA space to support the discoverability of content, like the Directory of Open Access Journals (DOAJ)[84] and the Directory of Open Access Books (DOAB),[85] which collate links to materials from various OA publishers.

We already discussed platformization,[86] but the topic is important to consider here as well. How do the platforms we use to share our materials support the goals and values of our organizations? Are the platforms we use limiting the openness of the content we wish to share or restricting our ability to engage in the actions permitted by its license? For example, content may only be reachable through an online portal, with no ability to download, edit, or engage with the content in any meaningful way. While platformization itself may not be a bad thing—we *can* choose to rely on platforms that are relevant to our communities and community-run—these questions and others should be at the forefront of our discussions about the platforms we select and continue to support.

When the platforms we use prove to be bad actors, we can choose to sever ties with them and implement alternative infrastructures that better reflect our values. In "The Platformization of Open," Andrews describes principles for "platform cooperativism," first developed by the International Cooperative Alliance, that open movements can pursue to encourage a better environment for sharing our work on platforms we can trust. The principles include "Voluntary and open membership, Democratic member control, Member economic participation, Autonomy and independence, Education, training, and information, Cooperation among cooperatives, and Concern for community."[87] When it comes to the OA and OE movements, there is much we can learn not just from one another but also from the wider movements for open participation, such as the cooperative theory that Andrews cites. Each movement is based in a unique group of people and content: stewarding and supporting those communities is not a one-off decision, and finding paths forward despite entrenched systems and platforms will take time.

Institutional Repositories

One place where OER and OA content frequently coexist is in digital or institutional repositories (IRs). IRs are "digital collections capturing and preserving the intellectual output of a single or multi-university community."[88] These places house pre-prints, versions of record,

and otherwise shared materials (e.g., CVs, handouts, and presentation slides) that meet the IR's specific guidelines for inclusion—which vary! Unlike externally managed repositories and referatories, IRs are more likely to be supported locally. Although most often seen at an individual college or university, IRs can be managed at the institutional,[89] consortial,[90] regional,[91] or even national level.[92] One main draw of IRs is their ability to increase access to research while also documenting and preserving works that might not otherwise be shared, such as cultural heritage artifacts or student-created works.

Placing OA content in IRs is a natural, fit-to-purpose solution for broadening access to research. Since the earliest days of the OA movement, the use of repositories like IRs and pre-print servers has been underscored as one path for making otherwise closed-access content available for readers,[93] with this practice being mandated at some institutions and at the national level in some countries.[94] The use of IRs stands apart from individual sharing methods because they are managed by an institution or group. Unlike referatories, IRs house and provide access to files directly, and unlike personal websites, IRs can function as digital archives with additional copies of content frequently held on back-up servers.

Other services often integrated in IRs include the use of stable URLs for long-term linking and digital object identifiers (DOIs), flexible identifiers that provide stable links that are associated with the specific piece of content being identified. Notably, DOIs "can be assigned to any digital object rather than a specific type of object, such as books (ISBNs), journals (ISSN), and articles (PMCIDs)."[95] The use of DOIs makes it easier for IRs to track and share analytics for their users, highlighting the impact, reach, and reuse of the content they have shared. Tracking analytics in this way is something that scholars can utilize to illustrate the impact of their work during promotion and tenure processes. As the use of alternative metrics for tracking the impact of scholarship continues to grow,[96] integrating OER into IR systems could be a path for those hoping to increase the discoverability of OER outside of OER referatories as well.

IRs also stand apart from other sharing options for their ability to ingest a wide range of material and file types, making space for nontraditional scholarship. In this way, IRs are particularly useful for sharing research from the humanities and social science disciplines. This could make an IR a natural place to share OER; however, the use of IRs for sharing OER is not always perceived positively. While IRs have been used to house OER successfully across several countries,[97] some practitioners have argued that OER should be hosted on platforms that are tailored for OER specifically, that research and teaching materials ought to be shared through different means than research, or that the need for quality control of OER could be a concern when bringing the teaching materials into their repositories.[98] However, there are good reasons why IRs are used to support the hosting and even publication of OER. IRs are well-suited to sharing content that is not made for traditional publishing, as in the case of "little OER,"[99] and IRs are more flexible at meeting diverse publishing needs than other popular OER publishing platforms, which might be tailored to textbooks specifically. For situations where funding for a publishing-specific platform isn't available, IRs can be used for publishing as well.[100]

There are some reasons why the hosting of static files may not mesh well with all OER projects. A unique aspect of OER is the need to remix and share back new versions of previously shared educational materials—making handling versions and updates a publishing challenge. Not every small edit or update warrants being treated as a new edition or version, a fact that is complicated by the lack of consistency in how the OE community represents new versions of OER. The lack of guidelines for versioning OER means that adaptations may not be shared back into the open, especially if content creators of an adaptation are worried about doing something "wrong" because copyright is involved. This phenomenon has been studied by Beaven, who found that while more than 40 percent of 151 materials used for creating a curriculum came from an OER repository, the majority of the educators did not reshare their adaptations of that content.[101] While they appreciated their colleagues who did so, they spoke of a lack of peer feedback and even questioned whether what they created would be useful to anyone else. This adoption, remixing, and use of OER while making a choice not to reshare has

been dubbed "dark reuse."[102] This is not an issue exclusive to or inherent in the use of IRs or sharing via other digital repositories. Rather, dark reuse has come about due to failures of our communities to build and adhere to standards and to educate each other across movements on their importance.

Dark reuse is not only found within the OE movement; however, the focus on reuse and adaptation for OER has made it necessary to discuss these practices. In fact, when the fifth "R" for open content, "the right to retain a copy," was first proposed, comments on the blog post discussed concerns about those who might reuse and adapt open content without giving back to the community:

> *...if copying is not made part of the system then it happens off the radar, and that hinders our ability even more. The beauty of a system like GitHub is it just assumes forking that happens should happen and then tries to deal with it technologically. The technical problem of a merge is so much more addressable than the culture problem of treating forking as a betrayal or bad practice. You can't have aggregate benefits without having a community first, and communities need to accept forks as normal.*[103]

Building a community that values keeping hold of our history, tracking changes, and embracing adaptation is a strong argument, and one that both the OA and OE movements could learn from. If we could accomplish both of these goals—developing community norms for the preservation of more "significant" versions of our work *and* being open to the creation of many remixed versions living in the open—we could do more to improve the discoverability of our materials overall. Perhaps our future will include a resharing ecosystem for scholarship and educational resources alike, not through the use of one platform for all our work, but through leveraging flexible platforms that fit individual groups' needs and budgets best, while connecting back to the works and spaces that are used by our peers.

Potential Connection Points

After years of dealing with accusations of OA being predatory, low-quality, or unnecessary, advocates within the OA movement have developed useful resources to support authors in understanding why they should share their work. Similarly, advocates within the OE movement have built a strong base of discussion around the affordability of OER and the benefits of access to information for a wide audience, with newer talking points addressing the value of equitable participation beyond access. **We can learn from one another's successes to inform the challenges we currently face.**

OA and OE both suffer from a reliance on platforms managed and supplied by third parties, and when we develop platforms custom-suited to our needs, they may be out-scoped or bought by commercial entities. In 2017, academic libraries had a rude awakening when a popular open source repository platform, bepress,[104] was acquired by the publishing giant, Elsevier; however, this change led to a useful discussion about our community's values and to the migration of several institutions away from the bepress platform to alternate, still open-source options.[105] This history provides a useful foundation for discussions in the OA and OE movements: **when platforms no longer serve our needs, we can pool our resources and facilitate paths forward through community-managed open source infrastructure.** Holding discussions within our movements about what we value is vital as our work continues to evolve in response to the presence of new technologies.

Conclusion

Throughout this chapter, the authors have covered a few areas where the OA and OE movements share similarities and discussed how we could do more together. Even though our movements have different areas of focus and our research has grown in different directions,

we still fill a similar niche and have the chance to pursue common goals. From funding norms and support systems to publishing and hosting open content, there are many places where members of the OA and OE movements could build upon one another's work.

These changes will not occur overnight, nor will they be accomplished by a single group. Open movements are not monolithic communities. Individuals and organizations that support OA or OE may not be interested in pursuing some of these projects, and others may not have the ability to pursue them. Recognizing these opportunities is the first step. From there, we can work within our movements to broaden our horizons, build on what has come before, and develop truly reflective movements with clear goals. Aside from funding, publishing, and sharing our work, we can also learn from the community-building happening across our movements, the national and international networks that have been established to support our goals, and our ties to the many other open movements that help us chart new paths forward.

As Tarkowski et al. quote from interviews they conducted with open movement leaders, "You need to bring a whole suite of people along with you on the journey, and that's a challenge. That's hard because you got to go out and you got to talk to people—also to those with whom you have not talked before."[106] Both within and outside of our movements, we tend to ignore the related histories and contributions of our peers outside of the open sphere.[107] If reaching outside of our communities is the next step for open movements as a whole, then perhaps we should start by reaching *across* to the open movements that are already walking alongside us.

Notes

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CHAPTER 28

Bridging the Divide: Examining the Intersections Between the Open and Digital Humanities Movements

Katharine Teykl

Introduction

Conceptually, digital humanities and open movements are natural partners, as both have similarly aligned values and goals. Facilitating opportunities for collaboration among researchers and seeking to make broader public impacts by improving the accessibility of research and information are broad goals that overlap between both movements. Despite efforts to bridge the gap and address shared goals and challenges, the relationship between open movements and digital humanities has been complicated and disjointed. Upon closer examination of the history between digital humanities and open movements, several questions arise. How has digital humanities actively supported the open movements, and how can these two groups come together to support their shared goals? What obstacles have hindered initiatives and explicit collaborations between digital humanities and Open? And finally, how have digital humanities practitioners, librarians, and others begun to rebuild from the pieces that have been left behind?

This chapter explores the disconnect between digital humanities and the open movements and how building connections between the two groups may further advance the desired end goal of both groups. The chapter explores how the common goals of both movements have intersected over time, identifies why explicit collaborations have not taken place as anticipated, identifies new opportunities for more direct collaboration between these movements through the formation of the open humanities movement, and explores how the social component of digital humanities can be leveraged to promote open initiatives specific to this niche space.

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Getting Off the Ground: Early Collaborations and the Social Component of Digital Humanities and Open

Digital humanities can be a difficult field to define due to its continuous evolution and incorporation of new ideas, technologies, and methodologies. For this writing, digital humanities is defined as the application of technology or computational approaches in the humanities. Broader methodologies that fall under the digital humanities umbrella include, but are not limited to, textual analysis, digital exhibitions, spatial analysis, and digital annotation and transcription. Digital humanities overlaps with other subdisciplines within the broader humanities movement, as one of the aims has been to improve the access of the humanities to an audience outside of the academy. As noted in Susan Smulyan's "Why Public Humanities?" digital humanities has a particularly strong overlap with public humanities and calls into question the scholar's role in addressing issues related to the accessibility of information in the digital age.[1]

Similarly, as defined by the Open Data Handbook, the Open movement "…seeks to work towards solutions of many of the world's most pressing problems in a spirit of transparency, collaboration, re-use and free access."[2] The Open Definition from the Open Knowledge Foundation is also succinct in its definition of open, stating that open content "can be freely used, modified, and shared by anyone for any purpose."[3] The energetic and collaborative nature of open movements and digital humanities makes them natural partners, potentially with a shared goal of facilitating openness within humanities research more broadly. Even within the earliest days of digital humanities, collaboration and community-driven efforts to create open available content were beginning to appear.

Digital humanities refined its own identity as a subdiscipline in the late 2000s and early 2010s, but glimmers of what appeared to be the incorporation of openness within digital humanities research began even with the earliest initiatives within digital humanities (or humanities computing, as it was known back then). In "The History of Humanities Computing" in *A Companion to Digital Humanities*, Susan Hockey notes that the advent of the internet in the early 1990s greatly expanded the reach of humanities computing and resulted in highly collaborative projects that were not previously possible. In describing the Orlando Project, a collaborative project in which researchers encoded text documents related to women in the British Isles,[4] Hockey explains the impact of the internet on collaboration and increased ability to share work related to humanities computing: "The Internet also made it possible to carry out collaborative projects in a way that was never possible before. The simple ability for people in different places to contribute to the same document collections was a great advance on earlier methods of working."[5]

Community-driven efforts within digital humanities have long been a staple of digital humanities. A component of Matthew Kirschenbaum's definition in the seminal publication "What is Digital Humanities, and What's It Doing in English Departments?" is well-suited: "… digital humanities is also a social undertaking. It harbors networks of people who have been working together, sharing research, arguing, competing, and collaborating for many years."[6] Kirschenbaum further notes an early success resulting from a community-driven desire to make texts available electronically: the Text Encoding Initiative (TEI).[7] Prior to the development of TEI in the late 1980s, there was no universal standard for encoding texts electronically; however, community-driven efforts through the TEI-Consortium allowed for the creation of a standard for text encoding, which continues to be utilized within humanities research today. Although TEI can be considered a major victory in collaboration and information sharing in the humanities, Hockey notes an early disconnect between digital humanities and library and information science more broadly as TEI began to be integrated into digital library platforms and digital publications, where the community-driven efforts of TEI presented additional challenges, noting that the flexible, changeable standards of TEI conflicted with the rigid standards of library and information science that aid in the findability of research.[8]

Another driving force behind openness in digital humanities is the emphasis on open sharing of information through digital means. Digital humanities initiatives, particularly after the boom in popularity in the 2010s, encouraged and incentivized the use of social media to share goings-on, exciting new projects, and to keep the community informed about emerging developments and new tech that may be useful. Larger digital humanities organizations, such as the Alliance for Digital Humanities Organizations (ADHO)[9] and the Association for Computers and the Humanities (ACH),[10] maintain active email listservs and social media pages to help others keep abreast of scholarly discussions and ongoing projects within the digital humanities space. Even in 2010, Kirschebaum noted the popularity of social media outlets Facebook and Twitter (now X) in facilitating discussions of a definition of digital humanities.[11] This has allowed digital humanities practitioners to readily discuss developments through informal channels, driving the fast growth of the field and the energetic pace of development that continues today. In a sense, even if projects are not "open," the ability to disseminate information about the projects increases general awareness of what is happening in the scholarly space, which may contribute to future collaboration opportunities. However, despite the historical use and encouragement of this open sharing of information within digital humanities spaces, channels of communication that exist through larger professional organizations have largely gone unmaintained and thus have remained stagnant and underutilized. Although special interest groups within ACH and ADHO remain active, discussion groups and forums facilitated by smaller organizations have slowly died out. As communities have chosen to leave social media or migrate to alternative platforms, maintaining connections to discussions within larger communities can inhibit the ability to effectively share information across a previously viable and popular channel.

Aligning with affordable and accessible technology and avenues to gain expertise, the use and promotion of open software and freely available tools has been keenly supported within the digital humanities space. Some examples of open-source tools that are popular in the digital humanities space include Gephi (a tool for network analysis), Jupyter Notebooks, Voyant (web-based text analysis), and Mukurtu CMS (digital exhibitions). Although proprietary and so-called "freemium" tools are available, there has been an explicit call to utilize open-source tools when available within research and to clearly encourage and promote the use of open-source tools whenever possible. One example of this direct call to action comes from The Programming Historian, an open access digital humanities journal that publishes and maintains an open repository of lessons on technical topics pertinent to digital humanities and requests potential contributors to utilize open-source software and data when possible, citing the desire to minimize costs and their commitment to "open source values."[12]

Additionally, groups that develop tools related to digital humanities research often share educational materials in open formats or use freely available formats, such as Jupyter Notebooks. HathiTrust Research Center (HTRC), which conducts research and educational outreach related to text mining from the HathiTrust Digital Library, utilizes Google Colab for educational materials related to text mining, which not only provides users with an openly available platform in which to enter their code in Python, but also eliminates a technical barrier by not requiring users to install and maintain a Python environment such as Jupyter Notebooks, Visual Studio Code, or similar free platforms.[13] Constellate, a "freemium" text data mining platform previously offered by Ithaka, maintains a GitHub repository with downloadable Jupyter Notebooks related to text data mining from various text sources, including JSTOR and Portico.[14] Although this presents additional challenges in maintaining and updating the code files and GitHub repositories, these groups substantially increase access to educational resources related to text mining, a popular methodology within the digital humanities space.

Similarly, a significant challenge for digital humanities practitioners is acquiring openly available data to use for analysis due to copyright constraints, file size, and the decreasing ability to acquire data of an adequate quantity and quality from proprietary sources. This is particularly an issue for practitioners of text mining and analysis, as the process for acquiring data is often marred with challenges (detailed later in this section). Efforts to improve access to

open data sources in the humanities have improved dramatically through community-driven movements. In particular, there has been significant attention from larger digital humanities organizations, such as the Association for Computers and the Humanities (ACH), to develop openly available multilingual corpora, which aim to address the dual challenges of underrepresentation of non-English-language corpora in the text analysis space as well as difficulties accessing these corpora.[15]

A hallmark of the digital humanities movement is the embracing of freely available digital publishing outside of traditional publishing media, such as through websites and digital exhibitions. The popularity of using alternative publication formats within digital humanities, such as project-specific websites, has allowed researchers to display the results, methods, and visualizations outside of the traditional publishing landscape and provides opportunities for improved engagement with audiences outside of academe. Digital annotated editions, for example, are longstanding within digital humanities and align with the Open movement in improving access to information in openly available formats. One example that exemplifies community-driven efforts in this area is the Music Encoding Initiative (MEI), which supports multiple projects related to encoding musical notation in machine-readable formats, creating a defined space for musicologists engaging in work related to digital annotated editions in music.[16] Using these formats also allows more freedom for storytelling and the ability for authors to share that may be constrained by traditional publishing means.

Examining the Disconnect between Digital Humanities and the Open Movement

Despite existing initiatives and collaborations between the Open movement and digital humanities, the lack of cohesive efforts between the two groups continues to hinder progress toward true openness within digital humanities. This section explores some potential explanations for this disconnect and possible ways to bring these two movements together.

Technical barriers remain a significant hurdle in achieving true "openness" within digital humanities. Although community-driven efforts, such as those through Programming Historian, attempt to alleviate some of this burden, participating in existing avenues of openness within digital humanities requires practitioners to already possess knowledge of a given technical skill or platform. For example, to participate in a text-encoding project, users are often required to learn encoding standards, such as TEI. In conjunction with digital humanities scholarship, researchers often elect to make their code available through a code repository, which allows other researchers to analyze and potentially adapt their work. For example, dh-tech maintains the GitHub repository "Awesome Digital Humanities," which promotes free and open-source tools related to digital humanities and tracks the development of new tools.[17] While this promotes openness in one sense, it presents additional barriers depending on the level at which a user needs to engage with code or a given tool. Popular code repository platforms, including GitLab and GitHub, can have a bit of a learning curve for new users, due to their complex hierarchical structures and fairly strict community recommendations for maintaining code in them.

Those wishing to engage with the full potential of the platform will also need to learn the basics of version control and file revision using Git, which ultimately requires users to feel comfortable using command-line utilities. It often ends up becoming a slippery slope of needing to learn more and more, contributing to ongoing systemic issues surrounding access to digital humanities knowledge.

Even with early examples of community-driven collaborations, their widespread use has resulted in additional knowledge requirements for digital humanities practitioners attempting to publish their work openly. For example, *Digital Humanities Quarterly* has previously required potential contributors to submit their manuscripts entirely encoded in TEI. Although TEI has been perennial within the digital humanities landscape, it has fallen "out of vogue" within

digital humanities spaces, and finding updated resources to learn TEI can be difficult for newcomers. In addition, as time has progressed and newer methods take precedence, learning an XML standard and spending additional time and effort for the purposes of publishing in an open access journal is an additional burden that many researchers new to digital humanities can be reluctant to take on. At the time of this writing, *Digital Humanities Quarterly* now permits potential contributors to submit manuscripts without having them encoded in TEI, requesting them to be submitted as a .docx file or in a standard XML encoding schema.[18]

There is also a pattern of "active inaction." Digital humanities practitioners have consistently identified the need to support various facets of the larger Open movement, specifically open access publishing and open data, but moving to implement specific actions and initiatives has been slow or incomplete. As Marcel Knöchelmann notes in "Open Science in the Humanities, or: Open Humanities?" there has been significant discourse related to implementing practices from open science within digital humanities work, such as preprints, open peer review, and more flexible copyright licensing for humanities research.[19] However, Knöchelmann notes that there is a disconnect with implementing these practices within the broader context of digital humanities: "Though questions of publishing data, their infrastructures, and degrees of openness are partly integrated into discourses of the digital humanities, it is not connected paradigmatically to a broader open humanities."[20] The slowness to adopt open initiatives within the broader digital humanities movement is not wholly unprecedented; in fact, it follows larger trends in the broader work of Open. In tandem with this "active inaction," Longley and Arthur state that humanities disciplines have historically been slower to adapt open research initiatives, having long focused on traditional publishing means (e.g., monographs).[21] Consequently, despite active efforts to support openness within digital humanities, efforts to get things off the ground have been hindered.

Increasing interest from vendors in supporting digital humanities projects has also contributed to issues surrounding data access, further hindering the ability to open data access for digital humanities projects. For example, vendor contracts now include explicit language prohibiting downloading copious amounts of data from their servers, which may be prohibitive for projects that require larger datasets. Another popular option provided by vendors is the "walled garden," providing access to an API to download substantial amounts of data after writing up a lengthy project proposal and submitting a substantial fee determined by the amount and specific kind of content a researcher requests to access. In the case study "Negotiating a Text Mining License for Faculty Researchers," attempting to negotiate the "right to mine" copyrighted data outside of a vendor contract is a lengthy process that creates bottlenecks in the research, and that the monetary costs and substantial wait times for obtaining the data can be a deterrent for researchers and institutions to support text mining projects altogether.[22] Vendor-provided datasets can be costly and difficult to acquire for individual researchers, and in some cases, a vendor-provided tool is mandatory to access data for use with projects. ProQuest TDM Studio, for example, allows users to access millions of documents for text analysis but requires users to conduct analyses on their proprietary instance of Jupyter Notebooks (an open-source application) and their limited collection of Python and R scripts.[23] These barriers put in place by vendors hinder the ability of digital humanities practitioners to obtain data for their research and make it more challenging to promote the use of open data within digital humanities research.

Although the use of alternative publishing means, such as project websites and digital exhibitions, continues to be popular within digital humanities, digital humanities practitioners who aspire to publish in discipline-specific open access journals can run into a few hurdles in the process. Research in digital humanities is limited to a few specialized journals, which naturally inhibits the reach of a researcher's work. *Digital Humanities Quarterly* and *Digital Scholarship in the Humanities* trend toward exclusively showcasing projects or new technologies, while explicitly prohibiting submissions of "work-in-progress reports, descriptions of projects, tools, techniques or methodologies."[24] This can limit avenues for digital humanities practitioners to publish, pushing them to explore alternative publication formats outside of journals, particularly to share ongoing work.

Publication costs can be a deterrent for open access publishing in digital humanities as well. As with many open access journals, *Digital Scholarship in the Humanities*, managed by Oxford University Press, requires an article processing charge (APC) in order to be published. As of 2024, the APC was £2,623 (approximately $3,317), with discounts permitted for institutions with a qualifying agreement and for early-career researchers.[25] Although these discounts exist, this high APC can be cost-prohibitive to researchers with limited funding or from smaller institutions, contributing to systemic issues of representation within the larger digital humanities space.

Finally, some of the disconnect can be explained by the insular and exclusive nature of digital humanities, remnants of which can be seen even in well-intentioned efforts to promote openness within the field. Access to training and education around digital humanities and related technologies remains a significant challenge for digital humanities practitioners. Although efforts to improve educational opportunities for those wishing to engage in digital humanities have increased and have become increasingly available to those wishing to develop the necessary technical knowledge, there is a perpetual feeling of "needing to catch up" as new tools, methods, or programming languages become widely used. As new technology progresses, other tools, programs, and languages break or become obsolete, presenting additional challenges for researchers in terms of obsolescence and sustainability.

Often, getting involved with digital humanities starts with attempting to learn skills to accomplish a designated project or task, which is not always useful for users who wish to learn a specific tool but do not have an active project suitable for the tool. In the chapter "Opening Up Digital Humanities Education," Lisa Spiro also notes the inequities of this existing model of training within the digital humanities, writing, "In the past, many have entered the digital humanities by serving as informal apprentices on projects. But this approach is not scalable or equitable, since it essentially requires one to be in the right place at the right time."[26] Allowing for more equitable access to digital humanities education would, in turn, alleviate some of the barriers discussed within this section. Spiro offers some suggestions for improving equitable access to digital humanities education in the context of curriculum development, such as making materials "remixable" for different courses and adopting open licenses so that course material can be reused and adapted.[27] In addition, including open-source software and supplementing course materials with tutorials related to digital humanities could join together ideas and practices from the open education and open-source movements.

Though these barriers have prevented digital humanities from truly being open, the emerging open humanities movement in the digital humanities and broader humanities movements may illuminate a path forward and offer a more explicit opportunity for collaboration between the two groups.

Looking Forward: Open Digital Humanities

With direct action being taken among the digital humanities and Open movements to include one another, the next step in furthering the alliance between the two movements is to combine efforts in a cohesive and effective manner. A movement within the broader humanities is open humanities, which aims to incorporate the principles and goals of the Open movement within the context of humanities research and publishing. Following this precedent, improving the identity of "openness" within digital humanities and striving to incorporate ideas from other open movements within the practice of digital humanities is essential in moving toward "open digital humanities."

Open Access Publishing

Expanding existing efforts to promote open access publishing would be a simple first step. As discussed previously, there are many open access journals and publications available within the digital humanities space, and there remains a strong culture of utilizing alternative

publishing formats, such as websites and digital exhibitions. In addition to these existing journals, encouraging open humanities journals to publish digital humanities work may be another option, and some publications have explicitly included "open" in their names to denote their commitment to open access and open scholarship in the humanities. The Open Humanities Press, for example, supports "open access scholarship in journals, books and exploring new forms of scholarly communication."[28] Because many open journals and presses in the broader humanities movements are independently operated and community-driven, these journals may be of particular interest to contributors who may want to publish in a more traditional format but do not necessarily want to go through a large publisher.

Open-Source Software and Open Data

In addition, more decisive action on collaborating with established movements within the larger open movement, such as open data and open-source software, can lift some of the knowledge and technical barriers discussed previously. While there has been a significant amount of attention on these within digital humanities spaces, it can be difficult to find open datasets or well-supported communities for open-source software and applications. In the area of open data, the *Journal of Humanities Data* has published about datasets in the humanities with "high reuse potential," as well as connecting users with Harvard Dataverse and Dryad, which helps start to consolidate these datasets to be used for other work.[29] Sustainable preservation of humanities research data is also of paramount importance to ensure that the results of digital humanities research can remain for years to come. One such initiative in this area is DARIAH (Digital Research Infrastructure for the Arts and Humanities), which aims to promote interoperable, long-term data storage solutions for humanities.[30] Similarly, more active initiatives similar to these will help digital humanities practitioners access data and software without having to endure the onerous process of vendor contract negotiations.

Communication

Finally, leveraging existing channels of communication and community with the digital humanities spaces, such as social media and communication from professional organizations in digital humanities and is another avenue for openness to be supported within digital humanities. Keeping these channels operational and well-maintained is the most significant challenge, as existing channels of communication can be disjunct and have largely gone unmaintained or, in some cases, have disappeared entirely. Using these channels to highlight open initiatives within digital humanities, including projects that use open data and open-source software, can help fuel additional research in areas that would have been limited previously by the dearth of open data in a given area, such as multilingual corpora.

Conclusion

Similar to other movements within Open, such as open science, incorporating and improving openness within the practice of digital humanities will continue to remain reliant on its community to promote open access publishing and find ways to support initiatives in open data and open-source software specific to the humanities. Encouraging open access publishing and fostering initiatives in open data and open-source software tailored to humanities research and opening and expanding existing avenues of communication and alternative publishing can help support the aligned goals of the digital humanities and open movements. Looking forward, incorporating elements of the existing open humanities movement with the digital humanities space can allow digital humanities to become more explicitly involved within the larger Open movement and address challenges for the future, including equitable access and sustainable preservation of humanities-related data, and can continue the open sharing of information related to digital humanities projects.

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Chapter Authors

A

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B

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Michael Boock, Professor Emeritus at Oregon State University, brings over 30 years of experience in developing digital library services, with expertise in digital repositories, collections, preservation, data management, open access, and copyright in research and teaching. For the past two decades, as Scholarly Communication Librarian and Head of the Center for Digital Scholarship at Oregon State University, he has led efforts to expand and innovate open access initiatives that enhance the visibility, stewardship, and preservation of scholarly work. He has been recognized with a Fulbright Scholarship in Bulgaria, has published numerous research articles, and has served as chair of the MetaArchive Cooperative.

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C

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D

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E

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F

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G

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H

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Marla Hertz, PhD, is an associate professor and research data management librarian in the Office of Scholarly Communication at the University of Alabama at Birmingham. She provides support and instruction on responsible conduct of research topics, including data management, journal selection, open access, and avoidance of predatory publishers. Her research focuses on how to tailor scholarly support services to different user bases. Her practice is informed by her former role as an academic researcher in the microbiology field for fifteen years prior to entering librarianship.

Amber Hoye, formerly Boise State University, is the director of programs at The Rebus Foundation, a Montreal-based Canadian Registered Charity and Non-Profit. With a master's in educational technology and a graduate certificate in online teaching from Boise State University, she combines instructional design expertise with project management to support Rebus's vision that the power of learning and education creates better spaces for us all. Amber has successfully led the development of numerous openly licensed textbooks and supplemental materials. Her research focuses on K-12 instructor engagement with open education and developing best practices for incorporating all types of content into OER.

Scott Hudnall is the coordinator of reference services and emerging technologies at Indiana University Northwest. His primary duties include leading research support through student worker training and librarian consultations, and implementing tech solutions for library needs. Further tasks involve library instruction for classes, collection development, and gathering library statistics for assessments. As an associate librarian on a regional campus, his activities dip into other areas such as acquisitions, interlibrary loan, and circulation as needed. Scott's scholarship to date has explored potential uses of animated GIFs for library instruction. Outside of libraries, he serves as past president and board member of the Lake County Historical Society in northwest Indiana, which has operated continuously since 1875.

Cynthia Hughes is a cataloger at the University of Wyoming. Her research focus is on rural libraries, and she is a member of ARSL. She first became interested in rural libraries when her first library job out of graduate school was helping establish a new public library.

Sherena Huntsman is an associate professor at Boise State University. Her ongoing research focuses on the investigation of marginalizing power structures in digital institutional spaces. This research and her teaching philosophy interconnect to inform her current investigation into normalizing the accessibility of digital course materials as part of pedagogical best practices in an effort to transform workspaces. Her research and pedagogical pursuits exist at the intersection of feminist theory, disability rhetoric, social justice, accessibility, and participatory design.

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J

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Brianna Johns has years of experience working in open-source and community science spaces, previously serving as the community coordinator for the Gathering for Open Science Hardware (GOSH). In this role, Bri supported a global network of artists, researchers, activists, and hardware developers with a clear mission: to make open science hardware ubiquitous. With a background in both zoology and international studies, Bri is interested in examining the intersections of science, technology, and society within the context of conservation. Bri is currently a Fulbright student researcher and SEEKCommons fellow studying the Mothbox, a low-cost and open-source insect monitor. In doing so, they're taking a deeper look at the role of emerging open-source technologies in biodiversity monitoring efforts.

K

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M

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P

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R

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S

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Mitchell Scott is the coordinator of collections strategies at the University of Kentucky. His scholarship and professional practice explore the intersections of library collections, affordability, access, and academic publishing—particularly as they relate to library-led textbook affordability programs, open educational resources (OER), and open access publishing models. He has contributed to numerous peer-reviewed journals and edited volumes, and is a frequent presenter at national conferences and webinars on topics ranging from library-led affordability initiatives to collection strategy and public library partnerships.

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T

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W

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